

Local quantitative refinements of Gowers’s proof of Szemerédi’s theorem

Density transfer and phase-flat partitions; energies, restriction and dependent random choice;
cube and progression counts; floor patterns and the Proposition 17.7 phase extraction

Five research manuscripts prepared for Vladimir Reshetnikov,
merged into one ProveIt research report (sources 01–05)

All sources: 6 October 2026 merged: 6 October 2026

Abstract

Gowers’s 2001 proof of Szemerédi’s theorem passes through many local estimates: phase-flat partitions and density increments (Section 2 and Lemma 5.15 of the paper), the energy and restriction steps of the inverse theorem (Proposition 6.1, Proposition 7.3, Lemma 7.4, Lemmas 9.2–9.4, 12.4, 14.7–14.8), cube and progression counts (Corollaries 3.3–3.6, Lemma 3.10) and floor-pattern counts (Corollaries 17.5–17.6, Proposition 17.7). This report merges five independent research manuscripts, all dated 6 October 2026 and prepared for Vladimir Reshetnikov, that sharpen these estimates against the corrected statements of ProveIt’s transcription and Lean development. The material is arranged in five thematic Parts. *Part I*: a sharp one-sided density–mass bound, proved in four sources; a sharp one-cell phase envelope; the lower constant $1/3$ (optimal) in Lemma 2.3; Corollary 2.4 with 36π and $3\alpha/4$; three density-sensitive Fourier increments, of which the increment $3\delta(1-\delta)\kappa/(4-(4-3\delta)\kappa)$, $\kappa = \eta/(2\delta(1-\delta))$, of one source is at least $3/2$ times those of the other two at their stated parameters and lengths of the same order (proved in the write); simultaneous phase-flat partitions with the optimal exponent $1/(m+1)$. *Part II*: a mixed, weighted extension of Proposition 6.1 on every finite abelian group with coefficient one; a cubic dependent-random-choice certificate giving $7\delta^3 n/(3\sqrt{6})$ in Lemma 7.4; an elementary Balog–Szemerédi–Gowers bound $(c/8, 2^{19}c^{-9})$; exact moment interpolation and a polynomial restriction theorem over prime fields. *Part III*: the exact fourth-order term of the centered cube count on every finite abelian group; the sharp gap $\|f\|_{U^3}^8 \geq 2\|f\|_{U^2}^8$ for real centered functions on odd-order groups, hence the sharp leading constant $6\sqrt{2}$ and, in the write, the answer $c_3^{\text{odd}} = 2^{-1/2}$ to one source’s question; the fractional correction exponent $16/3$; density-weighted progression counting. *Part IV*: the floor-pattern count C_k (two sources) with $|\mathcal{F}(\mathbf{r})| = (\prod_i 2r_i)C_k$, $2^{k(k-1)/2} \leq C_k$, $\log_2 C_k \sim k^2$ (using the classical threshold-function count), certified $C_1, \dots, C_4 = 1, 2, 10, 154$, and a local phase-extraction coefficient $2^{-k^2 - k \log_2 k - O(k)}$ at Proposition 17.7. *Part V*: the crosswalk to the formal ledger, the sources’ formalization plans, and their 46 research questions, deduplicated with credits. No source claims a new global bound for $r_k(N)$. The report is unrefereed. Of its statements only one, a lifting lemma from modular to integer progressions, is formalized, and it was formalized before the sources arrived; the report gains no formal status.

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F.1 Guide to this report

This report merges five research manuscripts, numbered 01–05 in arrival order and then in ASCII order of their archive names. All five refine local estimates of W. T. Gowers’s proof of Szemerédi’s theorem [1], and all five compare their statements with ProveIt’s edited transcription of the paper (`Combinatorics/Ramsey/Papers/sz-thm-gowers-proof`) and its Lean development (`Combinatorics/Ramsey/Lean/GowersSzemerédi`). The report lives beside both, in `Combinatorics/Ramsey/Research/GowersSzemerédi/local-quantitative-refinements`, by Vladimir’s direction; that placement confers no formal status (Section F.5).

No.	Delivered title	Pin	Parts
01	Sharp Local Refinements of Gowers’s Szemerédi Argument: density transfer, carry-pattern entropy, and formalization interfaces	3a25020a6	I, IV, V
02	Energy-Sensitive Inverse Steps and Sharp Density Extraction	66c02f8a1*	I, II, V
03	Sharper local estimates in Gowers’s proof of Szemerédi’s theorem: density increments, exact cube expansions, and dependent random choice	3a25020a6	I, II, III, V
04	Sharper Local Bounds in Gowers’s Proof of Szemerédi’s Theorem: polynomial homomorphism restriction, sharp cube estimates, and floor-pattern complexity (the base)	327773149	II, III, IV, V
05	Sharp Finite Phase Partitions and Density-Sensitive Increments	327773149	I, V

* source 02 records no pin; this is the branch reference it observed (its Appendix A).

Arrangement. Every manuscript spans two to four themes, so the report is arranged by theme, not by manuscript. Part I treats density transfer and phase-flat partitions; Part II the inverse step (energies, moments, restriction, dependent random choice); Part III cube and progression counts; Part IV floor patterns and the phase extraction of Proposition 17.7; Part V the formal interface, the crosswalk to the ledger, and the research questions. Inside a Part the sources follow in numerical order. Each source’s sections are printed whole, as delivered, with their own subsections; a section heading carries its source tag (for example “[01]”) and a line naming the delivered section. Each manuscript’s abstract and opening section are printed in the front matter (Sections F.6–F.10), where its title block is quoted.

Numbering. Sections restart in each Part and carry its numeral: Section I.9 is the ninth section of Part I, and the front matter’s sections are F.1–F.10. Statements and equations are numbered within sections, so source 02’s Theorem 7.1 is Theorem I.9.1 here. A statement or equation $k.j$ of a source keeps its index j within its section, except that sources 02 and 03 numbered their equations consecutively through the whole manuscript, whereas here they are numbered within sections. Source 04’s tags (M1)–(M7), (R1)–(R32), (A1)–(A14), (Q1), (Q2) and (*) are kept. *Bare numbers such as “Lemma 5.15” or “Corollary 2.4” always refer to Gowers’s paper* (in the corrected transcription when the text says so), as in all five sources; this report’s own numbers always carry a Part numeral or an F. Plain-text references of a source to its own sections were made live in one place (source 02, Section II.4); everywhere else the sources refer to their own numbers through cross-references, which now print this report’s numbers.

Concordance. Delivered sections and the sections here:

Source	Delivered section → section here
01	1→F.6, 2→I.1, 3→I.2, 4→I.3, 5→I.4, 6→I.5, 7→I.6, 8→I.7, 9→IV.1, 10→IV.2, 11→IV.3, 12→IV.4, 13→IV.5, 14→V.3, 15→V.4, 16→V.5, App. A→V.6, App. B→V.7
02	1→F.7, 2→II.1, 3→II.2, 4→II.3, 5→II.4, 6→I.8, 7→I.9, 8→I.10, 9→I.11, 10→V.8, 11→V.9, 12→V.10, App. A→V.11, App. B→I.12
03	1→F.8, 2→I.13, 3→I.14, 4→I.15, 5→III.1, 6→III.2, 7→II.5, 8→II.6, 9→V.12, 10→V.13, 11→V.14, App. A→V.15
04	1→F.9, 2→II.7, 3→II.8, 4→II.9, 5→II.10, 6→III.3, 7→III.4, 8→IV.6, 9→V.16, 10→V.17, 11→V.18, App. A→V.19, App. B→V.20
05	1→F.10, 2→I.16, 3→I.17, 4→I.18, 5→I.19, 6→I.20, 7→V.21, 8→V.22, 9→V.23, 10→V.24

Notes and edits. Notes added in the write are set in small type, tagged [**write**] and dated 6 October 2026. They credit results that several sources prove, mark second routes, correct stale statements about the repository against HEAD, record questions answered by another source, and say what the write checked. The write proves four small statements itself, each inside a note with its proof: the comparison of the three Fourier increments (after Table 3), the identification of the four forms of the one-sided bound (after Theorem I.4.1), the value $c_3^{\text{odd}} = 2^{-1/2}$ (after Theorem III.3.4), and the comparison of the two elementary lower bounds for C_k (after Theorem IV.6.2). No delivered statement or proof was changed or dropped. The write’s interventions in delivered text are all of the following kind:

- labels prefixed (`gsr:part:source:`); a label on every delivered section (`...:sec:dk`) and on two remarks of source 03 (`gsr:dt:03:rem:repo`, `gsr:cc:03:rem:rectify`); citation keys unified into one bibliography;
- section headings tagged with the source; theorem styles unified (remarks of sources 04 and 05 were set in the definition style); the sources’ page layout (title pages, running heads, colours, fonts) not reproduced;
- source 02’s plain “Section 3” made a live reference (above), the two figure paths renamed to the shipped files, and plain-text bookmark forms (`\texorpdfstring`) given to two headings with mathematics, invisible in print;
- source 04’s missing backslash in `_{\epsilon\in\Omega_k}` restored, twice, with a dated note (Section IV.6).

The final section of Part V that the sources’ own questions feed, Section V.2, applies Vladimir’s standing rule of 4 October 2026.

Five further sources. Five more manuscripts on the same subject (sources 06–10, arrival 62e21161f) were placed in this directory by ad37962c4; their prefixed code, data and figures are present. They are not used in this text. A later write will add them to these Parts as dated additions, without renumbering anything printed here.

F.2 What is proved, and where

“Proved” means proved in the cited source by an ordinary written proof, unrefereed. “External” marks a published theorem used and not reproved. Nothing here is checked by a proof assistant, except the one formalized statement marked **formal**.

Statement	Where	Sources
<i>Part I. Density transfer and phase-flat partitions</i>		

Statement	Where	Sources
Sharp one-sided mass bound	Thm. I.9.1, Thm. I.14.1; case	02, 03; 01, 05
$\Pr(X > t) \geq ((a+t)T - at)/(a(b-t))$ for mean-zero $X \in [-a, b]$, $T = \mathbb{E}X_+$; forced maximum $aT/(a-T)$; extremizers	$a = \delta$: Thm. I.4.1, Lem. I.17.3	
Short-arc density tail with phase parameter s , occupied-mass exceptions, optimal forced maximum a_* , the trade-off $s_{\max}(t)$; sharp in the finite weighted model, not claimed for arithmetic characters	Thms. I.2.1, I.2.3, I.3.1	01
One-cell phase envelope $C^2 \leq (1-\rho^2)F^2 + \rho^2L^2$ and the linear law $C \leq \rho L + (1-\rho)S$ (real f only)	Lem. I.1.2, I.1.1; Thm. I.8.1, Cor. I.8.2	01, 02
Finite-knot linear-programming reduction; multiphase score	Thm. I.4.2, Cor. I.4.3	01
Lemma 5.15 with $ P \geq \alpha(2+\alpha)N/(2(4-\alpha)(M-1))$ (needs $\alpha > 0$, nonempty cells)	Cor. I.9.3, (I.14.4)	02, 03
Simultaneous phase-flat partition, $L = \lfloor (N \prod_i \varepsilon_i / 2^m)^{1/(m+1)} \rfloor$, every cell in $[L, 2L-1]$	Thm. I.5.1, Thm. I.18.2	01, 05
Fourier increment $\delta + 3\delta b\kappa/(4 - (4-3\delta)\kappa) \geq \delta + 3\eta/8$, $b = 1 - \delta$, length $\lfloor \sqrt{(N/2\pi) \arcsin(\eta/4\delta b)} \rfloor$; real frequency, bounded weights, any N	Thm. I.6.1	01
Fourier increment $\delta + \delta\alpha/(4\delta - \alpha)$, length $\geq \sqrt{\alpha N/(16\pi\delta(1-\delta))}$, exact integer length, popularity, trade-off	Thm. I.17.4	05
Fourier increment with adjustable loss λ	Thm. I.15.2	03
Increment comparison: source 01's is at least 3/2 times source 05's (and source 03's at $\lambda = 1/2$) at stated parameters	[write] note after Table 3	write
Lemma 2.3 with lower constant 1/3 (optimal, also under the exact Lean hypotheses)	Thm. I.16.4	05
Corollary 2.4 with 36π and $3\alpha/4$, same upper length	Cor. I.16.6	05
Optimal simultaneous localization order $1 + (\varepsilon^m N)^{1/(m+1)}$; one-phase barrier $\ell \leq 1 + \sqrt{3r(N-1)}$	Thm. I.18.5; Thm. I.7.2	05; 01
Spectral extraction $E \leq \sqrt{EV} + 2R(v-V)$, Gram correction, optimal width allocation, short-cell discard	Thm. I.19.2, Cor. I.19.4, Prop. I.19.3, Prop. I.20.1	05
Robust phase-to-density transfer; refinement with one global phase budget and its optimum, <i>conditional on</i> $\text{PP}(K, R_0)$ (Corollary 5.6, now formal for intervals)	Thms. I.10.1, I.11.1, Cor. I.11.2, Prop. I.11.3	02
<i>Part II. The inverse step</i>		
$\mathcal{T}_w(f, g)^4 \leq \min\{m(f)^4\mathcal{U}(g), m(g)^4\mathcal{U}(f)\}\mathcal{E}(w)$ on every finite abelian group, coefficient 1 optimal; $\ f\ _{U^3} \leq \ f\ _2$; graph and balanced forms; spectral certificate; deficits; alignment; rigidity	Thm. II.2.1 and Sections II.2–II.4	02
Lemma 7.4 with $ K \geq 7\delta^3 n/(3\sqrt{6})$, same threshold and 90%; optimal scalar tangent κ_* ; weighted tuples; one-witness trade-off	Thm. II.5.2, Section II.5	03
Elementary Proposition 7.3: $ A' \geq c A /8$, $ A' - A' \leq 2^{19}c^{-9} A $ (weaker than Reiher–Schoen)	Thm. II.6.3	03
Exact energy interpolation (M1)–(M7); Lemma 9.2 strengthened to $\gamma^7\beta^{15}p^{15}$	Thm. II.8.1, Cor. II.8.2, Prop. II.8.3	04
Polynomial restriction over $\mathbb{F}_p$; order eight: $2^{-95}\gamma^{112}\eta^{15} B ^{15}$ under $p \geq 2^{102}\gamma^{-112}\eta^{-16}\beta^{-1}$; method limit (prime fields only)	Thm. II.9.1, Cor. II.9.5, Prop. II.9.6	04

Statement	Where	Sources
Arrangement interpolation with profile σ ; Lemmas 14.8, 12.4 with $\beta^{7 \cdot 4^{k+1}-6}, \beta^{106}$	Thm. II.10.1	04
<i>Part III. Cube and progression counts</i>		
No centered terms of degree 1–3; error $\sum_{j \geq 4} \binom{m}{j} \delta^{m-j} \eta^j$	Lem. III.2.1, Cor. III.2.2; Lem. III.3.1	03, 04
Exact quartic term $\delta^{m-4}(R_d \mathcal{E}(f) + T_d \mathcal{E}_{(2)}(f))$ on every finite abelian group; K_d optimal over all groups; power 4 optimal for sets; fifth-order remainder necessary; an order-three correlation	Thm. III.2.6, Section III.2	03 (odd order also 04)
$\ f\ _{U^3}^8 \geq 2\ f\ _{U^2}^8$ for real centered f on odd-order groups, equality, stability	Thm. III.3.4, Cor. III.3.5	04
Three-cube bound with sharp leading term $6\sqrt{2}\delta^4 u^4$; correction $u^{16/3}$ with optimal exponent for $\gcd(G , 6) = 1$; sharpness for sets	Thms. III.3.6, III.3.8, Props. III.3.7, III.3.9, III.3.10	04
$c_3^{\text{odd}} = 2^{-1/2}$, attained by real parts of characters (source 03's question at $d = 3$)	[write] note after Thm. III.3.4	write, from 04
Generalized von Neumann bound for invertible (admissible) slopes; telescoping error $(k-2)\alpha$ in place of $2^k \alpha$; optimal elimination order	Lem. III.1.1, Thm. III.1.2; Lem. III.4.2, Thm. III.4.3	03, 04
$T_k^*(A) \geq N^2(\delta^k - uS_{k-2}(\delta)) - \delta N$ and its thresholds	Cor. III.1.3, Cor. III.4.4	03, 04
Lifting a modular progression from a short interval	Lem. III.4.5, Rem. III.1.4	04, 03; formal
<i>Part IV. Floor patterns</i>		
$ \mathcal{F}(\mathbf{r}) = (\prod_i 2r_i)C_k$; arbitrary integer intervals and the modular count	Thm. IV.1.2; Thm. IV.6.3	01, 04
$C_k \leq \mathcal{R}(H_k, k)$, $\log_2 C_k \leq k^2 + (\log_2 e - 1)k$; $A_k \leq \mathcal{R}(k2^{k-1}, k+1)$	Thms. IV.1.4, IV.3.1; Thm. IV.6.2	01, 04
$C_k \geq 2^{\lfloor k^2/4 \rfloor}$ (01); $C_k \geq 2^{k(k-1)/2}$, $A_k \geq \prod_{j < k} (2^j + 1)$ (04; stronger for $k \geq 3$)	Thm. IV.2.1; Thm. IV.6.2	01; 04
$\log_2 C_k \sim k^2$, $\log_2 A_k \sim k^2$ (external: the classical threshold-function count)	Thms. IV.2.3, IV.3.1	01
Translation-class budgets independent of the modulus Polynomial floor entropy $(1 + o(1))n^{d+1}/d!$ (lower half external: Baldi–Vershynin)	Thm. IV.3.2; Cor. IV.6.4 Thm. IV.4.1	01, 04 01
$C_1, \dots, C_4 = 1, 2, 10, 154$; $A_1, A_2, A_3 = 2, 6, 38$ (exact rational certificates, Python verifiers)	Prop. IV.5.1; Section IV.6	01; 04
Local phase extraction at Proposition 17.7 with coefficient $2^{-k^2 - k \log_2 k - O(k)}$ in place of $2^{-2(k+1)^3}$ (not propagated to Section 18)	Thm. IV.6.7, Cor. IV.6.8; Prop. IV.3.3	04; 01
<i>Part V</i>		
Density and length bookkeeping for an abstract iteration	Prop. V.12.1	03
<i>Open, or not certified</i> (Section V.2)		
$C_5 = 8410$ and $A_4 = 770$ (intake recount, floating-point infeasibility; <i>not certified</i>)	Item IV.2	intake
The second coefficient of the $u^{16/3}$ term (between $(9/4)^{1/3}$ and 6); c_d^{odd} for $d \geq 4$; the $k \log k$ term of $\log_2 C_k$; 46 delivered questions in all	Section V.2	all

What no source claims. No source claims a new bound for $r_k(N)$; all five cite Leng, Sah and Sawhney's $r_k(N) \ll N \exp(-(\log \log N)^{c_k})$ for $k \geq 5$ [2] as the benchmark, and say their results

are local. No source claims literature priority for its formulations or an exhaustive priority search. No source compiled Lean code or claims kernel verification. Each source's further limitations are printed in place and collected in Section V.2.

F.3 Notation across the five sources

No delivered symbol was renamed; the same letter often means different things in different sources. Each Part opens with a short table for its own sections; this is the whole dictionary. The tempting false readings are in italics.

Symbol	Meanings, by source and Part
$\mathbb{E}, \mathcal{E}, E$	$\mathbb{E}$ is always an average (expectation). $\mathcal{E}(w)$: 02, normalized additive energy of a weight on $G \times \hat{G}$ (factor N^{-3} ; Part II). $\mathcal{E}_j(w)$: 04, $M^{-1} \sum \hat{w} ^{2j}$ with the <i>unnormalized</i> transform, counting $2j$ -tuples (Part II). $\mathcal{E}(f) = \ f\ _{U^2}^4$ and $\mathcal{E}_{(2)}(f)$, the order-two energy: 03 (Part III). $\mathcal{E}(\lambda)$: 04, the additive energy of $\lambda_r = a_r ^2$ (Part III). E : 05, captured Fourier energy $\sum c_i ^2$ (Part I); 01, an exceptional set (Part I); 02 and 03, the additive energy $E(\Gamma)$, $E(A)$ of a graph or a set (Part II); 04, E and E_j in Lemma IV.6.6 (Part IV). E_{osc} : 03, an oscillation error. <i>None of these is a normalization of another.</i>
Q	03, 04: Q_d the normalized cube count (the same quantity, Part III). 01: Q_i box counts (Part I). 05: $Q_i(L)$ box counts and Q in Lemma I.16.1 (Part I). 02: Q and $Q_{a,b}(p, t)$, good-mass certificates; Q_i input progressions (Part I). 04: Q the kernel ratio sum (R14), Q_K (R32), $Q_k = k2^{k-1}$, intervals Q_i (Parts II, IV); questions Q1–Q12 and displays (Q1), (Q2) (Part V). 03: Q in Lemma I.15.1.
$R, \mathcal{R}$	03: R_d the number of ordinary parallelograms of $\{0, 1\}^d$, $\mathcal{R}_d(F)$ a remainder (Part III). 04: P_d the same number; R_d, R_6, R_7 remainders; R_j counts of respected arrangements (Part II); $R_d(H) = \sum_{j \leq d} \binom{H}{j}$ (Part IV); R_i cells. 01: $\mathcal{R}(h, d) = \sum_{j \leq d} \binom{h}{j}$ (Part IV), <i>so 01's $\mathcal{R}(H, d)$ is 04's $R_d(H)$, arguments reversed.</i> 02: $R = \mathbb{E} K_u(s) ^2$ (Part II), R_0 the partition threshold, R_j refined cells (Part I). 05: R an oscillation budget (Part I). 03: R the number of steps (Part V). <i>04's remainder R_d is not 03's count R_d.</i>
$T, \mathcal{T}$	02: $\mathcal{T}_w(f, g)$ and T , weighted derivative correlations (Part II). 03: T_d binary tetrahedra (Part III); $T = \mathbb{E}X_+$, T_0 (Part I). 04: $T_d(B)$, $T_d^\phi(B)$ tuple counts (Part II); $T(s, t)$ Fourier cube coefficients, $T_k^*(A)$, $T_i(a)$ (Parts III, IV). 01: T_n , $T(n, d)$ threshold-function counts (Part IV); $T = \prod_i Q_i$ (Part I).
κ	01: $\eta/(2\delta(1 - \delta))$ (Part I). 02: $\mathcal{U}(f)/m(f)^4$ (Part II). 03: $7/(3\sqrt{6})$ and κ_* (Part II). 04: $\kappa_d = 1/(2(400d)^d)$ (Part II). 05: a Gram-matrix bound (Part I).
ρ	02: half the chordal diameter of a phase set (Part I). 03: a normalized Fourier coefficient (Part I); $c/(2 - c)$ (Part II). 04: m/M (Part II); the largest Fourier modulus (Part III). 01: $\max\{0, \lambda_* - \beta_E\}$ (Corollary I.2.4).
s	01: $\sin \theta$ for an arc of half-angle θ . 05: a positive mass, and $s(\varepsilon)$. 02: a weighted absolute mean. 03: $\sqrt{1 - \gamma/\varepsilon}$ (Part II). <i>01's s is 02's ρ, not 05's s.</i>
C_k, C_d, C	01: C_k fractional carry patterns (= 04's F_k , Part IV). 04: $C_d = 1 + \binom{2d}{2}$ (Part II). 02: C a local correlation, C_0 (Part I).
B, B_k	01: $B_k = (2(k+1))^k \mathcal{R}(k2^{k-1}, k+1)$ and $B_{\delta,s}(a)$. 04: $B_k = (2k)^k R_{k+1}(k2^{k-1})$, $B_5(f)$, the set $B \subset \mathbb{F}_p$. 02: B_0, B_ρ , a set B . 05: $B = \mathbb{E} f $, a width budget B . 03: $B(S)$ (= 04's $D(S)$). <i>01's and 04's B_k are different numbers.</i>
L	02: the total absolute mass $\sum f $. 03: $L = D $ (Part II), a length (Part I). 04: kernel bandwidth (Part II), number of labels (Part IV), cube forms L_v (Part III, also 03). 01, 05: cell lengths.

Symbol	Meanings, by source and Part
M, m, q	M : 01 $M_\delta = 2\delta(1 - \delta)$ and a modulus; 02, 03 a number of cells; 04 $ H $, $\sum \lambda^4$, $M_\delta(u)$, a modulus; 05 the modulus of Lemma 2.3. m : 03 2^d (cubes), $ V $ (DRC), $ A $ (energy); 04 $ B $, $ A $, $2h + 1$; 01 anchor count, monomial count; 05 the number of phases. q : 04 2^d (Part III), kernel degree (Part II), $\lceil h/k \rceil$ (Part IV); 01, 03, 05 a common difference; 01 q_j cell densities; 02 q_t good mass.
α, η, u	Gowers’s “ α -uniform of degree $d - 1$ ” means $\ f\ _{U^d}^{2^d} \leq \alpha$: α is a norm power, not a norm (03, 04). Elsewhere α is a correlation (02, 05, Gowers’s Corollaries 2.4–2.5) or a retained proportion (04, Part II). η : 01 a normalized correlation; 03 a U^d norm (Part III) or a cell discrepancy bound; 04 the unrespected proportion (Part II). u : 04 a U^d norm.
arcs	01: an arc of r turns, half-angle $\theta = \pi r$, $s = \sin \theta$. 05: an arc of width ε turns, disk radius $\sin(\pi\varepsilon)$. 02: chordal diameter 2ρ , so $\rho = \sin \theta$. 03: chordal error ε . <i>A width in turns is not an angle.</i>
Fourier	02, 03, 04 (outside its Section II.8), 05: normalized, $\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}$. 04 Section II.8: unnormalized. 01: normalized correlations $\mathbb{E}(h - \delta)e(\xi n)$, and the unnormalized cyclic transform in Corollary I.6.2. The Lean development: unnormalized (<code>ZMod.dft</code>), so its hypothesis $ \widehat{1_A}(r) \geq \alpha N$ is a normalized coefficient at least α .
Gowers norms	02: $\mathcal{U}(f) = \ f\ _{U^3}^8$, normalized; its repository counterpart is $N^4 \mathcal{U}(f)$, and the degree parameter of <code>UniformOfDegree</code> is 2. 03, 04: $\ f\ _{U^d}$ normalized; Lean’s <code>gowersNorm</code> $df = N^{(d+1)/2^d} \ f\ _{U^d}$.
numbering	“Lemma 5.15”, “Corollary 2.4”, “Section 17” and other bare numbers are Gowers’s. This report’s numbers carry a Part numeral (I–V) or F. Source 04’s display tags (Mj), (Rj), (Aj) are not this report’s equation numbers; <i>(A1) is not Appendix A, and (R3) is not a remainder.</i>

F.4 Provenance and merge decisions

Archives. The five archives arrived in commit 66f24b0d0 (“New research reports”, 6 October 2026) and were placed in this directory by aefb0b44a (“Place batch 115 (1/2)”, batch 115 of ProveIt’s incoming-reports intake), which also retired them from docs/incoming/. They survive in the arrival commit (git show 66f24b0d0:docs/incoming/<archive>).

No.	Archive	Bytes	Files	Pin	Manuscript
01	Gowers_Quantitative_Refinements(1).zip	541,437	15	3a25020a6	1,980 lines, 31 pp.
02	Gowers_Szemerédi_Refinements.zip	492,008	7	(66c02f8a1)	823 lines, 20 pp.
03	gowers_local_refinements.zip	408,961	10	3a25020a6	2,392 lines, 36 pp.
04	gowers_quantitative_refinements.zip	788,239	14	327773149	2,987 lines, 40 pp.
05	gowers_strengthening.zip	399,349	8	327773149	847 lines, 18 pp.

The pins are three commits of 6 October 2026, in the order 3a25020a6 $\rightarrow$ 327773149 $\rightarrow$ 66c02f8a1 $\rightarrow \dots \rightarrow$ HEAD. Every repository blob that a source cites was checked at intake against its pin and found correct.

Authorship and disclosures. Sources 02, 03 and 04 are “prepared for Vladimir Reshetnikov” (title block and PDF author field), source 05 is a “research note prepared for Vladimir Reshetnikov” with an empty PDF author field. *Source 01 credits ChatGPT*: its author line reads “Research report

prepared for Vladimir Reshetnikov / Mathematical development and computational checks with ChatGPT”, and its PDF author field reads “Research report prepared with ChatGPT”. No source names a human author. Source 03 mentions “independent mathematical reviews” that are not part of its delivery (note in Section V.13).

What each source contributed. 01: the sharp weighted density-tail theory with a phase parameter, simultaneous partitions, the strongest single-coefficient Fourier increment, the scale barrier (Part I), and the carry-pattern count with $\log_2 C_k \sim k^2$ and $C_4 = 154$ (Part IV). 02: the mixed weighted extension of Proposition 6.1 (Part II), sharp real phase freezing, the general sharp tail bound, the Lemma 5.15 replacement and the refinement theorem under $PP(K, R_0)$ (Part I). 03: the selection theorem and a Fourier increment (Part I), dependent random choice and an elementary Proposition 7.3 (Part II), progression counting and the exact quartic cube term on every group (Part III), iteration bookkeeping (Part V). 04: moments and polynomial restriction (Part II), the sharp odd-order cube constants (Part III), floor patterns with the stronger lower bound and the Proposition 17.7 coefficient (Part IV). 05: the sharp constant $1/3$, Corollary 2.4 with 36π , a density-sensitive increment with popularity, simultaneous localization with its optimal exponent, spectral extraction (Part I).

Overlaps. The five are independent manuscripts: no two share more than 0.8% of their text (intake measurement, 8-word shingles), and sources 01 and 04 are not editions of one another (different titles, pins and files; 0.23% shared text). The results proved by several sources are listed in Section F.2 with all their sources.

Where the merge had to choose.

- *Base.* Source 04 is the base: it was the largest and broadest at intake; its manuscript was staged as `article.tex` and its delivery README as `README.md`, both now replaced by this report and its README.
- *Thematic Parts,* by the intake’s recommendation and Vladimir’s direction, since every source spans two to four themes. Each delivered section is kept whole in exactly one Part; each source’s abstract and Section 1 are in the front matter. Sources’ conventions sections go with their first use: source 03’s Section 2 opens its Part I chapter, source 02’s Section 2 and source 04’s Section 2 open their Part II chapters. Source 02’s Appendix B, a rational example for its tail theorem, is in Part I with that theorem; its Appendix A is in Part V.
- *Results proved by several sources* are printed in each source’s own form, because the forms and hypotheses differ (general $[-a, b]$ against $a = \delta$; strict against non-strict events; all groups against odd-order groups), and because the proofs are independent, mostly by different routes. A dated note at each later print names the first print and credits every source; where two proofs follow the same argument, the note says so. No proof was replaced by a pointer.
- *The sources’ question sections* are printed as delivered, and the write’s Section V.2 deduplicates them with credits.

F.5 Relation to the formal project

The formal project consists of the corrected transcription (`Combinatorics/Ramsey/Papers/sz-thm-gowers-proof`), the Lean development (`Combinatorics/Ramsey/Lean/GowersSzemerédi`, namespace `LeanProofs.GowersSzemerédi`) and the ledger `Combinatorics/Ramsey/gowers-proof-status.json`, which records each numbered statement of the paper as a Lean proposition and its exact companion theorem if there is one. The sources compare their results with *corrected*

statements of this project: Lemma 2.3 with denominator 16, Corollary 2.4 with 128π and $\alpha/2$, Corollary 2.5’s exported pair and the stronger pair inside its proof, and so on.

Ledger at the write. At HEAD 25df8755d (6 October 2026) the ledger records 98 statements with exact companions and 22 open, against 88 and 32 at all three pins. The ten statements closed since the pins are `corollary_5_6`, `corollary_5_7`, `lemma_5_9`, `lemma_5_10`, `corollary_5_11`, `lemma_5_14`, `lemma_16_1`, `lemma_16_3`, `lemma_16_5` and `lemma_16_6`. The development is being extended by another session as this report is written, so these counts may already have moved; the crosswalk (Section V.1) records every declaration it cites with file and line at 25df8755d. Two stale statements of the sources are corrected in dated notes: source 02’s description of Corollary 5.6 as an unverified import (it is now `corollary_5_6_holds`), and source 03’s (and implicitly source 01’s) count of 88 exact and 32 open statements.

What is formalized. One statement of the five sources is formalized: source 04’s Lemma III.4.5 (lifting a modular progression from a short interval), with source 03’s rectification remark, as `hasNatAP_of_short_modular_sequence` and `hasNatAP_of_hasModAP_image` (`Proofs18IntervalTransfer.lean`, lines 22 and 92). Those declarations were added in 327773149, source 04’s own pin, before the sources arrived. Some baselines the sources improve are proof-level facts of the development: the stronger Corollary 2.5 pair (`cor25_large_scale`) and the exact second moment of Lemma 9.2 (inside `lemma_9_2_holds`). The Gowers–Cauchy–Schwarz inequality (`gowers_cauchy_schwarz`) and the triangle inequality (`gowersNorm_add_le`) are formal tools that sources 03 and 04 use. The simultaneous partition of sources 01 and 05 is not the formal degree-one partition `section5_strong_linear_partition` (`Proofs05LinearPartition.lean`, line 86), which is a one-phase statement with target lengths up to $r^{1/16}$ and diameter $r^{-1/8}N$. *Nothing else in this report is formalized, and the report gains no formal status* by its placement beside the Lean development. Each source’s formalization plan is printed in Part V.

F.6 Source 01. The two interfaces and the main conclusions

Source 01, delivered Section 1.

Delivered title block. “Sharp Local Refinements of Gowers’s Szemerédi Argument / Density transfer, carry-pattern entropy, and formalization interfaces / Research report prepared for Vladimir Reshetnikov / Mathematical development and computational checks with ChatGPT / 6 October 2026”; PDF author field “Research report prepared with ChatGPT”. Archive `Gowers_Quantitative_Refinements(1).zip`, manuscript `gowers_refinements.tex` (1,980 lines, 31 pages), pin 3a25020a6. Printed below: the abstract, the status paragraph and Section 1; Sections 2–8 are in Part I, 9–13 in Part IV, 14–16 and Appendices A–B in Part V.

Abstract of source 01 (as delivered). We develop two quantitative refinements of components of Gowers’s 2001 proof. First, a convex-envelope calculation solves the finite weighted extremal problem of converting correlation with a cellwise short-arc phase into a density tail. The resulting bound is sharp in the stated information model, keeps track of occupied mass in exceptional cells, and yields an explicit Fourier density increment on a progression of length on the scale $\sqrt{\eta N / [\delta(1 - \delta)]}$, in place of the $\sqrt{\eta^3 N}$ length in the printed Corollary 2.5. Second, the number of Boolean subset-sum floor patterns with coefficients in $(-r, r)$ is exactly $(2r)^k C_k$, where C_k counts fractional carry patterns. We prove a quadratic-exponent upper bound, an elementary $2^{\lfloor k^2/4 \rfloor}$ lower bound, and, using the classical enumeration of linear threshold functions, $\log_2 C_k \sim k^2$. A polynomial-feature extension has matching leading entropy for fixed degree. An affine version bounds translation classes independently of the modulus, providing a smaller combinatorial loss at the phase-removal interface in Section 17. Exact rational certificates give $C_1 = 1$, $C_2 = 2$,

$C_3 = 10$, and $C_4 = 154$. All local results are proved here; the threshold-function input is explicitly external. These refinements are not asserted to improve the best global bounds for Szemerédi's theorem, and no Lean kernel verification is claimed.

Status and scope. This is a research manuscript, not a refereed publication. “Sharp” below always refers to a precisely specified extremal problem, not to every arithmetic realization of that problem. The work establishes strengthened statements relative to the cited components of Gowers's paper; an exhaustive priority search for every formulation has not been completed. The exact factorization, extremal transfer formulas, and their proofs can be assessed independently of any novelty claim.

Two recurrent losses in a density-increment proof are logically distinct. One converts phase correlation into a local increase of density. The other counts the discrete floor patterns created when a continuous phase is cut into progression cells. Improving one does not automatically improve every subsequent loss. We keep these interfaces separate.

F.6.1 An explicit density-increment replacement

Write $e(t) = \exp(2\pi it)$ and $[N]_0 = \{0, \dots, N-1\}$. Suppose $A \subseteq [N]_0$ has density $0 < \delta < 1$ and, for some real ξ ,

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} (\mathbf{1}_A(n) - \delta) e(\xi n) \right| \geq \eta > 0. \quad (\text{F.6.1})$$

Put $\kappa = \eta/[2\delta(1-\delta)]$. Necessarily $0 < \kappa \leq 1$. Theorem I.6.1 proves that a proper arithmetic progression $P \subseteq [N]_0$ exists with

$$|P| \geq \max \left\{ 1, \left\lfloor \sqrt{\frac{N}{2\pi}} \arcsin \frac{\eta}{4\delta(1-\delta)} \right\rfloor \right\}, \quad (\text{F.6.2})$$

$$\frac{|A \cap P|}{|P|} \geq \delta + \frac{3\delta(1-\delta)\kappa}{4 - (4-3\delta)\kappa} \geq \delta + \frac{3\eta}{8}. \quad (\text{F.6.3})$$

No primality hypothesis is needed. A nonzero cyclic Fourier coefficient is a special case of (F.6.1). The printed Corollary 2.5 of Gowers gives length $\sqrt{\eta^3 N/(128\pi)}$ and increment $\eta/8$ in that special case [1, Corollary 2.5].

The ingredients here are elementary. In particular, the fact that a minimum-cell-size partition avoids some subsequent length loss is already implicit in the preceding Corollary 2.4. Our point is the density-sensitive, integer-exact formulation and the optimal finite weighted transfer theorem behind it, not a claim that the Fourier density-increment strategy itself is new.

F.6.2 A quadratic rather than cubic floor-pattern exponent

For $\alpha \in \mathbb{R}^k$, define

$$F_\alpha(S) = \left\lfloor \sum_{i \in S} \alpha_i \right\rfloor, \quad S \subseteq [k].$$

For integers $r_i \geq 1$, let $\mathcal{F}(\mathbf{r})$ be the set of these functions when $-r_i < \alpha_i < r_i$. Define

$$C_k = \# \left\{ S \mapsto \left\lfloor \sum_{i \in S} x_i \right\rfloor : x \in [0, 1)^k \right\}.$$

Theorem IV.1.2 gives the exact identity

$$|\mathcal{F}(\mathbf{r})| = \left(\prod_{i=1}^k 2r_i \right) C_k. \quad (\text{F.6.4})$$

Let

$$H_k = (k-2)2^{k-1} + 1, \quad \mathcal{R}(h, d) = \sum_{j=0}^d \binom{h}{j},$$

where $\binom{h}{j} = 0$ for $j > h$. We prove

$$2^{\lfloor k^2/4 \rfloor} \leq C_k \leq \mathcal{R}(H_k, k). \quad (\text{F.6.5})$$

In the isotropic case this replaces the 2^{2rk^3} upper bound in Gowers’s Corollary 17.5 by $(2r)^k \mathcal{R}(H_k, k)$ [1, Corollaries 17.5–17.6]. The threshold-function theorem discussed in Section IV.2 then yields

$$\log_2 C_k = k^2 + o(k^2). \quad (\text{F.6.6})$$

Thus a subquadratic exponent cannot bound the full family, even when the coefficient range is fixed. This optimal leading exponent is obtained by combining a reduction proved here with a classical external theorem.

F.6.3 What is, and is not, concluded

The proofs below solve two specific auxiliary problems: the optimal correlation-to-density-tail problem with freely placed short-arc phases, and the exact dependence of floor-pattern counts on integral coefficient ranges. They also locate a smaller translation-class budget for a phase-removal step. They do not supply a new inverse theorem for higher Gowers norms or a new globally optimized iteration.

For comparison, Leng, Sah, and Sawhney prove, for $k \geq 5$ and some $c_k > 0$,

$$r_k(N) \ll N \exp(-(\log \log N)^{c_k}),$$

where $r_k(N)$ is the largest size of a k -term-progression-free subset of $[N]$ [2]. That checked benchmark is already much stronger than what is obtained by merely adjusting constants in the 2001 argument.

F.7 Source 02. Scope, sources, and the precise improvements

Source 02, delivered Section 1.

Delivered title block. “Energy-Sensitive Inverse Steps and Sharp Density Extraction / Quantitative refinements of Gowers’s Szemerédi argument / Research article prepared for Vladimir Reshetnikov / October 6, 2026”. Archive `Gowers_Szemerédi_Refinements.zip`, manuscript `gowers_refinements.tex` (823 lines, 20 pages; not the same file as source 01’s `gowers_refinements.tex`). No pin in its README; its Appendix A records the observed branch reference `66c02f8a1`. Printed below: the abstract and Section 1; Sections 2–5 are in Part II, 6–9 and Appendix B in Part I, 10–12 and Appendix A in Part V.

Abstract of source 02 (as delivered). We strengthen two interfaces in the proof of Szemerédi’s theorem: the passage from derivative Fourier correlations to additive energy, and

the passage from local phase correlations to a density increment. The first result is a sharp, mixed-function, weighted extension of Gowers’s Proposition 6.1, valid on every finite abelian group without a pointwise boundedness assumption. If $m(f) = \mathbb{E}|f|^2$, if $\mathcal{U}(f) = \|f\|_{U^3}^8$, and if $\mathcal{T}_w(f, g)$ is a weighted derivative-correlation statistic, then

$$\mathcal{T}_w(f, g)^4 \leq \min\{m(f)^4 \mathcal{U}(g), m(g)^4 \mathcal{U}(f)\} \mathcal{E}(w).$$

Here $\mathcal{E}(w)$ is explicitly normalized additive energy on $G \times \widehat{G}$. We prove the identities underlying this inequality, give equality examples, and quantify its two Cauchy–Schwarz deficits. For balanced indicators, the resulting graph-energy estimate retains the variance $\delta(1 - \delta)$ rather than discarding it.

The second result is a sharp real-valued phase-freezing inequality. If the phases in every cell have chordal diameter at most 2ρ , local correlation C , signed cell mass S , and total absolute mass L satisfy $C \leq \rho L + (1 - \rho)S$. A sharp one-sided moment inequality then gives the total mass of cells with any prescribed positive density increment. This strengthens Lemma 5.15, yields an explicit refinement theorem using a single global phase budget, and permits closed-form optimization of that budget. The elementary results are proved in full; the polynomial partition specialization explicitly imports the repository’s corrected Corollary 5.6. These are local quantitative refinements, not a claimed improvement of the best global bound for Szemerédi’s theorem. Literature priority for the exact combined formulations is not established.

F.7.1 The two proof interfaces

Gowers’s proof [1] repeatedly turns a failure of uniformity into structure and then into increased density on an arithmetic progression. The supplied ProveIt transcription and Lean development [20] make the intermediate interfaces particularly useful targets: they have concrete finite statements, explicit normalizations, and independently reusable conclusions.

The starting point for our first improvement is Proposition 6.1. In the paper’s unnormalized Fourier convention, if $f : \mathbb{Z}_N \rightarrow \mathbb{C}$ satisfies $|f| \leq 1$, $B \subseteq \mathbb{Z}_N$, and $\phi : B \rightarrow \mathbb{Z}_N$ satisfies

$$\sum_{h \in B} |\widehat{\Delta_h f}(\phi(h))|^2 \geq \alpha N^3, \tag{F.7.1}$$

then the graph of ϕ contains at least $\alpha^4 N^3$ additive quadruples. The proof uses Cauchy–Schwarz and Fourier orthogonality, but bounds two quantities depending on f by their worst-case values. We retain both quantities and allow a weighted, multivalued spectrum and two different input functions.

For our second improvement, Lemma 5.15 assumes a real, mean-zero, 1-bounded function and a partition $P_1, \dots, P_M$ such that

$$\sum_j \left| \sum_{x \in P_j} f(x) \right| \geq \alpha N.$$

It produces a cell with mean at least $\alpha/4$ and size at least $\alpha N/(4M)$. We obtain a full tradeoff between the requested cell mean and the mass of all cells attaining it. The estimate uses the asymmetric range of a balanced indicator, $[-\delta, 1 - \delta]$, instead of only $[-1, 1]$.

Between these two interfaces lie the phase partitions of Section 5. We sharpen the geometric step that removes a nearly constant phase and then track one global error budget through an arbitrary collection of input progressions. No lower bound on each individual local correlation is needed.

F.7.2 What is and is not claimed

The weighted Fourier-energy inequality, the real-phase inequalities, the sharp moment bound, and their stated consequences have complete proofs below. The coefficient in the main energy inequality and the constants in the abstract phase and moment inequalities are optimal in the senses explicitly proved. “Sharp” does not mean that every later progression-length bound, or the complete density-increment iteration, is optimal.

The estimate $\|f\|_{U^3} \leq \|f\|_2$ and the general subject of extremizers are established mathematics; the broader near-extremizer theory is treated by Eisner and Tao [7]. Our elementary proof of that estimate is included to make the constants and the treatment of torsion transparent. Similarly, we do not claim that retaining a Cauchy–Schwarz denominator is a new general technique.

These results do not supply a new best asymptotic estimate for $r_k(N)$. For comparison, Leng, Sah, and Sawhney prove, for $k \geq 5$, a bound of the form

$$r_k(N) \ll N \exp(-(\log \log N)^{c_k}), \quad c_k > 0,$$

using quasipolynomial inverse-theorem bounds and a different quantitative organization of the density-increment argument [2, 3]. We use that published preprint statement as a benchmark, not as a claim to have exhaustively determined the latest record in every progression length.

The exact combined formulations here are independently derived refinements of the supplied proof. The literature check does not establish publication priority, and no previously documented major open problem is declared solved. Section V.9 separates useful next research problems from results proved here.

F.8 Source 03. Scope, starting point, and principal results

Source 03, delivered Section 1.

Delivered title block. “Additive Combinatorics / Sharper local estimates in Gowers’s proof of Szemerédi’s theorem / Density increments, exact cube expansions, and dependent random choice / Research report prepared for Vladimir Reshetnikov / 6 October 2026”. Archive `gowers_local_refinements.zip`, manuscript `gowers_local_refinements.tex` (2,392 lines, 36 pages), pin 3a25020a6. Its title page also says: “Source basis: Gowers (2001) and the two requested `ProveIt` directories, pinned to the revision documented in Section V.13. The accompanying code checks finite instances; the proofs in the text establish the general statements.” Printed below: the abstract and Section 1; Sections 2–4 are in Part I, 5–6 in Part III, 7–8 in Part II, 9–11 and Appendix A in Part V.

Abstract of source 03 (as delivered). We develop explicit local strengthenings of several estimates used in Gowers’s proof of Szemerédi’s theorem and relate them to the accompanying `ProveIt` Lean development. First, an exact asymmetric selection inequality converts partition discrepancy into a sharp lower bound for the mass of cells above a chosen density threshold. Combined with an elementary progression partition, it gives a Fourier density increment whose length retains a factor $[\delta(1 - \delta)]^{-1/2}$. Second, a telescoping progression count replaces an exponential error coefficient by a density-weighted sum. Third, we compute the entire fourth-order term of the centered cube count on every finite abelian group. Its two contributions are ordinary Fourier energy and an explicit correction from the quotient by doubling. The resulting fourth-order error and its universal leading constant are sharp, including for sequences of actual sets. Finally, a cubic dependent-random-choice certificate improves the exact-threshold subset size in Lemma 7.4 from order $\delta^5 n$ to order $\delta^3 n$; a separate one-witness argument greatly improves the elementary constants in Proposition 7.3. All claimed refinements have complete proofs. We

identify their relation to established methods, distinguish them from stronger modern theorems, and give a concrete formalization and research agenda. No improvement to the global record for Szemerédi’s theorem or solution of a major outstanding conjecture is asserted.

The aim of this report is to strengthen concrete parts of the proof in [1], with statements precise enough to be independently checked and incorporated into the supplied formal development [21]. The two supplied directories have different roles. The paper directory contains an edited transcription of the original article, including finite-scale repairs and explanatory comments. The Lean directory contains a catalogue of numbered propositions together with many companion proofs. We compare both with the published article rather than silently identifying their statements.

The work concentrates on three places where numerical losses can be removed without inventing a new inverse theorem: selection from a partition, expansion of a multilinear count, and extraction of a structured subfamily by random sampling. These are useful mathematical components in their own right. Their improvements are also easier to audit than a claim to have sharpened the final tower of parameters after an unexamined substitution into a long proof.

F.8.1 What is established

The following are the central conclusions. The conventions and all finite-scale assumptions appear in the cited statements within this report.

1. **Optimal selection from asymmetric discrepancy data.** For a mean-zero random variable $X \in [-a, b]$, its positive first moment T determines the exact best universal lower bound

$$\Pr(X > t) \geq \max \left\{ 0, \frac{(a+t)T - at}{a(b-t)} \right\}.$$

Theorem I.14.1 includes extremizers. Specializing to cell averages strengthens Lemma 5.15 and allows the density and size requirements to be chosen simultaneously.

2. **A Fourier increment with the density factor preserved.** For a normalized nonzero Fourier coefficient of size ρ , an explicit proper integer progression has the bounds in (I.15.3). The construction works for composite moduli and handles singletons and integer rounding directly. It improves on the stronger estimate already present inside the repository’s proof, not merely its weaker exported conclusion.
3. **A smaller progression-count error.** Theorem III.1.2 gives a density-weighted sum with $k - 2$ terms. Under the hypotheses of Corollary 3.3, in its original index order, this replaces $2^k \alpha$ by at most $(k - 2)\alpha$. Theorem III.1.2 also permits any family of slopes whose pairwise differences act invertibly on the group.
4. **An exact fourth-order cube expansion.** Theorem III.2.6 computes the leading possible error rather than only bounding it. There are no terms of degrees one, two, or three. Among degree-four supports, only ordinary parallelograms and a second family visible to order-two characters contribute. Corollary III.2.7 changes the displayed error scale in Lemma 3.10 from $\alpha^{1/2^d}$ to $\alpha^{4/2^d}$. The fourth power and the universal leading coefficient are optimal in the precise senses proved below.
5. **A larger common-neighborhood core at the same threshold.** Theorem II.5.2 replaces $2^{-1/2} \delta^5 n$ by $7\delta^3 n / (3\sqrt{6})$, while retaining both the threshold $\delta^2 m / 2$ and the 90% requirement in Lemma 7.4. The certificate also gives weighted higher-tuple versions.

6. **An explicit elementary energy-to-structure improvement.** For $E(A) \geq c|A|^3$, Theorem II.6.3 gives

$$|A'| \geq (c/8)|A|, \quad |A' - A'| \leq 2^{19}c^{-9}|A|.$$

Both exponents improve on Proposition 7.3’s displayed constants. The proof works in every abelian group and uses only elementary counting. It is substantially weaker than modern BSG results.

Several ingredients here are established general methods: normalized Gowers norms, telescoping, dependent random choice, and the passage from many additive quadruples to a small difference set. The contribution of this report is the explicitly proved package of refinements, the exact quartic classification with its sharpness statements, the optimized scalar selection problems, and a source-aware route to formalization. We make no historical priority claim for these formulations without a separate, exhaustive literature investigation.

F.8.2 Relation to contemporary quantitative results

Let $r_k(N)$ denote the largest size of a subset of $\{1, \dots, N\}$ containing no nonconstant k -term arithmetic progression. Gowers’s quantitative theorem gave a bound of the shape $r_k(N) \ll_k N(\log \log N)^{-c_k}$. Leng, Sah, and Sawhney prove the stronger estimate

$$r_k(N) \ll_k N \exp(-(\log \log N)^{c_k}) \quad (k \geq 5) \quad (\text{F.8.1})$$

for suitable $c_k > 0$ [2]. Their argument uses quasipolynomial inverse-theorem bounds [3] and a density increment strategy related to [5]. None of the estimates established here improves (F.8.1).

The historical questions at the end of Gowers’s paper must also be read in their date. For example, Jaber, Liu, Lovett, Ostuni, and Sawhney prove that a corner-free set in $G \times G$, with G finite abelian, has size at most

$$|G|^2 \exp(-(\log |G|)^{\Omega(1)})$$

[13]. Thus simply restating the 2001 request for a reasonable two-dimensional corners bound would not identify an unsolved problem. Our research questions in Section V.14 instead specify which finer optimization, extension, or formalization remains to be done.

F.9 Source 04. Purpose, results, and scope

Source 04, delivered Section 1.

Delivered title block. “Additive Combinatorics / Sharper Local Bounds in Gowers’s Proof of Szemerédi’s Theorem / Polynomial homomorphism restriction, sharp cube estimates, and floor-pattern complexity / Prepared for Vladimir Reshetnikov / 6 October 2026”. Archive `gowers_quantitative_refinements.zip`, manuscript `gowers_quantitative_refinements.tex` (2,987 lines, 40 pages; the base of the merge, staged as `article.tex`), pin 327773149. Its first lines read “Complete manuscript source. See README.txt for build instructions. Prepared on 2026-10-06. No new Lean kernel verification is claimed.”, and its title page “Companion materials: editable L^AT_EX source, compiled PDF, exact rational pattern certificates, verification programs, and a source manifest.” Printed below: the abstract and Section 1; Sections 2–5 are in Part II, 6–7 in Part III, 8 in Part IV, 9–11 and Appendices A–B in Part V.

Abstract of source 04 (as delivered). We give explicit refinements of local estimates in Gowers’s proof of Szemerédi’s theorem and identify interfaces for their formalization. Exact Fourier second moments sharpen the lifting of respected additive relations. A powered-cosine selection argument, with Fourier-weighted control of exceptional relations and separate treatment of repetitions, gives a polynomial restriction theorem at every fixed tuple order. For real centered functions on odd-order groups, we prove the sharp gap $\|f\|_{U^3}^8 \geq 2\|f\|_{U^2}^8$, classify equality, and obtain stability. The resulting three-cube bound has sharp leading coefficient $6\sqrt{2}$. Without 2- or 3-torsion, its first correction has optimal exponent $16/3$, witnessed by an explicit two-frequency family. Further results retain density factors in progression counting and determine the quadratic exponential scale of fractional floor patterns, improving the associated local phase-extraction loss. The article includes complete proofs, exact examples, supplementary verification programs, a map to the inspected Lean files, and twelve research questions. Improvements are relative to the specified local source estimates; worldwide priority, a new best global Szemerédi bound, and new kernel-checked Lean proofs are not claimed.

This article develops explicit improvements to several local estimates in Gowers’s proof of Szemerédi’s theorem [1]. The starting points are the paper and its companion Lean development in the ProveIt repository [23]. The improvements come from retaining quantities that a convenient coarse estimate discards: the size of a Fourier support measure, cancellation in a centered cube expansion, the simultaneous dependence of floor values on their parameters, and the number of patterns that occur at a fixed parameter.

The most substantial quantitative change concerns the passage from a map respecting many additive quadruples to a restriction respecting almost all additive tuples of a higher order. Here is a concrete consequence, stated at the same order as Gowers’s Corollary 9.4. Suppose that $B \subset \mathbb{F}_p$, $|B| = \beta p$, and $\phi : B \rightarrow \mathbb{F}_p$ respects at least $\gamma|B|^3$ additive quadruples. If $0 < \beta, \gamma, \eta \leq 1$ and

$$p \geq 2^{102} \gamma^{-112} \eta^{-16} \beta^{-1},$$

then there is $B' \subset B$ containing at least

$$2^{-95} \gamma^{112} \eta^{15} |B|^{15}$$

ordered additive 16-tuples, at least a proportion $1 - \eta$ of which are respected by ϕ . Both the exponents and the size hypothesis are explicit. The proof uses one powered cosine filter and bounds the exceptional relations with their Fourier weights. It generalizes to every fixed tuple order $2d$, with a retention bound proportional to $\alpha^{2d} \eta^{2d-1}$ when the initial respected-tuple proportion is normalized by $|B|^{2d-1}$.

The second set of results concerns additive cubes. For a real centered function on a finite abelian group of odd order, we prove the sharp inequality

$$\|f\|_{U^3}^8 \geq 2 \|f\|_{U^2}^8.$$

Equality, apart from the zero function, is characterized by support on one conjugate pair of Fourier characters. A quantitative stability statement identifies an almost extremizing Fourier pair. Combined with an exact enumeration of the surviving terms in a centered three-cube expansion, this gives a sharp leading coefficient $6\sqrt{2}$ for the cube excess as a function of $\|f\|_{U^3}$. In groups whose order is coprime to 6, a further refinement replaces a fifth-order remainder by a remainder of order $u^{16/3}$. An explicit two-frequency family shows why that fractional exponent occurs.

The remaining results provide useful quantitative interfaces for the larger proof. A telescoping expansion reduces the exponential counting factor in Corollary 3.3 to a density-weighted sum with

$k - 2$ terms. The statement extends to composite cyclic groups and other finite abelian groups under exact invertibility hypotheses. In Section 17, separating integer parts from fractional parts gives polynomial dependence on the slope range. Matching quadratic-order upper and lower bounds determine the exponential scale of the fractional floor-pattern count. A distinction between global patterns and the patterns for a fixed slope improves a subsequent Cauchy–Schwarz loss.

F.9.1 What is established, and what is not claimed

All results asserted as theorems below have proofs. The supplementary programs check finite instances, exact support counts, coefficient inequalities, and rational certificates. These computations are additional evidence and reproducibility aids; the general mathematical statements do not rest on numerical experimentation.

The claim of improvement is relative to the explicitly identified estimates in Gowers’s paper and the inspected repository files. Hölder interpolation, generalized von Neumann inequalities, and hyperplane-arrangement estimates are established tools. A targeted literature review did not determine priority for every exact refinement assembled here. Accordingly this article does not present the collection as a certified first discovery in the literature.

There is also a material distinction between improving a local lemma and improving the best bound in Szemerédi’s theorem. Leng, Sah, and Sawhney prove, for fixed $k \geq 5$, a bound of the form

$$r_k(N) \ll_k N \exp(-(\log \log N)^{c_k}), \quad c_k > 0,$$

using their quantitative inverse theorem [2, 3]. The local arguments here do not improve that result. Green and Tao’s four-term work [6] provides an additional example of why approximate-homomorphism restriction is a useful component of a density increment proof. Manners’s polynomial true-complexity theorem [14] places the counting arguments in a broader established theory. No unresolved global Szemerédi or inverse-theorem conjecture is declared solved in this article.

F.9.2 The inspected repository

The repository was inspected on 6 October 2026. Its paper transcription already contains editorial corrections, and the Lean directory separates formal statement definitions from companion proof modules. The referenced files agree with the snapshot

327773149bd272239f1f6c0015f76d5097512c88.

In particular, the relevant files include `Proofs09Moments.lean`, `Proofs03CubeUpper.lean`, `Proofs03ProgressionAverage.lean`, and `Proofs03ProgressionCount.lean`. The appendix gives a concrete plan for adding the strengthened results. No new Lean proof is claimed to have been compiled or kernel checked in the present project, and no repository changes are part of this deliverable.

F.10 Source 05. Scope, baseline, and main contributions

Source 05, delivered Section 1.

Delivered title block. “Sharp Finite Phase Partitions and Density-Sensitive Increments / Strengthenings of Gowers’ localization lemmas, simultaneous phases, and spectral-energy extraction / Research note prepared for Vladimir Reshetnikov / 6 October 2026”; empty PDF author field. Archive `gowers_strengthening.zip`, manuscript `article.tex` (847 lines, 18 pages), pin 327773149. Printed below: the abstract and Section 1; Sections 2–6 are in Part I, 7–10 in Part V.

Abstract of source 05 (as delivered). We investigate the finite localization step in Gowers’ proof of Szemerédi’s theorem, using the corrected paper and the corresponding Lean modules in the ProveIt repository as the comparison baseline. Three improvements are established with complete proofs. First, the corrected lower constant $1/4$ in Lemma 2.3 can be replaced by $1/3$ without changing its upper length bound; $1/3$ is the largest possible universal constant in that formulation. Second, a nonzero Fourier coefficient of normalized magnitude at least α for a set of density δ produces an arithmetic progression of length at least

$$\sqrt{\frac{\alpha N}{16\pi\delta(1-\delta)}} \quad \text{and density at least} \quad \delta + \frac{\delta\alpha}{4\delta - \alpha}.$$

An exact integer formula is stronger, and a positive proportion of the partition has a controlled increment. Third, simultaneous localization of m real linear phases has optimal order $1 + (\varepsilon^m N)^{1/(m+1)}$, including a partition with a lower bound on every cell and an explicit algebraic obstruction to a better exponent. A density-sensitive spectral-energy inequality links these partitions to multiplicative increments, and unequal phase tolerances can be optimized explicitly. The methods are elementary refinements of classical arguments; claims of literature priority, a new global Szemerédi bound, and completed Lean verification are not made. Exact-arithmetic partition tests and high-precision Fourier diagnostics accompany the article.

F.10.1 What is being strengthened

Gowers’ proof [1] separates a counting argument from the extraction of structure and then a density increment. Our subject is the last conversion: turning a structured correlation into density on a sufficiently long ordinary arithmetic progression. Even at the linear-phase level, this conversion involves three distinct sources of loss: integer rounding in a partition, the error from replacing an oscillating phase by a constant, and the passage from absolute block sums to a positive block average.

The requested repository contains an edited transcription of the paper and a substantial Lean development [24, 25, 26]. The transcription is not identical to the printed paper. In particular, it explicitly repairs finite rounding issues in Lemma 2.3 and Corollary 2.4. The relevant comparison is therefore with the *corrected repository statements*, not merely with the historical printed constants.

Write $t = rs/M$, where r is the length of the domain interval, M the modulus, and s the allowed image diameter. The repository’s version of Lemma 2.3 asserts a partition into progressions with lengths between $\sqrt{t}/4$ and $\sqrt{t}$. The printed lower constant $1/2$ cannot hold uniformly: when $r = s = M = 5$ and the phase is constant, it would require a partition of five points into sets all of size two. We prove that $1/3$, not $1/4$, is the sharp universal lower constant when the upper constant is fixed at one.

The repository’s Corollary 2.5 states a progression length of at least $\sqrt{\alpha^3 N / (128\pi)}$ and a density increment $\alpha/8$. Its large-scale proof already obtains the stronger pair $\sqrt{\alpha N / (128\pi)}$, $\alpha/4$ before weakening the conclusion [26]. Thus simply noticing this stronger pair would not constitute an advance over the repository. Our density-sensitive length estimate improves beyond that proof-level baseline as well.

F.10.2 A map of the results

The exact finite strengthening is Theorem I.16.4, including the optimality of $1/3$. Corollary I.16.6 simultaneously improves the corrected Corollary 2.4 while keeping its upper length bound. The density-sensitive replacement for Corollary 2.5 is Theorem I.17.4. The simultaneous-phase theorem and its matching exponent obstruction are Theorems I.18.2 and I.18.5. The spectral statement, Theorem I.19.2, depends on the residual-variance estimate

$$E \leq \sqrt{EV} + 2R(v - V), \tag{F.10.1}$$

where E is captured Fourier energy, V is the variance of the block densities, $v = \delta(1 - \delta)$, and R is a weighted phase-oscillation budget.

F.10.3 What is not asserted

The elementary ideas used below—pigeonholing on a torus, grouping residue classes, algebraic norms, and conditional expectation—are established mathematical tools. Our contribution is a set of explicit, proved finite statements and their quantitative combination. We have not established that every displayed refinement is absent from the literature.

Nor do we claim an improvement to the best global bounds for $r_k(N)$. Already in 2024, Leng, Sah, and Sawhney proved, for $k \geq 5$, an estimate of the form $r_k(N) \ll N \exp[-(\log \log N)^{c_k}]$, using substantially deeper inverse and localization results [2]. This is a comparison point, not an assertion of an exhaustive October 2026 literature ranking. Their work also emphasizes simultaneous localization and multiplicative increments. The results here concern an explicit linear-phase interface and do not replace the higher-order part of either proof.

Part I

Density transfer and phase-flat partitions

Contents. The last step of a density increment: a correlation with a phase that is nearly constant on the cells of a partition into progressions is turned into a dense cell, and such partitions are built. Sections I.1–I.7: source 01, delivered Sections 2–8 (the finite weighted phase problem, the sharp density-tail theorem with occupied-mass exceptions, the optimal forced maximum, finite convex optimization, integer-exact simultaneous partitions, the Fourier increment, the scale barrier). Sections I.8–I.12: source 02, delivered Sections 6–9 and Appendix B (sharp real phase freezing, the sharp density–mass trade-off, the replacement for Lemma 5.15, the refinement theorem under the partition input $\text{PP}(K, R_0)$). Sections I.13–I.15: source 03, delivered Sections 2–4 (its conventions, the exact selection theorem, a density-sensitive Fourier increment). Sections I.16–I.20: source 05, delivered Sections 2–6 (the sharp constant $1/3$ in Lemma 2.3, Corollary 2.4 with 36π and $3\alpha/4$, the density-sensitive Fourier increment, simultaneous phases with the optimal exponent, spectral energy, short-cell discards).

Printed more than once, credited in dated notes. The sharp one-sided mass bound (sources 01, 02, 03, 05; note after Theorem I.4.1); the one-cell phase envelope (01, 02; note after Lemma I.1.3); the Lemma 5.15 replacement (02, 03); the simultaneous phase-flat partition (01, 05); the scale barrier (01 for one phase, 05 for m phases). The three Fourier increments (01, 03, 05) differ; source 01’s increment is at least $3/2$ times the others at the same length scale (note after Table 3).

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
arc / phase width	01: an arc of r turns has half-angle $\theta = \pi r$ and $s = \sin \theta$; 02: chordal diameter 2ρ , so $\rho = s$; 03: chordal error $ e(\theta x) - z \leq \varepsilon$; 05: arc width ε turns, disk radius $\sin(\pi\varepsilon)$. <i>01’s r and 05’s ε are widths in turns, not angles.</i>
η, α, ρ	01: η = normalized correlation with $e(\xi n)$; 05: α = normalized Fourier coefficient; 03: ρ = normalized Fourier coefficient. For one coefficient 01’s η = 05’s α = 03’s ρ . 02: αN bounds a <i>sum over cells</i> of local correlations, and ρ is a phase half-diameter.
D, p, T, S, s, L	01: $D = \sum_j w_j q_j - \delta $ (twice the positive mass); 02: p positive mass, S sum of absolute cell sums, L total absolute mass; 03: $T = \mathbb{E}X_+$; 05: $S = \mathbb{E}g_+$, $s \leq S$, and $s(\varepsilon)$; 01: $s = \sin \theta$. <i>05’s s is a mass, 01’s s a sine.</i>
κ	01: $\kappa = \eta/(2\delta(1 - \delta))$; 05: a Gram-matrix bound (Corollary I.19.4).
Q, M, B, E	01: Q_i box counts, $M_\delta = 2\delta(1 - \delta)$, $B_{\delta,s}(a)$ largest correlation, E an exceptional set; 02: Q , $Q_{a,b}(p, t)$ good-mass certificates, M number of cells, B_0, B_ρ size constants; 05: $Q_i(L)$ box counts, M the modulus, $B = \mathbb{E} f $ and a width budget B , E captured Fourier energy, R an oscillation budget; 03: E_{osc} an oscillation error.

I.1 The finite weighted phase problem

Source 01, delivered Section 2.

I.1.1 Conventions

Let (Ω, μ) be a finite probability space with positive atom weights. Let $h : \Omega \rightarrow [0, 1]$, and put

$$\delta = \mathbb{E}h \in (0, 1), \quad b = 1 - \delta, \quad f = h - \delta, \quad M_\delta = 2\delta b. \quad (\text{I.1.1})$$

A partition $\mathcal{P} = (P_j)$ consists of nonempty cells. Set

$$w_j = \mu(P_j), \quad q_j = \mathbb{E}(h \mid P_j).$$

Then $\sum_j w_j = 1$ and $\sum_j w_j q_j = \delta$. Every assertion for this model includes ordinary subsets of a finite set, by taking uniform weights and $h = \mathbf{1}_A$.

A unit-modulus function ψ is *short-arc on a cell* if there is a real γ_j such that

$$\psi(x) \in \{\exp(i(\gamma_j + t)) : -\theta \leq t \leq \theta\} \quad (x \in P_j), \quad 0 \leq \theta < \pi/2. \quad (\text{I.1.2})$$

Write $s = \sin \theta \in [0, 1)$. Arc centers may differ between cells. We assume

$$|\mathbb{E}f\psi| \geq \eta \geq 0. \quad (\text{I.1.3})$$

The elementary chord inequality

$$|u - \delta| \leq \delta + (1 - 2\delta)u \quad (0 \leq u \leq 1) \quad (\text{I.1.4})$$

gives $\mathbb{E}|f| \leq M_\delta$, and hence $\eta \leq M_\delta$. Equality holds in the first inequality for indicator functions.

I.1.2 A cancellation-sensitive preliminary bound

Lemma I.1.1 (Cancellation and phase diameter). *Let f be any real function on a finite probability space, and suppose $|\psi| \leq 1$ and the diameter of $\psi(P_j)$ is at most $2s$, where $0 \leq s \leq 1$. If*

$$M = \mathbb{E}|f|, \quad D = \sum_j \left| \mathbb{E}(f \mathbf{1}_{P_j}) \right|,$$

then

$$|\mathbb{E}f\psi| \leq sM + (1 - s)D. \quad (\text{I.1.5})$$

Proof. In one cell let A and B be the positive and negative masses of f , measured with the original probability weights. Choose the corresponding phase averages u, v (arbitrarily within the phase hull if a mass is zero). They belong to the convex hull of the cell's phase values, so $|u|, |v| \leq 1$ and $|u - v| \leq 2s$. If $A \geq B$, then

$$|Au - Bv| = |(A - B)u + B(u - v)| \leq A - B + 2sB.$$

This equals $s(A + B) + (1 - s)|A - B|$. Interchanging the two signs proves the same estimate when $B > A$. Sum over cells and apply the triangle inequality. $\square$

For indicators, the coefficient M in (I.1.5) is $2\delta(1 - \delta)$, not the cruder bound 1. Consequently,

$$D \geq \frac{\eta - 2\delta(1 - \delta)s}{1 - s} \quad (\text{I.1.6})$$

whenever the right-hand side is positive. The next calculation retains more information than D alone.

I.1.3 The exact one-cell envelope

Define, for $0 \leq q \leq 1$,

$$\Phi_{\delta,s}(q) = \sqrt{(q - \delta)^2 + 4\delta bq(1 - q)s^2}. \quad (\text{I.1.7})$$

Lemma I.1.2 (Sharp cell bound). *Under (I.1.2),*

$$|\mathbb{E}((h - \delta)\psi \mid P_j)| \leq \Phi_{\delta,s}(q_j). \quad (\text{I.1.8})$$

For each $q \in [0, 1]$, equality can be realized on a finite weighted cell with mean q and an indicator h .

Proof. First suppose h is an indicator. Relative to the cell probability measure, the positive mass is $A = bq$ and the negative mass is $B = \delta(1 - q)$. Let U, V be the phase averages on the occupied and unoccupied portions. Taking independent samples of their phase values,

$$|AU - BV| \leq \max_{|u|, |v| \leq \theta} |Ae^{iu} - Be^{iv}|.$$

The omitted common rotation has modulus one. Since $|u - v| \leq 2\theta < \pi$,

$$|Ae^{iu} - Be^{iv}|^2 = (A - B)^2 + 4AB \sin^2 \frac{u - v}{2} \leq (A - B)^2 + 4ABs^2.$$

Here $A - B = q - \delta$, giving (I.1.8). Place all occupied mass at one endpoint of the arc and all unoccupied mass at the other to attain it. Cases $q = 0, 1$ use only one phase value.

For general h , replace an atom of weight w and value $h(x)$ by an occupied atom of weight $wh(x)$ and an unoccupied atom of weight $w(1 - h(x))$, with the same phase. Delete zero-weight atoms. This preserves both the cell mean and the correlation, so the indicator argument applies. $\square$

Lemma I.1.3 (Convexity and the endpoint chord). *The function $\Phi_{\delta,s}$ is convex on $[0, 1]$, with $\Phi_{\delta,s}(0) = \delta$ and $\Phi_{\delta,s}(1) = b$. Moreover,*

$$\Phi_{\delta,s}(q) < \delta + (1 - 2\delta)q \quad (0 < q < 1, \ 0 \leq s < 1). \quad (\text{I.1.9})$$

Proof. For $s = 0$, this is the convex function $|q - \delta|$. For $0 < s < 1$, differentiation gives

$$\Phi''_{\delta,s}(q) = \frac{4\delta^2 b^2 s^2 (1 - s^2)}{\Phi_{\delta,s}(q)^3} > 0. \quad (\text{I.1.10})$$

For completeness, if $G = \Phi_{\delta,s}^2$ then $(\sqrt{G})'' = (2GG'' - (G')^2)/(4G^{3/2})$, and the numerator divided by four is $4\delta^2 b^2 s^2 (1 - s^2)$. The endpoint values are immediate. Finally, the square of the right-hand side of (I.1.9), minus $\Phi_{\delta,s}(q)^2$, is $4\delta b q(1 - q)(1 - s^2) > 0$. The chord is positive, so taking square roots proves the strict inequality. $\square$

[write] **6 October 2026.** *One bound, two parametrizations.* Lemma I.1.2 and source 02's Theorem I.8.1 are the same bound. In a cell with positive mass $A = bq$ and negative mass $B = \delta(1 - q)$ (cell-normalized), source 02's signed and absolute masses are $F = A - B = q - \delta$ and $L = A + B$, and its bound $C^2 \leq (1 - \rho^2)F^2 + \rho^2 L^2 = F^2 + 4\rho^2 AB$ is $\Phi_{\delta,s}(q)^2$ with $\rho = s$. The arc conventions agree: an arc of half-angle θ has chordal diameter $2 \sin \theta$, so source 02's ρ (half the chordal diameter) is this $s = \sin \theta$. Both proofs sample the phases of the positive and the negative mass independently; source 02's statement is for arbitrary real f , and this one reduces bounded weights to indicators by splitting atoms. The linear consequence, Lemma I.1.1, is source 02's (I.8.2) and Corollary I.8.2, proved here directly and there from the quadratic bound. Credited to both sources. Source 05's Proposition I.17.2 (a disk centre inside the arc) and source 03's Lemma I.14.3 (arbitrary unit constants) are different, complex-valued transfer estimates.

I.2 A sharp density-tail theorem

Source 01, delivered Section 3.

Fix a target density $a \in [\delta, 1)$. Define

$$u_a = \frac{\Phi_{\delta,s}(a) - \delta}{a}, \quad (\text{I.2.1})$$

$$B_{\delta,s}(a) = \delta + \delta u_a, \quad (\text{I.2.2})$$

$$\Gamma_{\delta,s}(a) = 1 - 2\delta - u_a = \frac{M_\delta - B_{\delta,s}(a)}{\delta} > 0. \quad (\text{I.2.3})$$

Positivity in (I.2.3) follows from (I.1.9). The quantity $B_{\delta,s}(a)$ is the largest correlation permitted by these constraints if every cell density is at most a .

Theorem I.2.1 (Sharp excess and density-tail transfer). *Under the hypotheses of Section I.1, put*

$$\lambda_*(\delta, s, a, \eta) = \frac{(\eta - B_{\delta,s}(a))_+}{\Gamma_{\delta,s}(a)} = \frac{\delta (\eta - B_{\delta,s}(a))_+}{M_\delta - B_{\delta,s}(a)}. \quad (\text{I.2.4})$$

Then

$$\sum_{q_j > a} w_j \geq \sum_j w_j \frac{(q_j - a)_+}{1 - a} \geq \lambda_*(\delta, s, a, \eta). \quad (\text{I.2.5})$$

For every $\eta \in [0, M_\delta]$, the last bound and the corresponding lower bound on the mass of cells with $q_j > a$ are optimal among all finite weighted spaces and phases satisfying the stated constraints.

Proof. Convexity bounds $\Phi_{\delta,s}$ by the chord joining 0 to a on the first interval, and by the chord joining a to 1 on the second. Written as one formula, this is

$$\Phi_{\delta,s}(q) \leq \delta + u_a q + \Gamma_{\delta,s}(a) \frac{(q - a)_+}{1 - a}. \quad (\text{I.2.6})$$

To check the second chord, its slope minus u_a is

$$\frac{b - \Phi_{\delta,s}(a)}{1 - a} - u_a = \frac{b - \delta - u_a}{1 - a} = \frac{\Gamma_{\delta,s}(a)}{1 - a}.$$

Lemmas I.1.2 and I.1.3, the triangle inequality, and $\sum_j w_j q_j = \delta$ now give

$$\eta \leq \sum_j w_j \Phi_{\delta,s}(q_j) \leq B_{\delta,s}(a) + \Gamma_{\delta,s}(a) \sum_j w_j \frac{(q_j - a)_+}{1 - a}. \quad (\text{I.2.7})$$

Rearrange and use $0 \leq (q - a)_+/(1 - a) \leq \mathbf{1}_{q > a}$.

For sharpness, first take $\eta \geq B_{\delta,s}(a)$ and write $\lambda = (\eta - B_{\delta,s}(a))/\Gamma_{\delta,s}(a)$. Since $\eta \leq M_\delta$, we have $0 \leq \lambda \leq \delta$. Give cells of densities 1, a , 0 respective weights

$$w_1 = \lambda, \quad w_a = \frac{\delta - \lambda}{a}, \quad w_0 = 1 - \lambda - \frac{\delta - \lambda}{a}. \quad (\text{I.2.8})$$

They are nonnegative, sum to one, and have weighted mean δ . Use the equality construction of Lemma I.1.2 in each cell, and rotate the arcs independently so that all cell correlations point along the positive real axis. The resulting correlation is

$$w_0 \delta + w_a \Phi_{\delta,s}(a) + w_1 b = B_{\delta,s}(a) + \Gamma_{\delta,s}(a) \lambda = \eta.$$

Only the density-1 cells exceed a , and their normalized excess is also λ . Thus both bounds in (I.2.5) are equalities.

For $0 \leq \eta \leq B_{\delta,s}(a)$, use the $\lambda = 0$ configuration. If $B_{\delta,s}(a) > 0$, take two disjoint copies with opposite global phase rotations and relative weights $(1 + \eta/B_{\delta,s}(a))/2$ and $(1 - \eta/B_{\delta,s}(a))/2$. They still have mean δ , no cell exceeds a , and their total correlation is η . If $B_{\delta,s}(a) = 0$, only $\eta = 0$ arises in this case. Zero-weight cells are omitted throughout. $\square$

Remark I.2.2 (The scope of sharpness). The extremizers allow arbitrary real atom weights and independent choices of arc centers. With rational weights they can be converted to ordinary finite sets by replacing an atom with equally weighted copies. General weights can be approximated by rational ones. This is not a proof that the same extremizers occur for a fixed character on $\mathbb{Z}/N\mathbb{Z}$, a fixed progression partition, or a prescribed list of cell sizes. Such arithmetic constraints may permit further improvements.

I.2.1 Exceptional cells cost occupied mass

A usual exceptional-set estimate pays for the whole exceptional set. Here its occupied mass is the relevant quantity.

Theorem I.2.3 (Occupied-mass robust transfer). *Let $E \subseteq \Omega$, and partition $\Omega \setminus E$ into nonempty regular cells P_j . Impose the short-arc condition only on regular cells, and set*

$$\beta_E = \mathbb{E}(h\mathbf{1}_E).$$

Retain the global mean δ and correlation lower bound η . For $\delta \leq a < 1$,

$$\sum_{\substack{j \text{ regular} \\ q_j > a}} w_j \geq \sum_{j \text{ regular}} w_j \frac{(q_j - a)_+}{1 - a} \geq \max\{0, \lambda_*(\delta, s, a, \eta) - \beta_E\}. \quad (\text{I.2.9})$$

Here $w_j = \mu(P_j)$ uses the original measure, not the measure conditioned on the regular set.

Proof. Write $\varepsilon = \mu(E)$. On regular cells use (I.2.6); their total weight and occupied mass are $1 - \varepsilon$ and $\delta - \beta_E$. On E , use (I.1.4). Thus

$$\begin{aligned} \eta &\leq \delta(1 - \varepsilon) + u_a(\delta - \beta_E) + \Gamma_{\delta,s}(a) \sum_{j \text{ regular}} w_j \frac{(q_j - a)_+}{1 - a} + \delta\varepsilon + (1 - 2\delta)\beta_E \\ &= B_{\delta,s}(a) + \Gamma_{\delta,s}(a) \left(\beta_E + \sum_{j \text{ regular}} w_j \frac{(q_j - a)_+}{1 - a} \right). \end{aligned}$$

Rearranging proves the result, including the case $\eta \leq B_{\delta,s}(a)$. $\square$

The estimate is especially useful when the excluded cells have little of A , even if their total length is not particularly small. An exceptional set consisting of unoccupied points has $\beta_E = 0$: it incurs no loss in this formula. There is no contradiction with global cancellation, since the density and constant term were never renormalized after deletion.

Corollary I.2.4 (A large good cell). *For a uniform space of N points with m regular cells, let $\rho = \max\{0, \lambda_* - \beta_E\}$. If $\rho > 0$, some regular cell has*

$$q_j > a, \quad |P_j| \geq \frac{\rho N}{m}.$$

If all regular cells have size at least L , that cell also has size at least L . If instead all have size at most U , there are at least $\rho N/U$ good cells.

Proof. Their union has at least ρN points by Theorem I.2.3. Pigeonhole over at most m cells for the first assertion; divide their total size by the upper size bound for the last. $\square$

I.3 The optimal forced maximum and its tradeoff

Source 01, delivered Section 4.

The density-tail theorem contains a closed-form answer to a simpler question: how large must at least one cell density be?

Theorem I.3.1 (Optimal maximum density from a short-arc correlation). *Assume there are no exceptional points, and put $\kappa = \eta/M_\delta$. If $0 \leq s < \kappa \leq 1$, then*

$$\max_j q_j \geq a_* = \frac{\delta(\kappa - s^2)}{\kappa(1 - b\kappa) - \delta s^2}. \quad (\text{I.3.1})$$

Equivalently, the forced increment is

$$t_* = a_* - \delta = \frac{\delta b(\kappa^2 - s^2)}{\kappa(1 - b\kappa) - \delta s^2}. \quad (\text{I.3.2})$$

These bounds are sharp in the finite weighted model. If $\kappa \leq s$, no positive density increment is forced by this information alone.

Proof. The function $B_{\delta,s}(a)$ is continuous and strictly increasing for $\delta \leq a \leq 1$, after defining its endpoint value at 1 by continuity. Indeed, its nonconstant part is the secant slope $(\Phi_{\delta,s}(a) - \Phi_{\delta,s}(0))/a$. Strict convexity proves the assertion for $s > 0$; for $s = 0$ and $a \geq \delta$ it follows directly from $B_{\delta,s}(a) = 2\delta(a - \delta)/a$. Its endpoint values are

$$B_{\delta,s}(\delta) = M_\delta s, \quad B_{\delta,s}(1) = M_\delta.$$

If all $q_j \leq a$, the first chord gives $\eta \leq B_{\delta,s}(a)$. Thus the unique solution of $B_{\delta,s}(a_*) = M_\delta \kappa$ is a lower bound on $\max_j q_j$, including the endpoint $a_* = 1$ when $\kappa = 1$.

The equation for a_* is

$$\Phi_{\delta,s}(a_*) = \delta + a_*(2b\kappa - 1). \quad (\text{I.3.3})$$

Squaring, cancelling the common constant term, and removing the positive factor a_* gives a linear equation in a_* . It yields (I.3.1). There is no extraneous square-root branch. To see this explicitly, let

$$D_0 = \kappa(1 - b\kappa) - \delta s^2.$$

Since $s < \kappa$, $D_0 > \kappa(1 - b\kappa) - \delta \kappa^2 = \kappa(1 - \kappa) \geq 0$. Also

$$D_0 - \delta(\kappa - s^2) = b\kappa(1 - \kappa) \geq 0,$$

so $a_* \leq 1$, and (I.3.2) shows $a_* > \delta$. The right side of (I.3.3), after substitution, equals

$$\frac{\delta b(\kappa^2 + (1 - 2\kappa)s^2)}{D_0} > 0.$$

For $\kappa \leq 1/2$ positivity is immediate; for $\kappa > 1/2$, use $s^2 < \kappa^2$ and $1 - 2\kappa < 0$, with the case $\kappa = 1$ giving $1 - s^2 > 0$. Hence the squared solution satisfies the original root equation.

Sharpness uses densities 0 and a_* with weights $1 - \delta/a_*$ and δ/a_* and the aligned equality phases from Lemma I.1.2. Their correlation is $B_{\delta,s}(a_*) = \eta$. Finally, if $\kappa \leq s$, cells all of density δ can have correlation $M_\delta s$; opposite rotated copies reduce it to any $\eta \leq M_\delta s$ without changing any cell density. $\square$

I.3.1 Solving for the allowed arc width

At $s = 0$, the optimal increment is

$$t_0 = \frac{\delta b \kappa}{1 - b \kappa}.$$

For $0 < \kappa < 1$ and a requested increment $0 < t \leq t_0$, solving $t_*(s) = t$ gives

$$s_{\max}^2(t) = \frac{\kappa[\delta b \kappa - t(1 - b \kappa)]}{\delta(b - t)}. \quad (\text{I.3.4})$$

Here $t_0 < b$, so the denominator is positive. The right side decreases from κ^2 to 0 as t increases from 0 to t_0 . For $0 \leq s \leq s_{\max}(t)$ the forced increment is at least t . This follows either from (I.3.2), or from monotonicity of $\Phi_{\delta,s}$ in s , which makes a fixed correlation easier to obtain when the allowed arc is wider. At $\kappa = 1$, maximal correlation instead forces a density-1 cell for every $s < 1$; formula (I.3.4) need not be used at its removable endpoint.

The tradeoff has a practical interpretation. A wider allowed arc permits longer phase-flat progressions, but weakens the density increase. Formula (I.3.4) separates this exact probabilistic tradeoff from the Diophantine approximation that constructs the progressions.

I.4 Partition discrepancy and finite convex optimization

Source 01, delivered Section 5.

I.4.1 A sharp transfer using only cell discrepancy

For a partition define its normalized balanced discrepancy by

$$D = \sum_j w_j |q_j - \delta|. \quad (\text{I.4.1})$$

The equality $\sum_j w_j (q_j - \delta) = 0$ implies $D = 2 \sum_j w_j (q_j - \delta)_+$, and convexity gives $D \leq M_\delta$.

Theorem I.4.1 (Discrepancy-to-density tail). *Let $0 \leq t < 1 - \delta$. Then*

$$\sum_{q_j > \delta + t} w_j \geq \max \left\{ 0, \frac{D(\delta + t) - 2\delta t}{2\delta(1 - \delta - t)} \right\}. \quad (\text{I.4.2})$$

The lower bound is sharp among all finite weighted partitions of mean δ and discrepancy D . In addition,

$$\max_j q_j \geq \delta + \frac{\delta D}{2\delta - D}. \quad (\text{I.4.3})$$

Proof. Choose a phase constant on each cell, equal to 1 or -1 according to the sign of $q_j - \delta$. Then its global correlation is exactly D . Apply Theorem I.2.1 with $s = 0$, $\eta = D$, and $a = \delta + t$. In this case

$$B_{\delta,s}(a) = \frac{2\delta t}{\delta + t}, \quad \Gamma_{\delta,s}(a) = \frac{2\delta(1 - \delta - t)}{\delta + t},$$

which gives (I.4.2). The three-density equality construction in (I.2.8) has exactly this discrepancy, proving sharpness. For (I.4.3), use Theorem I.3.1 with $s = 0$ and $\kappa = D/M_\delta$; the case $D = 0$ is just the mean bound. The denominator is positive because $D \leq 2\delta(1 - \delta) < 2\delta$. $\square$

Suppose a uniform N -point space has a partition into m cells and a lower bound on its *average* cell size, $\bar{L} = N/m$. Let the positive right-hand side of (I.4.2) be Λ . Then some cell simultaneously has density greater than $\delta + t$ and size at least $\Lambda\bar{L}$. This is a direct local output for any supplied partition; it neither replaces an average by a minimum nor assumes that the good cell is among the largest cells in advance.

[write] **6 October 2026.** *The sharp one-sided bound, in four sources.* The bound (I.4.2) is the case $a = \delta$, $b = 1 - \delta$ of a sharp one-sided mass bound that four of the five sources prove independently, each in its own form: source 02's Theorem I.9.1 (cell means in $[-a, b]$, with the nonzero-mean extension (I.9.5)), source 03's Theorem I.14.1 (a random variable in $[-a, b]$, with the bound (I.14.1) on its essential supremum), and source 05's Lemma I.17.3 ($a = \delta$, $b = 1 - \delta$, with the non-strict event $g \geq t$, which makes its bound weaker, and a popularity corollary). The identification (checked at the write): the discrepancy is twice the positive mass, so with $T = D/2$, $a = \delta$ and $b = 1 - \delta$,

$$\frac{(a+t)T - at}{a(b-t)} = \frac{D(\delta+t) - 2\delta t}{2\delta(1-\delta-t)}, \quad \frac{aT}{a-T} = \frac{\delta D}{2\delta - D},$$

which are (I.4.2) and (I.4.3). The proof above is a genuinely different route, the case $s = 0$ of the broken-chord envelope of Theorem I.2.1; sources 02, 03 and 05 count the negative support directly. All four statements and proofs are kept. Only source 01 proves a tail bound that is sharp for a positive arc parameter $s > 0$ (Theorems I.2.1, I.2.3 and I.3.1); source 02's Theorem I.10.1, source 03's Theorem I.15.2 and source 05's Theorem I.17.4 first remove the phase and then apply the $s = 0$ bound.

I.4.2 A master reduction for additional moment information

The preceding broken-chord argument is a small exact linear program. The same mechanism handles more density statistics without repeated pigeonholing.

Theorem I.4.2 (Finite-knot extremal reduction). *Let $0 = t_0 < t_1 < \dots < t_m = 1$, let $F : [0, 1] \rightarrow \mathbb{R}$ be convex, and let $g_1, \dots, g_r$ be continuous functions affine on each interval $[t_i, t_{i+1}]$. Fix feasible values of $\mathbb{E}g_\ell(Q)$ for a finitely supported random variable $Q \in [0, 1]$. The maximum possible $\mathbb{E}F(Q)$ is attained by a distribution supported on $\{t_0, \dots, t_m\}$. It is therefore the value of a finite linear program. An optimizer may be chosen with at most the rank of the equality-constraint matrix (including total mass) many positive weights.*

Proof. If $q \in [t_i, t_{i+1}]$, write $q = (1 - \lambda)t_i + \lambda t_{i+1}$. Replace its probability mass by masses in the proportions $1 - \lambda$ and λ at the two endpoints. Every prescribed g_ℓ -moment is unchanged, since

g_ℓ is affine on this interval. The objective does not decrease, by convexity of F . Do this for every atom. The resulting optimization has finitely many nonnegative weights, one mass equation, and the given linear moment equations. Its feasible set is a closed subset of the probability simplex, so an optimum exists.

If an optimum has more positive weights than the equality matrix rank, there is a nonzero vector supported on those weights in the kernel of that matrix. Move in one of its two directions without decreasing the objective, until a weight first becomes zero. The mass equation ensures that the kernel vector has both signs, so such a move is possible. Iterate to reach the claimed support size. $\square$

Suitable moments include the mean, $|Q - \delta|$, and the stop-loss quantities $(Q - a_i)_+$, provided their breakpoints are included among the knots. The discontinuous function $\mathbf{1}_{Q > a_i}$ is not among the permitted continuous moments; its lower bound is obtained afterward from $(Q - a_i)_+ / (1 - a_i) \leq \mathbf{1}_{Q > a_i}$. This distinction matters at equality.

I.4.3 Several phases on one common partition

Corollary I.4.3 (A simultaneous-phase score). *Suppose phases $\psi_1, \dots, \psi_d$ are short-arc on the same partition, with half-angles θ_i and $s_i = \sin \theta_i < 1$. For weights $\omega_i \geq 0$ with $\sum_i \omega_i = 1$, put*

$$F(q) = \sum_i \omega_i \Phi_{\delta, s_i}(q), \quad \eta = \sum_i \omega_i |\mathbb{E}(h - \delta)\psi_i|.$$

Theorem I.2.1 and its occupied-mass version hold with $\Phi_{\delta, s}$ replaced by F in the definitions of u_a , B , and Γ .

Proof. The cell bound and the triangle inequality give $\eta \leq \sum_j w_j F(q_j)$. The function F is convex, has endpoint values δ, b , and lies strictly below their chord in the interior. The proofs of Theorems I.2.1 and I.2.3 use exactly these facts. $\square$

This is an interface, not an unconditional gain over choosing the largest Fourier coefficient. Requiring one partition to control many phases has its own dimension cost. Additional relations between phases could also rule out the independent equality constructions used for the scalar weighted model.

I.5 Integer-exact simultaneous progression partitions

Source 01, delivered Section 6.

We now construct short-arc cells without asymptotic rounding conventions. An arithmetic progression here has the form $\{a, a + q, \dots, a + (\ell - 1)q\}$ with positive integer q and $\ell \geq 1$. All its terms are required to lie in $[N]_0$; no modular wrapping is used.

Theorem I.5.1 (Anisotropic phase-flat partition). *Let $\xi_1, \dots, \xi_d \in \mathbb{R}$, $d \geq 1$, and let $0 < r_i \leq 1/2$. Choose an integer $L \geq 2$ and set*

$$Q_i = \left\lceil \frac{2(L-1)}{r_i} \right\rceil, \quad T = \prod_{i=1}^d Q_i. \quad (\text{I.5.1})$$

If $N \geq LT$, there is a partition of $[N]_0$ into arithmetic progressions, all with the same positive step $q \leq T$, such that every cell has length between L and $2L - 1$ and, on every cell, each $e(\xi_i n)$ lies in an arc of length at most r_i turns.

Proof. Partition $[0, 1)^d$ into T half-open boxes with side lengths $1/Q_i$. Among the $T + 1$ points

$$(\{j\xi_1\}, \dots, \{j\xi_d\}), \quad 0 \leq j \leq T,$$

two belong to the same box. Subtracting their indices gives an integer $1 \leq q \leq T$ and integers p_i with

$$q\xi_i = p_i + \varepsilon_i, \quad |\varepsilon_i| < 1/Q_i. \quad (\text{I.5.2})$$

Partition $[N]_0$ first into residue classes modulo q . Each class has at least $\lfloor N/q \rfloor \geq L$ elements. If a class has $n = vL + w$ elements, where $v \geq 1$ and $0 \leq w < L$, divide its ordered list into $v - 1$ consecutive blocks of length L and one final block of length $L + w$. Thus every block has length in $[L, 2L - 1]$ and retains step q .

On a block of length ℓ , a real lift of the i th phase has successive differences ε_i . Its total span in turns is at most

$$(\ell - 1)|\varepsilon_i| \leq \frac{2L - 2}{Q_i} \leq r_i.$$

Projection of this real interval to the circle gives the required arc. Every point belongs to exactly one block, proving the partition assertion. $\square$

Corollary I.5.2 (A convenient explicit length). *For every positive integer N and the same frequencies and widths, one may take*

$$L = \max \left\{ 1, \left\lfloor \left(\frac{N \prod_i r_i}{2^d} \right)^{1/(d+1)} \right\rfloor \right\} \quad (\text{I.5.3})$$

and obtain cell lengths between L and $2L - 1$ with the stated arc widths. When $L = 1$, use the singleton partition with step 1.

Proof. For $L \geq 2$, the definition of Q_i gives

$$Q_i \leq \frac{2(L - 1)}{r_i} + 1 \leq \frac{2L}{r_i}.$$

Consequently $L \prod_i Q_i \leq 2^d L^{d+1} / \prod_i r_i \leq N$ by (I.5.3). Apply Theorem I.5.1. Singletons satisfy the remaining case directly. $\square$

The exact condition $N \geq L \prod_i \lceil 2(L - 1)/r_i \rceil$ can sometimes permit a larger L than (I.5.3). It is monotone in L and can be checked by integer arithmetic once the widths have certified rational lower bounds. The proof requires no optimal Diophantine approximation algorithm: the pigeonhole construction suffices.

[write] **6 October 2026.** Theorem I.5.1 with Corollary I.5.2 and source 05's Theorem I.18.2 are the same construction, a rectangular pigeonhole followed by residue-class blocking into lengths $[L, 2L - 1]$, with the same box counts $\lceil 2(L - 1)/r_i \rceil$ and the same explicit length; credited to both. The hypotheses differ slightly: here the widths satisfy $r_i \leq 1/2$ and the theorem takes $L \geq 2$ (the corollary adds singletons), while source 05 allows widths $\varepsilon_i \leq 1$ and treats $L = 1$ inside the theorem. The one-phase case is also source 05's Proposition I.16.3 and, with a chordal error in place of an arc width, source 03's Lemma I.15.1.

I.6 A strengthened Fourier density-increment lemma

Source 01, delivered Section 7.

Theorem I.6.1 (An explicit replacement for Corollary 2.5). *Let $N \geq 1$, $h : [N]_0 \rightarrow [0, 1]$, and $\delta = N^{-1} \sum_n h(n) \in (0, 1)$. Suppose*

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} (h(n) - \delta) e(\xi n) \right| \geq \eta > 0$$

for some real ξ . Put $b = 1 - \delta$ and $\kappa = \eta/(2\delta b)$. There exists a proper arithmetic progression $P \subseteq [N]_0$ such that

$$|P| \geq L := \max \left\{ 1, \left\lfloor \sqrt{\frac{N}{2\pi}} \arcsin \frac{\eta}{4\delta b} \right\rfloor \right\}, \quad (\text{I.6.1})$$

$$\frac{1}{|P|} \sum_{n \in P} h(n) \geq \delta + \frac{3\delta b \kappa}{4 - (4 - 3\delta)\kappa} \geq \delta + \frac{3\eta}{8}. \quad (\text{I.6.2})$$

The progression can be chosen from a partition whose cell lengths all lie between L and $2L - 1$.

Proof. As in (I.1.4), $0 < \kappa \leq 1$. Set

$$\theta = \arcsin(\kappa/2), \quad s = \sin \theta = \kappa/2, \quad r = \theta/\pi.$$

Then $0 < r \leq 1/6$. Corollary I.5.2, with one frequency, supplies the claimed partition and length. Each arc of length r turns has half-angle $\pi r = \theta$, so Theorem I.3.1 applies, also for the singleton partition. Substituting $s = \kappa/2$ into (I.3.2) gives

$$t_* = \frac{\delta b(3\kappa^2/4)}{\kappa(1 - b\kappa) - \delta\kappa^2/4} = \frac{3\delta b \kappa}{4 - (4 - 3\delta)\kappa}.$$

The denominator is positive and at most 4. Since $\eta = 2\delta b \kappa$, this is at least $3\eta/8$. $\square$

Corollary I.6.2 (Nonzero cyclic Fourier coefficients, any modulus). *Let $N \geq 2$, $A \subseteq \mathbb{Z}/N\mathbb{Z}$ have density $0 < \delta < 1$, and let $r \not\equiv 0 \pmod{N}$. Use the unnormalized transform*

$$\widehat{\mathbf{1}_A}(r) = \sum_{x \in \mathbb{Z}/N\mathbb{Z}} \mathbf{1}_A(x) e(-rx/N).$$

If $|\widehat{\mathbf{1}_A}(r)| \geq \eta N > 0$, the set of standard representatives of A has the density-increment conclusion of Theorem I.6.1 on some ambient progression $P \subseteq [N]_0$. Primality of N is unnecessary.

Proof. The geometric sum $\sum_{x=0}^{N-1} e(-rx/N)$ vanishes for any nonzero residue r , whether or not r is a unit. Thus the Fourier coefficient of $\mathbf{1}_A - \delta$ is the same as that of $\mathbf{1}_A$ at r . Apply Theorem I.6.1 with $h = \mathbf{1}_A$ and $\xi = -r/N$. $\square$

I.6.1 A simpler asymptotic comparison with all rounding exposed

Since $\arcsin u \geq u$ for $0 \leq u \leq 1$, if $N\eta \geq 32\pi\delta(1-\delta)$ then the square root in (I.6.1) is at least 2, and

$$L \geq \sqrt{\frac{N\eta}{32\pi\delta(1-\delta)}}. \quad (\text{I.6.3})$$

Here we used only $\lfloor x \rfloor \geq x/2$ for $x \geq 2$. Compared with the printed lower bound $L_{\text{old}} = \sqrt{\eta^3 N / (128\pi)}$, this yields

$$\frac{L}{L_{\text{old}}} \geq \frac{2}{\eta\sqrt{\delta(1-\delta)}} \quad (\text{I.6.4})$$

in that parameter range. This is a comparison of two valid local conclusions, not a global extremal-set estimate.

δ	η	L_{old}	L from (I.6.1)	$\eta/8$	increment (I.6.2)
0.50	0.100	1576.96	126262	0.01250	0.042857
0.10	0.020	141.05	94055	0.00250	0.008359
0.01	0.002	4.46	89674	0.00025	0.000834
0.01	0.010	49.87	201570	0.00125	0.007519

Table 3: Numerical evaluations of the conditional bounds at $N = 10^{12}$. These are parameter comparisons, not constructions of sets with those Fourier coefficients. The new length is an integer; the old displayed bound is left as a real number.

[write] **6 October 2026.** *Two comparisons the source does not make.* (1) *The repository’s proof-level baseline.* The comparison above is with the printed Corollary 2.5, which is also the statement exported by the formal project (`corollary_2_5`). The repository’s proof of that corollary already obtains the stronger pair $\sqrt{\eta N / (128\pi)}$ and $\eta/4$ before weakening it (the private lemma `cor25_large_scale`, `Proofs02DensityIncrement.lean`, line 375 at HEAD); sources 03 and 05 say so (Remark I.15.3 and Section I.17, Subsection “Comparison with the actual repository baseline”). Against that baseline Theorem I.6.1 still gains: since $\arcsin u \geq u$, its length before the floor is at least $\sqrt{\eta N / (8\pi\delta(1-\delta))}$, a factor $4/\sqrt{\delta(1-\delta)}$ above $\sqrt{\eta N / (128\pi)}$, and its increment $3\eta/8$ exceeds $\eta/4$.

(2) *Cross-source comparison (intake finding, proved at the write).* Under the same hypothesis (this η is source 05’s α and source 03’s ρ), the increment (I.6.2) is at least $3/2$ times source 05’s increment $\delta\alpha/(4\delta-\alpha)$ (Theorem I.17.4), which is also source 03’s increment at $\lambda = 1/2$ (I.15.3). *Proof.* Put $b = 1 - \delta$ and $\kappa = \eta/(2\delta b) \in (0, 1]$. Then $\delta\eta/(4\delta - \eta) = 2\delta b\kappa/(4 - 2b\kappa)$, so the ratio of the two increments is

$$\frac{3\delta b\kappa}{4 - (4 - 3\delta)\kappa} \cdot \frac{4 - 2b\kappa}{2\delta b\kappa} = \frac{3}{2} \cdot \frac{4 - 2b\kappa}{4 - (4 - 3\delta)\kappa} = \frac{3}{2} + \frac{3}{2} \cdot \frac{(2 - \delta)\kappa}{4 - (4 - 3\delta)\kappa},$$

because $(4 - 2b\kappa) - (4 - (4 - 3\delta)\kappa) = (2 - \delta)\kappa$. The denominator is at least $4 - 4\kappa + 3\delta\kappa \geq 3\delta > 0$, so the ratio exceeds $3/2$. The lengths are on the same scale: for fixed δ, η and $N \rightarrow \infty$, with $u = \eta/(4\delta b)$, the length L here is asymptotic to $\sqrt{N \arcsin(u)/(2\pi)}$, source 05’s exact L to $\sqrt{Nu/(2\pi)}$, and source 03’s length at $\lambda = 1/2$ to $\sqrt{Nu/(4\pi)}$; so this length is not asymptotically shorter than either. Among the three statements at their stated parameters, Theorem I.6.1 is therefore the strongest. The comparison is of these stated pairs, not of trade-off curves: source 03’s parameter $\lambda \rightarrow 0$ and source 05’s width $\varepsilon \rightarrow 0$ in (I.17.9) buy increments approaching $\delta\eta/(2\delta - \eta) = \delta b\kappa/(1 - b\kappa)$, which exceeds the increment in (I.6.2) for every $\kappa < 1$ (the difference of $3(1 - b\kappa)$ and $4 - (4 - 3\delta)\kappa$ is $1 - \kappa$), at the price of much shorter progressions, and source 01’s own trade-off is (I.3.4). Sources 03 and 05 are kept for these and their other features (popularity bound, exact integer length). The intake’s scan of 5,000 random (δ, α) found minimum ratio 1.500014, in agreement.

I.6.2 Many increment cells instead of one

On the partition in Theorem I.6.1, fix any target $\delta < a < a_*$. Then $\lambda_*(\delta, \kappa/2, a, \eta) > 0$. Theorem I.2.1 gives a disjoint family of good progressions whose union has at least $\lambda_* N$ points and whose number is at least

$$\frac{\lambda_* N}{2L - 1}.$$

Each has density greater than a and length at least L . This branching conclusion is often more useful than selecting one progression. Any further count of longer arithmetic progressions inside these cells still requires an appropriate theorem at the new density and length.

I.7 A sharp scale barrier for the phase-flat method

Source 01, delivered Section 8.

The $\sqrt{rN}$ scale of the one-frequency partition is not just an artifact of the proof. A concrete badly approximable frequency gives a matching upper scale for every individual short-arc progression.

Lemma I.7.1 (A quadratic irrational estimate). *For $\xi = (\sqrt{5} - 1)/2$ and every integer $q \geq 1$,*

$$\|q\xi\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{1}{3q}. \quad (\text{I.7.1})$$

Proof. Let p be a nearest integer to $q\xi$, and set $\varphi = (1 + \sqrt{5})/2$. The nonzero integer $p^2 + pq - q^2$ is the product $(p - q\xi)(p + q\varphi)$. It is nonzero because ξ is irrational. Also $p \geq 0$ and

$$0 < p + q\varphi \leq q(\xi + \varphi) + \frac{1}{2} = q\sqrt{5} + \frac{1}{2} < 3q.$$

Taking absolute values of the product proves (I.7.1). $\square$

Theorem I.7.2 (An intrinsic square-root obstruction). *Let $0 < r < 1/2$ and $\xi = (\sqrt{5} - 1)/2$. If a proper progression $P = \{a + jq : 0 \leq j < \ell\} \subseteq [N]_0$ has all phase values $e(\xi n)$ in one arc of length r turns, then*

$$\ell \leq 1 + \sqrt{3r(N - 1)}. \quad (\text{I.7.2})$$

Proof. There is nothing to prove for $\ell = 1$. Otherwise lift every phase to the same real interval of length r . Consecutive differences of the lifts lie in $[-r, r] \subset (-1/2, 1/2)$ and are congruent to $q\xi$ modulo one. There is at most one such representative, so all these differences equal the same number ε , with $|\varepsilon| = \|q\xi\|_{\mathbb{R}/\mathbb{Z}}$. Hence

$$(\ell - 1)\|q\xi\|_{\mathbb{R}/\mathbb{Z}} \leq r.$$

Lemma I.7.1 implies $\ell - 1 \leq 3rq$. Containment in $[N]_0$ gives $q(\ell - 1) \leq N - 1$. Multiplying these two inequalities gives $(\ell - 1)^2 \leq 3r(N - 1)$. $\square$

This proves optimality of the exponent $1/2$ for a uniform theorem about single-frequency phase-flat progressions. It does not rule out longer *density-increment* progressions that are obtained by another mechanism, or longer phase-flat progressions for special frequencies.

[write] **6 October 2026.** Source 05's Theorem I.18.5 proves the same kind of obstruction for m simultaneous phases, with the algebraic vector of Lemma I.18.3; the unwrapping step is its Lemma I.18.4. At $m = 1$ its constant $c_1 = 1/20$ gives $\ell \leq 1 + \sqrt{20\varepsilon(N - 1)}$ for arcs of width ε turns, weaker than $\ell \leq 1 + \sqrt{3r(N - 1)}$ above (here $r = \varepsilon$). Credited to both; the golden-ratio constant is source 01's.

I.8 Sharp phase freezing for real functions

Source 02, delivered Section 6.

The balanced function of a set is real. This allows cancellation between its positive and negative parts to be treated more accurately than by a complex-valued triangle inequality.

Theorem I.8.1 (One-cell phase geometry). *Let P be a finite set, let $f : P \rightarrow \mathbb{R}$, and let $z : P \rightarrow \mathbb{C}$ satisfy $|z(x)| = 1$. Suppose*

$$|z(x) - z(y)| \leq 2\rho \quad (x, y \in P), \quad 0 \leq \rho \leq 1.$$

Writing $F = \sum_P f$, $L = \sum_P |f|$, and $C = |\sum_P fz|$, we have

$$C^2 \leq (1 - \rho^2)F^2 + \rho^2 L^2. \quad (\text{I.8.1})$$

Consequently

$$C \leq \rho L + (1 - \rho)|F|. \quad (\text{I.8.2})$$

The quadratic bound is sharp for any prescribed positive and negative masses and any $\rho \in [0, 1]$.

Proof. Let $p = \sum f_+$ and $q = \sum f_-$, so $F = p - q$ and $L = p + q$. If either mass is zero, the ordinary triangle inequality proves (I.8.1). Otherwise, sample a phase Z_+ with probabilities proportional to f_+ and independently sample Z_- with probabilities proportional to f_- . Then

$$\sum_P fz = \mathbb{E}(pZ_+ - qZ_-).$$

For every pair of sampled phases,

$$|pZ_+ - qZ_-|^2 = (p - q)^2 + pq|Z_+ - Z_-|^2 \leq F^2 + 4pq\rho^2 = (1 - \rho^2)F^2 + \rho^2 L^2.$$

Take expectations and use the triangle inequality. Equality is attained when all positive mass has one phase and all negative mass has another phase at distance 2ρ .

For (I.8.2), put $s = |F| \leq L$. Squaring the claimed linear majorant and subtracting the right side of (I.8.1) gives

$$[\rho L + (1 - \rho)s]^2 - [(1 - \rho^2)s^2 + \rho^2 L^2] = 2\rho(1 - \rho)s(L - s) \geq 0.$$

Both sides being nonnegative, the claim follows. $\square$

Corollary I.8.2 (Sharp aggregate freezing). *Let $P_1, \dots, P_M$ partition a finite set X , and permit a different unit-modulus phase z_j on each cell, with chordal diameter at most 2ρ . Set*

$$C = \sum_j \left| \sum_{P_j} fz_j \right|, \quad S = \sum_j \left| \sum_{P_j} f \right|, \quad L = \sum_X |f|.$$

For real f ,

$$C \leq \rho L + (1 - \rho)S. \quad (\text{I.8.3})$$

For $\rho < 1$ this gives $S \geq (C - \rho L)/(1 - \rho)$, whenever the latter lower bound is useful.

Proof. Sum (I.8.2). To see optimality of the aggregate linear law from knowledge of L, S, ρ alone, combine cells with $F = 0$ and two endpoint phases, for which $C = \rho L$, with one-sign cells having constant phase, for which $C = L = |F|$. Their contributions realize every ratio $S/L \in [0, 1]$. A zero global mean may also be imposed by using one positive and one negative one-sign cell of equal absolute mass, and rotating their phases if necessary. $\square$

For heterogeneous diameter bounds $2\rho_j$, the proof yields the more informative inequality

$$C \leq S + \sum_j \rho_j \left(\sum_{P_j} |f| - \left| \sum_{P_j} f \right| \right).$$

Thus only the part of a cell capable of positive–negative cancellation incurs a phase-freezing penalty.

Remark I.8.3 (Why the real-valued hypothesis is essential). Take phases $e^{i\theta}, e^{-i\theta}$ with $0 < \theta < \pi/2$ and take f to be their complex conjugates. Then $C = L = 2$ and $S = 2 \cos \theta$, while $\rho = \sin \theta < 1$. The right side of (I.8.3) is strictly below 2. Therefore the improved law must not be exported to arbitrary complex inputs. Gowers’s complex-disc-valued formulations still require a different estimate; the present refinement applies directly to real balanced functions.

[write] **6 October 2026.** Theorem I.8.1 is the bound printed first as source 01’s Lemma I.1.2 (note there: $\rho = s$), here for arbitrary real f and with sharpness for any prescribed masses; Corollary I.8.2 is source 01’s Lemma I.1.1. The heterogeneous inequality above and the complex-input counterexample of the Remark are source 02’s alone.

I.8.1 Balanced indicators and the correct error scale

For $f = \mathbf{1}_A - \delta$ on $|X| = N$,

$$\sum_X f = 0, \quad \frac{1}{N} \sum_X |f| = \lambda := 2\delta(1 - \delta). \quad (\text{I.8.4})$$

If $C \geq \alpha N$ and $\rho < \alpha/\lambda$, Corollary I.8.2 gives

$$\frac{S}{N} \geq \sigma(\rho) := \frac{\alpha - \rho\lambda}{1 - \rho} > 0. \quad (\text{I.8.5})$$

This improves both the use of $L \leq N$ and the loss incurred by freezing at an arbitrary phase representative. For example, at $\delta = 1/10$, $\alpha = 1/50$, and $\rho = 1/20$,

$$\lambda = \frac{9}{50}, \quad \sigma = \frac{11}{950}.$$

The crude representative-phase error bound $2\rho N = N/10$ is larger than the assumed correlation $N/50$ and is useless at these parameters, while (I.8.5) remains positive.

For a modular polynomial phase $z(x) = \exp(2\pi i \phi(x)/N)$, containment of $\phi(P)$ in a modular interval of length at most DN , with $D \leq 1/2$, gives chordal diameter at most $2 \sin(\pi D)$. The simpler estimate $\rho \leq \pi D$ will suffice in the refinement theorem.

I.9 A sharp density–mass tradeoff

Source 02, delivered Section 7.

We now isolate the selection step from all arithmetic structure.

Theorem I.9.1 (Sharp positive-tail bound). *Let $n_1, \dots, n_M > 0$ have sum N , and let $a, b > 0$. Suppose the cell means $x_j \in [-a, b]$ satisfy*

$$\sum_j n_j x_j = 0, \quad p = \frac{1}{N} \sum_j n_j \max(x_j, 0) > 0.$$

For $0 \leq t < b$, let $J_t = \{j : x_j > t\}$ and let $q_t = N^{-1} \sum_{j \in J_t} n_j$. Then

$$q_t \geq Q_{a,b}(p, t) := \frac{p(1 + t/a) - t}{b - t}. \quad (\text{I.9.1})$$

The bound is positive exactly when $t < ap/(a - p)$. Necessarily $p \leq ab/(a + b)$, so $a - p > 0$. Whenever the right side is nonnegative, the bound is attained in the weighted model by means in $\{-a, t, b\}$, allowing a zero-weight type to be omitted.

Proof. The negative mass also equals pN . Since the negative means have magnitude at most a , the total weight of negative cells is at least pN/a . Hence the total weight of positive cells is at most $N(1 - p/a)$. The cells in J_t contribute at most bq_tN positive mass, and the other positive cells contribute at most

$$t(N(1 - p/a) - q_tN).$$

Therefore $p \leq bq_t + t(1 - p/a - q_t)$, which rearranges to (I.9.1). The feasibility bound follows from $p \leq b(1 - p/a)$.

For sharpness, assign weight p/a to the mean $-a$, weight $q = Q_{a,b}(p, t)$ to b , and the remaining weight $r = 1 - p/a - q$ to t . Direct calculation gives $bq + tr = p$, so the overall mean is zero. Moreover

$$r = \frac{b(1 - p/a) - p}{b - t} \geq 0.$$

The strict threshold $x > t$ counts precisely the b -type. Thus equality holds. Rational data can be realized by integer cell sizes after scaling; arbitrary real data give a weighted sharpness statement or can be approximated by rational data. $\square$

Corollary I.9.2 (Long good cells). *Under the hypotheses of Theorem I.9.1, if $Q_{a,b}(p, t) > 0$, there is a cell with*

$$x_j > t, \quad n_j \geq \frac{Q_{a,b}(p, t)N}{M - 1}.$$

The version with M in the denominator is also valid. If every cell has size at least L_0 , there is a cell of size at least L_0 and mean at least $ap/(a - p)$.

Proof. There is at least one negative cell, and hence at most $M - 1$ good cells. Pigeonhole the good mass from (I.9.1). For the final assertion, divide the total positive mass pN by the upper bound $N(1 - p/a)$ for the total positive-cell weight. $\square$

The mass bound in Theorem I.9.1 is sharp. The particular $M - 1$ length corollary is not asserted to solve the finer extremal problem with a fixed number of cells and prescribed individual sizes.

[write] **6 October 2026.** Theorem I.9.1 is the sharp one-sided bound in its general form $[-a, b]$, proved independently in sources 02 (here), 03 (Theorem I.14.1, the same counting of the negative support in probabilistic language), 01 (the case $a = \delta$, $b = 1 - \delta$, by a convex envelope, Theorem I.4.1) and 05 (the same case with a non-strict event, Lemma I.17.3); see the note after Theorem I.4.1. The final assertion of Corollary I.9.2 (a cell of mean at least $ap/(a - p)$) is source 03's (I.14.1) and source 05's (I.17.3). The rational example of Section I.12 attains the bound.

I.9.1 A direct strengthening of Lemma 5.15

Suppose $f : X \rightarrow [-1, 1]$ has mean zero, and let x_j be its mean on a nonempty partition cell. If $S = \sum_j |\sum_{P_j} f| \geq \alpha N$ with $\alpha > 0$, then $p = S/(2N) \geq \alpha/2$ and necessarily $\alpha \leq 1$. The right side of (I.9.1) increases with p . Taking $a = b = 1$ and $t = \alpha/4$ gives the following explicit replacement.

Corollary I.9.3. *Under the hypotheses of Gowers's Lemma 5.15, after empty cells are discarded, some cell P satisfies*

$$\frac{1}{|P|} \sum_{x \in P} f(x) > \frac{\alpha}{4}, \quad |P| \geq \frac{\alpha(2 + \alpha)}{2(4 - \alpha)} \frac{N}{M - 1}. \quad (\text{I.9.2})$$

The coefficient of N/M , even without using $M - 1$, is larger than $\alpha/4$ by the factor

$$\frac{2(2 + \alpha)}{4 - \alpha} > 1.$$

For $\alpha = 2/5$ it is $2/15$, compared with the original $1/10$. The improvement follows from accounting for the negative support before spending the remaining mass on small positive means.

[write] **6 October 2026.** Source 03 states the same replacement, (I.14.4). *Comparison with the formal statement* (checked at HEAD): `lemma_5_15` (`Section05.lean`, line 352; exact companion `lemma_5_15_holds`, `Proofs05_10.lean`, line 735) allows $\alpha \geq 0$ and empty cells among its M cells, and concludes a cell with $\alpha|P_j|/4 \leq \sum_{P_j} f$ and $|P_j| \geq \alpha N/(4M)$. Corollary I.9.3 needs $\alpha > 0$ and counts only nonempty cells (at most M); under these conditions its mean bound is strict and its size bound is the larger one displayed. At $\alpha = 2/5$ the coefficient is $\alpha(2 + \alpha)/(2(4 - \alpha)) = 0.96/7.2 = 2/15$ (rechecked). Not formalized.

I.9.2 The asymmetric range of a balanced indicator

For $f = \mathbf{1}_A - \delta$, put $a = \delta$, $b = 1 - \delta$. If $S \geq \sigma N > 0$, then every

$$0 \leq t < \frac{\delta\sigma}{2\delta - \sigma}$$

satisfies

$$\frac{1}{N} \sum_{j: |A \cap P_j|/|P_j| > \delta + t} |P_j| \geq \frac{(\sigma/2)(1 + t/\delta) - t}{1 - \delta - t}. \quad (\text{I.9.3})$$

In particular there is a cell with density at least

$$\delta + \frac{\delta\sigma}{2\delta - \sigma}. \quad (\text{I.9.4})$$

The denominator is positive because $\sigma \leq 2\delta(1 - \delta)$. The expression in (I.9.4) never exceeds 1 under this feasibility condition.

I.9.3 A nonzero-mean extension

For incomplete covers or weighted variants, mean zero can fail. If the weighted mean is μ and the weighted absolute mean is s , define $p_+ = (s + \mu)/2$ and $p_- = (s - \mu)/2$. The same proof gives

$$q_t \geq \frac{p_+ - t(1 - p_-/a)}{b - t}, \quad 0 \leq t < b. \quad (\text{I.9.5})$$

Indeed the negative support has weight at least p_-/a , while the positive mass is p_+ . This formula avoids silently using the zero-mean identity on a proper subset of the original ambient set.

I.10 Combining correlation, phase control, and density extraction

Source 02, delivered Section 8.

Theorem I.10.1 (Robust phase-to-density transfer). *Let $A \subseteq X$, $|X| = N$, $\delta = |A|/N \in (0, 1)$, and $f = \mathbf{1}_A - \delta$. Let $P_1, \dots, P_M$ be a partition into nonempty cells. Suppose each cell has a unit-modulus phase of chordal diameter at most 2ρ , where $0 \leq \rho < 1$, and*

$$\sum_j \left| \sum_{P_j} f z_j \right| \geq \alpha N, \quad \alpha > 2\rho\delta(1 - \delta).$$

Define

$$\sigma = \frac{\alpha - 2\rho\delta(1 - \delta)}{1 - \rho}, \quad Q = \frac{(\sigma/2)(1 + t/\delta) - t}{1 - \delta - t}.$$

For every $0 \leq t < \delta\sigma/(2\delta - \sigma)$, the cells with density greater than $\delta + t$ have total size at least QN , and one of them has size at least $QN/(M - 1)$. If the cells are arithmetic progressions, so is the selected cell.

Proof. Use (I.8.4) and Corollary I.8.2 to obtain $S \geq \sigma N$. Apply (I.9.3) and Corollary I.9.2. No property of a progression is used in selection, so that property is preserved automatically. $\square$

For the numerical parameters at the end of Section I.8, $\sigma = 11/950$. Formula (I.9.4) gives a density increment of at least $11/1790$. Requiring the smaller increment $t = 1/400$ guarantees good-cell mass at least $261/68210$ of the ambient set. These are exact rational consequences, not estimates obtained from a simulation.

This theorem is a replacement interface for phase-to-density arguments, rather than an unconditional restatement of every printed version of Corollary 5.8. In particular, assertions about the length of a progression must be supplied by an actual partition theorem with its thresholds and rounding conditions made explicit.

I.11 A uniform refinement theorem and an optimized phase budget

Source 02, delivered Section 9.

I.11.1 The imported phase-partition input

Here we make the only non-elementary input in our progression-length specialization explicit. Suppose a class of polynomial phases on injectively parameterized modular arithmetic progressions has the following property, for constants $K \geq 1$ and $R_0 \geq 1$:

Partition input $\text{PP}(K, R_0)$. For a progression of size $m > R_0$, and any integer $1 \leq v \leq m^{1/K}$, there is a partition into nonempty arithmetic subprogressions of lengths $v-1$ or v , such that the phase values in each cell lie in a modular interval of length at most $Nm^{-1/K}$.

The supplied, edited transcription of Corollary 5.6 states this form for

$$K = (k!)^2 2^{(k+1)^2}, \quad R_0 = 2^{2^{40k^3+1}}, \quad k \geq 1, \quad (\text{I.11.1})$$

with the explicit requirement that the progression length is at most N [20]. Composing a polynomial with an injective progression chart $x \mapsto a + dx$ permits its use on each such input progression. We import this stated partition result; the lengthy Weyl-type recurrence proof is not reproved or independently Lean-verified here. The elementary theorem that follows is valid for any correctly established $\text{PP}(K, R_0)$, including stronger future inputs.

[write] **6 October 2026.** *Stale since the pin: Corollary 5.6 is now proved in Lean.* When source 02 was written (observed branch reference 66c02f8a1), Corollary 5.6 was an open statement of the formal project, and the two sentences above were accurate. At HEAD (25df8755d, 6 October 2026) `corollary_5_6` (`Section05.lean`, line 205) has the exact companion `corollary_5_6_holds` (`Proofs05FullPartition.lean`, line 85). Its statement is $\text{PP}(K, R_0)$ for a polynomial ϕ on $\mathbb{Z}_N$ of degree at most k and the initial interval $\{0, \dots, r-1\}$: $K = \text{polynomialPartitionConstant } k = (k!)^2 2^{(k+1)^2}$; the threshold `polynomialPartitionThreshold` $k = \text{weylThreshold } k^2 = 2^{2^{40k^3+1}} = R_0$, used with the strict inequality $R_0 < r$; $r \leq N$; $1 \leq v \leq r^{1/K}$; nonempty proper cells of length $v-1$ or v ; image diameter at most $r^{-1/K}N$. So the Weyl-type recurrence behind (I.11.1) is now proved in the Lean development (an exact companion in the ledger). Theorem I.11.1 also composes the polynomial with the chart $x \mapsto a + dx$ of each input progression. The degree half of that step is in Lean (`polynomialOn_affine_pullback`, `Proofs05Downstream.lean`, line 244), but Theorem I.11.1, Corollary I.11.2 and the transfer of cells back to the input progressions are not formalized. They gain no formal status. The related `corollary_5_8` is still open; `corollary_5_8_with_scale_holds` proves only its scale-qualified repair, and Section I.10 rightly declines to restate it.

Theorem I.11.1 (One global budget for many local phases). *Assume $\text{PP}(K, R_0)$. Let $Q_1, \dots, Q_M$ partition $G = \mathbb{Z}_N$ into nonempty injectively parameterized arithmetic progressions, and allow an independent polynomial phase z_i on every Q_i . Let $f : G \rightarrow \mathbb{R}$ and write*

$$C_0 = \sum_i \left| \sum_{Q_i} f z_i \right|, \quad L = \sum_G |f|.$$

For $0 < \rho < 1$, put

$$B_0 = \max\{4, (R_0 + 1)^{1/K}\}, \quad B_\rho = \max\{B_0, \pi/\rho\}. \quad (\text{I.11.2})$$

There is a refinement into J nonempty arithmetic progressions R_j such that

$$J \leq B_\rho M^{1/K} N^{1-1/K}, \quad (\text{I.11.3})$$

$$\sum_j \left| \sum_{R_j} f \right| \geq \frac{C_0 - \rho L}{1 - \rho}. \quad (\text{I.11.4})$$

Proof. Set $T_0 = \max\{4^K, R_0 + 1, (\pi/\rho)^K\} = B_\rho^K$. Treat an input cell of size m in two ways. If $m < T_0$, use singletons. Their count satisfies

$$m = m^{1-1/K} m^{1/K} \leq B_\rho m^{1-1/K},$$

and their phase diameter is zero.

If $m \geq T_0$, take $v = \lfloor m^{1/K} \rfloor$ in PP. Since $m^{1/K} \geq 4$, every resulting cell has length at least

$$v - 1 \geq m^{1/K} - 2 \geq \frac{1}{2} m^{1/K}.$$

The number of cells is at most $2m^{1-1/K}$. On each cell the half chordal diameter is at most $\pi m^{-1/K} \leq \rho$. The conditions $m > R_0$ and $m \leq N$ needed by the imported input are respected.

Summing the counts and applying concavity, or Hölder's inequality, gives

$$J \leq B_\rho \sum_i |Q_i|^{1-1/K} \leq B_\rho M^{1/K} N^{1-1/K}.$$

The sum of absolute local correlations on the refinement is at least C_0 , by the triangle inequality within each original cell. Apply Corollary I.8.2 to the refined cells and their inherited phases to obtain (I.11.4). $\square$

Unlike an argument applying a correlation threshold separately to each Q_i , the cost in (I.11.3) does not depend on a smallest positive local correlation. Cells with tiny or zero correlation cause no singular dependence. The theorem also allows different polynomials in different input cells. The exponent $1/K$ is inherited unchanged from PP.

Corollary I.11.2 (Explicit progression density increment). *In Theorem I.11.1, take $f = \mathbf{1}_A - \delta$, assume $C_0 \geq \alpha N$, and choose ρ, t as in Theorem I.10.1. With its value of Q , there is an arithmetic progression P such that*

$$\frac{|A \cap P|}{|P|} > \delta + t, \quad |P| \geq \frac{Q}{B_\rho} \left(\frac{N}{M} \right)^{1/K}. \quad (\text{I.11.5})$$

Proof. Apply Theorem I.10.1 to the refinement and use $J - 1 \leq J \leq B_\rho M^{1/K} N^{1-1/K}$. The positive lower bound for Q ensures that a negative cell and a good cell exist, so $J \geq 2$. $\square$

The word “progression” in this corollary means an injectively parameterized modular progression when the input cells are modular. Converting a wrapping modular progression to proper integer progressions is a separate cost. It must not be omitted when this interface is inserted into a proof over $[N]$.

I.11.2 Closed-form optimization

The free phase budget can be optimized exactly within the certificate just proved. Fix $\delta \in (0, 1)$, $0 < \alpha < \lambda = 2\delta(1 - \delta)$, and

$$0 \leq t < \frac{\delta\alpha}{2\delta - \alpha}.$$

Define

$$A_t = \frac{1 + t/\delta}{2}, \quad d = A_t \alpha - t, \quad e = A_t \lambda - t. \quad (\text{I.11.6})$$

The identity $e = \delta(1 - \delta - t)$ makes its positivity explicit. Then $0 < d < e$ and the good-mass certificate is

$$Q(\rho) = \frac{d - e\rho}{(1 - \rho)(1 - \delta - t)}, \quad 0 < \rho < d/e.$$

Proposition I.11.3 (Optimal uniform phase budget). *The coefficient $Q(\rho)/B_\rho$ in (I.11.5) is maximized at*

$$\rho_{\text{opt}} = \min \left\{ 1 - \sqrt{1 - d/e}, \frac{\pi}{B_0} \right\}. \quad (\text{I.11.7})$$

If the first entry is no larger than the second, the maximum coefficient is

$$\frac{e(1 - \sqrt{1 - d/e})^2}{\pi(1 - \delta - t)} = \frac{\delta}{\pi}(1 - \sqrt{1 - d/e})^2. \quad (\text{I.11.8})$$

Proof. When $\rho \leq \pi/B_0$, the denominator B_ρ equals π/ρ . Apart from the positive constant $1/[\pi(1 - \delta - t)]$, the objective is

$$H(\rho) = \frac{\rho(d - e\rho)}{1 - \rho}.$$

Its derivative has the sign of $d - 2e\rho + e\rho^2$. The unique critical point in $(0, d/e)$ is $r = 1 - \sqrt{1 - d/e}$; the derivative is positive before r and negative after it. At this point $d = e(2r - r^2)$, so $H(r) = er^2$.

When $\rho \geq \pi/B_0$, the denominator is constant B_0 , while $Q'(\rho) = (d - e)/[(1 - \delta - t)(1 - \rho)^2] < 0$. Hence no larger budget beyond that boundary can improve the certificate. Combining the two regions proves (I.11.7) and (I.11.8). If the boundary is outside the admissible interval, the first critical point is used automatically. $\square$

This optimization improves a particular rigorously derived lower bound; it is not a proof that the underlying polynomial partition itself is optimal. In the regime dominated by the enormous recurrence threshold R_0 , the boundary π/B_0 can control the optimum, so optimizing the elementary interface alone cannot erase the main recurrence loss.

I.12 A rational equality example for the moment bound

Source 02, delivered Appendix B.

Take $a = b = 1$, $p = 1/5$, $t = 1/10$, and assign weights

$$\frac{1}{5}, \quad \frac{2}{3}, \quad \frac{2}{15}$$

to means $-1, 1/10, 1$. Their weighted mean is

$$-\frac{1}{5} + \frac{1}{15} + \frac{2}{15} = 0,$$

the positive mass is $1/5$, and the mass of cells with mean strictly above $1/10$ is $2/15$. Formula (I.9.1) gives exactly

$$\frac{\frac{1}{5}(1 + \frac{1}{10}) - \frac{1}{10}}{1 - \frac{1}{10}} = \frac{2}{15}.$$

For $N = 150$, the three cell sizes are 30, 100, 20. Taking f constant on these cells realizes the abstract example on a finite set with integer cardinalities.

[write] **6 October 2026.** The example attains Theorem I.9.1 with $a = b = 1$, $p = 1/5$, $t = 1/10$ (arithmetic rechecked at the write); it is printed here, in Part I, next to the theorem it illustrates, rather than with source 02's other appendix (Section V.11).

I.13 Conventions and analytic background

Source 03, delivered Section 2.

Unless stated otherwise, G is a finite abelian group, written additively, and $\mathbb{E}_{x \in G} = |G|^{-1} \sum_{x \in G}$. Fourier transforms in this report are normalized:

$$\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}, \quad f(x) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(x).$$

Character orthogonality says that the average of a character is zero unless it is trivial. For $f: G \rightarrow \mathbb{C}$ and $d \geq 1$, define

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{v \in \{0,1\}^d} \mathcal{C}^{|v|} f \left(x + \sum_i v_i h_i \right), \quad (\text{I.13.1})$$

where $\mathcal{C}$ denotes complex conjugation. These expressions are nonnegative. One has $\|f\|_{U^1} = |\mathbb{E}f|$, and for $d \geq 2$ they are norms, satisfy the triangle inequality, and are monotone in d . The recursive identity

$$\|f\|_{U^{d+1}}^{2^{d+1}} = \mathbb{E}_h \|f(\cdot) \overline{f(\cdot + h)}\|_{U^d}^{2^d}$$

follows by splitting the last cube coordinate. Iterated Cauchy–Schwarz proves positivity, monotonicity, and the mixed cube inequality used below; these are the standard elementary analytic facts of the source's Section 3. Fourier orthogonality also gives $\|f\|_{U^2}^4 = \sum_{\chi} |\widehat{f}(\chi)|^4$.

For $A \subset G$, write $\delta = |A|/|G|$ and $f_A = \mathbf{1}_A - \delta$. The source's statement that A is α -uniform of degree $d - 1$ means

$$\|f_A\|_{U^d}^{2^d} \leq \alpha. \quad (\text{I.13.2})$$

In particular its parameter α is a norm power, not the norm itself. This distinction is essential when comparing exponents.

A cube is a parameter tuple $(x, h_1, \dots, h_d)$ with its 2^d labelled vertices. We count parameter tuples; repeated vertices, zero increments, and coincident cubes under symmetries are retained. Arithmetic progressions in a finite group are similarly counted by their parameters (x, t) unless an ordinary integer progression is specified. A positive integer step and containment in the representative interval are proved explicitly in Section I.15.

The symbols m, N, δ are local to each section. In the cube section $m = 2^d$ is the number of vertices. In the DRC section m is the size of the witness set, while in the energy section it is $|A|$. Every statement gives its own quantifiers. An empty sum is zero and a zeroth power, including 0^0 in finite polynomial identities, is one.

I.14 An exact selection theorem for density increments

Source 03, delivered Section 3.

The last step of a density increment argument is often a finite averaging problem. A partition has appreciable total discrepancy, and one wants a cell which is both large and unusually dense. Keeping the asymmetric bounds of a balanced indicator gives more information than replacing them by $[-1, 1]$. The result below solves the associated one-sided mass problem exactly.

Theorem I.14.1 (Sharp one-sided mass bound). *Let X be a real random variable on a probability space, with $-a \leq X \leq b$, $\mathbb{E}X = 0$, where $a, b > 0$. Set*

$$T = \mathbb{E}X_+ = \mathbb{E}X_- = \frac{1}{2}\mathbb{E}|X|.$$

Then $0 \leq T \leq ab/(a+b)$. If $T > 0$, then

$$\text{ess sup } X \geq \frac{aT}{a-T}. \quad (\text{I.14.1})$$

For every $0 \leq t < b$,

$$\Pr(X > t) \geq \max \left\{ 0, \frac{(a+t)T - at}{a(b-t)} \right\}. \quad (\text{I.14.2})$$

Whenever $0 \leq t < aT/(a-T)$, the positive lower bound in (I.14.2) is attained by a probability distribution supported on $\{-a, t, b\}$. Thus it cannot be increased using only the stated information. Bound (I.14.1) is sharp as well.

Proof. Writing $p_- = \Pr(X < 0)$ gives $p_- \geq T/a$. Likewise $\Pr(X > 0) \geq T/b$, so $T/a + T/b \leq 1$. If h is the essential supremum of X , then

$$T \leq h \Pr(X > 0) \leq h(1 - T/a),$$

which proves (I.14.1). Put $q = \Pr(X > t)$. The mass where $0 < X \leq t$ is at most $1 - p_- - q$; when $t = 0$ the same inequality below still holds. Consequently

$$T \leq t(1 - p_- - q) + bq \leq t(1 - T/a) + (b-t)q.$$

Rearranging proves (I.14.2).

For sharpness set

$$p_{-a} = T/a, \quad p_b = \frac{(a+t)T - at}{a(b-t)}, \quad p_t = 1 - p_{-a} - p_b.$$

The assumed range of t makes p_b positive, and

$$p_t = \frac{ab - (a+b)T}{a(b-t)} \geq 0.$$

These weights sum to one, the negative first moment is T , and the positive first moment is $tp_t + bp_b = T$. Only the atom at b lies above t , giving equality. For (I.14.1), use the two-point distribution at $-a$ and $aT/(a-T)$, with weights T/a and $1 - T/a$. $\square$

The sharpness assertion concerns arbitrary probability weights. Rational weights give literal finite models after clearing denominators; approximation gives the corresponding limiting optimality when prescribed cardinalities or indicator-valued functions impose integrality restrictions.

[write] **6 October 2026.** Theorem I.14.1 is printed first, in cell form, as source 02's Theorem I.9.1; the proof here is the same counting of the negative support, and is kept with the statement it proves (this form adds the essential-supremum bound (I.14.1), which source 02 states as the last assertion of its Corollary I.9.2). Credited to sources 01, 02, 03 and 05 (note after source 01's Theorem I.4.1).

Corollary I.14.2 (A large cell with a specified increment). *Let $a, b > 0$, let $P_1, \dots, P_M$ be a partition of a finite probability space into cells of positive measure, and let f be real, mean zero, and valued in $[-a, b]$. Write $w_j = \mu(P_j)$ and $x_j = \mathbb{E}(f \mid P_j)$. Suppose*

$$\sum_j w_j |x_j| \geq 2T_0 > 0.$$

If $0 \leq t < aT_0/(a - T_0)$, some cell satisfies

$$x_j > t, \quad w_j \geq \frac{(a + t)T_0 - at}{a(b - t)(M - 1)}. \quad (\text{I.14.3})$$

In particular $M \geq 2$ under these hypotheses.

Proof. Apply Theorem I.14.1 to the random variable taking value x_j with probability w_j . Its T is at least T_0 , and the bound in (I.14.2) is increasing in T . Since $T > 0$ and the mean is zero, at least one cell has negative average. There are therefore at most $M - 1$ cells above t . One of them carries at least $1/(M - 1)$ of their total mass. Also $T_0 \leq ab/(a + b) < a$, so the displayed denominators are positive. $\square$

For $f = 1_A - \delta$ take $a = \delta$, $b = 1 - \delta$. The resulting formula retains the density throughout the extraction. For the symmetric setting of Gowers's Lemma 5.15, put $a = b = 1$ and $2T_0 = \alpha$. At the same requested increment $t = \alpha/4$, it gives

$$\frac{|P_j|}{N} \geq \frac{\alpha(2 + \alpha)}{2(4 - \alpha)(M - 1)}, \quad \mathbb{E}(f \mid P_j) > \alpha/4, \quad (\text{I.14.4})$$

for $0 < \alpha \leq 1$. This improves the historical $\alpha/(4M)$ guarantee. More usefully, (I.14.3) allows the required increment and the retained mass to be chosen together. The exactness claim applies to the total mass formula; dividing by the number of eligible cells is a separate averaging step.

[write] **6 October 2026.** (I.14.4) is source 02's Corollary I.9.3; the comparison with the formal lemma_5_15 is in the note after it.

I.14.1 Phase removal with the actual error budget

Lemma I.14.3 (Partition phase transfer). *Let f be an integrable complex function and let P_j partition a finite probability space. On each cell let ψ_j be a phase, with $|\psi_j(x)| = 1$, and choose a constant z_j of modulus one. Put*

$$\rho = \sum_j \left| \int_{P_j} f \psi_j d\mu \right|, \quad E_{\text{osc}} = \sum_j \int_{P_j} |f| |\psi_j - z_j| d\mu.$$

Then

$$\sum_j \left| \int_{P_j} f d\mu \right| \geq \rho - E_{\text{osc}}. \quad (\text{I.14.5})$$

If f is real and mean zero, the total positive cell discrepancy is at least $(\rho - E_{\text{osc}})/2$.

Proof. For each j , subtract $z_j \int_{P_j} f$ from $\int_{P_j} f \psi_j$ and use the triangle inequality. Sum the resulting inequalities. For real f of mean zero, the positive and negative cell discrepancies have equal total mass. $\square$

This formulation allows different phases and different oscillations on different cells. Cells on which $|f|$ has small mass may tolerate a much larger phase error. If all oscillations are at most ε , then $E_{\text{osc}} \leq \varepsilon \mathbb{E}|f|$. In particular,

$$\mathbb{E}|1_A - \delta| = 2\delta(1 - \delta). \quad (\text{I.14.6})$$

For a weight $g: \Omega \rightarrow [0, 1]$ of mean δ , the same expression is an upper bound for $\mathbb{E}|g - \delta|$: pointwise, $(g - \delta)_+ \leq (1 - \delta)g$. Discarding this factor and using only $|f| \leq 1$ loses useful information, especially for small δ .

I.15 A density-sensitive Fourier increment on proper progressions

Source 03, delivered Section 4.

We give an independent progression construction, so the constants below do not depend on any unrounded version of the partition lemma in the source. All progressions in this section are ordinary progressions of integers inside $\{0, \dots, N - 1\}$, with positive integer step.

Lemma I.15.1 (Elementary phase flattening). *Let $N \geq 1$, $\theta \in \mathbb{R}$, and $0 < \varepsilon \leq 1$. Set*

$$L = \max \left\{ 1, \left\lfloor \sqrt{\frac{\varepsilon N}{4\pi}} \right\rfloor \right\}.$$

There is a partition of $\{0, \dots, N - 1\}$ into proper integer arithmetic progressions, each of length between L and $2L - 1$, on each of which $e(\theta x) = \exp(2\pi i \theta x)$ differs from a fixed unit complex number by at most ε .

Proof. If $L = 1$, take singletons. Otherwise set $Q = \lfloor N/L \rfloor$. The circular gaps between the $Q + 1$ points $0, \theta, \dots, Q\theta$ modulo one give an integer q with $1 \leq q \leq Q$ and

$$\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{Q + 1}.$$

This remains true if some points coincide. Split the integer interval into residue classes modulo q . Every class has at least $\lfloor N/q \rfloor \geq L$ elements. A chain of $s \geq L$ elements can be split into consecutive blocks of sizes in $[L, 2L - 1]$: write $s = hL + r$, $0 \leq r < L$, and use $h - 1$ blocks of size L and one of size $L + r$. Within any such block, the phase difference from its first point is bounded by

$$2\pi(2L - 2)\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{4\pi(L - 1)}{Q + 1} < \frac{4\pi L^2}{N} \leq \varepsilon.$$

Here $|e(u) - e(v)| \leq 2\pi\|u - v\|_{\mathbb{R}/\mathbb{Z}}$ was used together with the triangle inequality on the circle. The blocks inherit the positive step q and remain inside the original interval. $\square$

Theorem I.15.2 (Fourier increment with density retained). *Let $A \subset \mathbb{Z}/N\mathbb{Z}$ have density $0 < \delta < 1$. Suppose a nontrivial character has normalized Fourier coefficient of magnitude at least $\rho > 0$ on 1_A . For any $0 < \lambda < 1$, there is a proper integer arithmetic progression $P \subset \{0, \dots, N - 1\}$ such that*

$$|P| \geq \max \left\{ 1, \left\lfloor \sqrt{\frac{\lambda \rho N}{8\pi\delta(1 - \delta)}} \right\rfloor \right\}, \quad (\text{I.15.1})$$

$$\frac{|A \cap P|}{|P|} \geq \delta + \frac{\delta(1 - \lambda)\rho}{2\delta - (1 - \lambda)\rho}. \quad (\text{I.15.2})$$

No primality assumption on N is needed.

Proof. Let $f = 1_A - \delta$. Orthogonality of the nontrivial character implies $\rho \leq \min\{\delta, 1 - \delta\}$ and $|\mathbb{E}_x f(x)e(-rx/N)| \geq \rho$ for the relevant r . Set

$$\varepsilon = \frac{\lambda\rho}{2\delta(1-\delta)}.$$

Since $\max\{\delta, 1 - \delta\} \geq 1/2$, this satisfies $0 < \varepsilon \leq \lambda < 1$. Apply Lemma I.15.1 to $\theta = -r/N$. The triangle inequality over cells gives total cellwise phase correlation at least ρ . Equations (I.14.5) and (I.14.6) therefore give

$$\sum_P \mu(P) |\mathbb{E}(f \mid P)| \geq (1 - \lambda)\rho.$$

The positive discrepancy is at least $T_0 = (1 - \lambda)\rho/2$. Apply (I.14.1) with $a = \delta$, $b = 1 - \delta$ to obtain a cell of average at least $\delta T_0/(\delta - T_0)$. Every cell has the lower length bound (I.15.1), proving both conclusions. The denominator is positive because $\rho \leq \delta$ and $\lambda > 0$. $\square$

The balanced choice $\lambda = 1/2$ gives the particularly simple bounds

$$|P| \geq \max \left\{ 1, \left\lfloor \sqrt{\frac{\rho N}{16\pi\delta(1-\delta)}} \right\rfloor \right\}, \quad \frac{|A \cap P|}{|P|} \geq \delta + \frac{\delta\rho}{4\delta - \rho}. \quad (\text{I.15.3})$$

The increment is at least $\rho/4$. The length keeps the factor $[\delta(1-\delta)]^{-1/2}$, and all finite rounding is explicit. For example, if $\rho = \kappa\delta$ with $0 < \kappa \leq 1$ and $\delta \leq 1/2$, the lower length scale is $\sqrt{\kappa N/[16\pi(1-\delta)]}$, while the density grows by the relative factor $1 + \kappa/(4 - \kappa)$. This observation is conditional on a Fourier coefficient of that relative size; it is not an assertion that every sparse set has one.

Remark I.15.3 (What was already in the repository). The pinned implementation of Corollary 2.5 already obtains an increment $\rho/4$ and length $\sqrt{\rho N/(128\pi)}$ when $\rho N \geq 4\pi$, before weakening these to its public conclusion. Thus the $\rho/4$ constant alone is not a contribution of this article. The additional conclusions here are the asymmetric selection formula, the density factor in the progression length, the adjustable loss λ , and an independent construction valid at every finite scale.

[write] **6 October 2026.** Checked at HEAD (6 October 2026): unchanged. The private `cor25_large_scale` is at `Proofs02DensityIncrement.lean`, line 375, and the exported `corollary_2_5_holds` at line 517; the file's blob (85b5c3a) is the one at both pins. *Relation to the other sources.* At $\lambda = 1/2$, (I.15.3) is the simplified form (I.17.8) of source 05's Theorem I.17.4: the same length bound and the same increment, by a different construction (a chordal error bound in place of an arc width and a disk centre). Source 05's exact integer length is asymptotically $\sqrt{2}$ times longer, and source 01's increment (Theorem I.6.1) is at least $3/2$ times this one (note after Table 3).

I.16 Exact finite partitions and the sharp rounding constant

Source 05, delivered Section 2.

I.16.1 Notation and two elementary tools

Set $I_N = \{0, 1, \dots, N - 1\}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. An arithmetic progression is

$$P = \{a, a + q, \dots, a + (\ell - 1)q\} \subset I_N, \quad q \geq 1, \quad \ell \geq 1.$$

Its length is ℓ . All progressions considered here are proper as subsets of the integers, not just as cyclic sequences. An arc of width ε in $\mathbb{T}$ is the image of a real interval of length ε . Arc containment is a little stronger than a bound on pairwise circular distances, and it is the condition used throughout.

Lemma I.16.1 (Circular pigeonholing). *For every $\theta \in \mathbb{R}$ and integer $Q \geq 1$, there is an integer q with*

$$1 \leq q \leq Q, \quad \|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{Q+1}.$$

Proof. Place $0, \theta, \dots, Q\theta$ on the circle. A repeated point gives a zero return. Otherwise, the $Q+1$ cyclic gaps have total length one, so a smallest gap has length at most $1/(Q+1)$. The difference of the two corresponding indices has absolute value between one and Q , and the circular distance of its multiple of θ is no larger than that gap. $\square$

Lemma I.16.2 (Exact residue-class blocking). *Let N, q, L be positive integers with $qL \leq N$. The interval I_N can be partitioned into progressions of common difference q , each of length between L and $2L-1$. The number of cells is at most $\lfloor N/L \rfloor$.*

Proof. Each residue class modulo q has at least L elements. If one class has $n = aL + b$ elements with $a \geq 1$ and $0 \leq b < L$, take $a-1$ consecutive blocks of length L and one final block of length $L+b$. The resulting cells have the asserted lengths. Since the cells are disjoint and each has at least L points, their number is at most N/L . $\square$

Proposition I.16.3 (Flexible one-phase partition). *Let $N \geq 1$, $0 < \varepsilon \leq 1$, and let $L \geq 1$ be an integer such that*

$$2L(L-1) \leq \varepsilon N. \tag{I.16.1}$$

Then I_N has a partition into progressions of one common positive difference, all with lengths in $[L, 2L-1]$, on each of which $\theta x + \beta$ lies in an arc of width at most ε . This holds for every real θ, β .

Proof. Condition (I.16.1), with $\varepsilon \leq 1$, implies $L \leq N$. Put $Q = \lfloor N/L \rfloor$ and choose q using Lemma I.16.1. Then $qL \leq N$. Let $d \in [-1/2, 1/2]$ be a representative of $q\theta$ with $|d| \leq 1/(Q+1)$. Apply Lemma I.16.2. On a cell of length $\ell \leq 2L-1$, the phase has a real lift whose span is

$$(\ell-1)|d| \leq \frac{2L-2}{Q+1} \leq \frac{2L(L-1)}{N} \leq \varepsilon.$$

The middle inequality uses $Q+1 > N/L$; allowing a weak inequality also covers $L=1$. $\square$

The largest permitted integer in this proposition is

$$L_*(N, \varepsilon) = \left\lfloor \frac{1 + \sqrt{1 + 2\varepsilon N}}{2} \right\rfloor. \tag{I.16.2}$$

There is no exceptional small-scale regime: $L_* = 1$ automatically covers it. With $u = \varepsilon N/2$, maximality gives $u < L_*(L_*+1) \leq 2L_*^2$, and hence

$$L_* \geq \sqrt{\varepsilon N/4}, \quad L_* > \sqrt{\varepsilon N/2} - \frac{1}{2}. \tag{I.16.3}$$

The exact integer value, rather than either weakened estimate, is preferable in implementations.

I.16.2 The optimal constant with the original upper bound

Theorem I.16.4 (Sharp capped partition). *Suppose $0 < \varepsilon \leq 1$ and $\varepsilon N \geq 1$. Every real linear phase on I_N admits a partition into arithmetic progressions, with a common positive difference, such that each image lies in an arc of width at most ε and each cell satisfies*

$$\frac{1}{3}\sqrt{\varepsilon N} \leq |P| \leq \sqrt{\varepsilon N}. \quad (\text{I.16.4})$$

The lower constant $1/3$ is the largest possible universal constant if the upper constant remains one, even if different cells are allowed different common differences.

Consequently, under the hypotheses of the corrected Gowers Lemma 2.3, its lower bound can be improved from $\sqrt{rs/(16M)}$ to

$$\sqrt{rs/(9M)}, \quad (\text{I.16.5})$$

with the same image-diameter bound s and the same upper length bound $\sqrt{rs/M}$.

Proof. Write $t = \varepsilon N$, $U = \lfloor \sqrt{t} \rfloor$, and $K = \lceil U/2 \rceil$. Then $K \geq 1$ and

$$2K(K-1) \leq U^2 \leq t, \quad 2K-1 \leq U.$$

Proposition I.16.3, with $L = K$, gives cells of lengths between K and $2K-1$, and hence at most $\sqrt{t}$. Also

$$\sqrt{t} < U + 1 \leq 2K + 1 \leq 3K.$$

Thus every cell has length strictly greater than $\sqrt{t}/3$, which implies the stated weak bound.

For sharpness, take $N = 9$, a constant phase, and $\varepsilon = 1 - 1/M$ with $M \geq 10$. Then $t = 9(1 - 1/M) < 9$ and $t \rightarrow 9$ as $M \rightarrow \infty$. A putative lower constant $c > 1/3$ would eventually require every cell to have at least two points, while the upper bound $\sqrt{t} < 3$ forces every cell to have exactly two points. Nine points cannot be partitioned into pairs. This argument uses cardinalities alone and therefore applies to arbitrary allowed cell shapes.

For the modular specialization, use $N = r$, $\varepsilon = s/M$, and slope a/M for an affine map $x \mapsto ax + b \pmod{M}$. The constructed lift has integral span after multiplication by M , so its span at most s is exactly the required discrete diameter bound. The sharpness example becomes $r = 9$, $s = M - 1$, and constant affine map; it respects $r, s \leq M$ and $rs \geq M$. $\square$

Remark I.16.5. The improvement $16 \rightarrow 9$ in the denominator is an actual strengthening of the corrected statement, not a change of upper-bound convention. By contrast, using L_* in (I.16.2) gives a larger minimum but permits a larger maximum, approximately $\sqrt{2\varepsilon N}$. The two formulations serve different purposes. The sharpness of $1/3$ concerns the capped formulation only.

[write] **6 October 2026.** *Checked against the formal statement at HEAD. lemma_2_3 (Sections01_03.lean, line 141; exact companion lemma_2_3_holds, Proofs02Partition.lean, line 561) assumes $0 < N$, $0 < r$, $0 < s$, $r \leq N$, $s \leq N$, $N \leq rs$ and an affine map $x \mapsto ax + b$ from $\{0, \dots, r-1\}$ to $\mathbb{Z}_N$. It concludes a partition into proper progressions with image diameter at most s (`diameterAtMost`) and $\sqrt{rs/(16N)} \leq |P_j| \leq \sqrt{rs/N}$. Lean's N is the modulus M here. Under exactly these hypotheses $\varepsilon = s/M \leq 1$ and $\varepsilon r = rs/M \geq 1$, so Theorem I.16.4 gives the lower bound $\sqrt{rs/(9M)}$. The nine-point example ($r = 9$, $s = M - 1$, $M \geq 10$, constant map) satisfies them, since $9 \leq M$, $M - 1 \leq M$ and $M \leq 9(M - 1)$. So $1/3$ is optimal for the exact Lean formulation, as the theorem says. The improvement is not formalized.*

I.16.3 A strengthened Corollary 2.4 with the same upper length

Write $e(t) = \exp(2\pi it)$. If the values of $e(\phi(x))$ on a cell are contained in an arc of width ε , replacing the phase by its unit midpoint produces pointwise error at most $\pi\varepsilon$.

Corollary I.16.6. *Let $f : I_r \rightarrow \mathbb{C}$ satisfy $|f| \leq 1$, let $\phi(x) = \theta x + \beta$ be real linear, and suppose*

$$\left| \sum_{x \in I_r} f(x) e(-\phi(x)) \right| \geq \alpha r > 0, \quad \alpha r \geq 4\pi.$$

There is a partition into J arithmetic progressions for which

$$\sqrt{\alpha r / (36\pi)} \leq |P_j| \leq \sqrt{\alpha r / (4\pi)}, \quad (\text{I.16.6})$$

$$J \leq \sqrt{36\pi r / \alpha}, \quad (\text{I.16.7})$$

$$\sum_{j=1}^J \left| \sum_{x \in P_j} f(x) \right| \geq \frac{3}{4} \alpha r. \quad (\text{I.16.8})$$

Proof. We have $\alpha \leq 1$. Apply Theorem I.16.4 with $N = r$ and $\varepsilon = \alpha/(4\pi)$. The length and count bounds follow. On every cell choose a unit midpoint z_j for the conjugated phase arc. Then

$$\alpha r \leq \sum_j \left| \sum_{x \in P_j} f(x) \right| + \frac{\alpha}{4} \sum_x |f(x)| \leq \sum_j \left| \sum_{x \in P_j} f(x) \right| + \frac{\alpha}{4} r.$$

□

Thus the corrected Corollary 2.4 constants 128π and $\alpha/2$ can be replaced by 36π and $3\alpha/4$, respectively, without changing the upper length bound. Neither a prime modulus nor a rational slope is needed. This statement does not assert optimality of the new correlation constant or of 36π .

[write] **6 October 2026.** Compared with `corollary_2_4` (`Sections01_03.lean`, line 158; exact companion `corollary_2_4_holds`, `Proofs02PhasePartition.lean`, line 276), which assumes $r \leq N$, $4\pi \leq \alpha r$ and a linear phase on $\mathbb{Z}_N$, and gives lengths in $[\sqrt{\alpha r / (128\pi)}, \sqrt{\alpha r / (4\pi)}]$, at most $\sqrt{128\pi r / \alpha}$ cells and retained correlation $(\alpha/2)r$. The corollary above keeps the upper length, improves 128π to 36π and $\alpha/2$ to $3\alpha/4$, and drops $r \leq N$, allowing any real linear phase (checked at the write against the Lean statement; not formalized).

I.17 Density-sensitive localization and popular increments

Source 05, delivered Section 3.

I.17.1 Optimal disk geometry for an arc

Lemma I.17.1 (The center need not lie on the unit circle). *Suppose a subset of the unit circle is contained in an arc of width $\varepsilon < 1/2$, with midpoint $e(\gamma)$. Set $a = \pi\varepsilon$. The disk with center $c = \cos(a)e(\gamma)$ and radius $\sin(a)$ contains the arc. No smaller disk contains both endpoints of the full arc.*

Proof. Rotate to $\gamma = 0$. For $|u| \leq a$, with $a < \pi/2$,

$$|e^{iu} - \cos a|^2 = 1 + \cos^2 a - 2 \cos a \cos u \leq \sin^2 a.$$

The endpoints have distance $2 \sin a$, giving the matching radius lower bound. $\square$

All expectations below are normalized by the total number of points. For a partition $\mathcal{P}$, let $g = \mathbb{E}(f \mid \mathcal{P})$ denote the function constant on each cell and equal to its average there.

Proposition I.17.2 (Correlation transfer with an explicit error budget). *Suppose ψ has modulus one and its values on each cell lie in an arc of width $\varepsilon < 1/2$. Let*

$$\alpha \leq |\mathbb{E}f\psi|, \quad B = \mathbb{E}|f|, \quad W = \mathbb{E}|g|.$$

Then

$$W \geq \frac{\alpha - B \sin(\pi\varepsilon)}{\cos(\pi\varepsilon)}. \quad (\text{I.17.1})$$

In particular, $W \geq \alpha - \pi\varepsilon B$ whenever this latter expression is nonnegative.

Proof. Choose the center c_P from Lemma I.17.1 on each cell. With $w_P = |P|/N$,

$$|\mathbb{E}f\psi| \leq \sum_P w_P |c_P| |\mathbb{E}_P f| + \sin(\pi\varepsilon) \mathbb{E}|f| = \cos(\pi\varepsilon)W + \sin(\pi\varepsilon)B.$$

Rearrange. The simpler bound follows from $\sin u \leq u$ and $\cos u \leq 1$. $\square$

The dependence on B is essential for sparse sets. For $f = \mathbf{1}_A - \delta$, where $\delta = |A|/N$, one has exactly

$$\mathbb{E}f = 0, \quad -\delta \leq f \leq 1 - \delta, \quad \mathbb{E}|f| = 2\delta(1 - \delta) = 2v. \quad (\text{I.17.2})$$

Bounding $\mathbb{E}|f|$ by one discards information of order δ when δ is small.

I.17.2 A sharp positive-mass conversion

Lemma I.17.3 (Capacity-sensitive extraction). *Let g be real on a finite probability space, with $\mathbb{E}g = 0$ and $-\delta \leq g \leq b = 1 - \delta$, where $0 < \delta < 1$. Put $S = \mathbb{E}g_+ = \mathbb{E}(-g)_+$. If $0 < s \leq S$, then*

$$\max g \geq \eta(s) := \frac{\delta s}{\delta - s}. \quad (\text{I.17.3})$$

For $0 < t < \eta(s)$,

$$\Pr(g \geq t) \geq \frac{s(1 + t/\delta) - t}{b - t}. \quad (\text{I.17.4})$$

In particular,

$$\Pr(g \geq \eta(s)/2) \geq \frac{s}{2(b - \eta(s)/2)}. \quad (\text{I.17.5})$$

The bound (I.17.3) is sharp as a consequence of the stated data alone.

Proof. Let $M = \max g$ and $p = \Pr(g > 0)$. Then $S \leq Mp$ and $S \leq \delta(1 - p)$. Consequently $S/M + S/\delta \leq 1$, whence $M \geq \delta S/(\delta - S)$. The right-hand side is increasing in S . Here $S \leq \delta(1 - \delta) < \delta$, so all denominators are positive.

For popularity, write $w = \Pr(g \geq t)$. The positive set has mass at most $1 - S/\delta$, and hence

$$S \leq t(1 - S/\delta) + (b - t)w.$$

Rearranging and using $S \geq s$ gives (I.17.4). Setting $t = \eta(s)/2$ gives (I.17.5). Finally, for $s \leq \delta(1 - \delta)$, the two-point distribution taking values $-\delta$ and $\eta(s)$ with probabilities s/δ and $1 - s/\delta$ has mean zero and positive mass exactly s . It attains (I.17.3). $\square$

[write] **6 October 2026.** (I.17.3) and (I.17.4) are the case $a = \delta$, $b = 1 - \delta$ of the sharp bound printed first as source 02's Theorem I.9.1, here with the event $g \geq t$ in place of $g > t$; since $\Pr(g \geq t) \geq \Pr(g > t)$, this form is weaker. Credited to sources 01, 02, 03 and 05 (note after source 01's Theorem I.4.1). The popularity bound (I.17.5) is source 05's.

I.17.3 A strengthened Fourier density-increment theorem

We normalize the Fourier transform on $\mathbb{Z}/N\mathbb{Z}$ by

$$\widehat{h}(r) = \frac{1}{N} \sum_{x=0}^{N-1} h(x)e(-rx/N).$$

The repository uses an unnormalized transform; its hypothesis $|\widehat{A}(r)| \geq \alpha N$ is our hypothesis $|\widehat{\mathbf{1}}_A(r)| \geq \alpha$.

Theorem I.17.4 (Density-sensitive Fourier increment). *Let $A \subset \mathbb{Z}/N\mathbb{Z}$ have density $0 < \delta < 1$. Suppose that for a nonzero frequency r , $|\widehat{\mathbf{1}}_A(r)| \geq \alpha > 0$. Put*

$$v = \delta(1 - \delta), \quad \varepsilon_0 = \frac{\alpha}{4\pi v}, \quad L = \left\lceil \frac{1 + \sqrt{1 + \alpha N/(2\pi v)}}{2} \right\rceil, \quad \eta = \frac{\delta\alpha}{4\delta - \alpha}. \quad (\text{I.17.6})$$

There is a partition of the standard representatives I_N into progressions of a common positive difference, all of lengths in $[L, 2L - 1]$, with the following properties:

- (i) *Some cell P has density at least $\delta + \eta$.*
- (ii) *The union of cells having density at least $\delta + \eta/2$ has size at least*

$$\frac{\alpha}{8(1 - \delta - \eta/2)} N. \quad (\text{I.17.7})$$

In particular, a progression exists with

$$|P| \geq \sqrt{\frac{\alpha N}{16\pi\delta(1 - \delta)}}, \quad \frac{|A \cap P|}{|P|} \geq \delta + \frac{\delta\alpha}{4\delta - \alpha} \geq \delta + \frac{\alpha}{4}. \quad (\text{I.17.8})$$

No primality assumption or lower-scale hypothesis is needed.

Proof. The sum of a nontrivial character on a finite cyclic group is zero, even when N is composite. Therefore

$$\alpha \leq \min(\delta, 1 - \delta), \quad \widehat{\mathbf{1}_A - \delta}(r) = \widehat{\mathbf{1}_A}(r).$$

In particular $\alpha/v \leq 2$, so $0 < \varepsilon_0 \leq 1/(2\pi) < 1/2$. Proposition I.16.3 gives the partition with $L = L_*(N, \varepsilon_0)$. Apply Proposition I.17.2 to $f = \mathbf{1}_A - \delta$ and the conjugated character. By (I.17.2),

$$W \geq \alpha - \pi\varepsilon_0(2v) = \alpha/2.$$

The block averages g are real and mean zero, so $\mathbb{E}g_+ = W/2 \geq \alpha/4$. Lemma I.17.3, with $s = \alpha/4$, proves (i) and (ii). Finally (I.16.3) gives $L \geq \sqrt{\varepsilon_0 N}/4$, which is exactly the first bound in (I.17.8). Since $4\delta - \alpha \leq 4\delta$, the increment is at least $\alpha/4$. $\square$

Remark I.17.5 (There is no hidden singleton exception). When the input scale is small, the formula gives $L = 1$ and the construction uses singletons. The theorem remains valid: all estimates follow from the same proof. In particular, the upper bound $\alpha \leq 1 - \delta$ prevents an impossible required density above one.

I.17.4 The full oscillation–increment trade-off

The preceding choice of ε_0 is convenient rather than optimal. For any $0 < \varepsilon < 1/2$ with $2v \sin(\pi\varepsilon) < \alpha$, the same argument gives lengths at least $L_*(N, \varepsilon)$ and positive mass at least

$$s(\varepsilon) = \frac{\alpha - 2v \sin(\pi\varepsilon)}{2 \cos(\pi\varepsilon)}. \quad (\text{I.17.9})$$

The corresponding increment is $\delta s(\varepsilon)/(\delta - s(\varepsilon))$, and the popularity formula (I.17.4) remains available. Thus length and increment can be optimized against each other at the final finite step.

For example, using

$$\varepsilon = \frac{1}{\pi} \arcsin\left(\frac{\alpha}{4v}\right)$$

gives $s(\varepsilon) = \alpha/[4\sqrt{1 - (\alpha/(4v))^2}]$, while also giving an arc width at least ε_0 . This mildly improves both quantities in Theorem I.17.4. The gain is only a constant-factor refinement; it should not be confused with the density factor $v^{-1/2}$ in the progression length.

The proof also extends to any weight $a : I_N \rightarrow [0, 1]$ of average δ , since $\mathbb{E}|a - \delta| \leq 2\delta(1 - \delta)$. The latter follows pointwise from $|u - \delta| \leq u + \delta - 2u\delta$ for $0 \leq u \leq 1$. The interpretation is then a local increase of the average weight, not of the cardinality of a subset.

I.17.5 Comparison with the actual repository baseline

For large scales, the ratio of the exact new minimum length to the stronger length already present inside the repository proof is asymptotically

$$\frac{\sqrt{\alpha N/(8\pi v)}}{\sqrt{\alpha N/(128\pi)}} = \frac{4}{\sqrt{v}}. \quad (\text{I.17.10})$$

Even the uniform square-root bound in (I.17.8) improves that baseline by a factor $\sqrt{8/v}$. These are local length comparisons under the same Fourier hypothesis.

δ	Repository statement	Repository proof	New exact L	Length ratio
0.5	6233.47	24933.89	199471	8.00
0.1	49.87	4986.78	66490	13.33
0.01	0.0499	498.68	20048	40.20
0.001	0.0000499	49.87	6311	126.55

Table 4: The statement uses $\sqrt{\alpha^3 N/(128\pi)}$; the proof uses $\sqrt{\alpha N/(128\pi)}$. The last column compares the new exact integer bound with the proof-level expression.

Table 4 evaluates the formulas at $N = 10^{12}$ and $\alpha = \delta^2$. The numbers are conditional bounds, not a claim that a particular set with those parameters was constructed. Values below one in the older expression are simply vacuous real lower bounds on an integer length.

[write] 6 October 2026. The first row was recomputed at the write: 6233.47, 24933.89, $\lfloor (1 + \sqrt{1 + \alpha N/(2\pi v)})/2 \rfloor = 199471$ and ratio 8.00; the intake recomputed all four rows. Source 01's Theorem I.6.1 gives, under the same hypothesis, an increment at least $3/2$ times $\delta\alpha/(4\delta - \alpha)$ at a length that is not asymptotically shorter (note after Table 3); Theorem I.17.4 keeps its popularity bound (ii), its exact integer length and the trade-off (I.17.9).

I.18 Simultaneous phases and an optimal exponent

Source 05, delivered Section 4.

I.18.1 An anisotropic integer formulation

Sequentially applying a square-root localization m times can leave an exponent 2^{-m} . A simultaneous pigeonhole argument instead gives $1/(m+1)$. This is the elementary linear-phase analogue of the simultaneous localization problem that appears in more advanced density-increment arguments.

Lemma I.18.1 (Rectangular pigeonholing). *For real $\theta_1, \dots, \theta_m$ and positive integers $Q_1, \dots, Q_m$, there exists*

$$1 \leq q \leq \prod_{i=1}^m Q_i, \quad \|q\theta_i\|_{\mathbb{R}/\mathbb{Z}} < 1/Q_i \quad (1 \leq i \leq m).$$

Proof. Partition $[0, 1)^m$ into $\prod_i Q_i$ half-open rectangular boxes of side lengths $1/Q_i$. Two of the $\prod_i Q_i + 1$ points $j(\theta_1, \dots, \theta_m) \pmod{1}$ occupy the same box. Their index difference is the required q . $\square$

Theorem I.18.2 (Simultaneous linear-phase partition). *Let $N \geq 1$, $m \geq 1$, and $0 < \varepsilon_i \leq 1$. Define*

$$Q_i(L) = \max \left(1, \left\lceil \frac{2(L-1)}{\varepsilon_i} \right\rceil \right). \quad (\text{I.18.1})$$

For every positive integer L such that

$$L \prod_{i=1}^m Q_i(L) \leq N, \quad (\text{I.18.2})$$

there is a partition of I_N into progressions with one common positive difference, all of lengths in $[L, 2L - 1]$, such that every phase $\theta_i x + \beta_i$ lies in an arc of width ε_i on every cell. The number of cells is at most $\lfloor N/L \rfloor$.

In particular, one may always take

$$L = \max \left(1, \left\lfloor \left(\frac{N \prod_i \varepsilon_i}{2^m} \right)^{1/(m+1)} \right\rfloor \right). \quad (\text{I.18.3})$$

Proof. For $L = 1$ use singletons. Otherwise apply Lemma I.18.1 with $Q_i = Q_i(L)$. The resulting return q satisfies $qL \leq N$, so Lemma I.16.2 applies. Along a cell, a real lift of phase i has span at most $(2L - 2)/Q_i \leq \varepsilon_i$.

For the explicit bound, if the floor in (I.18.3) is at least two, then

$$Q_i(L) \leq \frac{2(L - 1)}{\varepsilon_i} + 1 \leq \frac{2L}{\varepsilon_i},$$

and hence $L \prod_i Q_i(L) \leq 2^m L^{m+1} / \prod_i \varepsilon_i \leq N$. The cases with $L = 1$ were handled separately. $\square$

The largest integer satisfying (I.18.2) is often appreciably better than (I.18.3). Its feasibility predicate is monotone, so it can be found by binary search. The theorem has no discarded exceptional set and gives a lower bound for *every* cell, rather than merely for the average length.

[write] **6 October 2026.** The same construction as source 01's Theorem I.5.1 and Corollary I.5.2 (note there); credited to both.

I.18.2 An explicit badly approximable vector

The optimality statement will be proved from scratch. This avoids importing a quantitative transference theorem with hidden constants.

Lemma I.18.3 (An algebraic obstruction). *Let $m \geq 1$, $\xi = 2^{1/(m+1)}$, and $\theta = (\xi, \xi^2, \dots, \xi^m)$. Then, for every integer $q \geq 1$,*

$$\max_{1 \leq i \leq m} \|q\xi^i\|_{\mathbb{R}/\mathbb{Z}} \geq c_m q^{-1/m}, \quad c_m = \frac{1}{2^{m+1} m (5m)^m}. \quad (\text{I.18.4})$$

Proof. Choose nearest integers p_i to $q\xi^i$ and put $h = \max_i |q\xi^i - p_i|$, $Q = \lceil q^{1/m} \rceil$. The $(Q + 1)^m > q$ vectors in $\{0, \dots, Q\}^m$ cannot give distinct residues of $\sum_i u_i p_i$ modulo q . Subtract two vectors with the same residue. This gives integers a_i , not all zero, with $|a_i| \leq Q$ and $q \mid \sum_i a_i p_i$. Define

$$a_0 = -q^{-1} \sum_i a_i p_i, \quad P(X) = a_0 + \sum_{i=1}^m a_i X^i.$$

Then P is a nonzero integer polynomial of degree at most m and

$$|P(\xi)| \leq \frac{mQh}{q}. \quad (\text{I.18.5})$$

Since $\xi^i < 2$ and $|p_i/q - \xi^i| \leq 1/(2q)$, we have $|a_0| \leq (5/2)mQ$.

The polynomial $X^{m+1} - 2$ is irreducible by Eisenstein's criterion at 2. Its roots are $\xi\zeta^j$, $0 \leq j \leq m$, where $\zeta = e(1/(m+1))$. The product

$$\prod_{j=0}^m P(\xi\zeta^j)$$

is a nonzero integer: it is the algebraic norm of the algebraic integer $P(\xi)$, equivalently an integer resultant. For $j \neq 0$,

$$|P(\xi\zeta^j)| \leq |a_0| + \sum_i |a_i| \xi^i \leq 5mQ.$$

Therefore $|P(\xi)| \geq (5mQ)^{-m}$. Combining with (I.18.5) gives $h \geq q/[m(5m)^m Q^{m+1}]$. Finally $Q \leq 2q^{1/m}$ yields (I.18.4). $\square$

Lemma I.18.4 (Unwrapping a flat progression). *Suppose $0 < \varepsilon < 1/2$ and the points $\beta, \beta + \theta, \dots, \beta + (\ell - 1)\theta$ lie in an arc of width ε , with $\ell \geq 2$. Then*

$$(\ell - 1) \|\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \varepsilon.$$

Proof. Choose the unique representatives of the points in a fixed real interval of length ε . Consecutive differences belong to $[-\varepsilon, \varepsilon]$ and are all congruent to θ modulo one. Since $2\varepsilon < 1$, they are the same real number. Thus the lifts form a real arithmetic progression of span $(\ell - 1) \|\theta\|_{\mathbb{R}/\mathbb{Z}}$, which is at most ε . $\square$

I.18.3 The sharp-order finite localization problem

For a vector $\theta \in \mathbb{R}^m$, let $F(\theta; N, \varepsilon)$ be the largest possible minimum cell length among all partitions of I_N into arithmetic progressions on which every $\theta_i x$ lies in an arc of width ε . Different cells may use different differences. Define

$$F_m(N, \varepsilon) = \inf_{\theta \in \mathbb{R}^m} F(\theta; N, \varepsilon).$$

This is a precise extremal formulation introduced for this note; we do not present it as a previously catalogued open problem.

Theorem I.18.5 (Optimal simultaneous localization exponent). *For $N \geq 1$ and $0 < \varepsilon < 1/2$,*

$$F_m(N, \varepsilon) \geq \max \left(1, \left\lfloor (2^{-m} \varepsilon^m N)^{1/(m+1)} \right\rfloor \right), \quad (\text{I.18.6})$$

$$F_m(N, \varepsilon) \leq 1 + (\varepsilon/c_m)^{m/(m+1)} (N - 1)^{1/(m+1)}. \quad (\text{I.18.7})$$

Consequently

$$F_m(N, \varepsilon) \asymp_m 1 + (\varepsilon^m N)^{1/(m+1)}. \quad (\text{I.18.8})$$

The exponent $1/(m+1)$ cannot be increased, even if one asks only for a single flat progression instead of a partition.

Proof. The lower bound is Theorem I.18.2 with equal widths. For the upper bound, take the explicit algebraic vector in Lemma I.18.3. If a flat progression has length $\ell \geq 2$ and difference q , then

$$q(\ell - 1) \leq N - 1, \quad c_m q^{-1/m} \leq \max_i \|q\theta_i\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{\varepsilon}{\ell - 1}.$$

Rearranging gives $\ell - 1 \leq (\varepsilon/c_m)^{m/(m+1)} (N - 1)^{1/(m+1)}$. This bounds every flat progression for that vector, hence every partition minimum. Singletons satisfy the same bound. The floor and additive one in the two estimates are absorbed into constants depending only on m , proving (I.18.8). $\square$

Remark I.18.6 (Scope of sharpness). This theorem proves sharpness for linear phases and uniform arc containment. It does not prove an optimal progression length for arbitrary density increments, which might be found without flattening a prescribed phase. It also gives a symmetric-width sharpness result, not a complete anisotropic lower-bound theory for independently varying ε_i .

[write] **6 October 2026.** At $m = 1$ source 01's Theorem I.7.2 gives the sharper bound $\ell \leq 1 + \sqrt{3\varepsilon(N-1)}$ with the golden-ratio frequency (note there).

I.19 Spectral energy, residual variance, and unequal tolerances

Source 05, delivered Section 5.

I.19.1 Why several correlations should be combined

A single Fourier coefficient gives an additive increment. A collection carrying energy comparable with δ^2 can instead give an increment comparable with δ . The useful issue here is not the existence of this classical principle, but how much phase precision it needs when the input is sparse.

Let $a : I_N \rightarrow [0, 1]$ have average $\delta \in (0, 1)$, set $f = a - \delta$ and $v = \delta(1 - \delta)$. Let $r_1, \dots, r_m$ be distinct nonzero frequencies, and write

$$\chi_i(x) = e(r_i x/N), \quad c_i = \widehat{f}(r_i), \quad E = \sum_i |c_i|^2 > 0, \quad \Sigma = \sum_i |c_i|. \quad (\text{I.19.1})$$

Zero coefficients may be removed. For a partition $\mathcal{P}$, put

$$a_P = \mathbb{E}(a \mid \mathcal{P}), \quad g = a_P - \delta, \quad V = \mathbb{E}g^2.$$

Orthogonality gives $E \leq \mathbb{E}|f|^2 \leq v$, and conditional expectation gives $0 \leq V \leq v$.

Lemma I.19.1 (Residual mass remembers the captured variance). *On every cell P , $\mathbb{E}_P|a - a_P| \leq 2a_P(1 - a_P)$. Consequently*

$$\mathbb{E}|a - a_P| \leq 2(v - V). \quad (\text{I.19.2})$$

For indicator functions, equality holds.

Proof. For $0 \leq u, b \leq 1$, $|u - b| \leq u + b - 2ub$. Average with $b = a_P = \mathbb{E}_P a$. Averaging over cells gives

$$\mathbb{E}a_P(1 - a_P) = \delta - \mathbb{E}a_P^2 = \delta - (\delta^2 + V) = v - V.$$

For $u \in \{0, 1\}$ the pointwise inequality is an equality. □

Theorem I.19.2 (Density-sensitive spectral extraction). *Suppose each χ_i is contained in an arc of width $\varepsilon_i < 1/2$ on every cell of $\mathcal{P}$, and set*

$$R = \sum_i |c_i| \sin(\pi \varepsilon_i). \quad (\text{I.19.3})$$

Then

$$E \leq \sqrt{EV} + 2R(v - V). \quad (\text{I.19.4})$$

If $R \leq E/(4v)$, then

$$V \geq V_0 := \frac{E}{(1 + \sqrt{1 - E/v})^2} \geq E/4, \quad (\text{I.19.5})$$

and some cell has average at least

$$\delta + V_0/\delta \geq \delta + E/(4\delta). \quad (\text{I.19.6})$$

Proof. Let $h = \sum_i c_i \chi_i$. With the normalized inner product $\langle u, v \rangle = \mathbb{E}(uv)$, orthogonality gives $\langle f, h \rangle = E$ and $\|h\|_2 = \sqrt{E}$. Decompose

$$E = \langle g, h \rangle + \langle a - a_P, h \rangle.$$

The first term has absolute value at most $\sqrt{VE}$. On each cell subtract from χ_i its disk center supplied by Lemma I.17.1. Since $a - a_P$ has zero mean on the cell, the subtraction changes no inner product. The residual term is therefore at most

$$\sum_i |c_i| \sin(\pi \varepsilon_i) \mathbb{E}|a - a_P| \leq 2R(v - V),$$

proving (I.19.4).

Assume $R \leq E/(4v)$. Since $v - V \geq 0$, divide the resulting inequality by E to obtain

$$1 \leq \sqrt{V/E} + \frac{1}{2} - \frac{V}{2v}.$$

With $x = \sqrt{V/E}$ and $\tau = E/v \in (0, 1]$, this is $\tau x^2 - 2x + 1 \leq 0$. Hence

$$x \geq \frac{1 - \sqrt{1 - \tau}}{\tau} = \frac{1}{1 + \sqrt{1 - \tau}},$$

which proves (I.19.5). Finally, if $D = \max_P a_P$, then $a_P^2 \leq D a_P$, and so $\delta^2 + V \leq D\delta$. This yields (I.19.6). $\square$

The conversion $D \geq \delta + V/\delta$ is sharp from the mean and variance alone: a random variable supported on 0 and D has variance $\delta(D - \delta)$. We do not claim that the complete phase-to-variance bound (I.19.5) is globally optimal.

I.19.2 A completely explicit progression corollary

Take equal widths

$$\varepsilon = \frac{E}{4\pi v \Sigma}. \quad (\text{I.19.7})$$

Since $E/\Sigma \leq \max_i |c_i| \leq \min(\delta, 1 - \delta)$, we have $\varepsilon \leq 1/(2\pi) < 1/2$. The inequality $\sin u \leq u$ gives $R \leq E/(4v)$. Apply Theorem I.18.2 with these widths and use the largest feasible L from (I.18.2). Then some ordinary arithmetic progression of length at least L has the increment (I.19.6). A simpler guaranteed choice is

$$L = \max \left(1, \left\lceil \left[\frac{N}{2^m} \left(\frac{E}{4\pi v \Sigma} \right)^m \right]^{1/(m+1)} \right\rceil \right). \quad (\text{I.19.8})$$

If $E \geq c\delta^2$, this gives an increment at least $c\delta/4$. Moreover $\Sigma \leq \sqrt{mE}$, so

$$\varepsilon \geq \frac{\sqrt{E}}{4\pi v\sqrt{m}} \geq \frac{\sqrt{c}}{4\pi(1-\delta)\sqrt{m}}.$$

Thus, under a fixed relative-energy hypothesis and fixed m , the required width need not shrink with δ . This is the density-sensitive feature lost by a generic unit-bound error estimate. It does not assert that every sparse progression-free set has such energy on a fixed-size spectrum.

I.19.3 Optimizing unequal phase widths

The sufficient linearized budget is

$$\sum_i b_i \varepsilon_i \leq B, \quad b_i = |c_i| > 0, \quad B = \frac{E}{4\pi v}. \quad (\text{I.19.9})$$

The guaranteed length in (I.18.3) increases with $\prod_i \varepsilon_i$. Fix a harmless cap $\varepsilon_i \leq 1/4$.

Proposition I.19.3 (Optimal allocation for the continuous bound). *If $B < \frac{1}{4} \sum_i b_i$, the product $\prod_i \varepsilon_i$ subject to (I.19.9) and $0 < \varepsilon_i \leq 1/4$ is maximized by*

$$\varepsilon_i = \min\left(\frac{1}{4}, \frac{\lambda}{b_i}\right), \quad \sum_i \min(b_i/4, \lambda) = B. \quad (\text{I.19.10})$$

If $B \geq \frac{1}{4} \sum_i b_i$, use all widths equal to $1/4$. When no cap is active,

$$\varepsilon_i = \frac{E}{4\pi m v |c_i|}, \quad \prod_i \varepsilon_i = \left(\frac{E}{4\pi m v}\right)^m \prod_i |c_i|^{-1}. \quad (\text{I.19.11})$$

Proof. Put $x_i = b_i \varepsilon_i$. Apart from a constant, the logarithm of the objective is $\sum_i \log x_i$. At a maximum with unused capacity excluded, $\sum_i x_i = B$. If two uncapped coordinates are unequal, moving a sufficiently small amount from the larger to the smaller strictly increases their product, preserves their sum, and respects both caps. Thus all uncapped x_i equal a common λ . A capped coordinate cannot exceed λ , since a small transfer from it to an uncapped coordinate would again improve the product. Consequently $x_i = \min(b_i/4, \lambda)$. Strict concavity of $\sum_i \log x_i$ makes these conditions sufficient as well; monotonicity of the displayed sum gives the unique relevant level. The uncapped formula follows immediately. $\square$

By the arithmetic–geometric mean inequality, the uncapped product improves the equal-width product by the factor

$$\frac{(\Sigma/m)^m}{\prod_i |c_i|} \geq 1.$$

This optimizes the continuous product bound, *not* the discontinuous exact ceiling problem (I.18.2). For finite implementations, evaluate the exact feasibility test after allocation; neighboring rational widths can sometimes do better. Also, adding frequencies increases energy but increases dimension. Choosing the best subset of frequencies remains a separate optimization problem.

I.19.4 An explicit extension to nonorthogonal phases

The orthogonality in Theorem I.19.2 is essential to its constants, but the argument admits a controlled extension. Let $\psi_1, \dots, \psi_m$ be arbitrary functions of modulus one on the finite probability space. Assume that $\kappa > 0$ satisfies

$$\left\| \sum_i z_i \psi_i \right\|_2^2 \leq \kappa \sum_i |z_i|^2 \quad \text{for every } (z_1, \dots, z_m) \in \mathbb{C}^m. \quad (\text{I.19.12})$$

Thus κ is a bound for the largest eigenvalue of the Gram matrix. One may always use $\kappa = m$, though a smaller certified bound is preferable. Set $c_i = \langle f, \psi_i \rangle$ and $E = \sum_i |c_i|^2 > 0$; retain v , V , and the arc-width budget R as before.

Corollary I.19.4 (Gram-controlled spectral extraction). *Under these hypotheses,*

$$E \leq \sqrt{\kappa E V} + 2R(v - V), \quad E \leq \kappa v. \quad (\text{I.19.13})$$

If $R \leq E/(4v)$, some cell has average at least

$$\delta + \frac{E}{\kappa \delta (1 + \sqrt{1 - E/(\kappa v)})^2} \geq \delta + \frac{E}{4\kappa \delta}. \quad (\text{I.19.14})$$

Proof. For $h = \sum_i c_i \psi_i$, the identity $\langle f, h \rangle = E$ still holds without orthogonality, while (I.19.12) gives $\|h\|_2 \leq \sqrt{\kappa E}$. Cauchy-Schwarz first yields $E^2 \leq \|f\|_2^2 \|h\|_2^2 \leq v \kappa E$, proving $E \leq \kappa v$. The same cellwise subtraction used in Theorem I.19.2 proves the first inequality in (I.19.13). Under the stated budget, put $x = \sqrt{\kappa V/E}$ and $\tau = E/(\kappa v)$. The resulting inequality is again $\tau x^2 - 2x + 1 \leq 0$. It gives $V \geq (E/\kappa)/(1 + \sqrt{1 - \tau})^2$. Apply $\max_P a_P \geq \delta + V/\delta$. $\square$

This includes polynomial phases and interval-restricted characters once their arc-flat partitions and a Gram bound are supplied. It does not supply either the required higher-degree partition or a useful small value of κ automatically.

I.20 Average-length partitions and exceptional-cell budgets

Source 05, delivered Section 6.

Higher-order localization statements sometimes control the number of cells rather than the minimum cell length. The following elementary interface keeps that distinction explicit and avoids unnecessarily replacing an average length by its square root.

Proposition I.20.1 (Discarding short cells quantitatively). *Let $a : I_N \rightarrow [0, 1]$ have average $\delta \in (0, 1)$, and let a partition with at most J cells satisfy*

$$\mathbb{E} |\mathbb{E}(a - \delta \mid \mathcal{P})| \geq 2s > 0.$$

For any real threshold $T > 0$ with $(1 - \delta)JT/N < s$, some cell of length at least T has average at least

$$\delta + s - (1 - \delta)JT/N. \quad (\text{I.20.1})$$

In particular, some cell has length at least

$$T = \frac{sN}{2(1 - \delta)J} \quad \text{and average at least } \delta + s/2. \quad (\text{I.20.2})$$

Proof. The total positive block mass is at least s . Cells of length less than T have total normalized size at most JT/N and therefore contribute at most $(1 - \delta)JT/N$ to that positive mass. On the remaining cells the positive mass is at least the difference in (I.20.1). Since their total normalized size is at most one, one remaining cell has positive average at least that difference. $\square$

If $N/J \geq N^\gamma$, this preserves length of order sN^γ , not merely $N^{\gamma/2}$, under the specified correlation hypothesis. This is an extraction statement only. It does not justify replacing short-cell steps inside a geometric or nilpotent factorization proof, where those steps may be needed for reasons other than density extraction.

More generally, Proposition I.17.2 and Lemma I.17.3 apply to any circle-valued phase once a valid arc-flat partition is supplied. The spectral extension to an arbitrary family requires the Gram correction in Corollary I.19.4, or an orthonormal family for the uncorrected theorem. The hard issue for higher degree is supplying the appropriate partition and controlling the Gram matrix; Theorem I.18.5 cannot simply be transferred to nonlinear phases.

Part II

The inverse step: energies, moments, restriction and dependent random choice

Contents. From many correlated derivatives to an approximate homomorphism. Sections II.1–II.4: source 02, delivered Sections 2–5 (a sharp weighted, mixed-function extension of Proposition 6.1 on every finite abelian group, its equality cases, deficits and rigidity). Sections II.5–II.6: source 03, delivered Sections 7–8 (a cubic dependent-random-choice certificate for Lemma 7.4, and an elementary replacement for Proposition 7.3). Sections II.7–II.10: source 04, delivered Sections 2–5 (its notation; exact second moments for Lemma 9.2; a polynomial restriction theorem for Lemma 9.3 and Corollary 9.4 over prime fields; arrangement moments for Lemmas 12.4, 14.7 and 14.8). The three sources treat different steps; no statement is printed twice. Source 04’s Section II.7 also fixes the cube notation used by its Part III sections.

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
$\mathcal{E}$	02: $\mathcal{E}(w)$, additive energy of a weight on $G \times \widehat{G}$, normalized by N^{-3} ; 04: $\mathcal{E}_j(w) = M^{-1} \sum_{\chi} \widehat{w}(\chi) ^{2j}$ with the <i>unnormalized</i> transform, counting $2j$ -tuples; 03: $E(A)$ the additive energy of a set. <i>These are three different normalizations.</i>
$\mathcal{U}, T$	02: $\mathcal{U}(f) = \ f\ _{U^3}^8$ (normalized), $\mathcal{T}_w(f, g)$ a weighted derivative correlation, T its graph case; 04: $T_d(B)$ and $T_d^\phi(B)$ counts of additive and respected $2d$ -tuples.
κ, ρ, m	02: $\kappa = \mathcal{U}(f)/m(f)^4$, $m(f) = \mathbb{E} f ^2$; 03: $\kappa = 7/(3\sqrt{6})$, $\kappa_*, \rho = c/(2-c)$, $m = V $ or $ A $; 04: $\kappa_d = 1/(2(400d)^d)$, $\rho = m/M$ with $m = A $.
R, L, Q	02: $R = \mathbb{E} K_u(s) ^2$; 04: R_j counts respected arrangements, L the kernel bandwidth, Q the kernel ratio sum (R14); 03: $L = D $. Source 04’s tags (M1)–(M7), (R1)–(R32), (A1)–(A14) label displays: <i>(R3) is a display, not a remainder R_3; (A1) is not Appendix A.</i>

II.1 Normalization and finite Fourier identities

Source 02, delivered Section 2.

Let G be a finite abelian group of order $N \geq 1$, written additively, and let $\widehat{G}$ be its character group, written multiplicatively. Thus $(h, \chi) + (k, \psi)$ in $G \times \widehat{G}$ means $(h + k, \chi\psi)$. All averages over G are normalized; sums over $\widehat{G}$ are not:

$$\mathbb{E}_x a(x) = \frac{1}{N} \sum_{x \in G} a(x), \quad \widehat{a}(\chi) = \mathbb{E}_x a(x) \overline{\chi(x)}.$$

Fourier orthogonality gives

$$\mathbb{E}_x |a(x)|^2 = \sum_{\chi \in \widehat{G}} |\widehat{a}(\chi)|^2. \quad (\text{II.1.1})$$

For normalized convolution $(a * b)(s) = \mathbb{E}_h a(s - h)b(h)$, we have $\widehat{a * b} = \widehat{a} \widehat{b}$. These facts follow by expanding the sums and using $\mathbb{E}_x \chi(x) = 0$ for a nontrivial character and 1 for the trivial character.

Define

$$\Delta_h f(x) = f(x) \overline{f(x - h)}, \quad m(f) = \mathbb{E}_x |f(x)|^2,$$

and define the eighth power of the normalized U^3 norm by

$$\mathcal{U}(f) = \mathbb{E}_h \sum_{\chi \in \widehat{G}} |\widehat{\Delta_h f}(\chi)|^4. \quad (\text{II.1.2})$$

It also equals the usual cube average

$$\mathbb{E}_{x,a,b,c} \prod_{\omega \in \{0,1\}^3} \mathcal{C}^{|\omega|} f(x + \omega_1 a + \omega_2 b + \omega_3 c), \quad (\text{II.1.3})$$

where $\mathcal{C}$ denotes complex conjugation. One proof first expands the fourth Fourier moment in (II.1.2); character orthogonality leaves the additive quadruple constraint, which is parameterized by three variables. The remaining derivative variable gives (II.1.3), after changes of signs in the increment variables. In particular $\mathcal{U}(f)$ is nonnegative.

For $f, g : G \rightarrow \mathbb{C}$, write

$$A_{f,g}(h, \chi) = \mathbb{E}_x f(x) \overline{g(x-h)\chi(x)}.$$

This is just a Fourier transform of a mixed derivative, not an additional assumption about f or g .

Definition II.1.1 (Weighted correlation and energy). For any $w : G \times \widehat{G} \rightarrow [0, \infty)$, define

$$\mathcal{T}_w(f, g) = \mathbb{E}_h \sum_{\chi} w(h, \chi) |A_{f,g}(h, \chi)|^2, \quad (\text{II.1.4})$$

$$\mathcal{E}(w) = \frac{1}{N^3} \sum_{\substack{h_1+h_2=h_3+h_4 \\ \chi_1\chi_2=\chi_3\chi_4}} \prod_{j=1}^4 w(h_j, \chi_j). \quad (\text{II.1.5})$$

The factor N^{-3} in (II.1.5) is deliberate: it normalizes the energy of a full additive graph to 1. It is not the uniform-probability normalization on the group of order N^2 . If $w = \mathbf{1}_\Gamma$, then $N^3 \mathcal{E}(w)$ is the ordinary ordered additive-quadruple count of Γ .

II.2 A sharp weighted extension of Proposition 6.1

Source 02, delivered Section 3.

Theorem II.2.1 (Mixed energy-sensitive inverse step). *For every finite abelian group G , every $f, g : G \rightarrow \mathbb{C}$, and every nonnegative weight $w : G \times \widehat{G} \rightarrow [0, \infty)$,*

$$\boxed{\mathcal{T}_w(f, g)^4 \leq \min\{m(f)^4 \mathcal{U}(g), m(g)^4 \mathcal{U}(f)\} \mathcal{E}(w).} \quad (\text{II.2.1})$$

There is no pointwise boundedness assumption on f, g , and no graph or bounded-fiber assumption on w . The coefficient 1 cannot be decreased.

II.2.1 Three exact identities

For $u \in G$ define

$$F_u(s) = f(s) \overline{f(s-u)}, \quad D_u(s) = g(s) \overline{g(s-u)}, \quad V_u(h) = \sum_{\chi} w(h, \chi) \chi(u),$$

and put $K_u = D_u * V_u$. Use the normalized inner product

$$\langle F, K \rangle = \mathbb{E}_{u,s} F_u(s) \overline{K_u(s)}.$$

Then

$$\mathcal{T}_w(f, g) = \langle F, K \rangle, \quad (\text{II.2.2})$$

$$R := \mathbb{E}_{u,s} |K_u(s)|^2 = \mathbb{E}_u \sum_{\psi} |\widehat{D}_u(\psi)|^2 |\widehat{V}_u(\psi)|^2, \quad (\text{II.2.3})$$

$$\mathbb{E}_u \sum_{\psi} |\widehat{V}_u(\psi)|^4 = \mathcal{E}(w). \quad (\text{II.2.4})$$

For (II.2.2), expand the square in (II.1.4), set the second spatial variable equal to $s - u$, and obtain

$$\mathbb{E}_{u,s} f(s) \overline{f(s-u)} \mathbb{E}_h \overline{g(s-h)} g(s-u-h) \sum_{\chi} w(h, \chi) \overline{\chi(u)}.$$

This is exactly $\langle F, K \rangle$. Nonnegativity of w also makes the original expression visibly real and nonnegative. Identity (II.2.3) is Parseval and the convolution identity.

For (II.2.4), expand

$$\widehat{V}_u(\psi) = \frac{1}{N} \sum_{h, \chi} w(h, \chi) \chi(u) \overline{\psi(h)}.$$

The fourth power contributes N^{-4} . Summing over ψ contributes N and imposes $h_1 + h_2 = h_3 + h_4$. Averaging over u imposes $\chi_1 \chi_2 = \chi_3 \chi_4$ without another factor. The remaining normalization is N^{-3} , proving the identity.

II.2.2 Proof of the theorem

The first exact norm is

$$\|F\|_2^2 = \mathbb{E}_{u,s} |f(s)|^2 |f(s-u)|^2 = m(f)^2.$$

Cauchy–Schwarz applied to (II.2.2) therefore gives

$$\mathcal{T}_w(f, g)^2 \leq m(f)^2 R. \quad (\text{II.2.5})$$

Applying Cauchy–Schwarz to (II.2.3), this time on the measure space with average in u and counting measure in ψ , gives

$$R^2 \leq \left(\mathbb{E}_u \sum_{\psi} |\widehat{D}_u(\psi)|^4 \right) \left(\mathbb{E}_u \sum_{\psi} |\widehat{V}_u(\psi)|^4 \right) = \mathcal{U}(g) \mathcal{E}(w). \quad (\text{II.2.6})$$

Squaring (II.2.5) and using (II.2.6) proves the first bound in (II.2.1).

For the other bound, let $w^\vee(h, \chi) = w(-h, \chi^{-1})$. Directly changing variables in the mixed Fourier coefficient gives

$$|A_{f,g}(h, \chi)| = |A_{g,f}(-h, \chi^{-1})|.$$

Consequently $\mathcal{T}_w(f, g) = \mathcal{T}_{w^\vee}(g, f)$. Inversion is an automorphism of the abelian group $G \times \widehat{G}$, so $\mathcal{E}(w^\vee) = \mathcal{E}(w)$. Apply the first bound with f and g exchanged. Equality examples establishing the final assertion are given in Section II.3. $\square$

Corollary II.2.2 (Graph version). *Let $B \subseteq G$, $\phi : B \rightarrow \widehat{G}$, and $\Gamma = \{(h, \phi(h)) : h \in B\}$. If*

$$T = \mathbb{E}_h \mathbf{1}_B(h) |\widehat{\Delta_h f}(\phi(h))|^2 > 0,$$

then, writing $E(\Gamma)$ for the unnormalized additive energy,

$$\frac{E(\Gamma)}{N^3} \geq \frac{T^4}{m(f)^4 \mathcal{U}(f)}. \quad (\text{II.2.7})$$

In particular, the hypothesis (F.7.1) implies

$$E(\Gamma) \geq \frac{\alpha^4}{m(f)^4 \mathcal{U}(f)} N^3.$$

Proof. Take $g = f$ and $w = \mathbf{1}_\Gamma$. The unnormalized Fourier coefficient is N times our normalized coefficient, so the statistic in (F.7.1), divided by N^3 , is precisely T . Positive T makes the denominator nonzero by Theorem II.2.1. $\square$

[write] **6 October 2026.** *The formal statement is recovered.* The formal `proposition_6_1` (Section `ns06_07.lean`, line 37; exact companion `proposition_6_1_holds`, `Proofs05_10.lean`, line 505) is the case $|f| \leq 1$ on $\mathbb{Z}_N$ with a single-valued graph: there $m(f) \leq 1$ and $\mathcal{U}(f) \leq m(f)^4 \leq 1$ (Lemma II.3.1), so (II.2.7) gives $E(\Gamma) \geq \alpha^4 N^3$. The intake also tested Theorem II.2.1 on the non-cyclic groups $\mathbb{Z}/2 \times \mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3$ and on $\mathbb{Z}/6$ (largest ratio of left to right side 0.084); the delivered suite uses cyclic groups only. *Notation:* this $\mathcal{E}(w)$ is not source 03's $\mathcal{E}(f)$ (Part III), source 04's $\mathcal{E}_j(w)$ (Section II.8) or source 05's captured energy E (Part I); see Section F.3.

II.3 The L^2 denominator and equality examples

Source 02, delivered Section 4.

Lemma II.3.1 (Two tetrahedra). *For every finite abelian group and every complex function on it,*

$$\mathcal{U}(f) \leq m(f)^4, \quad \text{equivalently} \quad \|f\|_{U^3} \leq \|f\|_2. \quad (\text{II.3.1})$$

Proof. Take the absolute value of the cube average (II.1.3). Divide its eight factors into the four vertices

$$x, \quad x+a, \quad x+b, \quad x+c$$

and the complementary four vertices

$$x+a+b, \quad x+a+c, \quad x+b+c, \quad x+a+b+c.$$

Cauchy–Schwarz bounds the cube average by the geometric mean of the averages of the squared products on these two lists. The first list is a bijective change of coordinates from G^4 to G^4 . The second is also bijective: if its coordinates are y_1, y_2, y_3, y_4 , then

$$c = y_4 - y_1, \quad b = y_4 - y_2, \quad a = y_4 - y_3, \quad x = y_1 + y_2 + y_3 - 2y_4.$$

Thus each squared-product average is $m(f)^4$, proving (II.3.1). No division by 2 occurs in the inverse map, so groups with 2-torsion are included. $\square$

Corollary II.3.2 (Variance-normalized energy). *Whenever the denominator is nonzero,*

$$\mathcal{E}(w) \geq \left(\frac{\mathcal{T}_w(f, g)}{m(f)m(g)} \right)^4.$$

For a graph and a balanced indicator $f = \mathbf{1}_A - \delta$, where $\delta = |A|/N \in (0, 1)$,

$$\boxed{\frac{E(\Gamma)}{N^3} \geq \frac{T^4}{[\delta(1-\delta)]^4 \mathcal{U}(f)} \geq \frac{T^4}{[\delta(1-\delta)]^8}} \quad (\text{II.3.2})$$

Proof. Use Lemma II.3.1 in Theorem II.2.1. For a balanced indicator, direct averaging gives $m(f) = \delta(1-\delta)$. $\square$

The improvement in (II.3.2) is not a change in the fourth power of T . It is the retention of an exact variance denominator, together with a possibly smaller U^3 denominator. It is therefore misleading to describe it as a new exponent in Szemerédi's theorem. It can nevertheless be a large improvement in a local energy ledger.

Example II.3.3 (Quadratic-phase graph). Let $G = \mathbb{Z}_p$ for an odd prime p , put $q(x) = \exp(2\pi i x^2/p)$, and let $f = cq$, $c \neq 0$. Then

$$\Delta_h f(x) = |c|^2 \exp(2\pi i(2hx - h^2)/p).$$

Choose the graph $h \mapsto \chi_{2h}$. We have

$$m(f) = |c|^2, \quad \mathcal{U}(f) = |c|^8, \quad T = |c|^4, \quad E(\Gamma) = N^3.$$

Equality holds in both (II.2.1) and the variance-normalized bound. In particular the coefficient 1 in the main theorem is optimal even for a single-valued graph.

Example II.3.4 (A genuinely multivalued sharp example). Let $H \leq G$ have density $d = |H|/N$, let $f = \mathbf{1}_H$, and take $w = \mathbf{1}_{H \times H^\perp}$, where $H^\perp$ is the annihilator of H . The set $H \times H^\perp$ is a subgroup of $G \times \widehat{G}$ of size N , hence $\mathcal{E}(w) = 1$. Also

$$m(f) = d, \quad \mathcal{U}(f) = d^4, \quad \mathcal{T}_w(f, f) = d^2.$$

The cube formula gives $\mathcal{U}(f) = d^4$ because all eight vertices belong to H exactly when the base and three increments do. The Fourier coefficient of $\mathbf{1}_H$ is d on $H^\perp$ and 0 elsewhere, giving the value of $\mathcal{T}$. Again equality holds in the main theorem.

Example II.3.5 (Sharpness within balanced indicators). On an even cyclic group, take A to be the even residues. Then $f = \mathbf{1}_A - 1/2$ is one half of a character. For the graph with trivial frequency at every h ,

$$m(f) = \frac{1}{4}, \quad \mathcal{U}(f) = \frac{1}{256}, \quad T = \frac{1}{16}, \quad E(\Gamma)/N^3 = 1.$$

The original T^4 bound is $1/65536$, whereas (II.3.2) gives the exact value 1. This demonstrates a strict strengthening on an admissible balanced input, not merely on arbitrary unbounded functions.

II.4 Keeping the full derivative spectrum and tracking rigidity

Source 02, delivered Section 5.

II.4.1 A support-and-energy certificate without selecting one frequency

Proposition II.4.1. *Let $f \neq 0$, set $m = m(f)$ and $\kappa = \mathcal{U}(f)/m^4 \in (0, 1]$, and let $0 < \tau \leq 1$. Define*

$$\Gamma_\tau = \{(h, \chi) : |A_{f,f}(h, \chi)| \geq \tau m\}, \quad t_\tau = \frac{\mathcal{T}_{\Gamma_\tau}(f, f)}{m^2}.$$

Then

$$|\Gamma_\tau| \leq N\tau^{-2}, \tag{II.4.1}$$

$$\frac{E(\Gamma_\tau)}{N^3} \geq \frac{t_\tau^4}{\kappa}, \tag{II.4.2}$$

$$\frac{E(\Gamma_\tau)}{|\Gamma_\tau|^3} \geq \frac{t_\tau^4 \tau^6}{\kappa} \quad \text{if } \Gamma_\tau \neq \emptyset. \tag{II.4.3}$$

Proof. Parseval, followed by averaging in h , gives the exact identity

$$\sum_{h, \chi} |A_{f,f}(h, \chi)|^2 = N \mathbb{E}_{h, x} |f(x)|^2 |f(x - h)|^2 = Nm^2.$$

This proves the support bound. Apply Theorem II.2.1 to the full set Γ_τ , not a chosen graph, and substitute $\mathcal{U}(f) = \kappa m^4$. Finally divide by the cube of the support bound. Nonzero f has $\mathcal{U}(f) > 0$: the $h = 0$ summand in (II.1.2) already has a nonzero Fourier coefficient. $\square$

The certificate (II.4.3) is suitable input for an additive-energy-to-structure theorem of Balog–Szemerédi–Gowers type. The subsequent structural step is not proved anew here. The point is that there is no preliminary loss from discarding all but one frequency in each fiber. Whether that benefit survives the later graph and progression constructions is a separate quantitative problem.

II.4.2 Exact deficits in the two Cauchy–Schwarz steps

With F, D, V, K, R as in Section II.2, set

$$D_1 = m(f)^2 R - \mathcal{T}_w(f, g)^2, \quad D_2 = \mathcal{U}(g)\mathcal{E}(w) - R^2.$$

Both are nonnegative, and direct algebra gives

$$m(f)^4 \mathcal{U}(g)\mathcal{E}(w) - \mathcal{T}_w(f, g)^4 = m(f)^4 D_2 + D_1(m(f)^2 R + \mathcal{T}_w(f, g)^2). \tag{II.4.4}$$

Thus the loss is decomposed exactly, rather than hidden in an unspecified constant.

Proposition II.4.2 (Quantitative alignment). *Suppose all quantities used to normalize below are nonzero and $0 \leq \varepsilon < 1$. If*

$$\mathcal{T}_w(f, g)^4 \geq (1 - \varepsilon)m(f)^4 \mathcal{U}(g)\mathcal{E}(w),$$

then

$$\left\| \frac{F}{\|F\|_2} - \frac{K}{\|K\|_2} \right\|_2^2 \leq 2(1 - (1 - \varepsilon)^{1/4}),$$

$$\left\| \frac{a}{\|a\|_2} - \frac{b}{\|b\|_2} \right\|_2^2 \leq 2(1 - \sqrt{1 - \varepsilon}),$$

where $a(u, \psi) = |\widehat{D}_u(\psi)|^2$ and $b(u, \psi) = |\widehat{V}_u(\psi)|^2$. The first norm uses normalized measure on G^2 ; the second uses average in u and counting measure in ψ .

Proof. Let $A = \mathcal{T}^2/(m(f)^2 R)$ and $B = R^2/(\mathcal{U}(g)\mathcal{E}(w))$. The proof of Theorem II.2.1 gives $A, B \in [0, 1]$, and the hypothesis says $A^2 B \geq 1 - \varepsilon$. Hence $A \geq \sqrt{1 - \varepsilon}$ and $B \geq 1 - \varepsilon$. The normalized inner products of the first and second pairs are $\sqrt{A}$ and $\sqrt{B}$, respectively. They are real and nonnegative, by (II.2.2) and (II.2.3). Apply $\|v - w\|^2 = 2 - 2 \operatorname{Re} \langle v, w \rangle$ for unit vectors. $\square$

This is an alignment theorem for the proof's intermediate functions, not a full classification of all near-extremal pairs (f, w) . The distinction matters: subgroup-localized examples and polynomial phases already occur, and broader extremizer theory is known [7].

II.4.3 An elementary exact rigidity endpoint

Proposition II.4.3. *Let $f \not\equiv 0$, and let $\phi : G \rightarrow \widehat{G}$ be defined on all of G . For its graph statistic, $T \leq m(f)^2$. Equality holds if and only if $f = cq$, with $|q| = 1$, and for each h*

$$\Delta_h q(x) = c_h \phi(h)(x), \quad |c_h| = 1.$$

In this situation ϕ is a homomorphism and all third multiplicative derivatives of q equal 1.

Proof. The chosen Fourier coefficient in each fiber has squared magnitude at most the sum of squared magnitudes in that fiber. The sum of all fibers is $m(f)^2$ after normalized averaging, by Parseval. Equality forces every $\Delta_h f$ to have Fourier support at the chosen single character. At $h = 0$, $\Delta_0 f = |f|^2$ is nonnegative and has positive mean. Its sole character must therefore be trivial, and $|f|^2 = m(f)$ everywhere. Write $f = cq$ with $|q| = 1$. Fourier inversion gives the stated derivative formula. Conversely that formula makes Parseval an equality in every fiber.

The identity

$$\Delta_{h+k} q(x) = \Delta_h q(x) \Delta_k q(x - h)$$

shows that $\phi(h+k) = \phi(h)\phi(k)$, since a nonzero scalar multiple of a character has a unique character frequency. A further derivative of a character is constant, and a derivative of a unit-modulus constant is 1; therefore third derivatives vanish multiplicatively. This definition of a quadratic phase does not impose an unjustified classical-polynomial representation in small characteristic. $\square$

II.5 Sharper common-neighborhood selection

Source 03, delivered Section 7.

Gowers' Lemma 7.4 is a convenient place to make a concrete quantitative improvement without changing the combinatorial objects in the argument. Its hypothesis controls the average intersection of a family of sets; its conclusion finds a subfamily in which almost every pair has a reasonably large intersection. The published proof samples five witnesses and estimates the useful and exceptional contributions separately [1]. Keeping those contributions together allows three witnesses to suffice.

The method remains dependent random choice, whose subsequent development is surveyed by Fox and Sudakov [11]. Our assertion here is a precise replacement for the numerical statement in the source paper. We do not assert that the resulting exponent is optimal among all possible selection arguments.

II.5.1 A cubic certificate

All pairs below are ordered, and diagonal pairs are included. For a finite family $A_1, \dots, A_n$ of subsets of a set V of cardinality $m > 0$, write

$$p_{xy} = \frac{|A_x \cap A_y|}{m}.$$

A pair will be called bad when $p_{xy} < a/2$, where $a > 0$ is the specified lower bound for the mean of the p_{xy} .

Lemma II.5.1 (Rational cubic certificate). *For every $a > 0$ and $p \geq 0$, define*

$$F_a(p) = \begin{cases} -9p^3, & 0 \leq p < a/2, \\ p^3, & p \geq a/2. \end{cases}$$

Then

$$F_a(p) \geq \frac{49}{12}a^2p - \frac{343}{108}a^3. \quad (\text{II.5.1})$$

Proof. On the good region $p \geq a/2$, the desired inequality is the tangent-line inequality for p^3 at $p = 7a/6$. More explicitly,

$$p^3 - \frac{49}{12}a^2p + \frac{343}{108}a^3 = \left(p - \frac{7a}{6}\right)^2 \left(p + \frac{7a}{3}\right) \geq 0.$$

On the interval $0 \leq p \leq a/2$, the function

$$-9p^3 - \frac{49}{12}a^2p + \frac{343}{108}a^3$$

is concave, so its minimum on that interval occurs at an endpoint. Its values at 0 and $a/2$ are, respectively,

$$\frac{343}{108}a^3 \quad \text{and} \quad \frac{-243 - 441 + 686}{216}a^3 = \frac{a^3}{108}.$$

Both are positive. At $p = a/2$, the actual function F_a takes the larger, good-region value, so the strict definition of a bad pair presents no boundary issue. $\square$

Theorem II.5.2 (Cubic replacement for Gowers' Lemma 7.4). *Let m, n be positive integers, let V have cardinality m , and let $0 < \delta \leq 1$. Suppose that $A_1, \dots, A_n \subseteq V$ satisfy*

$$\sum_{x,y=1}^n |A_x \cap A_y| \geq \delta^2 mn^2. \quad (\text{II.5.2})$$

There is a set $K \subseteq \{1, \dots, n\}$ such that

$$|K| \geq \frac{7}{3\sqrt{6}} \delta^3 n, \quad (\text{II.5.3})$$

and at least 90% of the pairs $(x, y) \in K^2$ satisfy

$$|A_x \cap A_y| \geq \frac{\delta^2 m}{2}. \quad (\text{II.5.4})$$

In particular, the conclusion holds when every A_x has cardinality at least δm .

Proof. Put $a = \delta^2$. Independently sample j_1, j_2, j_3 uniformly from V , allowing repetitions, and set

$$K = \{x : j_1, j_2, j_3 \in A_x\}.$$

Let $B(K)$ count the bad ordered pairs in K^2 . The probability that $(x, y) \in K^2$ is p_{xy}^3 . Consequently, linearity of expectation gives the exact identity

$$\begin{aligned} \mathbb{E}[|K|^2 - 10B(K)] &= \sum_{x,y=1}^n F_a(p_{xy}) \\ &\geq \frac{49}{12}a^2 \sum_{x,y=1}^n p_{xy} - \frac{343}{108}a^3n^2 \\ &\geq \left(\frac{49}{12} - \frac{343}{108}\right)a^3n^2 = \frac{49}{54}a^3n^2. \end{aligned}$$

The first inequality is Lemma II.5.1; the second uses (II.5.2). Some realization therefore satisfies

$$|K|^2 - 10B(K) \geq \frac{49}{54}\delta^6n^2 > 0.$$

It follows simultaneously that $|K| \geq \sqrt{49/54}\delta^3n$ and $B(K) \leq |K|^2/10$. These are the required conclusions.

For the final assertion, put $d(v) = |\{x : v \in A_x\}|$. Then

$$\sum_{x,y} |A_x \cap A_y| = \sum_{v \in V} d(v)^2 \geq \frac{1}{m} \left(\sum_{v \in V} d(v) \right)^2 \geq \delta^2 mn^2,$$

by Cauchy–Schwarz and the assumed lower bounds on $|A_x|$. □

The constant $7/(3\sqrt{6})$ is approximately 0.95258. Thus the original $2^{-1/2}\delta^5n$ is replaced by a larger bound with exponent 3, while both the half-mean intersection threshold and the 90% requirement are preserved. The improvement comes from a stronger estimate of the same penalized random variable, rather than from a change in the definition of success.

[write] 6 October 2026. Compared with the formal statement (checked at HEAD): `lemma_7_4` (`Section06_07.lean`, line 167) has the exact companion `lemma_7_4_holds` (`Proofs07DRC.lean`, line 536), which samples five witnesses indexed by `Fin 5`, as Section V.13 describes. Theorem II.5.2 keeps its hypothesis, the threshold $\delta^2m/2$ and the 90% requirement, and replaces the size $2^{-1/2}\delta^5n$ by $7\delta^3n/(3\sqrt{6})$. Not formalized.

II.5.2 Weights and higher tuples

The certificate is independent of the counting measure on the index set. This gives a useful extension when vertices carry unequal importance or when intersections of more than two sets are under consideration.

Corollary II.5.3 (Weighted tuple selection). *Let X be a finite nonempty set, let $w_x \geq 0$ with $\sum_{x \in X} w_x = 1$, and let $(A_x)_{x \in X}$ be measurable subsets of a probability space (Ω, ν) . Fix an integer $r \geq 1$, and suppose*

$$\sum_{(x_1, \dots, x_r) \in X^r} w_{x_1} \cdots w_{x_r} \nu(A_{x_1} \cap \cdots \cap A_{x_r}) \geq a > 0.$$

There is a set $K \subseteq X$ with

$$w(K) := \sum_{x \in K} w_x \geq \left(\frac{49}{54}\right)^{1/r} a^{3/r}$$

such that the total product weight of tuples in K^r with $\nu(A_{x_1} \cap \dots \cap A_{x_r}) < a/2$ is at most $w(K)^r/10$.

Proof. Sample three independent points of Ω and let K be the indices of the sets containing all three points. A fixed tuple lies in K^r with probability p^3 , where p is its intersection measure. Apply (II.5.1) to each tuple and sum with its product weight. The expected value of $w(K)^r$ minus ten times the bad-tuple weight is at least $49a^3/54$. Since X is finite, there are only finitely many possible sets K , and one attains at least this expectation. Taking an r th root gives the conclusion. $\square$

II.5.3 The exact scalar optimization

The simple rational point $7a/6$ was chosen to make the certificate easy to verify. One can determine the optimal supporting line for this argument. Let $z > 1$ be the unique solution of

$$2z^3 - 3z^2 = 9. \tag{II.5.5}$$

The line tangent to p^3 at $p = za/2$ passes through the limiting bad point $(a/2, -9a^3/8)$. Applying that line instead of (II.5.1) gives

$$|K| \geq \kappa_* \delta^3 n, \quad \kappa_* = \sqrt{\frac{3z^2 - z^3}{4}} = 0.9537881585\dots \tag{II.5.6}$$

Indeed, the tangent has slope $3(za/2)^2$ and intercept $-2(za/2)^3$. Equation (II.5.5) verifies its value at the bad endpoint. Concavity on the bad interval and convexity on the good interval again prove a global minorant. Its value at the mean a is $(3z^2 - z^3)a^3/4$. Existence and uniqueness of $z > 1$ follow from the fact that $2z^3 - 3z^2$ is strictly increasing on $(1, \infty)$, starts at -1 , and tends to infinity. Moreover $2 < z < 5/2$, so the expression under the square root is positive.

For clarity, the meaning of optimality here is limited and exact. When $za/2 \leq 1$, consider arbitrary scalar random variables $P \in [0, 1]$ with $\mathbb{E}P = a$. A two-point distribution, with one value tending to $(a/2)^-$ and the other equal to $za/2$, approaches the tangent-line lower bound for $\mathbb{E}F_a(P)$. Thus no larger bound follows from this scalar mean constraint alone. The infimum is approached because badness is defined by a strict inequality.

This observation also explains a restriction of a two-witness version of the same argument. Its penalized scalar function is p^2 on the good region and $-9p^2$ on the bad region. The analogous tangent point is ha , where $h = (1 + \sqrt{10})/2$. The tangent's value at a is

$$(2h - h^2)a^2 < 0.$$

For sufficiently small a , the corresponding two-point distribution lies in $[0, 1]$ and gives a negative limiting expectation. Consequently, two witnesses and this scalar first-moment certificate cannot produce a positive uniform lower bound at the original parameters. This does not exclude a better theorem exploiting properties of actual intersection matrices, nor does it prove optimality of the exponent 3.

II.5.4 A flexible one-witness alternative

For applications, keeping the threshold exactly at half the mean may be unnecessary. Allowing a smaller constant multiple of the mean gives a much larger selected set using just one witness. This is an explicit form of the established variant in Fox and Sudakov [11, Lemma 5.1].

Lemma II.5.4 (One-witness tradeoff). *Under (II.5.2), let $0 < \gamma < \varepsilon < 1$. There is a set $K \subseteq \{1, \dots, n\}$ satisfying*

$$|K| \geq \sqrt{1 - \gamma/\varepsilon} \delta n$$

such that at most an ε fraction of the ordered pairs in K^2 have intersection smaller than $\gamma\delta^2 m$.

Proof. Choose $j \in V$ uniformly and set $K = \{x : j \in A_x\}$. Then

$$\mathbb{E}|K|^2 = \sum_{x,y} p_{xy} \geq \delta^2 n^2.$$

If $B(K)$ counts the pairs in K^2 with $p_{xy} < \gamma\delta^2$, then

$$\mathbb{E}B(K) = \sum_{p_{xy} < \gamma\delta^2} p_{xy} \leq \gamma\delta^2 n^2.$$

Thus

$$\mathbb{E}[|K|^2 - \varepsilon^{-1}B(K)] \geq (1 - \gamma/\varepsilon)\delta^2 n^2 > 0.$$

A realization attaining at least this expectation gives both claims. $\square$

II.6 An elementary quantitative replacement for Proposition 7.3

Source 03, delivered Section 8.

The preceding selection statements have different uses. The cubic version strengthens the exact statement of Lemma 7.4. The one-witness version is more efficient for the later small-difference-set argument, because that argument can tolerate a smaller absolute constant in the common-neighbor threshold. We also remove a separate loss by counting the total mass of popular differences directly.

Throughout this section, A is a finite nonempty subset of an arbitrary abelian group and $m = |A|$. Put

$$r_A(d) = \left| \{(a, b) \in A^2 : a - b = d\} \right|, \quad E(A) = \sum_d r_A(d)^2.$$

We assume $E(A) \geq cm^3$, where $0 < c \leq 1$. The sums involve only finitely many nonzero terms, so no finiteness assumption on the ambient group is required.

II.6.1 Popular differences retain more edge mass

Lemma II.6.1 (Popular mass). *Let*

$$D = \{d : r_A(d) \geq cm/2\}, \quad M = \sum_{d \in D} r_A(d), \quad \rho = \frac{c}{2-c}.$$

Then $D = -D$, $|D| \leq 2m/c$, and $M \geq \rho m^2$.

Proof. The symmetry follows from $r_A(d) = r_A(-d)$. Since $\sum_d r_A(d) = m^2$, the lower bound $r_A(d) \geq cm/2$ on D gives $|D| \leq 2m/c$. On D we have $r_A(d)^2 \leq mr_A(d)$, while off D we have $r_A(d)^2 \leq (cm/2)r_A(d)$. Therefore

$$cm^3 \leq E(A) \leq mM + \frac{cm}{2}(m^2 - M).$$

Rearranging proves $M \geq cm^2/(2 - c)$. $\square$

This estimate retains the edge density of the popular-difference relation. It avoids the loss from first counting popular values and then replacing each of their multiplicities by the smallest allowed multiplicity. In particular, the relation has density at least $\rho \geq c/2$.

II.6.2 A general walk-counting statement

Proposition II.6.2 (Compression by four edge differences). *Let $G \subseteq A \times A$ be a symmetric relation, with loops allowed, satisfying $|G| \geq \rho m^2$, where $0 < \rho \leq 1$. Suppose that every $(a, b) \in G$ has $a - b \in D$ for a finite set D , and put $L = |D|$. For $0 < \gamma < \varepsilon < \beta < 1/2$, set $s = \sqrt{1 - \gamma/\varepsilon}$. There is $A' \subseteq A$ such that*

$$|A'| \geq (1 - \varepsilon/\beta)s\rho m, \tag{II.6.1}$$

$$|A' - A'| \leq \frac{L^4}{(1 - 2\beta)\gamma^2 s \rho^5 m^3}. \tag{II.6.2}$$

Proof. Write $N(a) = \{v : (a, v) \in G\}$. By symmetry, double-counting, and Cauchy–Schwarz,

$$\sum_{a, b \in A} |N(a) \cap N(b)| = \sum_{v \in A} |N(v)|^2 \geq \frac{|G|^2}{m} \geq \rho^2 m^3.$$

Lemma II.5.4 supplies $U \subseteq A$, of cardinality $u \geq s\rho m$, with at most εu^2 bad ordered pairs. Here a pair is bad when its common G -neighborhood has cardinality smaller than $\gamma\rho^2 m$.

Let A' consist of the vertices of U that belong, in the first coordinate, to at most βu bad pairs in U^2 . Counting those pairs shows that at most $(\varepsilon/\beta)u$ vertices are excluded. This proves (II.6.1). Each $a \in A'$ has at least $(1 - \beta)u$ good partners in U . Consequently, every $a, b \in A'$ have at least $(1 - 2\beta)u$ common good partners $k \in U$.

Fix such a, b . For every common good partner k , choose

$$x \in N(a) \cap N(k), \quad y \in N(b) \cap N(k).$$

There are at least $(1 - 2\beta)u(\gamma\rho^2 m)^2$ triples (k, x, y) . Associate to each the four edge differences

$$d_1 = a - x, \quad d_2 = k - x, \quad d_3 = k - y, \quad d_4 = b - y.$$

They lie in D and satisfy

$$a - b = d_1 - d_2 + d_3 - d_4.$$

For the fixed endpoints a, b , the association is injective: one recovers successively $x = a - d_1$, $k = x + d_2$, and $y = k - d_3$.

For each element of $A' - A'$, fix one pair of endpoints representing it. The associated label quadruples for distinct differences are disjoint, because a label quadruple determines $d_1 - d_2 + d_3 - d_4$. Since there are only L^4 possible label quadruples,

$$|A' - A'| (1 - 2\beta) u \gamma^2 \rho^4 m^2 \leq L^4.$$

Using $u \geq s \rho m$ proves (II.6.2). This counts walks with specified endpoints; repeated vertices and loops are allowed throughout, and cause no loss in the injection. $\square$

II.6.3 Explicit bounds and their scope

Theorem II.6.3 (Elementary historical improvement). *If $E(A) \geq c|A|^3$ with $0 < c \leq 1$, there is $A' \subseteq A$ satisfying*

$$|A'| \geq \frac{c}{8} |A|, \quad |A' - A'| \leq 2^{19} c^{-9} |A|. \quad (\text{II.6.3})$$

More precisely, with $\rho = c/(2 - c)$, one may take

$$|A'| \geq \frac{\rho m}{2\sqrt{2}}, \quad |A' - A'| \leq 6912\sqrt{2} \rho^{-5} c^{-4} m. \quad (\text{II.6.4})$$

Proof. Use D from Lemma II.6.1, and define $G = \{(a, b) \in A^2 : a - b \in D\}$. That lemma gives $|G| \geq \rho m^2$ and $L \leq 2m/c$. Apply Proposition II.6.2 with

$$\gamma = \frac{1}{12}, \quad \varepsilon = \frac{1}{6}, \quad \beta = \frac{1}{3}, \quad s = \frac{1}{\sqrt{2}}.$$

The size bound is $\rho m/(2\sqrt{2})$, and the difference bound becomes

$$|A' - A'| \leq \frac{432\sqrt{2} L^4}{\rho^5 m^3} \leq 6912\sqrt{2} \rho^{-5} c^{-4} m.$$

Finally $\rho \geq c/2$, so

$$|A'| \geq \frac{cm}{4\sqrt{2}} \geq \frac{cm}{8}, \quad |A' - A'| \leq 221184\sqrt{2} c^{-9} m < 2^{19} c^{-9} m.$$

This proves both versions. $\square$

For comparison, Proposition 7.3 in the source paper gives a subset of size at least $2^{-20} c^{12} m$ with difference set of size at most $2^{38} c^{-24} m$ [1]. Theorem II.6.3 improves both exponents using only one sampled vertex, Cauchy–Schwarz, and the explicit counting injection above. Even if one changes only Lemma 7.4 to Theorem II.5.2 and leaves the remaining original proof intact, its formulas yield the weaker but direct improvements

$$|A'| \geq 2^{-14} c^8 m, \quad |A' - A'| \leq 2^{31} c^{-20} m.$$

Indeed, the original parameter $\delta = c^2/8$ then gives selected density $\alpha = \kappa \delta^4$, where $\kappa = 7/(3\sqrt{6})$; the original cleanup and walk count give $\alpha m/2$ and $120m/(\alpha \delta^4 c^4)$, respectively. The displayed relaxations use $\kappa > 1/2$ and $120/\kappa < 128$.

These are improvements to the historical proof architecture, rather than to the strongest available Balog–Szemerédi–Gowers theorem. For example, Reiher and Schoen prove that, for every $0 < \eta < 1/2$, the same energy hypothesis gives a subset satisfying

$$|A'| \geq (1 - \eta)c^{1/2}m, \quad |A' - A| \leq 2^{33}\eta^{-9}c^{-4}|A'|$$

[12, Theorem 1.2]. Their result is substantially stronger. The purpose of the elementary version is to supply a short, fully explicit replacement that can be checked independently and integrated into the existing development with a small dependency set. A complete improvement to the final quantitative Szemerédi bound would also require propagating the new parameters through the later arguments; no such consequence follows from this section alone.

[write] **6 October 2026.** *Compared with the formal statement (checked at HEAD):* `proposition_7_3` (Sections06_07.lean, line 135; exact companion `proposition_7_3_holds`, Proofs07BalogSzemeredi.lean, line 996) is stated on $\mathbb{Z}^n$ with the bounds $2^{-20}c^{12}|A|$ and $2^{38}c^{-24}|A|$. For $0 < c \leq 1$, $c/8 \geq 2^{-20}c^{12}$ (as $c^{11} \leq 2^{17}$) and $2^{19}c^{-9} \leq 2^{38}c^{-24}$ (as $c^{15} \leq 2^{19}$), so Theorem II.6.3, which holds in every abelian group, implies the bounds of the formal statement. Not formalized.

II.7 Notation and normalization

Source 04, delivered Section 2.

Let G be a finite abelian group, written additively, and let $N = |G|$. We write $\mathbb{E}_{x \in G} g(x) = N^{-1} \sum_{x \in G} g(x)$. Characters are written multiplicatively. Except in the moment section, which explicitly introduces an unnormalized transform, Fourier coefficients are normalized:

$$\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}.$$

Thus

$$\mathbb{E}_x |f(x)|^2 = \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2.$$

Sums over the dual group in this convention are ordinary sums, not averages.

For $d \geq 1$, put

$$\|f\|_{U^d(G)}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\epsilon \in \{0,1\}^d} \mathcal{C}^{|\epsilon|} f(x + \epsilon_1 h_1 + \dots + \epsilon_d h_d),$$

where $\mathcal{C}$ is complex conjugation. We omit G when it is clear. For $d = 1$ this means $\|f\|_{U^1} = |\mathbb{E} f|$. For $d \geq 2$ it is the Gowers norm. If f is real, conjugations can be omitted.

For a real function w , define its normalized cube count

$$Q_d(w) = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\epsilon \in \{0,1\}^d} w(x + \epsilon \cdot h).$$

For $w = 1_A$, $N^{d+1}Q_d(w)$ counts all ordered additive d -cubes, including degenerate cubes, exactly as in the corresponding count in the source. If $\delta = |A|/N$, its balanced function is $f = 1_A - \delta$.

The conversion to Gowers’s parameter is important:

$$f \text{ is } \alpha\text{-uniform of degree } d-1 \iff \|f\|_{U^d}^{2^d} \leq \alpha.$$

In particular, the letter u below generally denotes a norm, while α denotes an upper bound on a power of that norm. Confusing these two quantities would conceal the improvement in the exponent.

We use the standard Gowers–Cauchy–Schwarz inequality in normalized form:

$$\left| \mathbb{E}_{x,h} \prod_{\epsilon} \mathcal{C}^{|\epsilon|} f_{\epsilon}(x + \epsilon \cdot h) \right| \leq \prod_{\epsilon} \|f_{\epsilon}\|_{U^d}.$$

It follows by applying Cauchy–Schwarz successively in the cube coordinates; this is Lemma 3.8 in the source. Its ordinary consequences include the triangle inequality and $\|f\|_{U^d} \leq \|f\|_{U^{d+1}}$. For clarity, the latter also follows by writing

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_{h_1, \dots, h_{d-1}} \left| \mathbb{E}_x \Delta_{h_1} \cdots \Delta_{h_{d-1}} f(x) \right|^2.$$

Square this identity and use Cauchy–Schwarz, then apply $|\mathbb{E}g|^4 \leq \|g\|_{U^2}^4$ to each derivative. The resulting right-hand side is $\|f\|_{U^{d+1}}^{2^{d+1}}$.

An ordered additive $2d$ -tuple of a set B is a tuple with $x_1 + \cdots + x_d = x_{d+1} + \cdots + x_{2d}$. A map $\phi : B \rightarrow H$ *respects* the tuple if the same equation holds after applying ϕ . All tuple counts in this article are ordered and allow repetitions unless expressly called nonexceptional.

II.8 Exact second moments and density-sensitive lifting

Source 04, delivered Section 3.

Several losses in the paper occur when a second moment is bounded using the size of the ambient space rather than the size of the set under consideration. Retaining the exact second moment is useful on its own, and becomes particularly effective when followed by the restriction argument in the next section. The underlying interpolation inequality is standard Hölder; the contribution here is the careful normalization, the resulting sharper statements, and their use in the subsequent selection theorem.

II.8.1 Weighted energies on finite abelian groups

Let H be a finite abelian group of order M , and let $\hat{H}$ be its character group. For $w : H \rightarrow \mathbb{C}$, use the unnormalized transform

$$\hat{w}(\chi) = \sum_{x \in H} w(x) \overline{\chi(x)}.$$

For an integer $j \geq 1$, define

$$\mathcal{E}_j(w) = \frac{1}{M} \sum_{\chi \in \hat{H}} |\hat{w}(\chi)|^{2j}.$$

Character orthogonality gives

$$\mathcal{E}_1(w) = \sum_{x \in H} |w(x)|^2.$$

For an indicator $w = 1_A$, the same orthogonality calculation shows that $\mathcal{E}_j(A)$ counts ordered tuples $(a_1, \dots, a_{2j}) \in A^{2j}$ satisfying

$$a_1 + \cdots + a_j = a_{j+1} + \cdots + a_{2j}.$$

Repeated entries are included throughout.

Theorem II.8.1 (Exact energy interpolation). *Let $j \geq 2$, and suppose that $\mathcal{E}_1(w) > 0$. Then*

$$\mathcal{E}_j(w) \geq \frac{\mathcal{E}_2(w)^{j-1}}{\mathcal{E}_1(w)^{j-2}}. \quad (\text{M1})$$

For $j > 2$, equality holds if and only if all nonzero Fourier coefficients of w have the same modulus. In particular, if $A \neq \emptyset$, then

$$\mathcal{E}_j(A) \geq \frac{\mathcal{E}_2(A)^{j-1}}{|A|^{j-2}}. \quad (\text{M2})$$

For $j > 2$, equality in (M2) holds exactly when A is a coset of a subgroup.

Proof. Put $x_\chi = |\widehat{w}(\chi)|^2$. If $j = 2$, the assertion is an identity. For $j > 2$, Hölder gives

$$\sum_{\chi} x_{\chi}^2 = \sum_{\chi} x_{\chi}^{(j-2)/(j-1)} x_{\chi}^{j/(j-1)} \leq \left(\sum_{\chi} x_{\chi} \right)^{(j-2)/(j-1)} \left(\sum_{\chi} x_{\chi}^j \right)^{1/(j-1)}.$$

Raising to the power $j - 1$ and dividing by M^{j-1} proves (M1). Equality in Hölder says that the positive values of x_{χ} are equal.

Now let $w = 1_A$, and put $m = |A|$. Its trivial Fourier coefficient is m . Thus equality forces every nonzero Fourier coefficient to have modulus m . Let K be the subgroup generated by $A - A$. Equality in the triangle inequality shows that $|\widehat{1_A}(\chi)| = m$ precisely when χ is constant on A , equivalently when $\chi \in K^{\perp}$. Consequently Parseval gives

$$Mm = |K^{\perp}| m^2 = \frac{M}{|K|} m^2.$$

Hence $m = |K|$. Since A is contained in a coset of K , it is that entire coset. Conversely, the Fourier transform of a coset indicator has constant modulus on its support, so it gives equality. $\square$

The proof applies without change to any finite family of spectra: the index χ can be replaced by a pair (i, χ) . This observation will be used for arrangements, where the extra index records common sidelengths.

Corollary II.8.2 (Lifting respected graph energies). *Let B be a nonempty subset of a finite abelian group, and let $\phi : B \rightarrow K$ take values in another finite abelian group. Write $m = |B|$, let $0 \leq \gamma \leq 1$, and suppose that at least γm^3 ordered additive quadruples of B are respected by ϕ . Then, for every integer $j \geq 2$, at least*

$$\gamma^{j-1} m^{2j-1} \quad (\text{M3})$$

ordered additive $2j$ -tuples are respected by ϕ .

Proof. Apply Theorem II.8.1 to the indicator of the graph $\Gamma = \{(x, \phi(x)) : x \in B\}$. Its first energy is m , and its j -th energy is exactly the number in question. $\square$

For $B \subset \mathbb{F}_p$ of size βp , the source proof of Lemma 9.2 uses

$$\sum_{u,r} \left| \widehat{f_u}(r) \right|^2 \leq p^3.$$

The exact identity is

$$\sum_{u,r} \left| \widehat{f}_u(r) \right|^2 = p^2 |B| = \beta p^3. \quad (\text{M4})$$

Thus, under its unnormalized hypothesis of at least $\gamma_0 p^3$ respected quadruples, its conclusion strengthens to $\gamma_0^7 \beta^{-6} p^{15}$ respected 16-tuples. Under the hypothesis of Corollary 9.4, where $\gamma_0 = \gamma \beta^3$, this is

$$\gamma^7 \beta^{15} p^{15}, \quad (\text{M5})$$

rather than $\gamma^7 \beta^{21} p^{15}$. These comparisons refer to the displayed statements in the source and its repository transcription [1, 23].

[write] **6 October 2026.** *Proof-level baseline* (checked at HEAD): in `Proofs09Moments.lean`, the private lemma `section9_second_moment_le` (line 480), used for `lemma_9_2_holds` (line 527), computes the exact value $N^2 |B|$ of this sum (line 495) before weakening it to N^3 . So (M4) is already a proof-level fact in the repository, as the stronger Corollary 2.5 pair is (Part I); the exported `lemma_9_2` keeps the weaker bound, and (M5) and Corollary II.8.2 are not formalized.

II.8.2 A finite-size correction from the trivial character

There is a further refinement when the ambient group size matters. Its proof also identifies a useful limitation of using only the full first and second moments.

Proposition II.8.3 (Removing the trivial Fourier coefficient). *Let $A \subset H$, where $0 < m = |A| < M = |H|$, and put*

$$\rho = \frac{m}{M}, \quad \gamma = \frac{\mathcal{E}_2(A)}{m^3}.$$

Then $\gamma \geq \rho$, and for every integer $j \geq 2$,

$$\frac{\mathcal{E}_j(A)}{m^{2j-1}} \geq \rho + \frac{(\gamma - \rho)^{j-1}}{(1 - \rho)^{j-2}}. \quad (\text{M6})$$

The right side is at least γ^{j-1} . If only $\mathcal{E}_2(A) \geq \gamma_0 m^3$ is known, one may replace $\gamma - \rho$ by $\max\{\gamma_0 - \rho, 0\}$.

Proof. Remove the trivial character before applying the Hölder argument. The sums of the remaining squared and fourth Fourier moduli are

$$M \left(m - \frac{m^2}{M} \right), \quad M \left(\mathcal{E}_2(A) - \frac{m^4}{M} \right),$$

respectively. The remaining $2j$ -th moment is $M(\mathcal{E}_j(A) - m^{2j}/M)$. Substitution in the same interpolation inequality proves (M6). The second displayed quantity is nonnegative, giving $\gamma \geq \rho$. Finally, write

$$\gamma = \rho \cdot 1 + (1 - \rho) \frac{\gamma - \rho}{1 - \rho}$$

and apply convexity of $t \mapsto t^{j-1}$. □

This correction can be sharp away from subgroup examples. For the parabola $\Gamma = \{(x, x^2) : x \in \mathbb{F}_p\}$, with p odd, the trivial Fourier coefficient has modulus p , the coefficients at $(r, 0)$ with $r \neq 0$

vanish, and all coefficients whose second frequency is nonzero have modulus $\sqrt{p}$. Indeed, completing the square reduces the latter assertion to

$$\left| \sum_x e_p(ux^2) \right|^2 = \sum_{x,y} e_p(u(x-y)(x+y)) = p \quad (u \neq 0),$$

where $e_p(t) = \exp(2\pi it/p)$; the change of variables $(x, y) \mapsto (x - y, x + y)$ is invertible. It follows that

$$\mathcal{E}_2(\Gamma) = 2p^2 - p, \quad \mathcal{E}_j(\Gamma) = p^{2j-2} + (p-1)p^{j-1}. \quad (\text{M7})$$

All nonzero residual squared Fourier moduli agree, so Proposition II.8.3 is an equality in this example.

II.9 Polynomial restriction to approximate homomorphisms

Source 04, delivered Section 4.

We now replace the repeated degree-one filters in the proof of Lemma 9.3 by one filter of higher degree. The degree is chosen from the desired accuracy. Although more coefficient relations appear, their nonnegative Fourier weights have total mass one. A weighted error bound therefore avoids a loss proportional to the number of those relations. The retained number of configurations has a polynomial dependence with exponent linear in the number of entries, instead of the much larger exponent in the source.

For $B \subset \mathbb{F}_p$, let $T_d(B)$ denote its number of ordered additive $2d$ -tuples, and let $T_d^\phi(B)$ denote the number also satisfying

$$\phi(x_1) + \cdots + \phi(x_d) = \phi(x_{d+1}) + \cdots + \phi(x_{2d}). \quad (\text{R1})$$

Thus $T_d^\phi(B) = \mathcal{E}_d(\Gamma)$. An elementary upper bound that will be used repeatedly is

$$T_d(B) \leq |B|^{2d-1}, \quad (\text{R2})$$

because the first $2d - 1$ entries uniquely determine the last.

Theorem II.9.1 (Polynomial restriction theorem). *Let $d \geq 2$ be an integer, and let $0 < \alpha, \beta, \eta \leq 1$. Define*

$$\kappa_d = \frac{1}{2(400d)^d}, \quad \Lambda_d = 2 \left(1 + \binom{2d}{2} \right) (400d)^d. \quad (\text{R3})$$

Suppose that p is a prime with

$$p \geq \Lambda_d(\alpha\eta)^{-2d}\beta^{-1}. \quad (\text{R4})$$

Let $B \subset \mathbb{F}_p$, with $|B| = \beta p$, and let $\phi : B \rightarrow \mathbb{F}_p$ satisfy

$$T_d^\phi(B) \geq \alpha\beta^{2d-1}p^{2d-1}. \quad (\text{R5})$$

Then there is $B' \subset B$ for which

$$T_d(B') \geq \kappa_d \alpha^{2d} \eta^{2d-1} \beta^{2d-1} p^{2d-1}, \quad (\text{R6})$$

and

$$\frac{T_d^\phi(B')}{T_d(B')} \geq \frac{1}{1+\eta} \geq 1-\eta. \quad (\text{R7})$$

Equivalently, the size hypothesis is $|B| \geq \Lambda_d(\alpha\eta)^{-2d}$. It depends on the number of available points rather than a separate density requirement. The theorem concerns preservation of additive configurations. It does not merely assert that B' has many points. The normalization by $|B|^{2d-1}$, rather than the ambient configuration count, is essential for its use after Corollary II.8.2.

II.9.1 The kernel and its coefficients

Lemma II.9.2 (A powered cosine with many useful coefficients). *Let d, L be positive integers, and put $q = 4dL^2$. Write*

$$K_q(t) = \cos^{2q}(\pi t) = \sum_{\ell=-q}^q b_\ell e^{2\pi i \ell t}, \quad b_\ell = 4^{-q} \binom{2q}{q+\ell}. \quad (\text{R8})$$

Then $0 \leq K_q \leq 1$, its coefficients are nonnegative and symmetric, and

$$b_0 \geq \frac{1}{2\sqrt{q}}, \quad \sum_{\ell=-q}^q \left(\frac{b_\ell}{b_0} \right)^{2d} \geq L + 1. \quad (\text{R9})$$

Proof. The expansion follows by applying the binomial theorem to $(e^{\pi i t} + e^{-\pi i t})^{2q}/2^{2q}$. For the central coefficient, write $u_q = 4^{-q} \binom{2q}{q}$. We have $u_1 = 1/2$ and

$$\frac{u_{q+1}}{u_q} = \frac{2q+1}{2q+2} \geq \sqrt{\frac{q}{q+1}}.$$

After squaring, the last inequality reduces to $(2q+1)^2 \geq 4q(q+1)$. Induction proves the stated lower bound for b_0 .

For $0 \leq \ell \leq L$,

$$\frac{b_\ell}{b_0} = \prod_{j=1}^{\ell} \frac{q-j+1}{q+j} = \prod_{j=1}^{\ell} \left(1 - \frac{2j-1}{q+j} \right) \geq 1 - \frac{\ell^2}{q}. \quad (\text{R10})$$

Here all factors lie in $[0, 1]$, and $\prod_j (1 - z_j) \geq 1 - \sum_j z_j$, while $\sum_{j=1}^{\ell} (2j-1)/(q+j) \leq \ell^2/q$. Bernoulli's inequality now gives

$$\left(\frac{b_\ell}{b_0} \right)^{2d} \geq 1 - \frac{2d\ell^2}{q} \geq \frac{1}{2}. \quad (\text{R11})$$

The central term contributes 1, and the $2L$ terms with $1 \leq |\ell| \leq L$ contribute at least L . $\square$

II.9.2 Repeated entries and weighted relation errors

A direct union bound over all the kernel's coefficient relations would be expensive, because its degree depends on the desired accuracy. Instead, we keep their Fourier weights. Positivity and the fact that the weights sum to one allow all the unwanted relations to be controlled together.

Lemma II.9.3 (A rank-two relation inside a finite set). *Let $d \geq 2$, let p be prime, let $B \subset \mathbb{F}_p$ have size $m > 0$, and put*

$$v = (\underbrace{1, \dots, 1}_d, \underbrace{-1, \dots, -1}_d) \in \mathbb{F}_p^{2d}.$$

For every $\epsilon \in \mathbb{F}_p^{2d} \setminus \mathbb{F}_p v$,

$$\#\{x \in B^{2d} : v \cdot x = \epsilon \cdot x = 0\} \leq m^{2d-2}. \quad (\text{R12})$$

The number of additive $2d$ -tuples in B with repeated entries is at most $\binom{2d}{2} m^{2d-2}$.

Proof. The matrix with rows v, ϵ has rank two, so it has an invertible 2-by-2 column minor. Fix the other $2d - 2$ coordinates arbitrarily in B . The two remaining coordinates are uniquely determined in $\mathbb{F}_p$; either both belong to B , or this choice gives no admissible completion. This proves (R12).

If $x_i = x_j$, the vector $e_i - e_j$ supplies a second relation. It is not a scalar multiple of v , since it has zero coordinates whereas every nonzero scalar multiple of v has none. Apply (R12) to each of the $\binom{2d}{2}$ pairs of positions and take a union bound. $\square$

Lemma II.9.4 (Weighted selection estimates). *Let $d \geq 2$, $q \geq 1$, and let $p > 2q$ be prime. Let $B \subset \mathbb{F}_p$ have size $m > 0$, and let $\phi : B \rightarrow \mathbb{F}_p$. Choose r, s independently and uniformly in $\mathbb{F}_p$. Conditional on this choice, retain each point $x \in B$ independently with probability*

$$K_q \left(\frac{rx + s\phi(x)}{p} \right). \quad (\text{R13})$$

Integer representatives do not matter because K_q is periodic. Put

$$P_0 = b_0^{2d}, \quad Q = \sum_{\ell=-q}^q \left(\frac{b_\ell}{b_0} \right)^{2d}, \quad C_d = 1 + \binom{2d}{2}. \quad (\text{R14})$$

Let X count all selected respected additive tuples, and let Y count all selected unrespected additive tuples. Then

$$\mathbb{E}X \geq P_0 Q T_d^\phi(B), \quad \mathbb{E}Y \leq P_0 (T_d(B) - T_d^\phi(B)) + C_d m^{2d-2}. \quad (\text{R15})$$

Proof. For a fixed tuple $x = (x_1, \dots, x_{2d})$, abbreviate its point probabilities by $P_{r,s}(x_j)$, and form the product with multiplicities

$$A_{r,s}(x) = \prod_{j=1}^{2d} P_{r,s}(x_j).$$

Its actual conditional survival probability is the product over the *distinct* entries of x . Since all point probabilities lie in $[0, 1]$, the actual probability is at least $A_{r,s}(x)$, equals it when the entries are distinct, and exceeds it by at most 1 otherwise. This observation accounts for repeated points without assuming that their inclusion events are independent.

Expand $A_{r,s}(x)$ using the nonnegative Fourier coefficients from (R8). For $\epsilon \in \{-q, \dots, q\}^{2d}$, put $w_\epsilon = \prod_j b_{\epsilon_j}$. Orthogonality gives

$$\mathbb{E}_{r,s} A_{r,s}(x) = \sum_{\epsilon} w_\epsilon 1_{\{\epsilon \cdot x = 0\}} 1_{\{\epsilon \cdot \phi(x) = 0\}}, \quad \sum_{\epsilon} w_\epsilon = \left(\sum_{\ell=-q}^q b_\ell \right)^{2d} = 1. \quad (\text{R16})$$

Here $\epsilon \cdot \phi(x)$ denotes $\sum_j \epsilon_j \phi(x_j)$. The last equality uses $K_q(0) = 1$. All weights in this identity are nonnegative.

If ϵ is a scalar multiple of v modulo p , the condition $p > 2q$ forces its integer entries to be $(\ell, \dots, \ell, -\ell, \dots, -\ell)$ for a single $\ell \in [-q, q]$. These terms have total weight $P_0 Q$. Every one contributes for a respected additive tuple. Thus its expected product with multiplicities is at least $P_0 Q$, and the same lower bound holds for its actual survival probability. Summing proves the assertion about $\mathbb{E}X$.

For an unrespected additive tuple, every nonzero such ℓ is invertible modulo p , so only $\ell = 0$ can contribute among these scalar multiples. Its weight is P_0 . For each remaining coefficient vector, Lemma II.9.3 bounds the number of domain tuples satisfying the additional relation by

m^{2d-2} . Requiring that they be unrespected, or imposing the additional image relation in (R16), can only reduce this number. Summing first over tuples and then over coefficient vectors, and using $\sum_{\epsilon} w_{\epsilon} = 1$, gives

$$\sum_{\substack{x \in B^{2d} \\ x \text{ additive and unrespected}}} \mathbb{E}_{r,s} A_{r,s}(x) \leq P_0(T_d(B) - T_d^{\phi}(B)) + m^{2d-2}.$$

Finally, the difference between actual survival and this product need only be charged to repeated tuples. Lemma II.9.3 bounds their number by $\binom{2d}{2} m^{2d-2}$. Adding that error proves the assertion about $\mathbb{E}Y$. $\square$

II.9.3 Proof of the polynomial restriction theorem

Proof of Theorem II.9.1. Set

$$a = \alpha\eta, \quad m = |B| = \beta p, \quad H = m^{2d-1}, \quad c = \kappa_d a^{2d}. \quad (\text{R17})$$

Choose

$$L = \left\lceil \frac{4}{a} \right\rceil, \quad q = 4dL^2. \quad (\text{R18})$$

Since $a \leq 1$,

$$L \leq \frac{5}{a}, \quad q \leq \frac{100d}{a^2}. \quad (\text{R19})$$

The size condition is $m \geq \Lambda_d a^{-2d}$. It implies $p > 2q$, since $p \geq m$, $\Lambda_d > 200d$, and $a^{-2d} \geq a^{-2}$. Furthermore, with $C_d = 1 + \binom{2d}{2}$, it gives

$$C_d m^{2d-2} \leq cH, \quad (\text{R20})$$

because $\Lambda_d = C_d / \kappa_d$.

Apply Lemma II.9.4 with this kernel. Lemma II.9.2 gives

$$Q \geq L + 1 \geq \frac{4}{a}, \quad P_0 \geq (4q)^{-d} \geq \frac{a^{2d}}{(400d)^d} = 2c. \quad (\text{R21})$$

The input hypothesis and the bound $T_d(B) \leq H$ therefore imply

$$\mathbb{E}X \geq P_0 Q \alpha H, \quad \mathbb{E}Y \leq P_0 H + C_d m^{2d-2}. \quad (\text{R22})$$

Consequently

$$\begin{aligned} \mathbb{E}[\eta X - Y] &\geq P_0(\eta Q \alpha - 1)H - C_d m^{2d-2} \\ &\geq 3P_0 H - C_d m^{2d-2} \geq 5cH \geq cH. \end{aligned} \quad (\text{R23})$$

Some realization B' thus satisfies

$$\eta X - Y \geq cH. \quad (\text{R24})$$

There is no omitted class of tuples here: X and Y include all additive tuples in B' , with repeated entries counted. Hence

$$T_d(B') = X + Y \geq X \geq \frac{cH}{\eta} = \kappa_d \alpha^{2d} \eta^{2d-1} H, \quad (\text{R25})$$

and

$$\frac{T_d^{\phi}(B')}{T_d(B')} = \frac{X}{X+Y} \geq \frac{1}{1+\eta} \geq 1-\eta. \quad (\text{R26})$$

This proves both conclusions. $\square$

The weighted estimate is what makes the size condition depend on $(\alpha\eta)^{-2d}$. Counting all coefficient vectors separately would introduce a power of the bandwidth q . Instead, each unwanted relation is bounded by the same m^{2d-2} , and its contribution is multiplied by its Fourier weight; the weights have total mass one. The separate error for repeated entries has the same lower power of m . No independence assumption is made for repeated coordinates.

The exponent η^{2d-1} in the conclusion also deserves attention. The survival calculation supplies a quantity c proportional to $(\alpha\eta)^{2d}$, but (R24) yields $X \geq cH/\eta$. Keeping this final factor improves the retained count by η^{-1} .

II.9.4 Order eight and the comparison with Corollary 9.4

Corollary II.9.5 (Explicit order-eight restriction). *Let $0 < \beta, \gamma, \eta \leq 1$, let p be prime, and let $B \subset \mathbb{F}_p$ have size βp . Suppose that $\phi : B \rightarrow \mathbb{F}_p$ respects at least $\gamma |B|^3$ additive quadruples. If*

$$\boxed{p \geq 2^{102} \gamma^{-112} \eta^{-16} \beta^{-1}}, \quad (\text{R27})$$

then there is $B' \subset B$ containing at least

$$\boxed{2^{-95} \gamma^{112} \eta^{15} \beta^{15} p^{15}} \quad (\text{R28})$$

ordered additive 16-tuples, at least a proportion $1 - \eta$ of which are respected by ϕ .

Proof. Corollary II.8.2 permits $\alpha = \gamma^7$ in Theorem II.9.1 with $d = 8$. Its exact constants are

$$\kappa_8 = \frac{1}{2 \cdot 3200^8}, \quad \Lambda_8 = 242 \cdot 3200^8. \quad (\text{R29})$$

The integer inequalities

$$2 \cdot 3200^8 < 2^{95}, \quad 242 \cdot 3200^8 < 2^{102} \quad (\text{R30})$$

give the convenient powers of two in the statement. $\square$

The displayed count in source Corollary 9.4 is

$$\left(\frac{\gamma^7 \beta^6 \eta}{4} \right)^{2^{19}} \beta^{15} p^{15}. \quad (\text{R31})$$

The improvement has two distinct causes. Exact Parseval removes the β^6 inside the large power. The powered cosine replaces the repeated weak enrichment by one stronger selection, giving the powers $\gamma^{112} \eta^{15}$. Both changes preserve the same configuration normalization. The source corollary also inherits the sufficiently-large- p condition from Lemma 9.3; the present statement gives an explicit sufficient condition instead of suppressing it [1, 23].

This is a local improvement in the transition from a positive proportion of respected relations to an almost-homomorphism. Its relevance is not confined to the original exposition: later quantitative work also uses transitions of this kind, followed by additional steps that organize and extend the approximate homomorphism [6]. The present calculations do not by themselves supply an improved global bound for Szemerédi's theorem. Such a claim would require propagating the parameters through all later restrictions and density increments.

[write] **6 October 2026.** Rechecked at the write: at $\gamma = \eta = \beta = 1/2$, $(\gamma^7 \beta^6 \eta / 4)^{2^{19}} = 2^{-16 \cdot 2^{19}} = 2^{-8\,388\,608}$, while $2^{-95} \gamma^{112} \eta^{15} = 2^{-222}$ relative to $|B|^{15}$, under $p \geq 2^{102} \cdot 2^{112} \cdot 2^{16} \cdot 2 = 2^{231}$ (Section V.16). The formal `lemma_9_3` and `corollary_9_4` (Sections08_09.lean, lines 77 and 91; exact companions in `Proofs09Restriction.lean`, lines 376 and 416) quantify over all moduli $N \geq N_0$ without primality and are proved by another route; Corollary II.9.5 is a prime-field interface and does not prove them, as Section V.19 says. Not formalized.

II.9.5 What the exponent says about this selection method

The use of a higher-degree kernel raises a natural optimization question. There is a simple bound that explains why the powers in Theorem II.9.1 are natural for the particular expectation argument just used.

Proposition II.9.6 (A limitation of single-kernel amplification). *Let K be a real trigonometric polynomial with $0 \leq K \leq 1$, mean $b_0 > 0$, and Fourier coefficients b_ℓ . For $d \geq 2$, put*

$$Q_K = \frac{\sum_\ell |b_\ell|^{2d}}{b_0^{2d}}.$$

Then

$$Q_K \leq \frac{1}{b_0}. \quad (\text{R32})$$

Consequently, if $Q_K \geq 4/(\alpha\eta)$, the base survival probability b_0^{2d} is at most $(\alpha\eta/4)^{2d}$, and the scalar-multiple contribution $\sum_\ell |b_\ell|^{2d}$ to respected-tuple survival is at most $(\alpha\eta/4)^{2d-1}$.

Proof. Since $K \geq 0$, $|b_\ell| \leq b_0$. Parseval and $K^2 \leq K$ imply

$$\sum_\ell |b_\ell|^{2d} \leq b_0^{2d-2} \sum_\ell |b_\ell|^2 = b_0^{2d-2} \int_0^1 K(t)^2 dt \leq b_0^{2d-1}.$$

Divide by b_0^{2d} , and then use the assumed lower bound for Q_K . □

For an input with exactly αH respected tuples, the contribution guaranteed by the scalar-multiple coefficient relations is bounded by a constant times $\alpha^{2d} \eta^{2d-1} H$, under the stipulated amplification requirement. Theorem II.9.1 attains these parameter powers up to constants depending on d . This is an observation about that part of the single-kernel expectation budget. Additional relations and repeated entries can increase the actual expectation. The calculation is not a lower bound for every possible restriction procedure, does not rule out unusually favorable realizations, and does not establish optimality of the theorem's size threshold.

II.10 Arrangement moments retain the cross-section profile

Source 04, delivered Section 5.

The same principle applies to Lemmas 12.3 and 14.7, but their second moments carry multiplicities. The correct quantity to retain is the number of respected pairs of congruent cubes within a cross-section, rather than simply $|B|$.

II.10.1 The weighted representation

Let $B \subset \mathbb{F}_p^{k+1}$, with $k \geq 1$, let $\phi : B \rightarrow \mathbb{F}_p$, and write

$$B_r = \{y \in \mathbb{F}_p^k : (y, r) \in B\}.$$

For $h, y \in \mathbb{F}_p^k$, the axis-parallel cube with base y and sidelengths h has vertices

$$y + \epsilon \cdot h = (y_1 + \epsilon_1 h_1, \dots, y_k + \epsilon_k h_k), \quad \epsilon \in \{0, 1\}^k.$$

If all these vertices lie in B_r , define its value by

$$\Phi_h(y, r) = \sum_{\epsilon \in \{0, 1\}^k} (-1)^{\epsilon_1 + \dots + \epsilon_k} \phi(y + \epsilon \cdot h, r). \quad (\text{A1})$$

For each h , define the nonnegative integer weight

$$a_h(r, v) = \#\{y : y + \{0, 1\}^k \cdot h \subset B_r, \Phi_h(y, r) = v\}. \quad (\text{A2})$$

Let R_j count the respected j -arrangements: ordered lists of $2j$ cubes with common h , satisfying equal sums of the first and second halves of their cross-section indices, and equal sums of their cube values. Zero sidelengths and repeated vertices are included in this count, as in the moment identities of the source. Expanding the Fourier transform of a_h on $\mathbb{F}_p^2$ gives

$$R_j = \frac{1}{p^2} \sum_{h \in \mathbb{F}_p^k} \sum_{\chi \in \widehat{\mathbb{F}_p^2}} |\widehat{a}_h(\chi)|^{2j}. \quad (\text{A3})$$

In particular,

$$R_1 = \sum_{h, r, v} a_h(r, v)^2 \quad (\text{A4})$$

counts ordered congruent cube pairs in the same cross-section with equal values.

Theorem II.10.1 (Arrangement interpolation with profile). *Suppose that $R_2 > 0$. For every integer $j \geq 2$,*

$$R_j \geq \frac{R_2^{j-1}}{R_1^{j-2}}. \quad (\text{A5})$$

Define

$$\beta_r = \frac{|B_r|}{p^k}, \quad \beta = \frac{|B|}{p^{k+1}}, \quad \sigma = \frac{1}{p} \sum_r \beta_r^3. \quad (\text{A6})$$

Then

$$R_1 \leq \sigma p^{3k+1}, \quad \sigma \leq \beta. \quad (\text{A7})$$

Consequently, if $\theta > 0$ and $R_2 \geq \theta p^{5k+3}$, then

$$\boxed{R_j \geq \theta^{j-1} \sigma^{-(j-2)} p^{(2j+1)k+2j-1}}. \quad (\text{A8})$$

Proof. Apply the Hölder calculation of Theorem II.8.1 to the spectrum indexed by (h, χ) , using the common normalization p^{-2} in (A3). This gives (A5). The hypothesis $R_2 > 0$ ensures $R_1 > 0$, so the denominator is legitimate.

To bound R_1 , temporarily ignore the equality of cube values. A pair of cubes in B_r , with bases y, z and common sidelengths h , determines a triple

$$(y, z, y + h) \in B_r^3. \quad (\text{A9})$$

The last point is the opposite vertex of the first cube. This map is injective: the triple determines both bases and $h = (y + h) - y$. Hence there are at most $|B_r|^3$ such pairs, even before the value condition is imposed. Summing over r gives

$$R_1 \leq \sum_r |B_r|^3 = \sigma p^{3k+1}.$$

Since $0 \leq \beta_r \leq 1$, and $p^{-1} \sum_r \beta_r = \beta$, we have $\sigma \leq \beta$. Finally substitute these estimates into (A5). The ambient exponent is

$$(5k + 3)(j - 1) - (3k + 1)(j - 2) = (2j + 1)k + 2j - 1,$$

which proves (A8). $\square$

For $j = 8$, the replacement for Lemma 14.7 is therefore

$$R_8 \geq \theta^7 \sigma^{-6} p^{17k+15} \geq \theta^7 \beta^{-6} p^{17k+15}. \quad (\text{A10})$$

The source uses the same fourth-moment hypothesis but discards the factor σ when bounding the second moment. If every cross-section has density β , then $\sigma = \beta^3$, so the gain is β^{-18} . More generally, $\beta_r \leq K\beta$ for every r implies

$$\sigma \leq K^2 \beta^3. \quad (\text{A11})$$

This records exactly which information about the cross-sections can be used by the interpolation step.

Using Corollary 14.6 as stated in the repository's corrected transcription, the density power in Lemma 14.8 improves from $7 \cdot 4^{k+1}$ to $7 \cdot 4^{k+1} - 6$:

$$R_8 \geq \beta^{7 \cdot 4^{k+1} - 6} \gamma^{21k \cdot 4^{k+1}} p^{17k+15}. \quad (\text{A12})$$

Keeping the profile gives the stronger factor $\beta^{7 \cdot 4^{k+1}} \sigma^{-6}$. The analogous use in Lemma 12.4 changes $\beta^{112} \gamma^{336} p^{32}$ to $\beta^{106} \gamma^{336} p^{32}$. In that section the relevant fibers are vertical; its two coordinate roles are exchanged relative to the notation above. These conclusions use only the improved moment step, with the preceding hypotheses and estimates retained [1, 23].

II.10.2 Equality examples and the scope of sharpness

The arrangement interpolation and the profile bound extend to axis-parallel cubes over any finite abelian group G ; the proof uses only its characters and additive operations. This extension supplies a sharp example for the factor β^{-6} . Let $|G| = N$, let $C \leq G$, and take

$$B = G^k \times C, \quad \phi = 0, \quad \beta = \frac{|C|}{N}.$$

The nonempty cross-sections are full. Direct counting gives

$$R_1 = \beta N^{3k+1}, \quad R_2 = \beta^3 N^{5k+3}, \quad R_j = \beta^{2j-1} N^{(2j+1)k+2j-1}. \quad (\text{A13})$$

Thus both the interpolation inequality and $\sigma \leq \beta$ are equalities. This proves sharpness for the finite-abelian formulation; it does not assert the existence of proper subgroups of a cyclic group of prime order.

The uniform-profile case also has equality examples over a prime field. Let V be a coordinate subspace of $\mathbb{F}_p^k$, and take $B = V \times \mathbb{F}_p$, with $\phi = 0$. Every cross-section has density $\beta = |V|/p^k$. A cube is contained in V precisely when its base and sidelength vector belong to V . Therefore

$$R_j = |V|^{2j+1} p^{2j-1}, \quad \sigma = \beta^3. \quad (\text{A14})$$

Again the moment and profile estimates are equalities. In particular, the improvement by β^{-18} in the uniform-profile order-eight bound is attained at the available densities of these examples.

These equality constructions identify which parts of the argument cannot be improved using only the displayed moments and profile. They leave open the use of additional geometric or distributional information in applications.

Part III

Cube and progression counts

Contents. Counting consequences of uniformity. Sections III.1–III.2: source 03, delivered Sections 5–6 (progression counting by a density-weighted telescope; the exact fourth-order term of the centered cube count on every finite abelian group, its torsion correction, sharpness for sets, higher coefficients). Sections III.3–III.4: source 04, delivered Sections 6–7 (the sharp real centered gap $\|f\|_{U^3}^8 \geq 2 \|f\|_{U^2}^8$ on odd-order groups, the sharp leading constant $6\sqrt{2}$ of the three-cube bound, the fractional correction exponent $16/3$, and density-weighted progression counting with the lifting lemma).

Overlaps (dated notes). Cancellation through degree three and the quartic expansion (03 on every group, 04 on odd-order groups; $P_d = R_d$); the $\mathbb{Z}/3\mathbb{Z}$ cosine polynomial (identical in both); the generalized von Neumann bound and the telescoping theorem (independently written in both). **Cross-source answers.** Source 04’s Theorem III.3.4 answers source 03’s Research question V.14.1 for $d = 3$, with $c_3^{\text{odd}} = 2^{-1/2}$. Source 04’s Lemma III.4.5 is the only statement of the five sources that is formalized (`hasNatAP_of_short_modular_sequence`).

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
cube count	03: $Q_d(F)$, $m = 2^d$ vertices, $\eta = \ f\ _{U^d}$; 04: $Q_d(W)$, $q = 2^d$, $u = \ f\ _{U^d}$, $v = \ f\ _{U^2}$. <i>04’s q here is not its kernel degree q of Part II.</i>
R_d, P_d, T_d	03: R_d = number of ordinary parallelograms, T_d = binary tetrahedra, $K_d = R_d + T_d$, $\mathcal{R}_d$ the remainder; 04: P_d = number of parallelograms (= 03’s R_d), R_d, R_6, R_7 remainders. <i>04’s R_d is not 03’s R_d.</i>
$\mathcal{E}, B$	03: $\mathcal{E}(f) = \ f\ _{U^2}^4$, $\mathcal{E}_{(2)}(f)$ the order-two (quotient) energy, $B(S)$ the telescoping recursion; 04: $\mathcal{E}(\lambda)$ the additive energy of $\lambda_r = a_r ^2$, $B_5(f)$ the quintic correlation, $D(S)$ the recursion (= 03’s $B(S)$).
slopes, α	03: slopes a_i with invertible differences, $\Lambda_{\mathbf{a}}, \eta = \ 1_A - \delta\ _{U^{k-1}}$; 04: admissible slopes $c_i, \Lambda_{\mathbf{c}}, u$. Both: Gowers’s α -uniformity means $\ f\ _{U^d}^{2^d} \leq \alpha$; <i>α is a norm power, not a norm.</i>

III.1 Progression counting with fewer error terms

Source 03, delivered Section 5.

For integers $a_1, \dots, a_k$ and functions $g_i: G \rightarrow \mathbb{C}$, define

$$\Lambda_{\mathbf{a}}(g_1, \dots, g_k) = \mathbb{E}_{x, t \in G} \prod_{i=1}^k g_i(x + a_i t).$$

We say that the slopes have invertible differences on G if each map $x \mapsto (a_i - a_j)x$, $i \neq j$, is an automorphism. For $G = \mathbb{Z}/N\mathbb{Z}$ this means $\gcd(a_i - a_j, N) = 1$. Consecutive slopes satisfy the condition whenever $\gcd(N, (k-1)!) = 1$, which is weaker than primality of N .

Lemma III.1.1 (Generalized von Neumann inequality). *Suppose the slopes have invertible differences and $k \geq 2$. For $|F_i| \leq 1$,*

$$|\Lambda_{\mathbf{a}}(F_1, \dots, F_k)| \leq \|F_k\|_{U^{k-1}(G)}.$$

The same bound holds with any one of the functions distinguished.

Proof. For $k = 2$, the map $(x, t) \mapsto (x + a_1 t, x + a_2 t)$ is a bijection, so the average factors and is at most $|\mathbb{E}F_2| = \|F_2\|_{U^1}$. For the inductive step replace x by $y - a_1 t$, apply Cauchy–Schwarz in y , and remove the bounded factor $F_1(y)$. On expanding the square, write the two step variables as t and $t + u$. With $D_h F(z) = F(z)F(z + h)$, the resulting expression is

$$\mathbb{E}_u \mathbb{E}_{y,t} \prod_{i=2}^k D_{(a_i - a_1)u} F_i(y + (a_i - a_1)t).$$

Its absolute value is at most $\mathbb{E}_u \|D_{(a_k - a_1)u} F_k\|_{U^{k-2}}$ by induction. Hölder and the invertibility of $a_k - a_1$ bound this by

$$\left(\mathbb{E}_h \|D_h F_k\|_{U^{k-2}}^{2^{k-2}} \right)^{1/2^{k-2}} = \|F_k\|_{U^{k-1}}^2.$$

This bounds the square of the original average. Permuting the indices allows any function to be distinguished. $\square$

Theorem III.1.2 (Density-weighted telescoping). *Assume the slopes have invertible differences. Let $0 \leq g_i \leq 1$, $\delta_i = \mathbb{E}g_i$, and $f_i = g_i - \delta_i$. For any permutation π of $\{1, \dots, k\}$,*

$$\left| \Lambda_{\mathbf{a}}(g_1, \dots, g_k) - \prod_i \delta_i \right| \leq \sum_{j=3}^k \left(\prod_{\ell=j+1}^k \delta_{\pi(\ell)} \right) \|f_{\pi(j)}\|_{U^{j-1}}. \quad (\text{III.1.1})$$

For $k = 2$ the error is zero.

Proof. Expand by replacing the functions with their means one at a time, starting with the last function in the chosen order. The exact difference is a sum of k terms. Term j has coefficient $\prod_{\ell > j} \delta_{\pi(\ell)}$ and an average containing $g_{\pi(1)}, \dots, g_{\pi(j-1)}, f_{\pi(j)}$. The first term vanishes because $f_{\pi(1)}$ has mean zero. The second vanishes by the two-variable bijection used in Lemma III.1.1. Apply that lemma to each remaining term, with its j slopes. $\square$

In the setting of Corollary 3.3, use the original index order. The normalized uniformity norms that occur in (III.1.1) are then at most its parameter α . Therefore its bound $2^k \alpha$ may be replaced by

$$\alpha \sum_{j=3}^k \prod_{i=j+1}^k \delta_i \leq (k-2)\alpha.$$

The operation is a telescoping expansion, which has one discrepancy per replacement. It avoids the unnecessary 2^k -term expansion of all balanced factors simultaneously. This is a standard multilinear technique; its role here is to record the stronger bound with the exact inhomogeneous parameters used by the source.

III.1.1 Optimizing the order of replacement

If all available norms in (III.1.1) are bounded by the same η , arrange the densities in nonincreasing order. This minimizes the displayed upper bound: every tail product then consists of the smallest densities of the required size. When distinct functions have different norms at different levels, define for a set of indices S

$$B(S) = 0 \quad (|S| \leq 2), \quad B(S) = \min_{i \in S} \{ \|f_i\|_{U^{|S|-1}} + \delta_i B(S \setminus \{i\}) \} \quad (|S| \geq 3).$$

This finite recursion computes the best bound among these telescoping orders in $O(k2^k)$ arithmetic operations once the norms are supplied. It is an optimization of the bound, not a claim that the bound is the exact counting error. For fixed small k , this separates analytic estimates from a finite, reproducible choice of order.

Corollary III.1.3 (An explicit modular progression criterion). *Let $k \geq 3$, let $|G| = N$, and suppose multiplication by each of $1, \dots, k-1$ is an automorphism of G . Let $A \subset G$ have density $0 < \delta \leq 1$, and set*

$$\eta = \|1_A - \delta\|_{U^{k-1}}, \quad S_{k-2}(\delta) = 1 + \delta + \dots + \delta^{k-3}.$$

The number of ordered pairs (x, t) with $t \neq 0$ and $x, x+t, \dots, x+(k-1)t \in A$ is at least

$$N^2(\delta^k - \eta S_{k-2}(\delta)) - \delta N. \quad (\text{III.1.2})$$

In particular, if

$$\eta \leq \frac{\delta^k}{2S_{k-2}(\delta)}, \quad N \geq 4\delta^{1-k}, \quad (\text{III.1.3})$$

there are at least $\delta^k N^2/4$ such pairs. Each has k distinct vertices. In the notation of the source, it suffices that

$$\alpha \leq \left(\frac{\delta^k}{2S_{k-2}(\delta)} \right)^{2^{k-1}}$$

when A is α -uniform of degree $k-2$.

Proof. Gowers norm monotonicity gives $\|1_A - \delta\|_{U^{j-1}} \leq \eta$ for $3 \leq j \leq k$. Apply Theorem III.1.2 to equal functions and consecutive slopes. Exactly $|A| = \delta N$ parameter pairs have $t = 0$. For $t \neq 0$, equality of vertices with indices i, j would imply $(i-j)t = 0$, contrary to the automorphism assumption. Under (III.1.3), the right side of (III.1.2) is at least $\delta^k N^2/4$. $\square$

Remark III.1.4 (Modular and integer progressions). Corollary III.1.3 concerns progressions in the group. They do not automatically become ordinary integer progressions under the representative map to $\{0, \dots, N-1\}$. One sufficient rectification condition is that every representative vertex lies in an interval of length H with $2(H-1) < N$. Its consecutive second differences are zero modulo N and have absolute value below N , so they are zero as integers. This proves rectification, but obtaining the necessary interval control is additional information. The general modular counting criterion by itself supplies no such localization.

[write] 6 October 2026. This rectification argument is formalized, in the form of source 04's Lemma III.4.5 (note there): Lean's `hasNatAP_of_short_modular_sequence` takes representatives in $\{0, \dots, n\}$ with $2n < N$. The remark's interval of H consecutive integers reduces to that case by a translation, a step not stated in Lean. Sections III.1 and III.4 prove the same telescoping bound independently; see the note after Theorem III.4.3.

Remark III.1.5 (Why small torsion cannot simply be ignored). For $G = \mathbb{Z}/2\mathbb{Z}$, slopes 0, 1, 2 have a noninvertible difference. Taking the nontrivial real character f , the average $\mathbb{E}_{x,t} f(x)f(x+2t)$ equals one, although $\mathbb{E}f = 0$. The two centered factors are evaluated at the same point. Thus the pair cancellation used above really can fail when the slope hypothesis is removed.

III.2 The exact fourth-order term in the cube count

Source 03, delivered Section 6.

Gowers' Lemma 3.10 bounds a cube count by applying the triangle inequality to the Gowers norm. The resulting estimate is convenient, but it discards the mean-zero condition on the balanced function after that condition has been used to define uniformity. Retaining it has a substantial local effect: the first three error terms vanish, and the fourth can be computed exactly. The computation also identifies precisely what changes when the ambient finite abelian group has a nontrivial quotient of exponent two.

All results in this section concern the same labelled cubes as [1, Lemmas 3.7–3.10]; coincident vertices are allowed. The refinements are elementary consequences of the cube expansion, character orthogonality, and Gowers–Cauchy–Schwarz. Their proofs are included in full. They are useful sharpenings of the displayed lemma; we do not claim a new inverse theorem or a change in the best known quantitative bounds for Szemerédi's theorem.

III.2.1 Normalization and the centered expansion

Let G be a finite abelian group, with all averages normalized by the cardinality of their domain. Write $V_d = \{0, 1\}^d$, $m = 2^d$, and

$$L_v(x, h) = x + \sum_{i=1}^d v_i h_i \quad (v \in V_d).$$

For a real function $F : G \rightarrow \mathbb{R}$, define

$$Q_d(F) = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{v \in V_d} F(L_v(x, h)). \quad (\text{III.2.1})$$

Thus $Q_d(F) = \|F\|_{U^d}^m$; the usual conjugations in the definition of U^d have no effect for real functions. In particular, $Q_d(F) \geq 0$. If $F = \mathbf{1}_A$, the number of labelled cubes in A is $|G|^{d+1} Q_d(F)$.

Put $\delta = \mathbb{E}F$ and $f = F - \delta$. For $S \subseteq V_d$, put

$$M_S(f) = \mathbb{E}_{x, h} \prod_{v \in S} f(L_v(x, h)), \quad M_\emptyset(f) = 1. \quad (\text{III.2.2})$$

The exact finite expansion is

$$Q_d(\delta + f) = \sum_{S \subseteq V_d} \delta^{m-|S|} M_S(f). \quad (\text{III.2.3})$$

No limiting argument or regularity assumption is involved.

For real functions $g_v : G \rightarrow \mathbb{R}$, the normalized Gowers–Cauchy–Schwarz inequality gives

$$\left| \mathbb{E}_{x, h} \prod_{v \in V_d} g_v(L_v(x, h)) \right| \leq \prod_{v \in V_d} \|g_v\|_{U^d}. \quad (\text{III.2.4})$$

This is [1, Lemma 3.8] with normalized sums; its proof by successive Cauchy–Schwarz inequalities works on every finite abelian group. Taking $g_v = f$ on S and $g_v = 1$ elsewhere shows that, if $\eta = \|f\|_{U^d}$,

$$|M_S(f)| \leq \eta^{|S|}. \quad (\text{III.2.5})$$

We use also the standard monotonicity $\|f\|_{U^2} \leq \|f\|_{U^d}$ for $d \geq 2$.

Lemma III.2.1 (Cancellation through degree three). *If $\mathbb{E}f = 0$ and $1 \leq |S| \leq 3$, then $M_S(f) = 0$ on every finite abelian group.*

Proof. For $|S| = 1$, average over x . Otherwise choose a coordinate in which the distinct selected vertices are not all equal. For two or three vertices, one side of this coordinate contains exactly one selected vertex. Reflect that coordinate if necessary, so that this vertex is the only one with coordinate 1. A coordinate reflection preserves the average: it corresponds to the bijective change of variables $x' = x + h_i$, $h'_i = -h_i$, leaving the other h_j unchanged. After the reflection, h_i occurs in exactly one factor in the product defining M_S . Its average is $\mathbb{E}f = 0$. $\square$

Corollary III.2.2 (First refinement of Lemma 3.10). *For $d \geq 2$, $\delta \geq 0$, and real mean-zero f ,*

$$|Q_d(\delta + f) - \delta^m| \leq \sum_{j=4}^m \binom{m}{j} \delta^{m-j} \eta^j. \quad (\text{III.2.6})$$

Consequently the cube-count error is $O_{d,\delta}(\eta^4)$ as $\eta \rightarrow 0$ with d and positive δ fixed.

Proof. Expand using (III.2.3), discard the terms that vanish by Lemma III.2.1, and apply (III.2.5) to every remaining term. $\square$

The coefficient $\binom{m}{4}$ in (III.2.6) is itself far from necessary. Most four-vertex supports vanish as well.

III.2.2 Fourier energies and the quotient by doubling

Let $\widehat{G}$ be the character group, with characters written multiplicatively, and use normalized Fourier coefficients

$$\widehat{f}(\chi) = \mathbb{E}_{x \in G} f(x) \overline{\chi(x)}.$$

Write 1 for the trivial character. For real mean-zero f , set

$$\mathcal{E}(f) = \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^4 = \|f\|_{U^2}^4, \quad (\text{III.2.7})$$

$$\mathcal{E}_{(2)}(f) = \sum_{\substack{\chi \in \widehat{G} \setminus \{1\} \\ \chi^2 = 1}} |\widehat{f}(\chi)|^4. \quad (\text{III.2.8})$$

The omitted trivial-character term is zero. In particular,

$$0 \leq \mathcal{E}_{(2)}(f) \leq \mathcal{E}(f) \leq \eta^4. \quad (\text{III.2.9})$$

The second energy has a concrete quotient interpretation. Let $2G = \{2x : x \in G\}$ and

$$P_2 f(x) = \mathbb{E}_{z \in 2G} f(x + z).$$

This is the conditional average of f on the cosets of $2G$. A character survives this average exactly when it annihilates $2G$, equivalently when $\chi^2 = 1$. Fourier orthogonality therefore gives

$$\mathcal{E}_{(2)}(f) = \|P_2 f\|_{U^2(G)}^4. \quad (\text{III.2.10})$$

Thus the correction below measures the part of f visible in $G/2G$. If $|G|$ is odd, $2G = G$ and this correction is zero. If every element of G has order dividing two, then $P_2 f = f$.

III.2.3 A complete classification of four-vertex supports

Call a four-element subset $S \subseteq V_d$ an *ordinary parallelogram* if its vertices can be ordered a, b, c, e with $a + b = c + e$ as vectors in $\mathbb{Z}^d$. Call it a *binary tetrahedron* if

$$\bigoplus_{v \in S} v = 0$$

under coordinatewise addition modulo two, but S is not an ordinary parallelogram. This terminology describes the support in the Boolean cube, rather than a nondegeneracy condition on its image in G .

Lemma III.2.3 (The three possible fourth-order contributions). *For real mean-zero f and a four-element $S \subseteq V_d$,*

$$M_S(f) = \begin{cases} \mathcal{E}(f), & S \text{ is an ordinary parallelogram,} \\ \mathcal{E}_{(2)}(f), & S \text{ is a binary tetrahedron,} \\ 0, & \bigoplus_{v \in S} v \neq 0. \end{cases} \quad (\text{III.2.11})$$

These cases exhaust all four-element subsets.

Proof. If the XOR of the four vertices is nonzero, some coordinate has an odd number of ones, namely one or three. Reflect this coordinate when there are three. As in Lemma III.2.1, one difference variable then appears in exactly one centered factor, so the average vanishes.

Suppose that the XOR is zero. Reflect coordinates so that one chosen vertex becomes 0, and denote the other three vertices by u, v, w . The reflection preserves both the averages and the property of being an ordinary parallelogram. We have $u + v + w = 0$ over $\mathbf{F}_2$. Accordingly the only possible coordinate patterns across these three rows are

$$000, \quad 110, \quad 101, \quad 011.$$

At least two of the three nonzero patterns occur, because u, v, w are nonzero and pairwise distinct. Sum the independent variables h_i over coordinates of each of the three nonzero types, naming the sums r, s, t respectively. Every nonempty sum is uniform on G , and the nonempty sums are mutually independent.

If exactly two types occur, the four arguments, in an appropriate order, are

$$x, \quad x + r, \quad x + s, \quad x + r + s.$$

Their coefficient vectors form an ordinary parallelogram, and the average of the product of the four real f values is $\|f\|_{U^2}^4 = \mathcal{E}(f)$.

If all three types occur, the four arguments are

$$x, \quad x + r + s, \quad x + r + t, \quad x + s + t. \quad (\text{III.2.12})$$

There is no pairing with equal sums of coefficient vectors. For example, each possible pairing is ruled out by one of the three occupied coordinate types. Thus this is precisely the binary tetrahedron case.

It remains to evaluate (III.2.12). Substitute the Fourier expansion of f into all four factors, assigning characters $\chi_0, \chi_1, \chi_2, \chi_3$ in the displayed order. Averaging over r, s, t imposes

$$\chi_1 \chi_2 = 1, \quad \chi_1 \chi_3 = 1, \quad \chi_2 \chi_3 = 1.$$

Hence $\chi_1 = \chi_2 = \chi_3 = \chi$ with $\chi^2 = 1$. The average over x then forces $\chi_0 = \chi$. The correlation is therefore

$$\sum_{\chi^2=1} \widehat{f}(\chi)^4.$$

For such characters $\chi(x) \in \{-1, 1\}$, so $\widehat{f}(\chi)$ is real. The trivial character has coefficient zero, and the last display equals (III.2.8). $\square$

Remark III.2.4 (Why integer rank alone is insufficient). The four vectors in (III.2.12), including the coefficient of x , form a matrix of determinant ± 2 . They are independent over $\mathbb{Q}$ but need not give independent uniform values on G . Discarding all rationally independent supports would incorrectly omit $\mathcal{E}_{(2)}(f)$. The three Smith types for a four-vertex support are $(1, 1, 1, 0)$, $(1, 1, 1, 2)$, and $(1, 1, 1, 1)$, corresponding respectively to an ordinary parallelogram, a binary tetrahedron, and the final case of Lemma III.2.3. In the first type the final zero records rank three; for $d = 2$ the rectangular matrix has only three nonzero invariant factors.

Lemma III.2.5 (Exact multiplicities). *The numbers of ordinary parallelograms and binary tetrahedra in V_d are respectively*

$$R_d = \frac{6^d - 2 \cdot 4^d + 2^d}{8}, \quad (\text{III.2.13})$$

$$T_d = \frac{8^d - 3 \cdot 6^d + 3 \cdot 4^d - 2^d}{24}. \quad (\text{III.2.14})$$

Their sum is

$$K_d = R_d + T_d = \frac{m(m-1)(m-2)}{24}. \quad (\text{III.2.15})$$

Proof. Count ordered quadruples satisfying $a + b = c + e$ in $\mathbb{Z}^d$. There are six possible coordinate patterns, so the total is 6^d . The degeneracies $a = c, b = e$ and $a = e, b = c$ each contribute 4^d , and their intersection contributes 2^d . These are all degeneracies: for example, $a = b$ forces $c = e = a$ coordinatewise. An ordinary parallelogram with four distinct vertices has exactly one pairing into two equal-sum pairs. Ordering both pairs and ordering the pairs gives eight ordered solutions. This proves (III.2.13).

For the XOR count, choose distinct ordered vertices a, b, c arbitrarily and put $e = a \oplus b \oplus c$. The fourth vertex differs from each of the first three. Every unordered four-element XOR-zero subset is counted 24 times. The total is thus $m(m-1)(m-2)/24$. Every ordinary parallelogram has XOR zero, and the other XOR-zero subsets are exactly the binary tetrahedra. Subtracting R_d gives (III.2.14). $\square$

For orientation, the first coefficients are

d	m	R_d	T_d	K_d
2	4	1	0	1
3	8	12	2	14
4	16	100	40	140
5	32	720	520	1240

For comparison, $\binom{m}{4}$ equals 70, 1820, 35960 when $d = 3, 4, 5$. Even after the first three terms have been removed, counting every fourth-order support by its absolute upper bound wastes most of the available cancellation.

III.2.4 The sharpened cube-count theorem

Theorem III.2.6 (Exact fourth-order expansion with explicit remainder). *Let $d \geq 2$, $m = 2^d$, let G be a finite abelian group, and let $F : G \rightarrow \mathbb{R}$ have mean $\delta \geq 0$. Put $f = F - \delta$ and $\eta = \|f\|_{U^d}$. Then*

$$Q_d(F) = \delta^m + \delta^{m-4}(R_d\mathcal{E}(f) + T_d\mathcal{E}_{(2)}(f)) + \mathcal{R}_d(F), \quad (\text{III.2.16})$$

where

$$|\mathcal{R}_d(F)| \leq \sum_{j=5}^m \binom{m}{j} \delta^{m-j} \eta^j. \quad (\text{III.2.17})$$

An empty sum is zero. In particular, the formula for $d = 2$ is exact with no remainder.

Proof. Use (III.2.3). The constant term is δ^m , and the terms of degrees one, two, and three vanish by Lemma III.2.1. Lemmas III.2.3 and III.2.5 give the total fourth-order term. The terms of degrees at least five satisfy (III.2.17) by (III.2.5) and the triangle inequality. $\square$

The formula holds in particular for arbitrary weights $0 \leq F \leq 1$, as well as for characteristic functions. The energies in the fourth term are nonnegative, even though the higher-order remainder can have either sign.

Corollary III.2.7 (Uniformity-parameter form). *Let $A \subseteq G$ have density δ , and suppose $\|\mathbf{1}_A - \delta\|_{U^d}^m \leq \alpha$. Then*

$$Q_d(\mathbf{1}_A) \leq \delta^m + K_d \delta^{m-4} \alpha^{4/m} + \sum_{j=5}^m \binom{m}{j} \delta^{m-j} \alpha^{j/m}. \quad (\text{III.2.18})$$

If $|G|$ is odd, the coefficient K_d in the fourth-order term can be replaced by R_d . The latter replacement also holds on an arbitrary group whenever $P_2(\mathbf{1}_A - \delta) = 0$.

Proof. Apply (III.2.9) and Theorem III.2.6. The vanishing of the quotient energy follows from (III.2.10). $\square$

In the notation of Gowers' Lemma 3.10, the old upper bound is $(\delta + \alpha^{1/m})^m$. For fixed d and positive δ , its displayed excess is of order $\alpha^{1/m}$; the new bound has excess of order $\alpha^{4/m}$. This comparison concerns the strength of the cube-count estimate under exactly the same norm hypothesis. It does not replace the arithmetic-progression counting or inverse arguments elsewhere in the proof.

For example, on an odd-order group and with $d = 3$,

$$Q_3(F) = \delta^8 + 12\delta^4 \|f\|_{U^2}^4 + \mathcal{R}_3(F), \quad (\text{III.2.19})$$

$$|\mathcal{R}_3(F)| \leq 56\delta^3 \eta^5 + 28\delta^2 \eta^6 + 8\delta \eta^7 + \eta^8. \quad (\text{III.2.20})$$

On an arbitrary group the exact fourth-order term has the additional $2\delta^4 \mathcal{E}_{(2)}(f)$.

If one prefers a single remainder expression, Taylor's theorem applied to $(\delta + t)^m$ gives, for $m \geq 5$,

$$\sum_{j=5}^m \binom{m}{j} \delta^{m-j} \eta^j \leq \binom{m}{5} \eta^5 (\delta + \eta)^{m-5}. \quad (\text{III.2.21})$$

The polynomial bound (III.2.17) usually retains more useful information when the density is small.

III.2.5 Sharpness and a necessary torsion correction

Proposition III.2.8 (The fourth power cannot be increased). *Fix $d \geq 2$ and $0 < \delta < 1$. Even if one restricts to odd-order finite abelian groups and weights $F : G \rightarrow [0, 1]$ of mean δ , there is no bound*

$$Q_d(F) - \delta^m = O_{d,\delta}(\|F - \delta\|_{U^d}^q) \quad (q > 4)$$

valid as the centered norm tends to zero. Among all finite abelian groups, K_d is the optimal leading coefficient in (III.2.18), expressed in terms of $\eta = \alpha^{1/m}$.

Proof. On $G = \mathbb{Z}/3\mathbb{Z}$, let $g(x) = \cos(2\pi x/3)$ and $F_t(x) = \delta + tg(x)$. For positive sufficiently small t , the weights lie in $[0, 1]$ and have mean δ . The two nonzero Fourier coefficients of g are $1/2$, so $\mathcal{E}(g) = 1/8$. Theorem III.2.6 gives

$$Q_d(F_t) - \delta^m = \frac{R_d}{8} \delta^{m-4} t^4 + O_{d,\delta}(t^5), \quad (\text{III.2.22})$$

where the remainder is identically zero when $d = 2$. Since $\|F_t - \delta\|_{U^d} = t \|g\|_{U^d}$ and $\|g\|_{U^d} > 0$, the exponent $q > 4$ is impossible.

For the coefficient over unrestricted groups, take a nontrivial character $\chi : G \rightarrow \{-1, 1\}$ and $F_t = \delta + t\chi$. Then $\|t\chi\|_{U^d} = t$ and $\mathcal{E}(t\chi) = \mathcal{E}_{(2)}(t\chi) = t^4$. Therefore

$$Q_d(F_t) - \delta^m = K_d \delta^{m-4} t^4 + O_{d,\delta}(t^5). \quad (\text{III.2.23})$$

Any smaller universal leading coefficient would fail for small t . □

The same example proves that the torsion correction cannot be dropped. For a particularly transparent case, on $\mathbb{Z}/2\mathbb{Z}$ and with $d = 3$ one can compute the whole polynomial:

$$Q_3(\delta + t\chi) = \delta^8 + 14\delta^4 t^4 + t^8. \quad (\text{III.2.24})$$

Indeed, a subset S contributes precisely when both $|S|$ and every coordinate sum of its vertices are even. For the three-dimensional cube the only nonempty such supports have sizes four and eight: there are fourteen supports of size four and one of size eight. For size six, taking the complement would give two distinct vertices with equal XOR, which is impossible. Formula (III.2.24) directly distinguishes the correct coefficient 14 from the odd-order coefficient 12.

Remark III.2.9 (What sharpness has and has not shown). The identity determines the exact fourth-order coefficient in terms of the two Fourier energies. The constant K_d is also sharp when those energies are bounded only by η^4 over all groups. We have not asserted that R_d is the optimal leading constant in terms of η^4 when all groups are required to have odd order. That separate optimization involves the possible ratios $\|f\|_{U^2}^4 / \|f\|_{U^d}^4$ for real centered functions.

[write] 6 October 2026. *Settled for $d = 3$ by source 04.* By source 04's Theorem III.3.4, $\|f\|_{U^2}^4 \leq 2^{-1/2} \|f\|_{U^3}^4$ for real centered f on odd-order groups, with equality for $f = 2\text{Re}(c\xi)$. Hence on odd-order groups the best leading constant in terms of $\eta^4 = \|f\|_{U^3}^4$ is $R_3 \cdot 2^{-1/2} = 6\sqrt{2} < 12 = R_3$ (source 04's Theorem III.3.6 and Proposition III.3.7); the proof is in the note after Theorem III.3.4. For $d \geq 4$ the question remains open (Research question V.14.1).

III.2.6 Sharpness can be transferred to actual sets

The use of small real weights in the previous proposition is not an essential restriction on the exponent. The following rounding argument records the quantitative reason.

Lemma III.2.10 (Rounding a bounded weight in a cube norm). *Let $p : G \rightarrow [0, 1]$ and $d \geq 2$, and choose a random subset A by including the elements of G independently with probabilities $p(x)$. For $e = \mathbf{1}_A - p$ and $m = 2^d$,*

$$\mathbb{E}_A \|e\|_{U^d}^m \leq \frac{\binom{m}{2}}{|G|}. \quad (\text{III.2.25})$$

Here $\mathbb{E}_A$ denotes expectation over the choice of the set, distinct from the averages over group elements.

Proof. Expand the Gowers norm as a cube average. When its m vertex values are pairwise distinct, independence and $\mathbb{E}_A e(x) = 0$ show that the expected product is zero. Every other product has absolute value at most one. For distinct vertices $u, v \in V_d$, the difference $L_u - L_v$ has a coefficient 1 or -1 on at least one of the independent variables h_i , and is therefore uniform on G . The probability of a collision between this pair is $|G|^{-1}$. The union bound over pairs proves (III.2.25). $\square$

Corollary III.2.11 (Optimal order and universal constant for sets). *For every fixed $d \geq 2$ and $0 < \delta < 1$, there are finite abelian groups G_n with $|G_n| \rightarrow \infty$ and subsets $A_n \subseteq G_n$ whose densities δ_n tend to δ , such that, writing $\eta_n = \|\mathbf{1}_{A_n} - \delta_n\|_{U^d}$, we have $\eta_n \rightarrow 0$ and*

$$\frac{Q_d(\mathbf{1}_{A_n}) - \delta_n^m}{\delta_n^{m-4} \eta_n^4} \rightarrow K_d. \quad (\text{III.2.26})$$

If the groups are required to have odd order, the power four is still optimal, although this argument does not identify the best leading constant in that restricted setting.

Proof. First take $G_n = (\mathbb{Z}/2\mathbb{Z})^n$, $N = |G_n|$, and a nontrivial real character χ_n . Put $t = N^{-1/(10m)}$ and $p = \delta + t\chi_n$, which lies in $[0, 1]$ for large n . Lemma III.2.10 and Markov's inequality give a realization of A_n with

$$\|\mathbf{1}_{A_n} - p\|_{U^d} \leq N^{-1/(2m)} = t^5$$

for all sufficiently large n , since the failure probability is at most $\binom{m}{2}N^{-1/2}$. The inequality $|\mathbb{E}e| \leq \|e\|_{U^d}$ implies $|\delta_n - \delta| \leq t^5$. After centering, the error in U^d is at most $2t^5$, so $\eta_n = t + O(t^5)$.

Both $\mathbf{1}_{A_n}$ and p have U^d norm at most one. The norm triangle inequality and $|a^m - b^m| \leq m|a - b|$ for $a, b \in [0, 1]$ show that

$$|Q_d(\mathbf{1}_{A_n}) - Q_d(p)| \leq mt^5.$$

The change from δ^m to δ_n^m is at most mt^5 as well. Combining these estimates with (III.2.23) proves (III.2.26).

For odd-order groups use $G_n = (\mathbb{Z}/3\mathbb{Z})^n$ and let g_n be the cosine of a nontrivial character. Its U^d norm is the same fixed positive number as the function g on $\mathbb{Z}/3\mathbb{Z}$ in Proposition III.2.8. Round $p = \delta + N^{-1/(10m)}g_n$ in exactly the same way. Equation (III.2.22) gives a strictly positive limit after division by η_n^4 . Consequently no exponent greater than four can hold uniformly even for characteristic functions on odd-order groups. $\square$

Corollary III.2.12 (Exactly prescribed compatible rational densities). *If $0 < \delta < 1$ is rational, the sharp universal limit in (III.2.26) can be achieved with $\delta_n = \delta$ for every n . If the denominator of δ is odd, the odd-order sharpness of the fourth power can likewise be achieved at exactly that density.*

Proof. Write $\delta = a/q$ and use $G_n = (\mathbb{Z}/q\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})^n$, so $N = |G_n|$ makes δN an integer. Let χ_n be a nontrivial real character, put $t = N^{-1/(10m^2)}$, and round $p = \delta + t\chi_n$. The same Markov argument as before gives a realization A with

$$\|\mathbf{1}_A - p\|_{U^d} \leq N^{-1/(2m)} = t^{5m}.$$

Its density differs from δ by at most t^{5m} . Add or remove at most Nt^{5m} memberships to obtain a set B of cardinality exactly δN . If $e = \mathbf{1}_B - \mathbf{1}_A$ is supported on a fraction θ of the group, then $|e| \leq 1$ and

$$\|e\|_{U^d}^m \leq \mathbb{E}_{x,h} \left| \prod_v e(L_v(x, h)) \right| \leq \mathbb{E}_x \mathbf{1}_{\text{supp } e}(x) = \theta.$$

Thus the alteration costs at most t^5 in U^d norm. We obtain $\|\mathbf{1}_B - p\|_{U^d} = O(t^5)$, $\|\mathbf{1}_B - \delta\|_{U^d} = t + O(t^5)$, and $Q_d(\mathbf{1}_B) = Q_d(p) + O(t^5)$. Equation (III.2.23) gives the required limit. If q is odd, replace the second factor of G_n by $(\mathbb{Z}/3\mathbb{Z})^n$ and use the cosine of an order-three character. The same alteration argument and (III.2.22) prove optimality of the fourth power on odd-order groups at exactly the chosen density. $\square$

III.2.7 An exact formula for every higher coefficient

The fourth-order theorem is the first nontrivial instance of a complete Fourier description of the centered expansion. For a support $S \subseteq V_d$, define

$$\mathcal{K}_S(G) = \left\{ (\chi_v)_{v \in S} \in (\widehat{G} \setminus \{1\})^S : \prod_{v \in S} \chi_v = 1, \quad \prod_{v \in S} \chi_v^{v_i} = 1 \ (1 \leq i \leq d) \right\}.$$

All assigned characters are required to be nontrivial because f is centered. Character orthogonality gives the exact identity

$$M_S(f) = \sum_{(\chi_v) \in \mathcal{K}_S(G)} \prod_{v \in S} \widehat{f}(\chi_v). \quad (\text{III.2.27})$$

To prove it, expand each factor in (III.2.2); averaging over x gives the first character equation, and averaging over h_i gives the equation indexed by i . These equations are also sufficient for a Fourier term to survive. This is a finite exact formula, and it is valid without assumptions on the exponent or rank of G .

For groups of prime exponent, the equations can be studied as linear relations among the vectors $(1, v)$ over the prime field. For general finite abelian groups, their integer relation matrices and Smith invariant factors retain the additional torsion information. At four vertices, the only nontrivial invariant factor that can arise is two; this is why the single quotient energy in (III.2.8) suffices. Higher coefficients offer a concrete continuation: classify their support matrices, remove the supports for which (III.2.27) is empty, and estimate the remaining explicit Fourier correlations. The formula supplies an algorithmic route without assuming that the resulting classification will have a simple closed form.

III.2.8 The fifth-order remainder is also necessary

Proposition III.2.13 (Sharp order of the remainder). *Fix $d \geq 3$ and $0 < \delta < 1$. For weights $F : G \rightarrow [0, 1]$ on odd-order finite abelian groups, the remainder after the exact fourth-order term in Theorem III.2.6 cannot in general be bounded by $O_{d,\delta}(\|F - \delta\|_{U^d}^q)$ for any $q > 5$.*

Proof. Let $G = \mathbb{Z}/3\mathbb{Z}$, $g(x) = \cos(2\pi x/3)$, and $F_t = \delta + tg$ for positive sufficiently small t . Both nontrivial Fourier coefficients of g equal $1/2$, and its trivial coefficient is zero. Fourier expansion and character orthogonality show that every support moment $M_S(g)$ is nonnegative: each surviving term is a product of nonnegative Fourier coefficients.

In dimension at least three, consider the five-vertex support

$$S = \{0, e_1, e_2, e_3, e_1 + e_2 + e_3\}.$$

Its moment is

$$\mathbb{E}_{x,a,b,c} g(x)g(x+a)g(x+b)g(x+c)g(x+a+b+c) = \frac{1}{16}.$$

Indeed, assigning a nontrivial character χ to the last factor forces the three middle characters to be χ^{-1} and the first character to be χ^2 . Each of the two choices for χ therefore contributes $(1/2)^5$. Consequently

$$c_d := \sum_{|S|=5} M_S(g) \geq \frac{1}{16} > 0.$$

The exact finite expansion gives

$$\mathcal{R}_d(F_t) = c_d \delta^{m-5} t^5 + O_{d,\delta}(t^6),$$

whereas $\|F_t - \delta\|_{U^d} = t \|g\|_{U^d}$. Since $\|g\|_{U^d} > 0$, no exponent $q > 5$ can hold. $\square$

In dimension three the entire polynomial is especially simple:

$$Q_3(\delta + tg) = \delta^8 + \frac{3}{2}\delta^4 t^4 + \frac{1}{2}\delta^3 t^5 + \frac{1}{2}\delta^2 t^6 + \frac{1}{32}t^8. \quad (\text{III.2.28})$$

For a direct verification, put $a = \delta + t$ and $b = \delta - t/2$, the two possible weight values. Group the triples (h_1, h_2, h_3) by their number r of nonzero entries. Coordinate reflections allow each nonzero difference to be replaced by 1 when summing over x . For $r = 0, 1, 2, 3$ there are respectively 1, 6, 12, 8 difference triples, and their sums over x are respectively

$$a^8 + 2b^8, \quad 2a^4b^4 + b^8, \quad a^4b^4 + 2a^2b^6, \quad a^2b^6 + 2a^3b^5.$$

It follows that

$$81Q_3(F_t) = a^8 + 8b^8 + 24a^4b^4 + 32a^2b^6 + 16a^3b^5.$$

Substitution and expansion give (III.2.28). Its t^5 coefficient supplies a concrete witness to Proposition III.2.13.

[write] **6 October 2026.** (III.2.28) is identical to source 04's (III.3.19), computed there by counting the 81 cube parameters by the number of vertices at 0; credited to both. The five-vertex support S above is the representative $\{000, 100, 010, 001, 111\}$ of source 04's eight surviving quintic supports (Lemma III.3.3), so $M_S(g) = B_5(g)$; the value $1/16$ is (III.3.8) with $a_{\pm 1} = 1/2$ and $2\xi = -\xi$ on $\mathbb{Z}/3\mathbb{Z}$.

Replacing t by $-t$ in (III.2.28) makes its fifth-order term negative. For sufficiently small positive t , this gives

$$Q_3(\delta - tg) < \delta^8 + 12\delta^4 \|tg\|_{U^2}^4.$$

Thus the exact quartic term is not, by itself, an unconditional lower bound for the whole excess; the sign of the remainder matters.

III.2.9 A five-vertex correlation that detects order three

The next evaluated support illustrates why a higher-order classification will require more than the order-two energy. Consider the five vertices

$$S_3 = \{0000, 1110, 1101, 1011, 0111\} \subseteq \{0, 1\}^4.$$

For every real centered function f on a finite abelian group,

$$M_{S_3}(f) = \sum_{\substack{\chi \neq 1 \\ \chi^3 = 1}} \widehat{f}(\chi)^4 \overline{\widehat{f}(\chi)}. \quad (\text{III.2.29})$$

Here the four nonzero vertices are the all-ones vector with one entry removed. To prove (III.2.29), assign characters $\chi_0, \chi_1, \dots, \chi_4$ to the five factors, with χ_j corresponding to the vertex omitting coordinate j . Averaging over h_j imposes $\prod_{i \neq j, i \geq 1} \chi_i = 1$. Dividing any two such equations shows $\chi_1 = \dots = \chi_4 = \chi$; each equation then says $\chi^3 = 1$. The x -equation gives $\chi_0 = \chi^{-1}$. Fourier inversion and real-valuedness of f yield the formula.

This correlation vanishes if 3 does not divide $|G|$. On $\mathbb{Z}/3\mathbb{Z}$ with $f(x) = \cos(2\pi x/3)$ it equals $1/16$, and replacing f by $-f$ changes it to $-1/16$. In particular it is not a nonnegative energy of the fourth-order kind. We have evaluated this one support; we do not assert that it exhausts all possible fifth-order types.

III.3 Sharp centered refinements of the cube-counting lemma

Source 04, delivered Section 6.

Gowers' Lemma 3.10 bounds the number of d -dimensional cubes by applying the triangle inequality to the constant and balanced parts of an indicator. Centering contains additional information that this application of the triangle inequality discards. We retain it, first by identifying correlations that vanish, and then by proving a sharp comparison between the second and third uniformity norms of real centered functions. These results strengthen the stated cube-counting lemma; no historical priority beyond that comparison is asserted. As Gowers explains, this lemma is supplementary to his main density-increment argument, so an improvement here does not by itself imply an improved global bound in Szemerédi's theorem.

Throughout this section, averages on a finite abelian group G are normalized. For $d \geq 2$, put $q = 2^d$, $V_d = \{0, 1\}^d$, and

$$L_v(x, h) = x + v \cdot h, \quad Q_d(W) = \mathbb{E}_{x, h_1, \dots, h_d \in G} \prod_{v \in V_d} W(L_v(x, h)).$$

For real W , conjugation has no effect, so $Q_d(W) = \|W\|_{U^q_d}^q$. For an indicator, $|G|^{d+1}Q_d(1_A)$ is exactly Gowers' ordered cube count, including degenerate cubes. Write $W = \delta + f$, where $\delta \geq 0$, f is real, and $\mathbb{E}f = 0$; these are standing assumptions throughout the section. Set $u = \|f\|_{U^d}$. No bound on f is needed for the inequalities below.

III.3.1 Cancellation, parallelograms, and support counts

Lemma III.3.1 (Cancellation through degree three). *Any at most three distinct forms L_v are jointly uniform on the corresponding power of G , for every finite abelian group G . Consequently, if*

$\delta \geq 0$,

$$|Q_d(\delta + f) - \delta^q| \leq \sum_{r=4}^q \binom{q}{r} \delta^{q-r} u^r. \quad (\text{III.3.1})$$

Proof. Reflect coordinates of the Boolean cube to move the first selected vertex to 0; these reflections correspond to invertible changes of the averaging variables. For three vertices the others are distinct nonzero Boolean vectors a, b . Two coordinates give a 2×2 minor of the matrix with rows a, b equal to 1 or -1 : use two coordinates distinguishing their supports, or one in their intersection and one in their difference if the supports are nested. Thus the map to $(x, x + a \cdot h, x + b \cdot h)$ is surjective, with equally sized fibers, on every abelian group. The cases of one or two vertices are immediate.

Expand the cube product according to the support on which f is chosen. Every term with one, two, or three factors of f vanishes. For a support of size r , the cube Cauchy–Schwarz inequality, with constant 1 at all other vertices, bounds its correlation in absolute value by u^r . Summation proves the assertion. $\square$

[write] **6 October 2026.** Lemma III.3.1 has the same statement and bound as source 03’s Lemma III.2.1 with Corollary III.2.2 (this u is 03’s η , this $q = 2^d$ is 03’s m). Second route: source 03 uses coordinate reflections, this proof joint uniformity through 2×2 minors. Credited to both.

Odd group order permits a much more precise fourth-order term.

Proposition III.3.2 (Exact quartic coefficient). *Suppose $|G|$ is odd, and put $v = \|f\|_{U^2}$. Then*

$$Q_d(\delta + f) = \delta^q + P_d \delta^{q-4} v^4 + R_d, \quad (\text{III.3.2})$$

$$|R_d| \leq \sum_{r=5}^q \binom{q}{r} \delta^{q-r} u^r, \quad (\text{III.3.3})$$

$$P_d = 2^{d-3} (3^d - 2^{d+1} + 1). \quad (\text{III.3.4})$$

In particular, $P_2 = 1$, $P_3 = 12$, $P_4 = 100$, and $P_5 = 720$.

Proof. Four distinct Boolean vertices are either affinely independent or form a parallelogram. Indeed, any three are affinely independent. If four are dependent, their affine span is a plane, and some projection onto two coordinate axes is injective on that plane. The four images must be all four vertices of the unit square. The affine inverse of this projection preserves its parallelogram relation.

If the four vertices are independent, a 3×3 Boolean minor after this translation has determinant ± 1 or ± 2 . The determinant bound follows by adjoining a row and column of ones, changing binary entries b to $1 - 2b$, and applying Hadamard’s inequality to the resulting 4×4 matrix. Multiplication by either determinant is invertible on an odd-order group. Thus the four forms are jointly uniform and their centered correlation is zero. On a parallelogram, any three forms are jointly uniform by Lemma III.3.1, while the fourth is their alternating sum. Its correlation is therefore exactly v^4 .

To count parallelograms, choose the k coordinates in which a pair of opposite vertices differs and fix the other coordinates. Around that center there are 2^{k-1} opposite pairs; choose two of them. The diagonal midpoint is unique, so

$$P_d = \sum_{k=2}^d \binom{d}{k} 2^{d-k} \binom{2^{k-1}}{2} = 2^{d-3} (3^d - 2^{d+1} + 1).$$

Bound the terms of degree at least five by cube Cauchy–Schwarz as before. $\square$

[write] **6 October 2026.** Proposition III.3.2 is the odd-order case of source 03's Theorem III.2.6, which holds on every finite abelian group with the extra term $T_d \delta^{q-4} \mathcal{E}_{(2)}(f)$; that term vanishes on odd-order groups by (III.2.10). This P_d is source 03's R_d : $2^{d-3}(3^d - 2^{d+1} + 1) = (6^d - 2 \cdot 4^d + 2^d)/8$ (both give 1, 12, 100, 720). Second route: affine independence and Boolean minors $\pm 1, \pm 2$ in place of source 03's XOR classification. Do not read this remainder R_d as source 03's count R_d , or as the binomial sum $R_d(H)$ of Section IV.6 (Section F.3).

A general organizing principle is useful. Over $G = \mathbb{F}_p^n$, consider the vector matroid of the coefficient vectors $(1, v)$. If a selected support has a coloop, its centered correlation vanishes. Indeed, linear algebra gives a direction on which that form has coefficient 1 and all other selected forms have coefficient 0. Averaging along that direction averages one factor of f to zero. Thus only supports with no coloop need to be counted.

For $d = 3$ this count can be made completely explicit, and the result also holds on every odd-order abelian group.

Lemma III.3.3 (The surviving supports of a 3-cube). *Suppose that G has odd order. The numbers of supports with no coloop, in sizes $0, 1, \dots, 8$, are*

$$1, 0, 0, 0, 12, 8, 28, 8, 1.$$

All twelve quartic correlations equal $\|f\|_{U^2}^4$. The eight quintic correlations are equal to

$$B_5(f) = \mathbb{E}_{x,a,b,c} f(x)f(x+a)f(x+b)f(x+c)f(x+a+b+c). \quad (\text{III.3.5})$$

Consequently, with $u = \|f\|_{U^3}$ and $v = \|f\|_{U^2}$,

$$Q_3(\delta + f) = \delta^8 + 12\delta^4 v^4 + 8\delta^3 B_5(f) + R_6 + R_7 + u^8, \quad (\text{III.3.6})$$

$$|R_6| \leq 28\delta^2 u^6, \quad |R_7| \leq 8\delta u^7. \quad (\text{III.3.7})$$

Proof. The twelve coplanar quadruples are the six faces and six diagonal rectangles. A plane contains at most four Boolean vertices: some projection onto two coordinate axes is injective on the plane, and its Boolean image has at most four points. A five-set therefore has rank four and has a coloop exactly when it contains a coplanar quadruple. Each of the twelve quadruples has four extensions to a five-set. No five-set contains two such quadruples, since distinct planes meet in a line and a line contains at most two Boolean vertices. Thus $\binom{8}{5} - 12 \cdot 4 = 8$ five-sets survive. Every set of size at least six remains rank four after a deletion.

The eight surviving five-sets consist of one parity tetrahedron and one vertex of the opposite parity. Cube symmetries act transitively on them; a representative is $\{000, 100, 010, 001, 111\}$. The corresponding changes of averaging variables prove (III.3.5). These arguments remain valid on odd-order groups: any four independent cube forms have a minor ± 1 or ± 2 , hence are jointly uniform; an extra vertex outside a parallelogram is therefore independent of its three determining forms. Finally apply cube Cauchy–Schwarz to the 28 sextic and 8 septic supports. $\square$

For comparison, in characteristic two the support counts are 1, 0, 0, 0, 14, 0, 28, 8, 1: the parity tetrahedra become two additional affine planes. The odd-order hypothesis is therefore a substantive restriction.

III.3.2 A sharp gap between the real centered U^2 and U^3 norms

Use normalized Fourier coefficients

$$a_\xi = \widehat{f}(\xi) = \mathbb{E}_{x \in G} f(x) \overline{\xi(x)}, \quad \xi \in \widehat{G},$$

and write the dual group additively. Fourier inversion in (III.3.5) gives

$$B_5(f) = \sum_{\xi \in \widehat{G}} a_{2\xi} a_{-\xi}^3 a_\xi. \quad (\text{III.3.8})$$

For real centered f on an odd-order group, $a_{-\xi} = \overline{a_\xi}$, $a_0 = 0$, and the nonzero coefficients occur in distinct pairs. Therefore, with $\rho = \max_\xi |a_\xi|$,

$$|B_5(f)| \leq \rho \|f\|_{U^2}^4 \leq 2^{-1/4} \|f\|_{U^2}^5. \quad (\text{III.3.9})$$

Theorem III.3.4 (Sharp real centered norm gap). *Let G have odd order and let $f : G \rightarrow \mathbb{R}$ have mean zero. Then*

$$\|f\|_{U^3}^8 \geq 2 \|f\|_{U^2}^8. \quad (\text{III.3.10})$$

For nonzero f , equality holds exactly when

$$f(x) = 2 \operatorname{Re}(c \xi(x))$$

for a nontrivial character ξ and a nonzero complex number c .

Proof. We first use the Fourier identity

$$\begin{aligned} \|f\|_{U^3}^8 &= \sum_{s, t \in \widehat{G}} |T(s, t)|^2, \\ T(s, t) &= \sum_{r \in \widehat{G}} a_r \overline{a_{r+s}} \overline{a_{r+t}} a_{r+s+t}. \end{aligned} \quad (\text{III.3.11})$$

The Fourier coefficients are normalized, but all sums in (III.3.11) are unnormalized. To verify the identity, expand the original cube in Fourier series. The four orthogonality equations say precisely that its frequency variables form an affine cube:

$$r_v = r_{000} + v_1(r_{100} - r_{000}) + v_2(r_{010} - r_{000}) + v_3(r_{001} - r_{000}).$$

These equations are solved without division. Grouping the two parallel faces gives (III.3.11).

Put

$$\lambda_r = |a_r|^2, \quad S = \sum_r \lambda_r^2 = \|f\|_{U^2}^4, \quad M = \sum_r \lambda_r^4.$$

Retain only the two coordinate axes in the nonnegative sum (III.3.11). Since $T(0, t) = \sum_r \lambda_r \lambda_{r+t}$, this gives

$$\|f\|_{U^3}^8 \geq 2\mathcal{E}(\lambda) - S^2, \quad \mathcal{E}(\lambda) = \sum_{r+s=t+w} \lambda_r \lambda_s \lambda_t \lambda_w. \quad (\text{III.3.12})$$

The subtraction corrects the double counting of the origin.

The energy summands are nonnegative. Retain the union of three families:

$$\begin{aligned} \mathcal{A}: & \quad r = t, \quad s = w, \\ \mathcal{B}: & \quad r = w, \quad s = t, \\ \mathcal{C}: & \quad s = -r, \quad w = -t. \end{aligned}$$

Each family contributes S^2 , because λ is even. Each pairwise intersection contributes M . Their triple intersection has $2r = 0$; odd group order and $\lambda_0 = 0$ make its contribution zero. Thus

$$\mathcal{E}(\lambda) \geq 3S^2 - 3M,$$

and hence

$$\|f\|_{U^3}^8 \geq 5S^2 - 6M. \quad (\text{III.3.13})$$

Group nonzero frequencies into pairs $\{r, -r\}$, and let b_j denote λ_r^2 for a representative of each pair. Then

$$S = 2 \sum_j b_j, \quad M = 2 \sum_j b_j^2 \leq \frac{S^2}{2}.$$

This proves (III.3.10).

Equality in (III.3.10) forces equality in the last inequality, so only one Fourier pair can be nonzero. Conversely, a parallelogram on a two-point subset of an odd-order group is degenerate in at least one direction. If the Fourier support is $\{\xi, -\xi\}$, all nonzero terms in (III.3.11) therefore lie on its coordinate axes. Directly, the two norm powers are $8|a_\xi|^8$ and $4|a_\xi|^8$, respectively. $\square$

[write] **6 October 2026.** *This answers source 03's Research question V.14.1 for $d = 3$ (cross-source finding of the intake; proof at the write).* Let G have odd order and $f : G \rightarrow \mathbb{R}$ be centered and nonzero; then $|G| \geq 3$, since on the trivial group a centered function vanishes. By (III.3.10),

$$\frac{\|f\|_{U^2}^4}{\|f\|_{U^3}^4} = \left(\frac{\|f\|_{U^2}^8}{\|f\|_{U^3}^8} \right)^{1/2} \leq 2^{-1/2},$$

with equality exactly for $f = 2 \operatorname{Re}(c\xi)$, ξ a nontrivial character and $c \neq 0$; such f exist on every odd-order group with $|G| \geq 3$. Hence $c_3^{\text{odd}} = 2^{-1/2}$, the supremum is attained, and the attaining functions are the real parts of characters. The best odd-order leading coefficient in source 03's reduction is $R_3 c_3^{\text{odd}} = 12 \cdot 2^{-1/2} = 6\sqrt{2}$, which is Theorem III.3.6 with Proposition III.3.7. This also settles source 03's Remark III.2.9 at $d = 3$: $R_3 = 12$ is not the optimal odd-order constant, $6\sqrt{2}$ is. For $d \geq 4$ the question is open; it is source 04's own Question 3 (Section V.2, item III.1).

Corollary III.3.5 (Stability of the norm gap). *Suppose that G has odd order. If $f \neq 0$, $0 \leq \varepsilon < 3$, and*

$$\|f\|_{U^3}^8 \leq (2 + \varepsilon) \|f\|_{U^2}^8,$$

then some nontrivial character ξ , with $c = \widehat{f}(\xi)$, satisfies

$$\|f - 2 \operatorname{Re}(c\xi)\|_{U^2} \leq (\varepsilon/3)^{1/4} \|f\|_{U^2}. \quad (\text{III.3.14})$$

Proof. Equation (III.3.13) implies $M/S^2 \geq 1/2 - \varepsilon/6$. For the pair masses $p_j = 2b_j/S$, we have $\sum_j p_j = 1$ and $\sum_j p_j^2 \geq 1 - \varepsilon/3$. Since $\max_j p_j \geq \sum_j p_j^2$, one pair carries at least $1 - \varepsilon/3$ of the Fourier fourth-power mass. Removing it and using the Fourier formula for U^2 proves the claim. $\square$

This is a statement of stability in U^2 , not an assertion of approximation in L^2 . A large amount of Fourier square mass can be spread over many small coefficients.

III.3.3 The sharp leading cube bound and its models

Theorem III.3.6 (Refined form of Lemma 3.10 in dimension three). *Let G have odd order, $f : G \rightarrow \mathbb{R}$ have mean zero, $\delta \geq 0$, and $u = \|f\|_{U^3}$. Then*

$$Q_3(\delta + f) \leq \delta^8 + 6\sqrt{2}\delta^4 u^4 + 2^{17/8}\delta^3 u^5 + 28\delta^2 u^6 + 8\delta u^7 + u^8. \quad (\text{III.3.15})$$

Consequently, if A is α -uniform of degree two in Gowers' convention,

$$Q_3(1_A) \leq \delta^8 + 6\sqrt{2}\delta^4 \alpha^{1/2} + 2^{17/8}\delta^3 \alpha^{5/8} + 28\delta^2 \alpha^{3/4} + 8\delta \alpha^{7/8} + \alpha. \quad (\text{III.3.16})$$

Proof. In (III.3.6), use $v^4 \leq u^4/\sqrt{2}$ from Theorem III.3.4 and

$$|B_5(f)| \leq 2^{-1/4}v^5 \leq 2^{-7/8}u^5.$$

The remainder bounds give (III.3.15). Gowers' parameter satisfies $u^8 \leq \alpha$, which yields (III.3.16). $\square$

The leading error at fixed density is now $\alpha^{1/2}$, with an optimal constant, in place of the $\alpha^{1/8}$ term in $(\delta + \alpha^{1/8})^8$. In higher dimensions, monotonicity and Theorem III.3.4 similarly improve the quartic term in Proposition III.3.2 to $P_d \delta^{d-4} u^4 / \sqrt{2}$ for $d \geq 3$; sharpness of that higher-dimensional constant is not established here.

Proposition III.3.7 (An exact model for the leading constant). *Let $p > 8$ be prime and $f(x) = t \cos(2\pi x/p)$ on $\mathbb{Z}/p\mathbb{Z}$. Then*

$$\|f\|_{U^2}^4 = t^4/8, \quad \|f\|_{U^3}^8 = t^8/32, \quad (\text{III.3.17})$$

$$Q_3(\delta + f) = \delta^8 + \frac{3}{2}\delta^4 t^4 + t^8/32 = \delta^8 + 6\sqrt{2}\delta^4 u^4 + u^8. \quad (\text{III.3.18})$$

For $0 < t \leq \min(\delta, 1 - \delta)$ this is a weight in $[0, 1]$.

Proof. The Fourier support is $\{1, -1\}$, with coefficients $t/2$. Odd-degree terms vanish, since an odd number of signs cannot sum to zero and $p > 8$ prevents modular aliasing. A six-set omits two distinct vertices. In a coordinate where those vertices differ, the selected set has exactly three ones, so no signed zero sum is possible there either.

In the full eight-vertex term, the positive-sign four-set must have two ones in each coordinate. There are exactly eight choices: six unions of two opposite-vertex pairs, and the two parity tetrahedra. To see completeness, if one opposite pair occurs, the remaining two vertices must also be opposite; otherwise, reflecting a selected vertex to 000 forces the other three to be 110, 101, 011. Thus the full term is $8(t/2)^8 = t^8/32$. The quartic term follows from Proposition III.3.2. $\square$

This proves that both the fourth power of u and its coefficient $6\sqrt{2}\delta^4$ are optimal. Moreover, a family attaining this leading coefficient as $u \rightarrow 0$ at fixed positive density must approach equality in Theorem III.3.4. Corollary III.3.5 then concentrates its Fourier fourth-power mass on one conjugate pair.

The next coefficient in (III.3.15) is also optimal over odd-order groups, but this assertion uses 3-torsion. On $\mathbb{Z}/3\mathbb{Z}$,

$$Q_3(\delta + t \cos(2\pi x/3)) = \delta^8 + \frac{3}{2}\delta^4 t^4 + \frac{1}{2}\delta^3 t^5 + \frac{1}{2}\delta^2 t^6 + t^8/32. \quad (\text{III.3.19})$$

For a direct verification, put $A = \delta + t$, $B = \delta - t/2$. Among the 81 cube parameters, the numbers with k vertices equal to zero are

k	0	2	3	4	8
number	8	32	16	24	1.

These counts follow by separating the cases of zero, one, two, or three nonzero increments. The cube count is therefore $(8B^8 + 32A^2B^6 + 16A^3B^5 + 24A^4B^4 + A^8)/81$; expansion gives (III.3.19). Since $u^8 = t^8/32$, its quintic term is exactly $2^{17/8}\delta^3u^5$. This optimality is not asserted for large prime cyclic groups, where there is no 3-torsion.

[write] **6 October 2026.** (III.3.19) is identical to source 03's (III.2.28), computed there by grouping difference triples; credited to both.

III.3.4 Without 3-torsion: a sharp fractional correction exponent

The distinction just observed leads to a further refinement. The fifth-order correlation requires a frequency and its double. A function near equality in the sharp norm gap concentrates on a pair; without 3-torsion, doubling leaves that pair. Balancing the resulting loss and gain produces a fractional power.

Theorem III.3.8 (A correction of order $u^{16/3}$). *Suppose $|G|$ is coprime to 6, $\mathbb{E}f = 0$, f is real, and $\delta > 0$. With $u = \|f\|_{U^3}$,*

$$Q_3(\delta + f) \leq \delta^8 + 6\sqrt{2}\delta^4u^4 + 6\delta^{8/3}u^{16/3} + 28\delta^2u^6 + 8\delta u^7 + u^8. \quad (\text{III.3.20})$$

The exponent 16/3 cannot be increased after subtracting the sharp leading term. The coefficient 6 is an explicit upper bound, not a claimed optimal constant. In the parameter $\alpha = u^8$, the correction has order $\alpha^{2/3}$.

Proof. The zero function is immediate. Otherwise put $S = \|f\|_{U^2}^4$, choose a largest Fourier pair $\{\xi, -\xi\}$, and set

$$\rho = |a_\xi|, \quad z = 1 - \frac{2\rho^4}{S}, \quad D = zS = \sum_{r \notin \{\xi, -\xi\}} |a_r|^4.$$

Here $0 \leq z < 1$. Since $M = \sum_r |a_r|^8 \leq \rho^4 S = (1 - z)S^2/2$, (III.3.13) gives $u^8 \geq (2 + 3z)S^2$. The convex function $g(z) = (1 + 3z/2)^{-1/2}$ lies below its chord on $[0, 1]$, so

$$g(z) \leq 1 - (1 - \sqrt{2/5})z \leq 1 - z/3.$$

Consequently,

$$12\delta^4 S \leq 6\sqrt{2}\delta^4 u^4 - 2\sqrt{2}\delta^4 u^4 z. \quad (\text{III.3.21})$$

There is no 3-torsion, so 2ξ and -2ξ are outside the selected pair. The two selected terms of (III.3.8) contribute at most $2\rho^4(D/2)^{1/4}$ in absolute value; the other terms contribute at most ρD . Thus

$$\begin{aligned} |B_5(f)| &\leq 2^{-1/4} S^{5/4} ((1 - z)z^{1/4} + (1 - z)^{1/4} z) \\ &\leq 2^{3/4} S^{5/4} z^{1/4} \leq 2^{1/8} u^5 z^{1/4}. \end{aligned}$$

Combine this with (III.3.21). For $A, K > 0$,

$$\sup_{z \geq 0} (Kz^{1/4} - Az) = \frac{3K^{4/3}}{4^{4/3}A^{1/3}}.$$

Taking $A = 2\sqrt{2}\delta^4 u^4$ and $K = 2^{25/8}\delta^3 u^5$ makes the right side $6\delta^{8/3}u^{16/3}$. Inserting the remainder estimates from (III.3.7) proves (III.3.20). $\square$

Proposition III.3.9 (A two-frequency obstruction). *Fix $0 < \delta < 1$ and a prime $p > 16$. There are weights W_u on $\mathbb{Z}/p\mathbb{Z}$, with mean δ and $\|W_u - \delta\|_{U^3} = u$, such that, as $u \downarrow 0$,*

$$Q_3(W_u) - \delta^8 = 6\sqrt{2}\delta^4 u^4 + (9/4)^{1/3}\delta^{8/3}u^{16/3} + o_\delta(u^{16/3}). \quad (\text{III.3.22})$$

Proof. For $f(x) = a \cos(2\pi x/p) + b \cos(4\pi x/p)$, with $a, b > 0$, Fourier inversion gives

$$\|f\|_{U^3}^8 = \frac{a^8 + 6a^4b^4 + b^8}{32}, \quad \|f\|_{U^2}^4 = \frac{a^4 + b^4}{8}, \quad B_5(f) = \frac{a^4b}{16}. \quad (\text{III.3.23})$$

In fact there is the complete identity

$$\begin{aligned} Q_3(\delta + f) &= \delta^8 + \frac{3}{2}\delta^4(a^4 + b^4) + \frac{1}{2}\delta^3a^4b + \frac{3}{2}\delta^2a^4b^2 \\ &\quad + \frac{a^8 + 6a^4b^4 + b^8}{32}. \end{aligned} \quad (\text{III.3.24})$$

Here is a finite counting proof of both this identity and the first formula in (III.3.23). Apply the Fourier cube identity (III.3.11) to $\delta + f$. Its frequency support is contained in $\{-2, -1, 0, 1, 2\}$, with coefficients $\delta, a/2, b/2$ at frequencies of absolute value 0, 1, 2, respectively. Since $p > 16$, the orthogonality equations have no modular aliasing, so the frequency variables are integer affine cubes in this five-point interval.

Reflecting directions with negative increments makes all increments nonnegative; their sum is at most four. Enumerate their nonzero magnitudes, choose their coordinate positions and signs, and then choose the initial vertex between -2 and 2 minus their sum. The possibilities are the empty list, the singletons (1), (2), (3), (4), the pairs (1, 1), (1, 2), (1, 3), (2, 2), and the triples (1, 1, 1), (1, 1, 2). If n_j is the number of vertices with frequency of absolute value j , this gives

(n_0, n_1, n_2)	number of frequency cubes
(8, 0, 0)	1
(4, 4, 0)	24
(4, 0, 4)	24
(3, 4, 1)	16
(2, 4, 2)	96
(0, 8, 0)	8
(0, 4, 4)	48
(0, 0, 8)	8

Each row contributes its count times $\delta^{n_0}(a/2)^{n_1}(b/2)^{n_2}$, proving (III.3.24). Setting $\delta = 0$ proves the stated U^3 formula. The remaining two formulas in (III.3.23) follow from the Fourier fourth moment and (III.3.8).

Let $c = (\sqrt{2}/6)^{1/3}$ and choose

$$b = c\delta^{-1/3}u^{4/3}, \quad a^4 = -3b^4 + \sqrt{32u^8 + 8b^8}.$$

For small u this is positive, $\delta + f$ takes values in $[0, 1]$, and (III.3.23) gives $\|f\|_{U^3} = u$ exactly. Since

$$a^4 = 4\sqrt{2}u^4 - 3b^4 + O(b^8/u^4),$$

equation (III.3.24) gives

$$Q_3(\delta + f) - \delta^8 = 6\sqrt{2}\delta^4u^4 - 3\delta^4b^4 + 2\sqrt{2}\delta^3u^4b + O_\delta(u^6).$$

Substitution of b produces the coefficient $-3c^4 + 2\sqrt{2}c = (9/4)^{1/3}$, proving (III.3.22). $\square$

For the local extremal problem over groups of order coprime to 6, these results determine the first coefficient and the next exponent. More precisely, subtract $6\sqrt{2}\delta^4u^4$ from the supremum of the cube excess over weights of mean δ with centered U^3 norm at most u , and divide by $\delta^{8/3}u^{16/3}$. The liminf of this normalized correction is at least $(9/4)^{1/3}$, and its limsup is at most 6. Proving convergence and determining the limit remain focused open questions; existence of an asymptotic second coefficient is not asserted here.

III.3.5 Sharpness for indicator sets

Proposition III.3.10 (Rounding the sharpness examples). *For every fixed $0 < \delta < 1$, there are subsets A_j of prime cyclic groups, with densities $\delta_j \rightarrow \delta$ and $\alpha_j = \|1_{A_j} - \delta_j\|_{U^3}^8 \rightarrow 0$, such that*

$$Q_3(1_{A_j}) - \delta_j^8 \sim 6\sqrt{2}\delta^4\alpha_j^{1/2}. \quad (\text{III.3.25})$$

The obstruction to improving the exponent $16/3$ also persists for indicator sets on growing prime cyclic groups.

Proof. Independently include each $x \in \mathbb{Z}/p\mathbb{Z}$ with probability $w(x)$, where w is a prescribed weight of mean δ . At most $\binom{2^d}{2}/p$ of the normalized cube parameters identify two distinct vertices: each equality is one nontrivial linear equation with probability $1/p$. Away from these parameters, independence makes the expected product of indicators equal the product of their probabilities. The same argument applies to the centered variables $1_A(x) - \delta$. Thus the expectation error in the centered cube polynomial is at most $2\binom{2^d}{2}/p$.

Changing one inclusion decision changes a normalized cube polynomial by at most $2^d/p$. Indeed, telescope the product over its vertex occurrences; each occurrence has p^d possible remaining parameters, and the other factors are bounded by one. The bounded-difference variance inequality therefore gives variance $O_d(1/p)$. The empirical density has variance at most $1/(4p)$. Replacing δ by the empirical density δ_A changes the centered cube polynomial by at most $2^d|\delta_A - \delta|$. Consequently, for sufficiently large p there is a realization simultaneously approximating the prescribed density, cube count, and centered cube power to any specified positive error.

Apply this to the cosine weights of Proposition III.3.7, with $t_j \rightarrow 0$, and choose primes p_j sufficiently large that all errors are $o(t_j^8)$. Then

$$\alpha_j = t_j^8/32 + o(t_j^8), \quad Q_3(1_{A_j}) - \delta_j^8 = \frac{3}{2}\delta^4t_j^4 + o(t_j^4),$$

which proves (III.3.25).

For the two-frequency example, choose approximation errors $o(u^{28/3})$. The error in replacing its prescribed u^4 by the rounded value is then $o(u^{16/3})$, since u^4 is the square root of the centered eighth power. The density and cube-count errors are smaller as well. Thus (III.3.22) transfers with its nonzero $u^{16/3}$ term. The 3-torsion quintic example can instead be rounded on $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$; it does not give quintic sharpness on prime cyclic groups. $\square$

III.4 Density-weighted progression counting

Source 04, delivered Section 7.

The expansion in Gowers's Corollary 3.3 uses every subset of the set of positions, producing a factor bounded by 2^k . An ordered telescoping expansion gives a smaller bound and keeps the density factors. This section first states the precise group hypothesis needed for the Cauchy–Schwarz argument.

Definition III.4.1. Integers $c_1, \dots, c_m$ are *admissible on G* if the map $x \mapsto (c_i - c_j)x$ is an automorphism of G for every $i \neq j$. For functions $a_i : G \rightarrow \mathbb{C}$, write

$$\Lambda_{\mathbf{c}}(a_1, \dots, a_m) = \mathbb{E}_{x,y} \prod_{i=1}^m a_i(x + c_i y).$$

For $G = \mathbb{Z}/N\mathbb{Z}$, admissibility is equivalent to $\gcd(c_i - c_j, N) = 1$ for all pairs. Consecutive slopes $0, 1, \dots, k-1$ are therefore admissible whenever $\gcd(N, (k-1)!) = 1$. Primality is sufficient but unnecessary.

Lemma III.4.2 (Generalized von Neumann bound). *Suppose $m \geq 2$, the slopes are admissible, and $|a_i| \leq 1$. For every j ,*

$$|\Lambda_{\mathbf{c}}(a_1, \dots, a_m)| \leq \|a_j\|_{U^{m-1}}.$$

Proof. Relabel so that the target is a_m . For $m = 2$, the map $(x, y) \mapsto (x + c_1 y, x + c_2 y)$ is a bijection: its inverse uses the automorphism $c_2 - c_1$. The average factors as $(\mathbb{E}a_1)(\mathbb{E}a_2)$, giving the assertion.

Assume $m > 2$ and the result for $m-1$. Replace x by $x - c_1 y$, and apply Cauchy–Schwarz in x , using $|a_1| \leq 1$. Writing $d_i = c_i - c_1$, this gives

$$|\Lambda_{\mathbf{c}}|^2 \leq \mathbb{E}_x \left| \mathbb{E}_y \prod_{i=2}^m a_i(x + d_i y) \right|^2 = \mathbb{E}_h \mathbb{E}_{x,y} \prod_{i=2}^m a_i(x + d_i y) \overline{a_i(x + d_i y + d_i h)}.$$

For each h , apply the inductive assertion to the $m-1$ derivative functions. Differences of the new slopes are still automorphisms. The triangle inequality bounds the last expression by

$$\mathbb{E}_h \|\Delta_{d_m h} a_m\|_{U^{m-2}}, \quad \Delta_t a(x) = a(x) \overline{a(x+t)}.$$

The map $h \mapsto d_m h$ is a bijection. Hölder's inequality and the derivative identity for Gowers norms give

$$\mathbb{E}_h \|\Delta_{d_m h} a_m\|_{U^{m-2}} \leq \left(\mathbb{E}_t \|\Delta_t a_m\|_{U^{m-2}}^{2^{m-2}} \right)^{1/2^{m-2}} = \|a_m\|_{U^{m-1}}^2.$$

Taking square roots completes the induction. □

Theorem III.4.3 (Telescoping with the density factors retained). *Let $k \geq 3$, let the slopes be admissible on G , and let $a_i : G \rightarrow [0, 1]$ have means δ_i . Put $b_i = a_i - \delta_i$ and $u_i = \|b_i\|_{U^{i-1}}$ for $i \geq 3$. Then*

$$\left| \Lambda_{\mathbf{c}}(a_1, \dots, a_k) - \prod_{i=1}^k \delta_i \right| \leq \sum_{j=3}^k \left(\prod_{i=j+1}^k \delta_i \right) u_j. \quad (*)$$

Every permutation of the positions gives a valid version, with the appropriate Gowers norm of the function in the j -th position.

Proof. Successively write $a_j = \delta_j + b_j$, starting with $j = k$ and ending with $j = 3$. Multilinearity gives the exact identity

$$\Lambda_{\mathbf{c}}(a_1, \dots, a_k) - \prod_i \delta_i = \sum_{j=3}^k \left(\prod_{i=j+1}^k \delta_i \right) \Lambda_{c_1, \dots, c_j}(a_1, \dots, a_{j-1}, b_j).$$

The remaining two-function average is exactly $\delta_1 \delta_2$ by pairwise admissibility. Each term is bounded by u_j , by Lemma III.4.2; all other functions in that term are 1-bounded. Taking absolute values proves the result. Relabeling preserves admissibility, proving the last assertion. $\square$

Under the source's hypotheses $u_j \leq \varepsilon$, the new error is

$$\varepsilon \sum_{j=3}^k \prod_{i=j+1}^k \delta_i \leq (k-2)\varepsilon.$$

This is a direct replacement for $2^k \varepsilon$, in the same normalization. When all densities equal δ , the factor is

$$S_{k-2}(\delta) := \sum_{r=0}^{k-3} \delta^r = \begin{cases} (1 - \delta^{k-2})/(1 - \delta), & \delta \neq 1, \\ k-2, & \delta = 1. \end{cases}$$

In particular it is at most $1/(1 - \delta)$ for $0 \leq \delta < 1$, independently of k , and the inequality is strict when $0 < \delta < 1$.

[write] **6 October 2026.** Source 03's Section III.1 proves the same generalized von Neumann bound (Lemma III.1.1; "slopes with invertible differences" there are "admissible slopes" here) and the same telescoping bound (Theorem III.1.2), by the same induction and expansion. Its order-optimization recursion $B(S)$ is $D(S)$ below. The two texts were written independently (the intake measured under 0.8% shared text); both are kept and credited. The first display of Corollary III.4.4 is source 03's (III.1.2). Source 03 concludes at least $\delta^k N^2/4$ progressions under $N \geq 4\delta^{1-k}$ (Corollary III.1.3), while source 04 concludes positivity under $N > 2\delta^{1-k}$. Source 04's exchange rule and source 03's real-character torsion example are each one source's alone. Source 04's $\mathbb{Z}/2$ example below has a complex a_3 and shows that the generalized von Neumann inequality itself fails.

III.4.1 Choosing the order of elimination

The preceding result has a small useful optimization principle. Suppose one has upper bounds $u_i(s) \geq \|b_i\|_{U^s}$, for all $2 \leq s \leq k-1$. For a subset S of positions define

$$D(S) = 0 \quad (|S| \leq 2), \quad D(S) = \min_{i \in S} (u_i(|S| - 1) + \delta_i D(S \setminus \{i\})) \quad (|S| \geq 3).$$

Induction on $|S|$ shows that $D(S)$ bounds the counting error for the subsystem indexed by S . Moreover it is precisely the smallest bound obtained by this family of telescoping estimates. Computing the recurrence requires at most $k2^k$ elementary comparisons, once the norm bounds have been supplied.

If a single upper bound v_i is used for each function at every remaining order, there is also a simple exchange rule. For two successive eliminations i, j , their contribution before the common later tail is either $v_i + \delta_i v_j$ or $v_j + \delta_j v_i$. Thus i should precede j exactly when

$$v_i(1 - \delta_j) \leq v_j(1 - \delta_i).$$

For a fixed pair of positions to retain, sort the other positions by $v_i/(1 - \delta_i)$, with the degenerate cases interpreted through the displayed inequality. One can compare the $\binom{k}{2}$ possible retained pairs. This optimizes the stated bound, and is not a claim of an optimal counting inequality among all methods.

III.4.2 An explicit supersaturation consequence

Corollary III.4.4 (A sharper uniformity threshold). *Let $A \subset G$ have density $0 < \delta \leq 1$, and suppose that the consecutive slopes $0, \dots, k-1$ are admissible, with $k \geq 3$. Put $u = \|1_A - \delta\|_{U^{k-1}}$. The number $T_k^*(A)$ of ordered k -term progressions with nonzero difference satisfies*

$$T_k^*(A) \geq N^2(\delta^k - uS_{k-2}(\delta)) - \delta N. \quad (\text{III.4.1})$$

In particular, if

$$u \leq \frac{\delta^k}{2S_{k-2}(\delta)} \quad \text{and} \quad N > 2\delta^{1-k},$$

then $T_k^*(A) > 0$.

Proof. Norm monotonicity bounds every U^{j-1} -norm in Theorem III.4.3 by u . There are N^2 ordered choices of start and difference. Exactly $|A| = \delta N$ of the counted progressions have difference zero, so subtracting them gives the asserted bound. The two displayed assumptions make its right-hand side positive. $\square$

Admissibility also ensures that a progression with nonzero difference has k distinct entries: a repetition would give $(i - j)y = 0$ with $0 < |i - j| < k$, contrary to the automorphism hypothesis. In the paper's uniformity parameter, the sufficient condition becomes

$$\alpha \leq \left(\frac{\delta^k}{2S_{k-2}(\delta)} \right)^{2^{k-1}}.$$

For fixed k and small δ , its density exponent is $k2^{k-1}$, compared with $k2^k$ in the sufficient condition $(\delta/2)^{k2^k}$ displayed in the inspected version of Corollary 3.6. This comparison concerns the modular progression step.

Lemma III.4.5 (Lifting from a short interval). *Let $A \subset \{0, \dots, n-1\}$ be viewed in $\mathbb{Z}/N\mathbb{Z}$, with $2(n-1) < N$. If A contains a nonconstant modular progression of length $k \geq 3$, then it contains an ordinary integer progression of length k and positive difference.*

Proof. Let $x_0, \dots, x_{k-1} \in \{0, \dots, n-1\}$ be the representatives. Each integer $x_i + x_{i+2} - 2x_{i+1}$ is divisible by N and has absolute value at most $2(n-1) < N$; it is therefore zero. Successive integer differences are equal. The common difference is nonzero because the modular progression is nonconstant. Reverse the sequence if that integer difference is negative. $\square$

The lemma specifies the extra hypothesis needed to draw an integer conclusion. A modular progression in an unrestricted subset of $\mathbb{Z}/N\mathbb{Z}$ can wrap around, so that conclusion should not be silently inserted into Corollary III.4.4.

[write] 6 October 2026. *Already formalized.* This lemma is proved in Lean as `hasNatAP_of_short_modular_sequence` (`Proofs18IntervalTransfer.lean`, line 22), with the set form `hasNatAP_of_hasModAP_image`

(line 92); both were added in commit 327773149, source 04's own pin. The Lean statement takes representatives $f(i) \leq n$ with $2n < N$ and a nonzero modular difference, for every length k : after $n - 1 \mapsto n$ it is this lemma, and it also covers $k < 3$. Source 04 does not mention the formal version. It is the only statement of sources 01–05 that is formalized; source 03's rectification remark (Remark III.1.4) is the same argument.

Remark III.4.6 (Why invertibility cannot simply be omitted). For $G = \mathbb{Z}/2\mathbb{Z}$, take slopes 0, 1, 2, $a_3(0) = 1$, $a_3(1) = i$, $a_1 = \overline{a_3}$, and $a_2 = 1$. The forms at positions 0 and 2 coincide, so $\Lambda = 1$. Both normalized Fourier coefficients of a_3 have modulus $2^{-1/2}$, whence $\|a_3\|_{U^2} = 2^{-1/4} < 1$. Thus the stated generalized von Neumann inequality itself fails without an algebraic hypothesis. The obstruction is an algebraic dependence, and the admissibility condition states exactly what the present proof uses.

Part IV

Floor patterns and the Proposition 17.7 phase extraction

Contents. The number of floor patterns of subset sums, which controls how a cube meets a partition into progressions (Corollaries 17.5–17.6 and Proposition 17.7). Sections IV.1–IV.5: source 01, delivered Sections 9–13 (exact range factorization, quadratic-exponent upper and lower bounds, $\log_2 C_k \sim k^2$ via the classical threshold-function count, affine patterns and translation classes, polynomial features, certified $C_1, \dots, C_4 = 1, 2, 10, 154$). Section IV.6: source 04, delivered Section 8 (the same counts as F_k , the stronger elementary lower bound $2^{k(k-1)/2}$, exact modular range dependence, few profiles at a fixed slope, and a local phase-extraction theorem with coefficient $2^{-k^2-k \log_2 k - O(k)}$ at the stage of Proposition 17.7).

Overlaps. Sources 01 and 04 are independent manuscripts; they share only this count ($C_k = F_k$), which they prove independently with different certificate formats. Dated notes credit the shared lemmas and record that source 01’s asymptotic answers source 04’s Question 8.

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
C_k, F_k, A_k	01: C_k fractional carry patterns, A_k affine patterns; 04: F_k (= 01’s C_k) and A_k (= 01’s A_k).
H_k, J_k, Q_k	Both: $H_k = (k-2)2^{k-1} + 1$; 01: $J_k = k2^{k-1}$; 04: $Q_k = k2^{k-1}$ (= 01’s J_k).
region bound	01: $\mathcal{R}(h, d) = \sum_{j \leq d} \binom{h}{j}$; 04: $R_d(H) = \sum_{j \leq d} \binom{H}{j}$. <i>The arguments are in the other order: 01’s $\mathcal{R}(H, d)$ is 04’s $R_d(H)$.</i>
B_k	01: translation budget $(2(k+1))^k \mathcal{R}(k2^{k-1}, k+1)$; 04: $(2k)^k R_{k+1}(k2^{k-1})$. <i>Different numbers</i> (note after Corollary IV.6.4).
ranges, M	01: coefficients in $(-r_i, r_i)$, M a modulus, T_n threshold counts, M_n nonnegative ones; 04: (a_i, b_i) with $m_i = b_i - a_i$, M a modulus, $\Omega_k = \{0, 1\}^k$, L labels and K local multiplicity in Lemma IV.6.6.

IV.1 Fractional carry patterns and exact range dependence

Source 01, delivered Section 9.

We turn to the floor-pattern interface. For $k \geq 1$ and $x = (x_1, \dots, x_k) \in [0, 1)^k$, define

$$c_x(S) = \left\lfloor \sum_{i \in S} x_i \right\rfloor, \quad S \subseteq [k], \quad C_k = \#\{c_x : x \in [0, 1)^k\}. \quad (\text{IV.1.1})$$

The values at the empty set and at singletons are always zero. We call c_x a *fractional carry pattern*. The terminology refers only to this finite function; no probabilistic assumption on x is made. For completeness set $C_0 = 1$.

IV.1.1 Why boundary points do not introduce extra patterns

A hyperplane-region count normally concerns strict inequalities. Floor functions also take values on the hyperplanes, so that issue must be settled before counting chambers.

Lemma IV.1.1 (Positive perturbation). *Every fractional carry pattern has a representative $y \in (0, 1)^k$ such that $\sum_{i \in S} y_i$ is nonintegral for every nonempty $S \subseteq [k]$.*

Proof. Given $x \in [0, 1]^k$, put $y_i = x_i + \varepsilon$. There are finitely many coordinates and subset sums. Choose $\varepsilon > 0$ so small that $x_i + \varepsilon < 1$ for every i and

$$|S|\varepsilon < \left\lfloor \sum_{i \in S} x_i \right\rfloor + 1 - \sum_{i \in S} x_i \quad (\emptyset \neq S \subseteq [k]).$$

Every right-hand side is strictly positive, including when the subset sum is an integer. The perturbation moves the sum positively but not to the next integer. Its floor is unchanged, and it is no longer an integer. All coordinates become strictly positive. $\square$

Theorem IV.1.2 (Exact separation of the coefficient range). *Let $r_1, \dots, r_k$ be positive integers and let $\mathcal{F}(\mathbf{r})$ be the family of functions*

$$S \mapsto \left\lfloor \sum_{i \in S} \alpha_i \right\rfloor, \quad -r_i < \alpha_i < r_i.$$

Then

$$|\mathcal{F}(\mathbf{r})| = \left(\prod_{i=1}^k 2r_i \right) C_k. \quad (\text{IV.1.2})$$

Proof. Write $\alpha_i = m_i + x_i$, where $m_i = \lfloor \alpha_i \rfloor$ and $x_i \in [0, 1)$. Necessarily

$$m_i \in \{-r_i, -r_i + 1, \dots, r_i - 1\}.$$

For every subset,

$$F_\alpha(S) = \sum_{i \in S} m_i + c_x(S). \quad (\text{IV.1.3})$$

The singleton values of F_α recover all m_i ; after subtracting their subset sums, they also recover c_x . Thus this decomposition is injective as a coding of the resulting functions.

Conversely, take any permitted integer vector m and any carry pattern. By Lemma IV.1.1, choose a representative $x \in (0, 1)^k$. Then $-r_i < m_i + x_i < r_i$ for every i , even when $m_i = -r_i$. Equation (IV.1.3) realizes the chosen pair. Hence there is a bijection between the pattern family and the Cartesian product of the integer vectors with the C_k carry patterns. $\square$

The endpoint convention is essential for the exact statement. The proof handles the open interval $(-r_i, r_i)$ precisely because each fractional pattern admits an interior representative. For nonintegral range bounds $R_i > 0$, replacing r_i by $\lceil R_i \rceil$ gives an immediate upper bound, but generally not the same exact factorization.

IV.1.2 A hyperplane-arrangement upper bound

Inside the open unit cube, a nonempty subset S produces integer threshold hyperplanes

$$\sum_{i \in S} x_i = j, \quad 1 \leq j \leq |S| - 1.$$

There are exactly

$$H_k = \sum_{\emptyset \neq S \subseteq [k]} (|S| - 1) = k2^{k-1} - (2^k - 1) = (k - 2)2^{k-1} + 1 \quad (\text{IV.1.4})$$

such hyperplanes. Different subsets have different normal vectors, and none of these hyperplanes is a duplicate of another.

Lemma IV.1.3 (Region bound). *The number of regions cut out by h affine hyperplanes in $\mathbb{R}^d$ is at most*

$$\mathcal{R}(h, d) = \sum_{j=0}^d \binom{h}{j}.$$

The same bound holds for their regions restricted to an open convex set.

Proof. Let $R(h, d)$ be the maximal possible number of regions. When one adds a hyperplane, each newly created region comes from splitting an old region that meets it. The old hyperplanes induce on the new hyperplane an arrangement of at most $h - 1$ hyperplanes in dimension $d - 1$; coincident or empty intersections create no additional regions. Therefore

$$R(h, d) \leq R(h - 1, d) + R(h - 1, d - 1).$$

The initial values $R(0, d) = 1$ and $R(h, 0) = 1$, together with Pascal's identity, prove the bound by induction. A region is an intersection of open half-spaces and is convex. Intersecting it with an open convex set cannot split it into several nonempty connected pieces, so restriction cannot increase the number. $\square$

Theorem IV.1.4 (Quadratic-exponent carry bound). *For every $k \geq 1$,*

$$C_k \leq \mathcal{R}(H_k, k). \quad (\text{IV.1.5})$$

For $k \geq 3$ this implies

$$\log_2 C_k \leq k^2 + (\log_2 e - 1)k. \quad (\text{IV.1.6})$$

Proof. Lemma IV.1.1 realizes every carry pattern off the threshold hyperplanes. Off those hyperplanes, the floor values determine, and are determined by, the sides of all the listed integer thresholds. A given sign pattern is an intersection of open half-spaces with the cube and is convex. Thus distinct carry patterns correspond exactly to nonempty regions of this arrangement inside the open cube. Apply Lemma IV.1.3.

For $h \geq k \geq 1$, set $t = k/h \leq 1$. Then

$$t^k \sum_{j=0}^k \binom{h}{j} \leq \sum_{j=0}^k \binom{h}{j} t^j \leq (1 + t)^h \leq e^k.$$

Hence $\mathcal{R}(h, k) \leq (eh/k)^k$. For $k \geq 3$, $H_k \geq k$ and $H_k \leq k2^{k-1}$. Substitute these inequalities and take binary logarithms. $\square$

Corollary IV.1.5 (A strengthened floor-pattern bound). *For positive integers r_i ,*

$$|\mathcal{F}(\mathbf{r})| \leq \left(\prod_i 2r_i \right) \mathcal{R}(H_k, k).$$

In particular, for $r_i = r$ and $k \geq 3$,

$$\log_2 |\mathcal{F}(r, \dots, r)| \leq k \log_2(2r) + k^2 + (\log_2 e - 1)k. \quad (\text{IV.1.7})$$

Proof. Combine Theorems IV.1.2 and IV.1.4. □

k	H_k	old log bound, $r = 1$	new log bound, $r = 1$
2	1	16	3.000
3	5	54	7.700
4	17	128	15.650
8	769	1024	69.358
12	20481	3456	155.025
16	458753	8192	272.667

Table 5: Binary logarithms of the bounds 2^{2k^3} and $2^k \mathcal{R}(H_k, k)$ for the coefficient range $(-1, 1)^k$. These upper bounds are not asserted to be exact counts.

IV.2 Lower bounds and the optimal entropy scale

Source 01, delivered Section 10.

IV.2.1 An elementary independent-choice construction

Theorem IV.2.1 (Elementary quadratic-exponent lower bound). *For $k \geq 1$,*

$$C_k \geq 2^{\lfloor k^2/4 \rfloor}. \quad (\text{IV.2.1})$$

Proof. The assertion for $k = 1$ is immediate. Write $k = m + n$ with $m, n \geq 1$. Use m anchor coordinates

$$x_j = \frac{2^{j-1}}{2^m} \quad (1 \leq j \leq m).$$

Their subset sums are precisely $t/2^m$, for $0 \leq t \leq 2^m - 1$. For each of the remaining n coordinates independently choose

$$y_i = \frac{h_i + 1/2}{2^m}, \quad 0 \leq h_i \leq 2^m - 1.$$

For an anchor subset with sum $t/2^m$,

$$\left\lfloor y_i + \frac{t}{2^m} \right\rfloor = \begin{cases} 0, & h_i + t < 2^m, \\ 1, & h_i + t \geq 2^m. \end{cases}$$

This threshold, as t ranges over all anchor subset sums, recovers h_i : for $h_i = 0$ there is no value 1, and otherwise the first value 1 occurs at $t = 2^m - h_i$. Therefore different choices of the vector $(h_1, \dots, h_n)$ give different carry patterns. There are 2^{mn} choices. Taking $m = \lfloor k/2 \rfloor$ and $n = \lceil k/2 \rceil$ proves the theorem. □

This construction is independent of any threshold-function enumeration. Together with (IV.1.6), it already determines the order $\log C_k = \Theta(k^2)$ by elementary arguments.

IV.2.2 Embedding Boolean threshold functions

Let T_n denote the number of Boolean linear threshold functions on $\{0,1\}^n$ admitting a strict representation

$$g(\varepsilon) = \mathbf{1}_{\sum_i w_i \varepsilon_i > t}, \quad \sum_i w_i \varepsilon_i \neq t \text{ for all } \varepsilon.$$

Let M_n count the nonconstant such functions with a representation using nonnegative weights. We need only a simple relation between these two counts and carry patterns.

Lemma IV.2.2 (Thresholds inject into one higher carry dimension). *For every $n \geq 1$,*

$$C_{n+1} \geq M_n \geq 2^{-n} T_n - 2. \quad (\text{IV.2.2})$$

Proof. If a coefficient w_i is negative, replace the Boolean variable ε_i by $1 - \varepsilon_i$ and adjust the constant threshold. This makes that coefficient nonnegative. Thus every threshold function is a coordinate-flipped version of a nonnegative-weight threshold function, with at most 2^n choices of coordinate flips. Including the two constant functions gives $T_n \leq 2^n(M_n + 2)$.

For the first inequality, fix a nonconstant nonnegative-weight threshold function. Strictness and nonconstancy imply $0 < t < \sum_i w_i$. Choose $c > 0$ such that $c \sum_i w_i < 1$, and set

$$x_0 = 1 - ct, \quad x_i = cw_i \quad (1 \leq i \leq n).$$

All coordinates lie in $[0, 1)$, and $x_0 \in (0, 1)$. For $S \subseteq [n]$, the sum $x_0 + \sum_{i \in S} x_i$ lies strictly between 0 and 2, is not 1, and satisfies

$$\left\lfloor x_0 + \sum_{i \in S} x_i \right\rfloor = \mathbf{1}_{\sum_{i \in S} w_i > t}.$$

Therefore the carry pattern restricted to subsets containing coordinate 0 recovers the original Boolean function. Distinct functions have distinct resulting carry patterns, regardless of the chosen strict representations and sufficiently small scaling constants. $\square$

The classical external input is

$$\log_2 T_n = n^2 - n \log_2 n + O(n). \quad (\text{IV.2.3})$$

The leading n^2 asymptotic is due to Zuev; the refinement in (IV.2.3) is attributed to Kahn–Komlós–Szemerédi. Baldi and Vershynin explicitly record both results and also recover the leading asymptotic as the degree-one case of their theorem [8, Section 1.4 and Theorem 1.1]. We use this input, rather than claiming to reprove it; the original references are [9, 10].

Theorem IV.2.3 (Optimal leading carry entropy). *Using the classical input (IV.2.3),*

$$k^2 - k \log_2 k - O(k) \leq \log_2 C_k \leq k^2 + (\log_2 e - 1)k. \quad (\text{IV.2.4})$$

In particular $\log_2 C_k \sim k^2$. Consequently, for arbitrary positive integer ranges $r_1, \dots, r_k$,

$$\log_2 |\mathcal{F}(\mathbf{r})| = \sum_{i=1}^k \log_2(2r_i) + k^2 + o(k^2), \quad (\text{IV.2.5})$$

where the error depends only on k , not on the ranges.

Proof. For $n = k - 1$, Lemma IV.2.2 and (IV.2.3) give

$$\log_2 C_k \geq (k - 1)^2 - (k - 1) \log_2(k - 1) - O(k).$$

The subtraction of two constant functions is negligible at this scale, and the factor 2^{-n} costs only $O(k)$ in the logarithm. The right side is $k^2 - k \log_2 k - O(k)$. Combine this with Theorem IV.1.4. Finally take logarithms in the exact identity (IV.1.2); its range term has no approximation error. $\square$

The leading coefficient 1 in (IV.2.4) is therefore settled by the stated classical input. The present estimates leave a possible $k \log k$ correction unresolved. For a completely elementary version of the article, omit (IV.2.3) and replace the asymptotic claim by the bounds in (F.6.5); every local density theorem is unaffected.

[write] 6 October 2026. This theorem answers source 04's Question 8 (dated note there, Section V.17): source 04's F_k is C_k and its A_k is the A_k of Section IV.3. Source 04's elementary lower bound $F_k \geq 2^{k(k-1)/2}$ (Theorem IV.6.2) is stronger than Theorem IV.2.1 for every $k \geq 3$ (note there).

IV.3 Affine patterns, translation classes, and general features

Source 01, delivered Section 11.

IV.3.1 Fractional affine offsets

For $t = (t_0, t_1, \dots, t_k) \in [0, 1)^{k+1}$ define

$$a_t(S) = \left\lfloor t_0 + \sum_{i \in S} t_i \right\rfloor, \quad A_k = \#\{a_t : t \in [0, 1)^{k+1}\}.$$

Here $a_t(\emptyset) = 0$. Let

$$J_k = \sum_{S \subseteq [k]} |S| = k2^{k-1} \quad (k \geq 1). \quad (\text{IV.3.1})$$

Theorem IV.3.1 (Affine fractional-pattern bound). *For $k \geq 1$,*

$$C_k \leq A_k \leq \mathcal{R}(J_k, k + 1). \quad (\text{IV.3.2})$$

In particular, with the classical input of Theorem IV.2.3, $\log_2 A_k \sim k^2$.

Proof. Taking $t_0 = 0$ shows $A_k \geq C_k$. For the upper bound, perturb all $k + 1$ coordinates by a common sufficiently small positive number. Each nonempty affine sum increases by $(|S| + 1)\varepsilon$ but does not reach the next integer, and t_0 remains less than one. Thus every pattern has an interior representative off all integer threshold hyperplanes. For each subset S the internal thresholds are

$$t_0 + \sum_{i \in S} t_i = j, \quad 1 \leq j \leq |S|.$$

Their number is J_k . Apply Lemma IV.1.3 in dimension $k + 1$. For $k \geq 2$, $J_k \geq k + 1$, and the binomial-sum estimate gives

$$\log_2 A_k \leq (k + 1) \log_2 \frac{ek2^{k-1}}{k + 1} = k^2 + O(k).$$

Use the lower bound from C_k for the leading asymptotic. $\square$

IV.3.2 Reduction modulo a modulus and quotienting by translations

For a positive integer M , let $\mathcal{A}_M(\mathbf{r})$ consist of

$$S \mapsto \left[\alpha_0 + \sum_{i \in S} \alpha_i \right] \pmod{M}, \quad \alpha_0 \in \mathbb{R}, \quad -r_i < \alpha_i < r_i.$$

The group $\mathbb{Z}/M\mathbb{Z}$ acts on these functions by adding a constant to all values. This action preserves the family, since α_0 is unrestricted.

Theorem IV.3.2 (Modulus-free translation-class budget). *For positive integers r_i and M ,*

$$|\mathcal{A}_M(\mathbf{r})| \leq M \left(\prod_i 2r_i \right) \mathcal{R}(J_k, k+1), \quad (\text{IV.3.3})$$

$$|\mathcal{A}_M(\mathbf{r})/(\mathbb{Z}/M\mathbb{Z})| \leq \left(\prod_i 2r_i \right) \mathcal{R}(J_k, k+1). \quad (\text{IV.3.4})$$

Proof. Write $\alpha_0 = m_0 + t_0$ and $\alpha_i = m_i + t_i$ with fractional parts in $[0, 1)$. Every pattern equals

$$m_0 + \sum_{i \in S} m_i + a_t(S) \pmod{M}.$$

There are M choices of m_0 modulo M , at most $\prod_i 2r_i$ integer coefficient vectors, and A_k fractional affine patterns. This proves (IV.3.3).

Every translation orbit has a unique representative with value zero at the empty set: subtract the original value there. This cancels m_0 while retaining the information in t_0 . The remaining possibilities are counted by at most $(\prod_i 2r_i) A_k$. Apply Theorem IV.3.1. Equivalently, the constant-translation action is free, as can be checked at the empty set, so every orbit has exactly M members. $\square$

For $M = 1$, the actual orbit count is one; the inequality remains valid. One must not model translation classes by simply discarding the fractional part of the affine offset. Canonical normalization at the empty set removes the *integer* constant modulo M , not its fractional part. Modular collisions may make either inequality strict, which is why no exact product formula is asserted in this affine modular setting.

IV.3.3 The combinatorial input for phase removal

At the relevant floor-pattern step of the proof of Proposition 17.7, the cell-index maps have affine subset-sum form, with coefficient range bounded by $k+1$ [1, proof of Proposition 17.7]. Consequently a valid translation-class budget is

$$B_k = (2(k+1))^k \mathcal{R}(k2^{k-1}, k+1). \quad (\text{IV.3.5})$$

Its logarithm is at most $k^2 + k \log_2 k + O(k)$, rather than a cubic polynomial. The following elementary statement makes precise what substituting this budget does, and what it does not do.

Proposition IV.3.3 (A transparent selection interface). *Suppose $z_1, \dots, z_L \in \mathbb{C}$ and $|\sum_{\ell=1}^L z_\ell| \geq U$. Then some ℓ satisfies $|z_\ell|^2 \geq U^2/L^2$. If the terms are indexed by translation classes from Theorem IV.3.2 with $r_i = k+1$, the denominator may be replaced by B_k^2 . More generally, any correctly established estimate $E \geq K/L^2$, with $K \geq 0$ and L the actual number of such classes, implies $E \geq K/B_k^2$.*

Proof. The triangle inequality gives $U \leq \sum_{\ell} |z_{\ell}| \leq L \max_{\ell} |z_{\ell}|$. Square, and then use $L \leq B_k$ from Theorem IV.3.2. The last assertion follows because $x \mapsto x^{-2}$ is decreasing for positive x . Zero classes cannot carry positive U ; the vacuous zero-signal case is handled separately. $\square$

Thus the combinatorial L^{-2} loss at this interface has a budget $2^{-2k^2-2k \log_2 k - O(k)}$. This is a local substitution conditional on the other estimates in the application being valid. We do not promote it to an improved version of every quantitative statement in Sections 17–18: localization, phase removal, uniformity normalization, and the earlier structural parameters must all be propagated separately.

IV.3.4 Nonnegative integer features and effective rank

The Boolean cube is not essential to the upper-bound mechanism. Let $V \subseteq \mathbb{N}^k \setminus \{0\}$ be a finite set of row vectors and consider

$$x \mapsto (\lfloor v \cdot x \rfloor)_{v \in V}, \quad x \in [0, 1]^k.$$

Put $H(V) = \sum_{v \in V} (\|v\|_1 - 1)$ and $d = \dim_{\mathbb{R}} \text{span } V$.

Proposition IV.3.4 (Rank-sensitive feature bound). *The number of these patterns is at most $\mathcal{R}(H(V), d)$. If V contains every coordinate unit vector, allowing $\alpha_i \in (-r_i, r_i)$ multiplies the number of fractional patterns by exactly $\prod_i 2r_i$.*

Proof. Every nonzero row has positive dot product with the all-ones vector. The common positive perturbation of Lemma IV.1.1 therefore preserves all floors while removing every integral equality and every boundary coordinate. Row v contributes at most $\|v\|_1 - 1$ internal integer hyperplanes; duplicates can only reduce the count. The linear map $x \mapsto (v \cdot x)_{v \in V}$ has rank d . Its image of the open cube is an open convex set relative to its d -dimensional image, and the relevant threshold hyperplanes restrict to affine hyperplanes in that image. Lemma IV.1.3 proves the bound. For the last assertion, $v \cdot m$ is an integer and the unit-vector coordinates recover m_i . Interior representatives prove surjectivity exactly as in Theorem IV.1.2. $\square$

Example IV.3.5 (Why arbitrary signed features need a boundary argument). Near $x = 0$, the map $x \mapsto (\lfloor x \rfloor, \lfloor -x \rfloor)$ has three patterns: $(0, -1)$ for $x > 0$, $(-1, 0)$ for $x < 0$, and $(0, 0)$ at zero. The one hyperplane has only two open regions. Thus a chamber count alone misses a floor pattern when opposite orientations occur. A sufficient replacement for nonnegative rows is a common direction u with $v \cdot u > 0$ for every row, together with room to perturb the parameter domain in that direction. Without such a condition one must count the relevant boundary strata as well.

IV.4 A polynomial generalization on the Boolean cube

Source 01, delivered Section 12.

The feature bound also gives an extension in which the input dimension and coefficient dimension are different. It is useful when a localization step produces low-degree polynomial floor functions rather than affine ones.

Fix $1 \leq d \leq n$ and a positive integer r . Let $\mathcal{P}_{n,d}(r)$ be the family of integer-valued functions on $\{0, 1\}^n$ of the form

$$\varepsilon \mapsto \left[\sum_{\substack{S \subseteq [n] \\ |S| \leq d}} \alpha_S \prod_{i \in S} \varepsilon_i \right], \quad -r < \alpha_S < r. \quad (\text{IV.4.1})$$

The constant monomial corresponds to $S = \emptyset$. Put

$$m = \sum_{j=0}^d \binom{n}{j}, \quad H_{n,d} = \sum_{j=1}^d \binom{n}{j} 2^{n-j}.$$

Let $T(n, d)$ count strict Boolean polynomial threshold functions of degree at most d ; on a Boolean cube a polynomial can be multilinearized without changing its values or increasing its degree.

Theorem IV.4.1 (Polynomial floor-pattern entropy). *For $1 \leq d \leq n$ and positive integer r ,*

$$T(n, d) \leq |\mathcal{P}_{n,d}(r)| \leq (2r)^m \mathcal{R}(H_{n,d}, m). \quad (\text{IV.4.2})$$

For fixed $d \geq 1$, using the polynomial threshold-function theorem of Baldi–Vershynin, if $r = r(n)$ is a positive integer with $\log r = o(n)$, then

$$\log_2 |\mathcal{P}_{n,d}(r)| = (1 + o(1)) n \sum_{j=0}^d \binom{n}{j} = (1 + o(1)) \frac{n^{d+1}}{d!}. \quad (\text{IV.4.3})$$

Proof. Index coefficient space by the m monomials of degree at most d . For an input ε , the evaluation vector has entries $\prod_{i \in S} \varepsilon_i \in \{0, 1\}$. It is nonzero because of the constant monomial. The sum of its coordinates counts the active monomials at ε . Summing the number of internal thresholds over all inputs gives

$$\sum_{\varepsilon \in \{0,1\}^n} (\#\{S : |S| \leq d, S \subseteq \text{supp } \varepsilon\} - 1) = \sum_{j=1}^d \binom{n}{j} 2^{n-j} = H_{n,d}.$$

Separate each coefficient into an integer part and a fractional part. There are at most $(2r)^m$ integer vectors. For each vector, fractional patterns are bounded by the hyperplane count in dimension m , using positive perturbation and Lemma IV.1.3. This proves the upper bound. It does not assert that different integer vectors always give disjoint pattern families.

For the lower bound, take a strict multilinear polynomial threshold function, represented by a polynomial p with no zero values on the Boolean cube. Multiply p by a sufficiently small positive c so that every coefficient of cp lies in $(-r, r)$ and $0 < |cp(\varepsilon)| < 1$ at every input. Then

$$\lfloor cp(\varepsilon) \rfloor = \begin{cases} -1, & p(\varepsilon) < 0, \\ 0, & p(\varepsilon) > 0. \end{cases}$$

Thus the integer-valued floor function encodes the threshold function. Distinct threshold functions give distinct floors.

For fixed d , the external theorem gives $\log_2 T(n, d) = (1 + o(1))nm$ [8, Theorem 1.1]. For sufficiently large n , $H_{n,d} \geq m$ and $H_{n,d} \leq 2^n m$. The binomial estimate from Theorem IV.1.4 therefore yields

$$\log_2 |\mathcal{P}_{n,d}(r)| \leq m \log_2(2r) + m \log_2(eH_{n,d}/m) \leq nm + m \log_2(2r) + m \log_2 e.$$

Since $\log r = o(n)$, this matches the lower bound to leading order. Finally $m = (1 + o(1))n^d/d!$ for fixed d . $\square$

The finite upper bound in (IV.4.2) is elementary; only the matching leading lower asymptotic uses the external threshold theorem. This generalization also shows why using the square of the coefficient dimension as a universal exponent can be wasteful: for $d > 1$, the sharp leading entropy is on the scale n^{d+1} rather than $m^2 \asymp n^{2d}$.

IV.5 Exact finite enumeration and reproducible checks

Source 01, delivered Section 13.

IV.5.1 Small carry counts

Proposition IV.5.1 (Certified small dimensions). *The first four carry counts are*

$$C_1 = 1, \quad C_2 = 2, \quad C_3 = 10, \quad C_4 = 154. \quad (\text{IV.5.1})$$

The first three values have the elementary proof below. The fourth is proved by an exhaustive finite rational-certificate enumeration supplied with this article; its certificate format and exact verification argument are given in the following subsection.

Proof of the first three values. For $k = 1$ every floor is zero. For $k = 2$, only $\lfloor x_1 + x_2 \rfloor \in \{0, 1\}$ varies, and both values occur.

For $k = 3$, Lemma IV.1.1 allows us to avoid integral sums. If $x_1 + x_2 + x_3 < 1$, every pair floor is zero. If the total is greater than 2, every pair sum is greater than 1, since the remaining coordinate is less than 1. These give two patterns. When the total lies in $(1, 2)$, all eight assignments of pair floors occur. Up to coordinate permutations, representatives with respectively zero, one, two, and three pair floors equal to one are

$$(0.4, 0.4, 0.4), \quad (0.7, 0.5, 0.1), \quad (0.8, 0.4, 0.4), \quad (0.6, 0.6, 0.6).$$

There are $1 + 3 + 3 + 1 = 8$ such pair patterns, all with total floor one. The two outer patterns are realized by $(0.1, 0.1, 0.1)$ and $(0.9, 0.9, 0.9)$. Hence $C_3 = 10$. $\square$

For example, Theorem IV.1.2 now gives exactly $C_4(2r)^4 = 154(2r)^4$ patterns with four coefficients in $(-r, r)$. At $r = 1$ the exact count is 2464, while the simple new arrangement upper bound is 51424. No assertion is made that these finite values or their sequence have not appeared elsewhere.

[write] 6 October 2026. $C_1, C_2, C_3 = 1, 2, 10$ are certified independently in source 04 (as F_1, F_2, F_3 , Section IV.6), together with $A_1, A_2, A_3 = 2, 6, 38$. $C_4 = 154$ is certified only here. At intake the sequence 1, 2, 10, 154 matched only OEIS A134088 (1, 1, 2, 10, 154, 7994, $\dots$, a power-series coefficient sequence); its next term 7994 differs from the uncertified recount $C_5 = 8410$ (note in Section IV.6), so the match looks like a coincidence. 2, 6, 38 gave no relevant hit. This is the intake's search of 6 October 2026, not repeated at the write; nothing was submitted to the OEIS.

IV.5.2 A solver-independent rational certificate

Order the H_k hyperplanes by increasing binary mask of S , and then by increasing integer threshold. Build a binary tree of strict signs. For each partial sign word, use variables $z = (x_1, \dots, x_k, \epsilon)$ and inequalities

$$\begin{array}{ll} -x_i + \epsilon \leq 0, & x_i + \epsilon \leq 1, \\ v \cdot x + \epsilon \leq j & \text{for the sign } v \cdot x < j, \\ -v \cdot x + \epsilon \leq -j & \text{for the sign } v \cdot x > j. \end{array}$$

Write this finite system as $Az \leq b$. A sign word is strictly feasible exactly when this system has a solution with $\epsilon > 0$.

There are two certificate types. A *primal certificate* is a rational vector z satisfying $Az \leq b$ and $z_{k+1} > 0$. A *dual certificate* is a rational vector $\lambda \geq 0$ satisfying

$$\lambda^T A = (0, \dots, 0, 1), \quad \lambda^T b \leq 0. \quad (\text{IV.5.2})$$

The latter proves that every solution of $Az \leq b$ has $\epsilon = \lambda^T Az \leq \lambda^T b \leq 0$, so the sign word is strictly infeasible. This implication needs no floating-point tolerance and no appeal to an optimizer's status message.

The generator uses a numerical linear program only to propose rational primal or dual vectors. Every proposal is checked with exact rational arithmetic before being accepted. It branches both ways at every feasible partial word and never branches below an infeasible word. The verifier reconstructs all inequalities from the sign words, checks every rational identity, checks that both children of every feasible nonterminal node are present, and rejects extra or unreachable nodes. It then counts feasible leaves. By induction on tree depth, this is an exhaustive count of all regions in the open cube. Theorem IV.1.4's region-pattern correspondence turns that count into C_k .

Dimension k	Hyperplanes H_k	Certificate records	Feasible leaves C_k
1	0	1	1
2	1	3	2
3	5	49	10
4	17	2061	154

Table 6: Exact certificate results in `certificates/carry.json`. The count of records includes both primal and dual certificates.

IV.5.3 Commands and the limits of computational evidence

From the package directory, verification uses only the Python standard library:

```
python code/carry_certificates.py verify certificates/carry.json
```

Regeneration additionally needs NumPy and SciPy:

```
python code/carry_certificates.py generate --max-k 4 \
--out certificates/carry.json
```

The supplementary checks use NumPy and SymPy:

```
python code/check_refinements.py
```

They verify six symbolic identities, 210 broken-chord parameter cases, 210 equality constructions, 175 maximum-density root cases, 2000 sampled weighted-cell inequalities, and 24 concrete progression-partition cases. The parameter grids and random seed are fixed in the script. These are useful error detectors, not substitutes for the general proofs. In contrast, the finite carry enumeration has exact certificates for every branch and a mathematical completeness argument, though its implementation is still ordinary Python rather than kernel-checked proof code.

The verifier is independent of the numerical optimizer, not an independent second implementation of every aspect of the encoding. Its small trusted computational core includes Python integer and rational arithmetic, the constraint generator, and the tree checks. This makes the remaining trust boundary explicit.

IV.6 Floor patterns and efficient extraction of a progression profile

Source 04, delivered Section 8.

Corollaries 17.5 and 17.6 of Gowers’s paper bound the number of functions obtained by taking integer parts of linear expressions on a Boolean cube [1]. These estimates control the number of ways that a cube can meet a partition into arithmetic progressions. We isolate the fractional parts of the coefficients, determine the exact dependence on their integer ranges, and prove matching quadratic orders of magnitude for the logarithm of the remaining complexity. A second observation reduces the loss incurred when selecting one progression profile.

The edited repository already uses a joint hyperplane count in its proof of Proposition 17.7 [23]. The refinements below go further by separating integer and fractional parameters and distinguishing the number of globally possible profiles from the number possible at a fixed slope. The arguments establish the stated bounds; they do not establish historical priority for these elementary refinements.

IV.6.1 Fractional patterns and their boundary values

Write $\Omega_k = \{0, 1\}^k$ and put

$$F_k = \# \left\{ (\lfloor \epsilon \cdot t \rfloor)_{\epsilon \in \Omega_k} : t \in [0, 1)^k \right\},$$

$$A_k = \# \left\{ (\lfloor t_0 + \epsilon \cdot t \rfloor)_{\epsilon \in \Omega_k} : (t_0, t) \in [0, 1)^{k+1} \right\}.$$

Set $F_0 = A_0 = 1$, and define

$$R_d(H) = \sum_{j=0}^d \binom{H}{j}, \quad H_k = (k-2)2^{k-1} + 1, \quad Q_k = k2^{k-1},$$

where $\binom{H}{j} = 0$ for $j > H$.

[write] 6 October 2026. *Source defect corrected in the write.* In both displays above the delivered source writes the subscript `_epsilon\in\Omega_k`, without the backslash before `epsilon`, so the delivered PDF prints the word “epsilon”; the intended $\epsilon \in \Omega_k$ is restored here. Notation: F_k is source 01’s C_k , A_k is source 01’s A_k , Q_k is source 01’s J_k , and $R_d(H)$ is source 01’s $\mathcal{R}(H, d)$, with the arguments in the other order.

Lemma IV.6.1 (Boundary removal). *Every pattern counted by F_k or A_k has a representative in the open parameter cube which lies on none of its nonconstant integer-threshold hyperplanes.*

Proof. Add a sufficiently small positive multiple of the all-ones vector to the parameters. Every nonconstant form under consideration increases strictly. An integer value moves into the interior of the same floor interval; a noninteger value remains in its floor interval if the step is sufficiently small. There are finitely many forms and every coordinate is initially less than 1, so one positive step works for all forms and coordinates. The identically zero form in the definition of F_k requires no adjustment. $\square$

Theorem IV.6.2 (Quadratic order of fractional complexity). *For every $k \geq 1$,*

$$2^{k(k-1)/2} \leq F_k \leq R_k(H_k), \quad \prod_{j=0}^{k-1} (2^j + 1) \leq A_k \leq R_{k+1}(Q_k). \quad (\text{IV.6.1})$$

Consequently $\log_2 F_k = \Theta(k^2)$ and $\log_2 A_k = \Theta(k^2)$. In particular,

$$F_k \leq e^k 2^{k(k-1)} \quad (k \geq 3), \quad A_k \leq e^{k+1} 2^{k^2-1} \quad (k \geq 2). \quad (\text{IV.6.2})$$

Proof. For a nonempty subset $S \subseteq \{1, \dots, k\}$ of size s , the form $\sum_{i \in S} t_i$ crosses precisely the possible thresholds $1, \dots, s-1$ inside the open cube. The resulting arrangement has

$$\sum_{s=1}^k \binom{k}{s} (s-1) = k2^{k-1} - (2^k - 1) = H_k$$

hyperplanes. A pattern is constant in each chamber. Its strict feasible region is an intersection of open halfspaces and is therefore convex, so a single pattern cannot occupy two chambers. Lemma IV.6.1 also accounts for all patterns originally attained on boundaries. Thus F_k is the number of chambers meeting the open cube. The standard arrangement bound, also stated as Lemma 17.4 in [1], gives $F_k \leq R_k(H_k)$.

With an intercept, the form $t_0 + \sum_{i \in S} t_i$ has thresholds $1, \dots, s$, and there are $Q_k = \sum_s s \binom{k}{s}$ relevant hyperplanes in dimension $k+1$. This gives the bound for A_k . For $H \geq d \geq 1$, $R_d(H) \leq (eH/d)^d$: apply $(1+u)^H \leq e^{Hu}$ with $u = d/H$, observing that $u^j \geq u^d$ for $j \leq d$. Applying this with $H_k/k \leq 2^{k-1}$ and $Q_k/(k+1) \leq 2^{k-1}$ proves (IV.6.2).

For the lower bound, fix one pattern counted by F_{k-1} . Its region contains a nonempty open set. Choose a representative in that set whose 2^{k-1} subset sums have pairwise distinct fractional parts. Such a choice is possible because the prohibited equalities are finitely many proper hyperplanes

$$\sum_i (\epsilon_i - \epsilon'_i) t_i = j, \quad \epsilon \neq \epsilon', \quad j \in \mathbb{Z}.$$

Only finitely many j can occur in the bounded parameter cube. In particular, no nonempty subset sum has fractional part zero. As t_k varies through $(0, 1)$, each nonempty old subset sum produces one jump, at $1 - \{\sum_{i \in S} t_i\}$. The $2^{k-1} - 1$ jump locations are distinct, and hence there are 2^{k-1} different extension patterns. Extensions of different old patterns remain different on the face $\epsilon_k = 0$. Therefore $F_k \geq 2^{k-1} F_{k-1}$, which proves the first lower bound.

For A_k , choose an old representative for which the 2^{k-1} affine sums $t_0 + \sum_{i \in S} t_i$ have distinct nonzero fractional parts. Varying t_k now gives 2^{k-1} distinct jumps and $2^{k-1} + 1$ extensions. Thus $A_k \geq (2^{k-1} + 1) A_{k-1}$, proving the second lower bound. $\square$

[write] **6 October 2026.** *Credits and a comparison.* The upper bound $F_k \leq R_k(H_k)$ is source 01's Theorem IV.1.4, and $F_k \leq e^k 2^{k(k-1)}$ is the same as 01's $\log_2 C_k \leq k^2 + (\log_2 e - 1)k$; the affine upper bound is 01's Theorem IV.3.1. The lower bound $2^{k(k-1)/2}$ is stronger than source 01's elementary $2^{\lfloor k^2/4 \rfloor}$ (Theorem IV.2.1) for every $k \geq 3$, and equal to it for $k = 1, 2$. *Proof:* $k(k-1)/2 - k^2/4 = k(k-2)/4$, which is positive for $k \geq 3$, and $\lfloor k^2/4 \rfloor \leq k^2/4$; for $k = 1, 2$ both exponents are 0 and 1. The product lower bound for A_k is source 04's alone. Using the classical threshold-function count, source 01 proves $\log_2 F_k \sim k^2$ and $\log_2 A_k \sim k^2$ (Theorems IV.2.3 and IV.3.1). This answers Question 8 below (dated note there).

IV.6.2 Exact dependence on integer ranges

Theorem IV.6.3 (Integer-part scaling). *Let $a_i < b_i$ be integers and $m_i = b_i - a_i$. The number of functions*

$$\epsilon \mapsto \left\lfloor \sum_{i=1}^k \epsilon_i \alpha_i \right\rfloor, \quad \alpha_i \in (a_i, b_i),$$

is exactly $F_k \prod_i m_i$. In particular, the count in Gowers's Corollary 17.5 is exactly $(2r)^k F_k$ for a positive integer r .

Let $F_{k,M}$ count the fractional patterns in the definition of F_k after reduction modulo a positive integer M . The corresponding number of functions with values in $\mathbb{Z}/M\mathbb{Z}$ is exactly

$$F_{k,M} \prod_{i=1}^k \min(m_i, M). \tag{IV.6.3}$$

Moreover, $F_{k,M} = F_k$ whenever $M \geq k$.

Proof. Write $\alpha_i = n_i + t_i$, where $n_i = \lfloor \alpha_i \rfloor$ and $0 \leq t_i < 1$. Then

$$\lfloor \epsilon \cdot \alpha \rfloor = \epsilon \cdot n + \lfloor \epsilon \cdot t \rfloor.$$

The singleton values recover n_i exactly, since $\lfloor t_i \rfloor = 0$. Subtracting their linear contribution recovers the fractional pattern. Conversely, every fractional pattern has an interior representative by Lemma IV.6.1; this representative works in every integer cell, including those with an excluded left endpoint. There are precisely $\prod_i m_i$ integer choices.

Modulo M , the same singleton argument recovers the residues of the n_i , and each interval of m_i consecutive integers has $\min(m_i, M)$ residues. Subtracting these residues recovers the fractional residue pattern, proving (IV.6.3). Finally, every fractional floor belongs to $\{0, \dots, k-1\}$; reduction modulo $M \geq k$ is injective on this range. $\square$

For arbitrary real endpoints, the same argument gives the upper bound $F_k \prod_i (\lceil b_i \rceil - \lfloor a_i \rfloor)$; equality need not hold because the extreme integer cells may be truncated.

[write] **6 October 2026.** With $a_i = -r_i$, $b_i = r_i$ the first assertion is source 01's Theorem IV.1.2, by the same decomposition and interior-representative argument (source 01's Lemma IV.1.1 is Lemma IV.6.1); the arbitrary integer intervals and the modular count (IV.6.3) are source 04's.

Corollary IV.6.4 (Affine modular bound). *With the integer endpoints of Theorem IV.6.3, let α_0 vary over all real numbers. The number of functions*

$$\epsilon \mapsto \lfloor \alpha_0 + \epsilon \cdot \alpha \rfloor \pmod{M}$$

is at most $MA_k \prod_i \min(m_i, M)$. The number of equivalence classes under addition of a common constant is at most $A_k \prod_i \min(m_i, M)$. Both assertions remain valid if every slope interval (a_i, b_i) is replaced by $[a_i, b_i]$.

Proof. Writing $\alpha_0 = n_0 + t_0$ gives

$$\lfloor \alpha_0 + \epsilon \cdot \alpha \rfloor = n_0 + \epsilon \cdot n + \lfloor t_0 + \epsilon \cdot t \rfloor.$$

Choose $n_0 \pmod{M}$, the allowed slope residues, and one of the A_k fractional patterns. This proves the upper bound. Translation by a constant acts freely, as seen at $\epsilon = 0$, and all M translates occur by changing n_0 .

To handle closed slope intervals, move each endpoint slope inward with velocity $d_i = 1$ at a_i and $d_i = -1$ at b_i , leaving other slopes fixed. Move t_0 upward with velocity $k + 1$. Every affine form has strictly positive velocity $k + 1 + \epsilon \cdot d \geq 1$. A sufficiently small movement preserves every floor and keeps $t_0 < 1$, while moving all slope endpoints into the open intervals. $\square$

In the symmetric case, Corollary IV.6.4 replaces the bound $M2^{2r(k+1)^3}$ in Gowers's Corollary 17.6 by

$$M \min(2r, M)^k R_{k+1}(k2^{k-1}).$$

The dependence on r is polynomial, and the exponent governing the fractional complexity is quadratic in k . Theorem IV.6.2 shows that this quadratic order cannot be reduced to $o(k^2)$.

[write] 6 October 2026. Source 01's Theorem IV.3.2 gives the two bounds of Corollary IV.6.4 with $\prod_i 2r_i$ and $\mathcal{R}(k2^{k-1}, k + 1)$ in place of $\prod_i \min(m_i, M)$ and A_k ; source 04's are never larger. *Notation clash* B_k . Source 01's translation budget $B_k = (2(k + 1))^k \mathcal{R}(k2^{k-1}, k + 1)$ (IV.3.5) and source 04's $B_k = (2k)^k R_{k+1}(k2^{k-1})$ (Theorem IV.6.7) differ. Source 01 uses Gowers's coefficient range $k + 1$ with open intervals; source 04 uses the closed slope interval $[-k, k]$ of its phase-extraction proof, through the closed-interval clause of this corollary. Both are valid under their hypotheses. Source 04's is smaller by the factor $((k + 1)/k)^k < e$.

IV.6.3 Small cases with exact certificates

The first values are

k	1	2	3
F_k	1	2	10
A_k	2	6	38

For example, in the definition of F_3 the three pair-sum threshold indicators attain all eight Boolean patterns. If all three are zero, the total floor can be 0 or 1; if all are one, it can be 1 or 2. For each of the other six patterns, the total floor is 1. This gives $F_3 = 2 + 2 + 6 = 10$. Similarly, the two singleton values for A_2 give two possibilities for the pair value when they agree and one when they disagree, giving $A_2 = 6$.

The accompanying files `pattern_census.py` and `pattern_census.json` certify all six counts. The arrangement is constructed one hyperplane at a time, and both possible children of every surviving chamber are checked. For a system of strict inequalities $Cx < b$, a feasible branch has a rational point satisfying all inequalities strictly. An infeasible branch has rational weights $y \geq 0$ satisfying

$$\sum_i y_i = 1, \quad C^\top y = 0, \quad b^\top y \leq 0.$$

Multiplying the strict inequalities by these weights would yield $0 < b^\top y \leq 0$, a contradiction. All 450 branch certificates are checked with exact fractions. Numerical linear programming proposes the certificates during generation; the verifier itself uses only Python’s standard library and runs with

```
python -S pattern_census.py --verify pattern_census.json
```

[write] **6 October 2026.** Source 01 certifies the same F_1, F_2, F_3 (its C_k) and also $C_4 = F_4 = 154$ (Proposition IV.5.1), with a different certificate format. The census is shipped as `code/04-restriction-cubes-pattern_census.py` with `data/04-restriction-cubes-pattern_census.json`; the command above runs on a copy with the delivered names (README). At intake an independent recount reproduced $C_1, \dots, C_4$ and A_1, A_2, A_3 , and found $C_5 = 8410$ and $A_4 = 770$. It used value-branching floating-point linear programs and re-evaluated every feasibility witness exactly, but the infeasibility side rests on floating-point solver status, so these two values are *not certified*; they are recorded in Section V.2, item IV.2. They are consistent with the bounds: $2^{10} = 1024 \leq C_5 \leq R_5(H_5) = R_5(49) = 2\,138\,410$ and $2 \cdot 3 \cdot 5 \cdot 9 = 270 \leq A_4 \leq R_5(Q_4) = R_5(32) = 242\,825$.

IV.6.4 Few profiles at a fixed slope

Lemma IV.6.5 (Intercept profiles). *For fixed $c \in \mathbb{R}^k$, the functions*

$$\epsilon \longmapsto \lfloor \beta + \epsilon \cdot c \rfloor, \quad 0 \leq \beta < 1,$$

take at most 2^k different values. The same bound holds after reduction modulo M . Over the integers the count is exactly one plus the number of distinct nonzero fractional parts among the nonempty subset sums $\epsilon \cdot c$.

Proof. Each noninteger subset sum gives one jump in $(0, 1)$, at $1 - \{\epsilon \cdot c\}$. Integer subset sums give no interior jump. There are at most $2^k - 1$ distinct jump locations. The floor function is right-continuous, so a jump does not create an extra isolated pattern. Reduction modulo M can only identify patterns. Generic slopes make all subset sums distinct modulo 1, showing that the integer bound is sharp. $\square$

Lemma IV.6.6 (Extraction from locally few classes). *Let $z_{s,a}$ be complex numbers on finite index sets, with each pair (s, a) assigned one of L labels. Suppose that at most K labels occur for each fixed a . Set*

$$E = \sum_a \left| \sum_s z_{s,a} \right|^2, \quad E_j = \sum_a \left| \sum_{s: j(s,a)=j} z_{s,a} \right|^2.$$

Then $E \leq K \sum_j E_j$, and some label satisfies $E_j \geq E/(KL)$.

Proof. For each a , apply Cauchy–Schwarz to its at most K class sums. Sum over a , then average over the L labels. $\square$

IV.6.5 A quantitative local phase-extraction statement

Here is a precise form of the improvement at the stage of Proposition 17.7. For a function u on $G = \mathbb{F}_p$, write

$$\mathcal{U}_s(u) = \sum_{x, h_1, \dots, h_s \in G} \prod_{\omega \in \Omega_s} \mathcal{C}^{|\omega|} u(x - \omega \cdot h),$$

where $\mathcal{C}$ denotes complex conjugation. This quantity is nonnegative; it is $p^{s+1}\|u\|_{U^s}^{2^s}$. We use the usual Gowers–Cauchy–Schwarz inequality for its mixed version [1, Lemma 3.8].

Theorem IV.6.7 (Local extraction with a quadratic exponential loss). *Let $k \geq 1$, let $p > k + 1$ be prime, and let $m = 2h + 1 > 3k$ satisfy $m \leq \sqrt{p}$. Put $I = \{-h, \dots, h\} \subseteq G$. Let $r \in G^k$, let $\tau : G^k \rightarrow G$ be a multi-affine polynomial, and let $f : G \rightarrow \mathbb{C}$. For $\epsilon \in \Omega_k$, put $f_\epsilon(s) = f(s - \epsilon \cdot r)$. Suppose $\alpha > 0$ and*

$$\sum_{a \in I^k} \left| \sum_{s \in G} e_p(-s\tau(a)) \prod_{\epsilon \in \Omega_k} \mathcal{C}^{|\epsilon|} f_\epsilon(s - \epsilon \cdot a) \right|^2 \geq \alpha p^2 m^k, \quad e_p(t) = \exp(2\pi it/p). \quad (\text{IV.6.4})$$

Set

$$q = \lceil h/k \rceil, \quad M = \lfloor p/q \rfloor, \quad B_k = (2k)^k R_{k+1}(k2^{k-1}).$$

There is a polynomial P of degree at most $k + 1$ and a partition $(R_i)_{i \in \mathbb{Z}/M\mathbb{Z}}$ into modular arithmetic progressions of lengths q or $q + 1$, each at least $m/(3k)$, such that

$$\sum_i \mathcal{U}_{k+1}(1_{R_i} f e_p(-P)) \geq \frac{\alpha p^2 m^k}{2^k B_k M} \geq \frac{\alpha}{9k^2 2^k B_k} m^{k+2} M. \quad (\text{IV.6.5})$$

Proof. Put $w = p/M$. Since $q \leq \sqrt{p}$, division $p = qM + r_0$ gives $0 \leq r_0 < q \leq M$. The intervals

$$Q_i = \{x \in \{0, \dots, p-1\} : iw \leq x < (i+1)w\}, \quad 0 \leq i < M,$$

therefore form a partition with lengths q or $q + 1$. Also $w \geq q \geq h/k \geq m/(3k)$, where the last inequality follows from $h = (m-1)/2$ and $m \geq 3$.

For each (s, a) , the unique progression profile of its cube is

$$j_{s,a}(\epsilon) = \lfloor s/w - \epsilon \cdot a/w \rfloor \pmod{M}.$$

This formula is valid across modular wraparound because $p/w = M$. Normalize by subtracting $j_{s,a}(0)$. The resulting profile has intercept $\{s/w\}$ and slopes $-a_i/w \in [-k, k]$. Corollary IV.6.4 gives at most $L \leq B_k$ normalized profiles globally. Lemma IV.6.5 gives at most 2^k for each fixed a .

Apply Lemma IV.6.6 to the integrand in (IV.6.4). There is a normalized profile $j : \Omega_k \rightarrow \mathbb{Z}/M\mathbb{Z}$ whose class sum has squared energy at least $\alpha p^2 m^k / (2^k L)$. Write this class sum as a sum over its M translates. A further Cauchy–Schwarz inequality gives

$$\sum_i \sum_{a \in I^k} |T_i(a)|^2 \geq \frac{\alpha p^2 m^k}{2^k L M}, \quad (\text{IV.6.6})$$

where

$$T_i(a) = \sum_s e_p(-s\tau(a)) \prod_{\epsilon \in \Omega_k} \mathcal{C}^{|\epsilon|} (1_{Q_{j(\epsilon)+i}} f_\epsilon)(s - \epsilon \cdot a).$$

The identity of this expression with the class sum uses the disjointness of the Q_i : exactly one product of their indicators is nonzero for any given (s, a) .

Extend the sum in (IV.6.6) to all $a \in G^k$, which increases it. We record the phase decomposition explicitly. For suitable polynomials P_ω of degree at most $k + 1$,

$$u\tau(a) = \sum_{\omega \in \Omega_{k+1}} (-1)^{|\omega|} P_\omega(s - \omega \cdot (a, u)). \quad (\text{IV.6.7})$$

Indeed, expand τ into squarefree monomials. For a monomial involving the coordinates indexed by $J \subseteq \{1, \dots, k+1\}$, use

$$\prod_{j \in J} b_j = \frac{1}{|J|!} \sum_{\nu \in \{0,1\}^J} (-1)^{|\nu|} \left(s - \sum_{j \in J} \nu_j b_j \right)^{|J|}.$$

Here J includes the coordinate of u ; all required factorials are invertible because $p > k+1$. Adding these identities proves (IV.6.7).

Expand $|T_i(a)|^2$ and write its two base points as s and $s - u$. Using (IV.6.7) makes the resulting expression the mixed $(k+1)$ -cube form of

$$h_{i,\omega} = 1_{Q_{j(\epsilon)+i}} f_{\epsilon} e_p(-P_{\omega}), \quad \omega = (\epsilon, b), \quad b \in \{0, 1\}.$$

Gowers–Cauchy–Schwarz and the arithmetic–geometric mean inequality give

$$\sum_{a \in G^k} |T_i(a)|^2 \leq \prod_{\omega} \mathcal{U}_{k+1}(h_{i,\omega})^{1/2^{k+1}} \leq 2^{-(k+1)} \sum_{\omega} \mathcal{U}_{k+1}(h_{i,\omega}).$$

For a fixed ω , summing over i permutes all blocks, so that $\sum_i \mathcal{U}_{k+1}(h_{i,\omega})$ is independent of $j(\epsilon)$. Averaging over ω and using (IV.6.6) produces one $\omega = (\epsilon, b)$ with

$$\sum_i \mathcal{U}_{k+1}(1_{Q_i} f_{\epsilon} e_p(-P_{\omega})) \geq \frac{\alpha p^2 m^k}{2^k L M}.$$

Finally put $t = \epsilon \cdot r$, replace $P_{\omega}(z)$ by $P_{\omega}(z + t)$, and translate every block by $-t$. Translation invariance of the cube form gives the first inequality in (IV.6.5). The second follows from $L \leq B_k$ and $p/M = w \geq m/(3k)$. $\square$

Corollary IV.6.8 (Explicit strict nonuniformity coefficient). *Under Theorem IV.6.7, the coefficient*

$$c_k = \frac{1}{18k^2 2^k (2k)^k R_{k+1}(k2^{k-1})}$$

gives strict failure of the inequality $\sum_i \mathcal{U}_{k+1}(1_{R_i} f_{\epsilon} e_p(-P)) \leq c_k \alpha m^{k+2} M$. For $k \geq 2$,

$$\log_2(1/c_k) \leq k^2 + k \log_2 k + (2 + \log_2 e)k + 2 \log_2 k + \log_2(9e).$$

Proof. The lower bound in (IV.6.5) is twice the proposed threshold and is positive. The logarithmic estimate follows from $R_{k+1}(Q_k) \leq (eQ_k/(k+1))^{k+1} \leq e^{k+1} 2^{k^2-1}$. $\square$

This gives a local coefficient of size $2^{-k^2 - k \log_2 k - O(k)}$ at the stage where Proposition 17.7 uses $2^{-2(k+1)^3}$. The statement above specifies the field characteristic and the multi-affine extension used in the phase argument. It does not recompute the later inverse-theorem induction or the final bound for $r_k(N)$. Those are separate problems, and the local improvement alone does not supply a new global quantitative Szemerédi theorem.

[write] **6 October 2026.** *Compared with the formal statement* (checked at HEAD): `proposition_17_7` (Sections17_18.lean, line 140; exact companion `proposition_17_7_holds`, `Proofs17LocalizedPhaseRemoval.lean`, line 2900) has the coefficient $2^{-2(k+1)^3}$. Its proof already bounds the profiles by a joint region count (`hyperplaneRegionCount`), as said at the start of this section. Source 01's Proposition IV.3.3 is a weaker selection interface at the same step: its loss is B_k^2 with source 01's B_k , against $2^k B_k$ here. Not formalized; nothing is propagated to Theorem 18.1, which is open in the ledger.

Part V

Lean interface, ledger crosswalk and research questions

Contents. Sections V.1 and V.2 are the write’s: the crosswalk of every comparison with the formal project, checked against the current ledger, and the deduplicated list of the sources’ 46 research questions with credits (Vladimir’s standing rule of 4 October 2026). Then each source’s remaining sections, in source order: 01’s formalization interface, questions, conclusions and appendices (delivered Sections 14–16, Appendices A–B); 02’s (Sections 10–12, Appendix A); 03’s iteration bookkeeping, Lean route, questions and reproducibility (Sections 9–11, Appendix A); 04’s quantitative comparisons, questions, closing perspective and appendices (Sections 9–11, Appendices A–B); 05’s Lean relation, computation, questions and conclusion (Sections 7–10). The sources’ question sections are printed as delivered, with dated notes where another source answers a question.

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
pins, counts	01, 03: pin 3a25020a6; 04, 05: pin 327773149; 02: observed branch reference 66c02f8a1. The ledger counts quoted by the sources (88 exact, 32 open) are those of the pins; at HEAD they are 98 and 22 (Section V.1).
Qk , (Q1), (Q2)	04: its questions are numbered Q1–Q12 and two displays are tagged (Q1), (Q2); 01: numbered Questions (Section V.4); 03: numbered Research questions (Section V.14); 02, 05: subsections and paragraphs. Item names in Section V.2 are the write’s.

V.1 The formal project: ledger crosswalk [write]

Added in the write, 6 October 2026.

Every comparison that a source makes with a corrected statement of the formal project was checked against the ledger and the Lean declarations at HEAD 25df8755d (6 October 2026). File and line numbers below are at that commit. Statements are **Sections*.lean** propositions; “exact” means the ledger lists an exact companion theorem. The ledger then records 98 exact companions and 22 open statements; at all three pins it recorded 88 and 32. The last column gives the formal status of the *improvement*. All of the comparisons are accurate at HEAD, except the stale ones the table names.

Gowers	Lean statement; companion (status)	Source claim here	Improvement
Lemma 2.3	<code>lemma_2_3</code> (Sections01_03:141); <code>lemma_2_3_holds</code> (Proofs02Partition:561) (exact)	constant $1/4 \rightarrow 1/3$, optimal under the Lean hypotheses (05, Thm. I.16.4)	not formalized
Cor. 2.4	<code>corollary_2_4</code> (:158); <code>corollary_2_4_holds</code> (Proofs02PhasePartition:276); also <code>corollary_2_4_before_rou</code> <code>nding_repair_false</code> (exact)	$128\pi \rightarrow 36\pi$, $\alpha/2 \rightarrow 3\alpha/4$, same upper length, $r \leq N$ dropped (05, Cor. I.16.6)	not formalized

Gowers	Lean statement; companion (status)	Source claim here	Improvement
Cor. 2.5	<code>corollary_2_5 (:180);</code> <code>corollary_2_5_holds (Proofs02</code> <code>DensityIncrement:517);</code> proof-level pair $\sqrt{\alpha N/128\pi}$, $\alpha/4$ in private <code>cor25_large_scale</code> <code>(:375)</code> (exact)	density-sensitive increments (01, 03, 05); 03 and 05 compare with the proof-level pair, 01 with the statement (note after Table 3)	not formalized
Thm. 3.2, Cor. 3.3	<code>theorem_3_2_holds (Proofs03Pr</code> <code>ogressionAverage:381);</code> <code>corollary_3_3 (:276),</code> <code>corollary_3_3_holds</code> <code>(Proofs03ProgressionCount:349)</code> (exact; N prime, $k \leq N$)	$2^k \alpha \rightarrow (k-2)\alpha$ in normalized terms; admissible slopes (03, 04, Part III)	not formalized
Cor. 3.6	<code>corollary_3_6 (:310);</code> <code>corollary_3_6_holds (Proofs03</code> <code>ProgressionExistence:157)</code> (exact)	density exponent $k2^{k-1}$ against $k2^k$ (04, Cor. III.4.4)	not formalized
Lemma 3.8	<code>lemma_3_8_holds</code> <code>(Proofs03GowersCS:513);</code> <code>gowers_cauchy_schwarz (:485);</code> <code>gowersNorm_add_le</code> <code>(Proofs03Minkowski:131)</code> (exact)	used as tools (03, 04)	—
Lemma 3.10	<code>lemma_3_10 (:340);</code> <code>lemma_3_10_holds</code> <code>(Proofs03CubeUpper:82)</code> (exact)	excess $\alpha^{1/2^d} \rightarrow \alpha^{4/2^d}$ (03); leading $6\sqrt{2}\delta^4 u^4$ for $d=3$ (04)	not formalized
Cor. 5.6	<code>corollary_5_6 (Section05:205);</code> <code>corollary_5_6_holds</code> <code>(Proofs05FullPartition:85)</code> (exact at HEAD, open at every pin)	02 imports it as $\text{PP}(K, R_0)$ with the same constants; <i>02's "not Lean-verified" is</i> <i>stale</i> (note in Section I.11)	02's Thm. I.11.1 not formalized
Cor. 5.8	<code>corollary_5_8 (Section05:242)</code> (open); <code>corollary_5_8_with_sca</code> <code>le_holds</code> <code>(Proofs05FullPartition:94)</code> proves a scale-qualified repair	02 declines to restate it (Section I.10)	—
Lemma 5.15	<code>lemma_5_15 (Section05:352);</code> <code>lemma_5_15_holds</code> <code>(Proofs05_10:735)</code> (exact; $\alpha \geq 0$, empty cells allowed)	size $\alpha N/(4M) \rightarrow$ $\alpha(2+\alpha)N/(2(4-\alpha)(M-1))$ for $\alpha > 0$, nonempty cells (02, 03)	not formalized
Prop. 6.1	<code>proposition_6_1</code> <code>(Sections06_07:37);</code> <code>proposition_6_1_holds</code> <code>(Proofs05_10:505)</code> (exact)	recovered by 02's Thm. II.2.1 with $m(f) \leq 1$, $\mathcal{U} \leq m^4$	not formalized
Prop. 7.3	<code>proposition_7_3 (:135);</code> <code>proposition_7_3_holds</code> <code>(Proofs07BalogSzemerédi:996);</code> also <code>proposition_7_3_group_holds</code> (exact)	$(2^{-20}c^{12}, 2^{38}c^{-24}) \rightarrow$ $(c/8, 2^{19}c^{-9})$ (03)	not formalized

Gowers	Lean statement; companion (status)	Source claim here	Improvement
Lemma 7.4	<code>lemma_7_4 (:167); lemma_7_4_holds (Proofs07DRC:536)</code> , five witnesses (exact)	$2^{-1/2}\delta^5 n \rightarrow 7\delta^3 n / (3\sqrt{6})$ (03)	not formalized
Lemma 9.2	<code>lemma_9_2 (Sections08_09:69); lemma_9_2_holds (Proofs09Moments:527)</code> ; exact $N^2 B $ at line 495 (exact)	(M4) exact Parseval is proof-level; (M5) $\gamma^7 \beta^{15} p^{15}$ (04)	not formalized
Lemma 9.3, Cor. 9.4	<code>lemma_9_3 (:77)</code> , <code>corollary_9_4 (:91)</code> ; companions <code>Proofs09Restriction:376, :416</code> (exact; <i>no primality</i>)	prime-field restriction, $2^{-95}\gamma^{112}\eta^{15} B ^{15}$ (04); does not prove the Lean statements, as 04 says	not formalized
Lemma 12.4	<code>lemma_12_4 (Sections12_13:158); lemma_12_4_holds (Proofs12Amplification:21)</code> (exact)	$\beta^{112} \rightarrow \beta^{106}$ (04, (A12))	not formalized
Cor. 14.6, Lemmas 14.7, 14.8	<code>corollary_14_6 (Sections14_15:266)</code> , <code>lemma_14_7 (:279)</code> , <code>lemma_14_8 (:292)</code> ; companions <code>Proofs14Arrangements:838</code> , <code>Proofs14HigherArrangements:712</code> , <code>Proofs14ProductToArrangements:18</code> (exact)	profile σ ; $7 \cdot 4^{k+1} \rightarrow 7 \cdot 4^{k+1} - 6$ (04, Thm. II.10.1)	not formalized
Lemma 17.4, Cors. 17.5, 17.6	<code>lemma_17_4_holds (Proofs17Regions:691); corollary_17_5 (Sections17_18:103)</code> , <code>corollary_17_6 (:117)</code> ; companions <code>Proofs17FinitePatterns:396, :501</code> (exact)	$2^{2rk^3} \rightarrow (2r)^k C_k$; $M 2^{2r(k+1)^3} \rightarrow M \min(2r, M)^k R_{k+1}(k 2^{k-1})$ (01, 04, Part IV)	not formalized
Prop. 17.7	<code>proposition_17_7 (:140); proposition_17_7_holds (Proofs17LocalizedPhaseRemoval:2900)</code> , joint <code>hyperplaneRegionCount</code> (exact)	$2^{-2(k+1)^3} \rightarrow 2^{-k^2 - k \log_2 k - O(k)}$ (04); weaker interface (01)	not formalized
Cor. 16.11, Thms. 1.3, 18.1, 18.2	all open ; <code>theorem_18_1</code> 's docstring warns about the tower estimate of Cor. 16.11	01's dependency warning (Section V.3)	—
Section 18 (interval transfer)	<code>hasNatAP_of_short_modular_sequence (Proofs18IntervalTransfer:22)</code> , <code>hasNatAP_of_hasModAP_image (:92)</code> , added in 327773149	04's Lemma III.4.5; 03's Remark III.1.4	formalized (same statement, $n - 1 \mapsto n$)

Normalizations. The Lean development uses the unnormalized Fourier transform and the unnormalized Gowers norm; the intake found the sources' conversion tables accurate (source 02's Section V.8 and Appendix, source 03's Section V.13); the write did not re-derive them. `Definitions.lean` has changed since the pins only by an added `Box.IsProper`, which no source uses; every

other file that a source cites is unchanged.

Formal status of this report. As stated in Section F.5: the lifting lemma is formalized and was formalized before the sources arrived; nothing else here is. The reciprocal pointer for `Combinatorics/Ramsey/FORMALIZATION_STATUS.txt` is left to the session that maintains the Lean development.

V.2 Further questions and research [write]

Added in the write, 6 October 2026.

This section applies Vladimir’s standing rule of 4 October 2026: no unproved or wrong claim is dropped; open claims are stated as questions with their source, the argument they came with and what is missing; wrong claims stay on record with a proof of their failure. The five sources ask 46 questions: source 01 ten (Section V.4, Questions 1–10 there in order), source 02 seven (Section V.9, delivered Subsections 11.1–11.7), source 03 nine (Section V.14, Research questions 1–9 there in order), source 04 twelve (Section V.17, Q1–Q12) and source 05 eight (Section V.23, paragraphs 1–8). They are printed there as delivered. Below they are grouped by theme, with every asking source credited; the counts per theme are 14, 9, 8, 7 and 8. Items are numbered within their theme, so later additions extend a list without renumbering it.

Unproved or wrong claims. The intake and the write found no claim of sources 01–05 that is wrong, and no claim asserted as proved without a proof. Three inputs are used but not proved in the sources. The classical threshold-function counts (Zuev; Kahn–Komlós–Szemerédi; Baldi–Vershynin) enter source 01’s asymptotics and are published theorems. The partition input $PP(K, R_0)$ of source 02 is now a formal theorem for polynomial phases on intervals (Section V.1). The “independent mathematical reviews” of source 03 are not shipped, and nothing here rests on them. Two values found at intake are not certified, $C_5 = 8410$ and $A_4 = 770$; they are item IV.2.

Answered inside the report (dated notes at the questions). Source 04’s Question 8, the leading constant of $\log_2 F_k/k^2$ and $\log_2 A_k/k^2$, is answered by source 01: it is 1, using the classical threshold count, and source 01 certifies $F_4 = C_4 = 154$. Source 03’s Research question on c_d^{odd} is answered for $d = 3$ by source 04’s Theorem III.3.4: $c_3^{\text{odd}} = 2^{-1/2}$, attained by real parts of characters. With it, source 03’s Remark III.2.9 is settled at $d = 3$. Source 04’s Question 3 is the same question as source 03’s for $d \geq 4$, and is merged with it (item III.1).

Theme I: density transfer and phase-flat partitions (14 questions)

- I.1** *Higher-degree phase partitions* (01 Question 8; 05 paragraph 6). The density transfer is proved for any short-arc phase (source 01’s Theorem I.2.1; source 05’s Proposition I.17.2 and Corollary I.19.4). Missing: polynomial or nilsequence partitions with a quantitative minimum cell length and small occupied exceptional mass (01), and density-sensitive oscillation budgets tracked through geometric complexity with all losses parameterized (05). Simultaneous linear approximation does not give them, and Theorem I.18.5 does not transfer to nonlinear phases.
- I.2** *The best transfer under arithmetic constraints* (01 Question 5; 05 paragraph 4). The extremizers of Theorem I.2.1 rotate cell arcs freely, and the disk radius and the capacity inequality are separately sharp (05). Asked: how much stronger the tail becomes for one fixed cyclic character on a common-step partition with nearly equal cells (01), and the exact feasible region of block positive mass, correlation, density and phase width (05).

- I.3** *Finite partition profiles and the size–increment trade-off* (05 paragraph 1; 01 Question 6). The universal constant $1/3$ is settled (Theorem I.16.4). Asked: the exact best minimum length for each (r, s, M) and slope, and the feasible pairs of minimum and maximum lengths (05); the best uniform pair of length and increment, combining (I.3.4) with an optimal finite partition, including the final incomplete blocks (01).
- I.4** *Retaining more than one density statistic* (01 Question 7). Theorem I.4.2 solves the optimization once stop-loss statistics are chosen. Missing: structural lemmas that propagate those statistics through repeated localization at acceptable cost.
- I.5** *Heterogeneous phase budgets* (02 Subsection 11.5; 03 Research question 5). The analytic constraint is exact (source 02’s heterogeneous inequality in Section I.8; source 03’s Lemma I.14.3). Missing: a refinement theorem whose cost depends on the oscillation allowed in each cell, so that budgets can be allocated by local mass. Any better $\text{PP}(K, R_0)$ transfers automatically to Theorem I.11.1.
- I.6** *The finite-cell extremal problem* (02 Subsection 11.6). Theorem I.9.1 solves the weighted total-mass problem. Open: the largest good cell when M and the allowed cell sizes are fixed, with integer cardinalities, which would sharpen the $M - 1$ corollary.
- I.7** *Constants in the simultaneous minimax theorem* (05 paragraph 2). The exponent $1/(m + 1)$ and the dependence on ε are sharp; the upper constant c_m is poor. Asked: the optimal leading constants, first for one or two phases (source 01’s Theorem I.7.2 improves the one-phase constant), the effect of a common difference, and the order of limits.
- I.8** *Anisotropic lower bounds* (05 paragraph 3). Is there a matching minimax upper bound in all anisotropic regimes, with the regime decomposition that loose tolerances require?
- I.9** *Adaptive selection of frequencies and tolerances* (05 paragraph 5). Proposition I.19.3 solves the continuous allocation. Missing: subset selection and the exact ceiling problem, relative to certified Fourier inputs.
- I.10** *Interval-localized spectral extraction* (05 paragraph 8). Corollary I.19.4 corrects for the Gram matrix by its top eigenvalue only. Missing: certified conditioning bounds for specific frequency configurations, using more of the Gram matrix.

The ten items carry the fourteen questions: 01 Questions 5–8, 02 Subsections 11.5–11.6, 03 Research question 5, and 05 paragraphs 1–6 and 8.

Theme II: the inverse step (9 questions)

- II.1** *A correlation-preserving structural theorem for multivalued spectra* (02 Subsection 11.1). Proposition II.4.1 keeps all large derivative frequencies. Missing: an energy-to-structure step that keeps a fixed fraction of t_τ and exploits κ , controlling both losses.
- II.2** *Near equality for the coupled pair (f, w)* (02 Subsection 11.2). Proposition II.4.2 gives two alignments, and the equality examples show distinct geometries. Asked: a dimension-free description of f and w jointly when Theorem II.2.1 is nearly an equality; results on near-maximal Gowers norms (Eisner–Tao) are relevant but do not classify the weight.

- II.3** *Carrying profiles through the proof* (02 Subsection 11.3; 04 Q7). Keep the distribution of $m(\Delta_{\mathbf{h}}f)$ and $\mathcal{U}(\Delta_{\mathbf{h}}f)$ across slices instead of their supremum (02), and the cross-section profile $\sigma = \mathbb{E}_r \beta_r^3$ through later restrictions (04). The difficulty in both: compatibility of structure between slices, and which operations preserve the profile.
- II.4** *The sharp fixed-threshold DRC profile* (03 Research question 3). The order of the best guaranteed $|K|/n$ under Lemma 7.4's exact criterion as $\delta \rightarrow 0$, and whether the exponent 3 of Theorem II.5.2 can be improved using constraints of actual intersection matrices; the scalar argument only rules out a two-witness first-moment certificate.
- II.5** *Optimizing the elementary extraction* (03 Research question 4). Optimize $\gamma, \varepsilon, \beta$, walk lengths and popularity thresholds in Proposition II.6.2, judging the pair of conclusions together and keeping the comparison with Reiher–Schoen explicit.
- II.6** *Lowering the size threshold of the restriction theorem* (04 Q4), from $|B| \geq \Lambda_d(\alpha\eta)^{-2d}$, by reflecting the additive structure of B in the exceptional relations or by a better kernel, with repeated entries handled.
- II.7** *The optimal retention law* (04 Q5). Proposition II.9.6 limits one expectation budget only. Asked: upper-bound examples or a better selection method, and simultaneous guarantees on $|B'|$, its energies and the respected density.
- II.8** *Restriction beyond prime cyclic groups* (04 Q6): versions for $\mathbb{Z}/N\mathbb{Z}$, $\mathbb{F}_q^n$ and other codomains, with hypotheses that quantify kernels of the coefficient maps.

The eight items carry the nine questions: 02 Subsections 11.1–11.3, 03 Research questions 3–4, and 04 Q4–Q7.

Theme III: cube and progression counts (8 questions)

- III.1** *The odd-order constant and the higher-dimensional norm gap* (03 Research question 1; 04 Q3). $c_d^{\text{odd}} = \sup \|f\|_{U^2}^4 / \|f\|_{U^d}^4$ over real centered f on odd-order groups (03), equivalently the best comparison of $\|f\|_{U^d}^{2^d}$ with $\|f\|_{U^2}^{2^d}$ for given group orders (04). *Settled for $d = 3$* (dated note after Theorem III.3.4): $c_3^{\text{odd}} = 2^{-1/2}$. Open for $d \geq 4$: monotonicity gives $\|f\|_{U^2}^4 \leq 2^{-1/2} \|f\|_{U^d}^4$ for $d \geq 3$ (04, after Theorem III.3.6), but its sharpness for $d \geq 4$ is not established. The best odd-order leading coefficient is $R_d c_d^{\text{odd}}$.
- III.2** *The fifth and sixth centered coefficients* (03 Research question 2). Classify the supports of sizes 5 and 6 for which (III.2.27) has solutions with no trivial character, with multiplicities and the quotient correlations they need; Lemma III.3.3 does this for $d = 3$ on odd-order groups, and (III.2.29) shows that order three enters.
- III.3** *The secondary extremal coefficient* (04 Q1). Between $(9/4)^{1/3}$ and 6 (display (Q1)); does the limit exist, and is the two-frequency value optimal?
- III.4** *Stability for cube extremizers* (04 Q2), beyond the U^2 stability of Corollary III.3.5, and whether near-optimal secondary behaviour concentrates on $\{\pm\xi, \pm2\xi\}$.
- III.5** *Quantitative rounding and derandomization* (03 Research question 8): the group size needed by the sharpness constructions, and an efficient deterministic rounding.

- III.6** *Counting that rectifies automatically* (03 Research question 6): a localized version of Theorem III.1.2 that counts ordinary integer progressions with all boundary errors. The lifting step itself is formal (Lemma III.4.5); its interval hypothesis is what a modular norm does not give.
- III.7** *Progression counting with more of the mixed information* (04 Q10): mixed norms, further cancellation, pairwise information, and non-admissible slopes with explicit subgroup averages.

The seven items carry the eight questions: 03 Research questions 1, 2, 6 and 8, and 04 Q1–Q3 and Q10.

Theme IV: floor patterns (7 questions)

- IV.1** *The second-order carry entropy* (01 Question 1; the remainder of 04 Q8). Is there a $-k \log_2 k$ term in $\log_2 C_k$, or is $\log_2 C_k = k^2 + O(k)$? Theorem IV.2.3 leaves exactly this scale; counting the hyperplane intersections that meet the open cube is the suggested route. The leading constant 1 is settled (dated note at 04’s Q8).
- IV.2** *Exact counts, arrangement structure, certified C_5 and A_4* (01 Question 2; 04 Q8, its computational part). A recurrence, an intersection-poset formula or a symmetry-reduced enumeration for C_k and A_k (01); certified counts for $k = 4$ (04): $C_4 = F_4 = 154$ is certified by source 01, A_4 is not. *Intake values, not certified*: $C_5 = 8410$ and $A_4 = 770$, from value-branching floating-point linear programs with every feasibility witness re-evaluated exactly. The infeasibility branches rest on solver status, not on rational certificates. Missing: rational certificates, for instance from source 01’s or source 04’s certificate generator, whose formats both handle these dimensions in principle. The values satisfy the proved bounds (note in Section IV.6).
- IV.3** *The modulus dependence and the realizable profiles* (01 Question 3; 04 Q9). When modular collisions stabilize, and the exact number of translation classes as a function of M (01); which global and fixed-slope profiles actually occur for slopes from a short discrete interval, whether an average multiplicity can replace the worst case 2^k , and small moduli (04).
- IV.4** *Reducing the selection loss* (01 Question 4). Which orthogonality or positivity hypotheses allow a phase-removal interface to lose L^{-1} instead of L^{-2} ; for arbitrary complex summands the L^{-2} loss is sharp. Source 04’s Lemma IV.6.6 already replaces B_k^2 by $2^k B_k$ through local multiplicity.
- IV.5** *Polynomial floor entropy beyond the leading term* (01 Question 9): secondary terms of $\log_2 |\mathcal{P}_{n,d}(r)|$ and the transition when $\log r \asymp n$.

The five items carry the seven questions: 01 Questions 1–4 and 9, and 04 Q8–Q9.

Theme V: integration and formalization (8 questions)

- V.1** *A complete, audited propagation through one argument* (01 Question 10; 02 Subsection 11.4; 03 Research question 7; 04 Q11). All four sources ask for a dependency graph of the quantitative parameters, into which the local improvements are substituted. Every later threshold would then be recomputed under the same hypotheses, ending in an explicit final bound, compared with the original argument and with Leng–Sah–Sawhney. Source 02 proposes the length-four proof as a first target; source 03’s Proposition V.12.1 supplies the final bookkeeping. Source 01

adds that the issue flagged at Corollary 16.11 must be resolved first. That issue is still open at HEAD (`corollary_16_11`, `theorem_18_1`).

V.2 Formal libraries, and formalizing sharpness (02 Subsection 11.7; 03 Research question 9; 04 Q12; 05 paragraph 7). A characteristic-independent library: weighted Fourier energy on $G \times \widehat{G}$ and real phase freezing (02); integer linear forms, centered multilinear averages and character constraints with torsion data (03); the low-risk interfaces first, namely moment interpolation, the telescope and the support enumeration, then the selection theorem (04); and sharpness as well as existence, starting with the parity obstruction for $1/3$ and the residual inequality (I.19.4) (05). The write adds one concrete gap. Source 02’s Theorem I.11.1 could become formal now that `corollary_5_6_holds` and `polynomial0n_affine_pullback` exist; what is missing is the transfer of cells back to the input progressions and the selection step. The lifting lemma (Lemma III.4.5) is already formal.

The two items carry the eight questions: 01 Question 10, 02 Subsections 11.4 and 11.7, 03 Research questions 7 and 9, 04 Q11–Q12, and 05 paragraph 7.

Further limitations stated by the sources (kept in place; listed here for the standing rule). Sharpness of the density transfer holds in the weighted model, not for arithmetic characters (01, Remark I.2.2). The constants 36π and $3/4$ are not claimed optimal (05). Sharpness of the simultaneous exponent is for linear phases, uniform arcs and symmetric widths (05). The coefficient 6 of the $u^{16/3}$ term is not claimed optimal, and the higher-dimensional constant $P_d/\sqrt{2}$ is not claimed sharp (04). The restriction theorem is prime-only and does not prove the Lean Lemma 9.3 or Corollary 9.4 (04). $\text{PP}(K, R_0)$ is an imported input (02). The elementary Balog–Szemerédi–Gowers bound is far weaker than Reiher–Schoen (03). R_d is not claimed to be the optimal odd-order constant (03; it is not, for $d = 3$). Finite and numerical checks prove nothing general (all five). No source has a new $r_k(N)$ bound, a priority claim, or Lean verification.

V.3 Relationship to the ProveIt formalization

Source 01, delivered Section 14.

V.3.1 The inspected snapshot and normalization

The source inspection is pinned to commit

3a25020a654d196f7f1aa2e37649752c000d0e65

of VladimirReshetnikov/ProveIt, rather than to a moving branch [15]. The original paper was read through the public PDF mirror listed in the bibliography; selected repository files, not the entire repository build, were inspected.

The inspected `Definitions.lean` uses the unnormalized Fourier transform, a balanced indicator, finite partitions, natural-number progressions, and average cell size [16]. Consequently a hypothesis $|\widehat{\mathbf{1}}_A(r)| \geq \alpha N$ corresponds to $\eta = \alpha$ in this article, not to $\eta = \alpha/N$. A sum of absolute balanced cell sums becomes our D after division by N .

The companion `Proofs02DensityIncrement.lean` develops the positive/negative mass split for the Fourier density increment [17]. The companion `Proofs17FinitePatterns.lean` uses a finite

sign-pattern/Sauer–Shelah approach for the inherited floor-pattern bounds [18]. These are natural locations for stronger reusable interfaces; no assertion that they have been replaced or recompiled is made.

[write] **6 October 2026.** At *HEAD* (25df8755d, 6 October 2026) the inspected files are unchanged except `Definitions.lean`, which gained a `Box.IsProper` that no source uses. The ledger now records 98 exact companions and 22 open statements, against 88 and 32 at this source’s pin. Of the statements compared here, `corollary_2_5`, `corollary_17_5`, `corollary_17_6` and `proposition_17_7` have exact companions, and `theorem_18_1` is open. Crosswalk: Section V.1.

V.3.2 Proposed formal interfaces

The dependencies can be organized in the following order.

Interface	Mathematical content	Reusable prerequisite
Real cell mass	$\sum g_j = 2 \sum (g_j)_+$ under $\sum g_j = 0$; weighted mean and occupied mass	Finite sums and order arithmetic
Cell phase envelope	Lemma I.1.2, first for an indicator, then for bounded weights	Complex norm, sine, finite convex combinations
Broken-chord transfer	Theorems I.2.1 and I.2.3 with explicit denominators	Convexity and mean identities
Integer partition	Theorem I.5.1, including the case of one block in a residue class	Finite pigeonhole and quo- tient/remainder
Fractional patterns	Positive perturbation and exact range factorization	Integer floor, finite minima of positive gaps
Pattern count	Hyperplane regions or an equivalent fi- nite sign-family bound	Dimension and finite set car- dinality
Translation quotient	Normalize a pattern by its empty-set value	Finite group action or an ex- plicit canonical representative
Certificate checker	Rational primal/dual checks and exhaus- tive tree validation	Exact arithmetic and struc- tural recursion

A practical first target is the $s = 0$ discrepancy theorem: its proof needs no trigonometry, no Fourier transform, and no structural additive combinatorics. The next target is the cancellation inequality, followed by the full envelope. Proving the statements over arbitrary finite weighted families prevents repeated normalization and counting arguments later.

For the carry side, one should formalize interior realization before using a strict sign-pattern theorem. The signed-feature counterexample in Section IV.3 is a small regression test against a false boundary shortcut. The exact coefficient-range factorization also needs the open-endpoint surjectivity argument, not merely an upper-bound map.

For progression lengths, retain natural-number inequalities $N \geq L \prod_i Q_i$ and $L \leq \ell \leq 2L - 1$ as long as possible. Introduce square roots and floors only in the corollary. This isolates small-scale cases rather than trying to recover them from real asymptotic estimates. If actual code is added,

it should prove a new strengthened statement and derive the old statement as a corollary where appropriate; changing a catalogue definition alone is not a proof.

V.3.3 A dependency warning, not a claimed global repair

The inspected `Sections17_18.lean` records the numbered statements as proposition-valued definitions. Its note attached to Theorem 18.1 flags an unsupported tower estimate in the published Corollary 16.11 as a quantitative dependency [19]. This is a report of the repository’s warning, not an independent conclusion here that a published headline theorem is false. Our local theorems do not use that estimate. In particular, applying Theorem I.4.1 to a supplied partition is unconditional, but asserting the existence of that partition with a given tower exponent would require a separate validated source.

No Lean or Lake executable was available in the working environment, and no new Lean theorem was compiled in producing this package. The document contains a formalization plan, conventional mathematical proofs, and separately described computational checks. It does not present a proposition-valued specification as a kernel-verified theorem.

[write] **6 October 2026.** Checked at HEAD: `theorem_18_1` is still open in the ledger, and its docstring in `Sections17_18.lean` still warns about the unsupported tower estimate in the published Corollary 16.11 (line 168). `FORMALIZATION_STATUS.txt` still lists resolving “the documented Corollary 16.11 bound” among its open obligations, and `corollary_16_11` is open. The dependency warning above is therefore accurate.

V.4 Further research questions and concrete next steps

Source 01, delivered Section 15.

The questions below are posed by this report. They are unresolved within this development; this is not a certification that every formulation is new or currently open in the entire literature.

Question V.4.1 (The second-order carry entropy). *Does $\log_2 C_k$ have a $-k \log_2 k$ correction, or is $\log_2 C_k = k^2 + O(k)$? More generally, determine the coefficient of $k \log k$ if it exists.*

Theorem IV.2.3 leaves exactly this scale between its upper and lower estimates. A promising route is to count intersections of the internal subset-sum hyperplanes that actually meet the open cube, rather than use the unrestricted region bound. The elementary anchor construction is too sparse to decide the leading coefficient, let alone this correction. Threshold-function embeddings give a strong baseline but need not capture all simultaneous carry levels.

Question V.4.2 (Exact counts and arrangement structure). *Can one obtain a recurrence, a useful intersection-poset formula, or a symmetry-reduced exact enumeration for C_k and A_k ?*

The supplied certificate format scales by adding more branches, but its size grows rapidly. Coordinate permutations and complement transformations suggest reductions; any quotient must retain enough information to certify that all orbit representatives are covered. An exact-arithmetic checker in a proof assistant would make such computations useful as formal regression tests, not merely numerical evidence.

Question V.4.3 (The modulus dependence of affine patterns). *For fixed k and coefficient range, when do modular collisions stabilize, and what is the exact number of translation classes as a function of M ?*

Theorem IV.3.2 is uniform in M but deliberately discards these collisions. A finite list of integer-valued normalized patterns exists; modular coincidence is then controlled by divisibility of finitely many coordinate differences. Determining those difference vectors could give an exact piecewise description, including a threshold beyond which no new collisions occur. The offset's fractional part must remain visible.

Question V.4.4 (Can the selection loss be reduced?). *What additional orthogonality or positivity assumptions allow a phase-removal interface to lose only L^{-1} rather than L^{-2} in energy, without changing the desired structural conclusion?*

For arbitrary complex summands the squared pigeonhole loss in Proposition IV.3.3 is sharp: equal aligned summands attain it. A stronger theorem therefore needs additional hypotheses, not just more careful algebra. One possible approach is to retain the sum of squared contributions across translation classes until after a structural selection step. Such a mechanism has not been established here.

Question V.4.5 (Sharpness under arithmetic phase constraints). *How much stronger can the density-tail bound become when ψ is one fixed cyclic character and the cells are proper progressions with a common step and nearly equal lengths?*

The extremizers of Theorem I.2.1 freely rotate cell arcs and assign endpoint phases to occupied points. A genuine character imposes spacing and compatibility restrictions. A quantitative analysis of those restrictions could improve constants or density dependence beyond the abstract sharp bound; it would not contradict that bound's optimality in the larger weighted model.

Question V.4.6 (The optimal finite size-increment tradeoff). *Can one combine the exact density tradeoff (I.3.4) with an optimal finite partition algorithm, including the final incomplete blocks, to characterize the best uniform pair of progression length and increment?*

Our integer partition is transparent but not claimed to have optimal constants. Theorem I.7.2 fixes the square-root exponent for one-frequency phase-flat methods. Remaining improvements may come from choosing unequal block lengths, exploiting rational frequencies, or paying for selected short cells through their occupied mass rather than their ambient size.

Question V.4.7 (Retaining more than one density statistic). *Which finite collection of stop-loss moments most efficiently preserves useful density-tail information through repeated localization steps?*

Theorem I.4.2 supplies an exact optimization method once the statistics are chosen. The research issue is to prove structural lemmas that propagate those statistics at acceptable cost. A two- or three-knot state could be substantially more informative than a single discrepancy number, but a global gain must include the cost of obtaining that state.

Question V.4.8 (Higher-degree phase partitions). *Can polynomial or nilsequence localization furnish partitions with both quantitative minimum cell sizes and small occupied exceptional mass, so that the sharp tail theorem applies without a large selection loss?*

The density transfer itself is already proved for any short-arc phase; what is missing is the required structural partition. Simultaneous linear approximation alone does not solve the higher-degree problem. The polynomial floor-pattern bound in Theorem IV.4.1 controls one finite complexity term but does not construct those partitions.

Question V.4.9 (Polynomial floor entropy beyond the leading term). *For fixed degree d , determine the secondary terms of $\log_2 |\mathcal{P}_{n,d}(r)|$, and identify the transition when $\log r$ is comparable to n .*

The finite upper bound separates a coefficient-range term from a region count, but unlike Theorem IV.1.2 it need not be an exact product. Understanding overlaps between different integer coefficient vectors is essential. The threshold-function lower bound alone does not settle the large-range contribution.

Question V.4.10 (A fully audited quantitative integration). *After independently checking the earlier structural parameter estimates, what explicit global improvement survives if the new density-transfer and translation-class bounds are propagated through one complete argument?*

The first deliverable should be a dependency graph with normalized quantities and verified recurrences, including the issue flagged at Corollary 16.11 in the inspected repository. Only then is it meaningful to compare a resulting global bound with modern benchmarks. This prevents a smaller local factor from being mistaken for a new global theorem.

V.5 Conclusions and result-status ledger

Source 01, delivered Section 16.

The work yields a reusable sharp transfer principle and a more economical floor-pattern count. The density calculation is sensitive to the actual balanced mass $2\delta(1 - \delta)$, to the full distribution of cell densities, and to occupied exceptional mass. The counting calculation separates integral coefficient freedom from fractional carries, treats boundary points correctly, and exhibits a threshold-complexity obstruction to a subquadratic exponent. Both refinements have explicit hypotheses and can be inserted at well-defined proof interfaces.

Claim	Status in this package
Sharp finite weighted density-tail theorem, including equality examples	Proved in Theorem I.2.1; arithmetic extremizers are not claimed.
Occupied-mass exceptional-set bound and optimal forced maximum	Proved in Theorems I.2.3 and I.3.1.
Explicit Fourier density increment with integer-exact length	Proved in Theorem I.6.1, for arbitrary real frequency and bounded weights; no prime-modulus restriction.
Exact range factorization and quadratic-exponent carry upper/lower bounds	Proved by elementary arguments in Theorems IV.1.2, IV.1.4, and IV.2.1.
$\log_2 C_k \sim k^2$ and polynomial-floor leading entropy	Deduced from explicitly identified classical threshold-function inputs; those inputs are not reproved here.
$C_4 = 154$	Exhaustive exact rational certificates with a Python verifier and a completeness argument; not Lean-verified.

Claim	Status in this package
Smaller translation-class budget	Proved in Theorem IV.3.2; subsequent global parameter propagation is not supplied.
Improved record bound for Szemerédi’s theorem	Not claimed.
Priority of every formulation	Not certified by an exhaustive literature search.

V.6 Boundary and normalization checklist

Source 01, delivered Appendix A.

The formulas in this article are intended to be used with the following conditions kept explicit.

1. The phase results assume $0 < \delta < 1$. At $\delta = 0$ or 1 , a bounded function with that mean is constant and the balanced correlation is zero; formulas dividing by $\delta(1 - \delta)$ should not be used.
2. The short-arc theorem uses half-angle $\theta < \pi/2$. The progression partition uses full arc length r in turns, so $\theta = \pi r$, not $2\pi r$.
3. A normalized Fourier coefficient is the unnormalized coefficient divided by N . The normalized discrepancy is likewise the sum of cell balanced masses divided by N .
4. The density-tail event is $q > a$, strictly. Its stop-loss lower bound may vanish at the optimal forced maximum even though a cell with $q \geq a$ is guaranteed.
5. Exceptional occupied mass is $\mathbb{E}(h\mathbf{1}_E)$ in the original probability measure. It is not a conditional density on E .
6. The coefficient-range factorization assumes integral $r_i \geq 1$ and open intervals $(-r_i, r_i)$. Fractional coordinates can lie on a boundary only because Lemma IV.1.1 removes that boundary without changing the pattern.
7. Natural progressions have positive step and no repeated terms. Singletons handle the small-scale case; no nonintegral number is treated as a possible cell length.
8. The affine translation quotient retains the fractional offset. Its modulus-free upper bound does not assert absence of modular collisions.
9. The threshold-function asymptotic is an external theorem. The elementary finite pattern bounds and all density-transfer statements remain valid without it.

V.7 Source and artifact provenance

Source 01, delivered Appendix B.

The repository snapshot was read on 6 October 2026. The paper directory contains both a PDF and a TeX representation. The research arguments use the original paper’s relevant PDF pages and selected Lean source files, not an assumption that every statement in the repository has already

been proved. The pinned commit identifies precisely which source comments and normalizations were inspected. No repository files were modified.

The package includes the article source, compiled PDF, certificate data, verification and supplementary-test scripts, result logs, and a README. The article’s source uses an internal bibliography and does not require BibTeX or downloaded repository files to compile. No copies of the copyrighted source papers or font files are redistributed.

V.8 Formalization interface and reproducibility

Source 02, delivered Section 10.

V.8.1 What was inspected

The repository’s `Definitions.lean` defines `fourier` using the unnormalized `ZMod.dft`, defines `difference` with $x - h$, and defines `balanced`, `IsPartition`, `UniformOfDegree`, and the unnormalized `gowersNorm`. The inspected `ProofInfrastructure.lean` provides, among other helpers, `IsPartition.sum_card` and `sum_balanced`. The inspected part of `Proofs02DensityIncrement.lean` contains private helpers for partition sums and the exact equality between total absolute mass and twice positive mass [20].

The normalized U^3 quantity in this article translates to the repository’s degree-two uniformity numerator by

$$\mathcal{U}(f) = N^{-4} \sum_{a,b \in G} \left| \sum_{x \in G} \Delta_a \Delta_b f(x) \right|^2.$$

Thus the degree parameter of `UniformOfDegree` is 2 here, not 3. For the repository’s unnormalized order-three Gowers norm, the eighth power is $N^4 \mathcal{U}(f)$. These normalizations should be established as explicit bridge lemmas before the energy theorem is imported.

V.8.2 Suggested dependency order

The most economical formalization starts with finite weighted real sums: positive/negative mass, the negative-support bound, and Theorem I.9.1. It then proves the real-phase lemma using positive and negative weighted averages. These results have no Fourier dependencies and can be added independently of a full higher-order inverse theorem.

For Theorem II.2.1, first formalize the three identities (II.2.2)–(II.2.4) over `ZMod N`. Keep all statements polynomial and multiplication-based, rather than dividing by $m(f)$ or $\mathcal{U}(f)$; this handles zero cases without extra hypotheses. A later generalization can replace `ZMod N` by an abstract finite abelian group and its character group.

Next prove the two-tetrahedron bijections in Lemma II.3.1, then add graph indicators and balanced indicators as specializations. The final arithmetic-progression wrapper should accept a proved $\text{PP}(K, R_0)$ hypothesis rather than hiding it behind an unverified claim about exact cell counts.

Care is needed with empty cells: the repository’s general partition predicate does not itself require every cell to be nonempty. Empty cells must be removed before using cell means, $M - 1$, or

a minimum cell length. Likewise, injectivity of a modular progression parameterization is not the same as nonwrapping in its standard integer representatives.

V.8.3 Verification status

No Lean compiler or Lake executable was available in the working environment. Accordingly, none of the new results is represented as kernel-checked, and no untested Lean proof is included. The source reading was targeted rather than a certification of every theorem in the supplied development. The mathematical proofs in this article are the primary deliverable.

The accompanying `verify.py` independently checks the weighted Fourier identities and both asymmetric forms of the energy inequality on finite cyclic groups, compares spectral and direct convolution energy for small groups, tests phase and tail inequalities, and evaluates exact sharpness examples. It uses the fixed seed 20261006. The run recorded in `verification_results.json` includes 490 random energy cases, 2,500 random phase cases, 3,000 zero-mean moment cases, 1,500 nonzero-mean moment cases, and 1,500 phase-budget optimization cases, together with three exact-form equality families evaluated numerically and a rational moment equality example. These experiments are regression checks, not proofs or substitutes for formal verification.

[write] **6 October 2026.** The suite is shipped as `code/02-energy-freezing-verify.py`; it overwrites `verification_results.json` beside itself, so the README's rerun route runs it on a copy. At intake it reproduced the recorded run identically (8,990 random cases and the exact families). The corrected Corollary 5.6 that Section I.11 imports is now formalized (note there).

V.9 Further research questions and proposed work

Source 02, delivered Section 11.

The following are proposed research targets suggested by the proven interfaces. Their inclusion is not a claim that each is a previously published open problem.

V.9.1 A correlation-preserving structural theorem for multivalued spectra

Proposition II.4.1 retains all significant derivative frequencies. Can the subsequent energy-to-structure and graph-selection steps preserve a fixed fraction of t_τ while exploiting the denominator κ ? The decisive issue is not merely obtaining a small-doubling subset: it is ensuring that the subset continues to carry the correlation needed for integration. A useful first goal is a theorem whose hypotheses include both additive energy and a specified nonnegative correlation weight, and whose conclusion controls both losses explicitly.

V.9.2 Classify near equality for the coupled pair (f, w)

Proposition II.4.2 gives two quantitative alignments, while Examples II.3.3–II.3.4 show distinct sharp geometries. Existing results about near-maximal Gowers norms [7] are relevant but do not by themselves state a classification of the additional spectral weight in our mixed inequality. One can ask for a dimension-free theorem describing w and f jointly when the main inequality is nearly saturated, with appropriate treatment of subgroups and quadratic phases.

V.9.3 Retain norm profiles through higher-order integration

The graph-energy step is repeatedly applied to derivatives in higher-degree arguments. A natural target is a weighted family theorem that keeps the distribution of $m(\Delta_{\mathbf{h}}f)$ and $\mathcal{U}(\Delta_{\mathbf{h}}f)$ instead of replacing every slice by a supremum. The immediate difficulty is compatibility: good structural pieces from different slices must agree on overlapping configurations. A local improvement cannot be counted as a global gain until this compatibility cost is controlled.

V.9.4 A complete quantitative ledger for one progression length

An attainable next project is to propagate the present denominators, the sharp real-phase law, and the optimized budget through every step of the length-four proof, with one explicit choice of parameters. This would determine whether a real improvement survives the remaining inverse and recurrence estimates. Only after that full ledger is complete would it be justified to state an improved numerical threshold for that proof. Comparisons should distinguish improved constants within Gowers's route from stronger bounds obtained by later methods [4, 2].

V.9.5 Sharper recurrence inputs and heterogeneous budgets

Theorem I.11.1 treats PP as a replaceable component. Any better value of K or R_0 transfers automatically to the progression certificate. A separate optimization problem assigns different diameter budgets to cells of different sizes and correlations, using the heterogeneous inequality in Section I.8. The local quadratic bound contains still more information than its linear envelope; exploiting it may improve nonuniform cases without altering the recurrence theorem.

V.9.6 The finite-cell extremal problem

Theorem I.9.1 solves the weighted total-good-mass problem sharply. It does not solve the problem of the largest good cell when M and all allowed cell sizes are fixed. The three-type extremizer explains why a negative cell and a low-positive cell can consume separate parts of the cell budget. A precise finite-dimensional optimization, including integer cardinalities, could strengthen the $M - 1$ corollary and produce exact finite- N certificates.

V.9.7 A characteristic-independent Lean library

The main analytic results work on every finite abelian group and the two-tetrahedron argument uses no field division. A useful library contribution would formalize weighted Fourier energy, its normalization on $G \times \widehat{G}$, and real-valued phase freezing independently of the prime-cyclic presentation. Regression examples should include even cyclic groups, subgroup spectral rectangles, empty weights, zero functions, and wrapping versus nonwrapping progression charts.

V.10 Conclusion

Source 02, delivered Section 12.

The strongest unconditional analytic conclusion of this article is the mixed weighted inequality (II.2.1), with its exact norm denominators and coefficient-one equality examples. The strongest

elementary density conclusion is the sharp tail bound (I.9.1), coupled with the sharp aggregate real-phase law (I.8.3). Together they strengthen identifiable interfaces in Gowers’s argument, including Proposition 6.1 and Lemma 5.15, and yield an explicitly optimized progression certificate under a clearly stated partition input.

The remaining challenge is structural, not merely algebraic: local gains must survive extraction, integration, recurrence, and iteration. The article makes those gains available without presenting that uncompleted propagation as a new global theorem.

V.11 Source audit and normalization checklist

Source 02, delivered Appendix A.

The repository branch reference observed during this session was

66c02f8a1d26f553deac5a91dada1739e0d15064.

The specific inspected blobs are recorded below, so the source comparison is not dependent only on a moving branch name.

File below the supplied repository paths	Git blob SHA
<code>sz-thm-gowers-proof.tex</code>	41085a0efde8c1932e86e80c791984f52841c22e
<code>Definitions.lean</code>	202260e0d7ec6f033c1758347381c34a6f0d9212
<code>ProofInfrastructure.lean</code>	ebd9e7ed02d81de2e17b97b4223f861b785ed5d8
<code>Proofs02DensityIncrement.lean</code>	85b5c3ad8769b9769e1939ae5c05f0757f0678b5

[write] **6 October 2026.** Checked at HEAD (25df8755d): the four blob identifiers are those of commit 66c02f8a1. The transcription, `ProofInfrastructure.lean` and `Proofs02DensityIncrement.lean` are unchanged at HEAD; `Definitions.lean` has changed (blob 97113b7), by an added `Box.IsProper` that source 02 does not use. Source 02’s delivery README records no pin; this branch reference is its provenance (Section F.4).

The targeted paper passages were the corrected Corollary 5.6 and nearby phase-partition statements, Lemma 5.15, and Proposition 6.1. The transcription includes editorial changes; a theorem imported from that edited source is labeled as such above rather than silently attributed to the exact original printed formulation.

Quantity	This article	Paper/Lean convention
Fourier coefficient	$\mathbb{E}_x f(x) \overline{\chi(x)}$	$N \widehat{f}(\chi)$
Derivative-graph statistic	T	$N^3 T$
Order-three cube quantity	$\mathcal{U}(f)$	$N^4 \mathcal{U}(f)$
Graph additive energy	$\mathcal{E}(\mathbf{1}_\Gamma)$	$N^3 \mathcal{E}(\mathbf{1}_\Gamma)$
Balanced L^2 mass	$m(f) = \delta(1 - \delta)$	$Nm(f)$ before averaging

V.12 How local improvements affect an iteration

Source 03, delivered Section 9.

The estimates above remove real losses, but they occur at different places in the proof. The cube theorem improves a counting consequence of uniformity. The DRC theorem improves extraction of a subfamily from intersection data. The Fourier theorem converts an already established correlation into a density increment on an ordinary progression. Their parameters are not interchangeable. In particular, a larger common-neighborhood core does not by itself provide an improved correlation theorem for arbitrary high Gowers norms.

Figure 1 displays two of the local comparisons using the literal formulas proved in the text. For cube counts, retaining the centering changes the first possible error power, while classifying its supports further reduces its coefficient. For DRC, retaining the original success criterion and changing the exponent from five to three gives a growing advantage as the density parameter decreases.

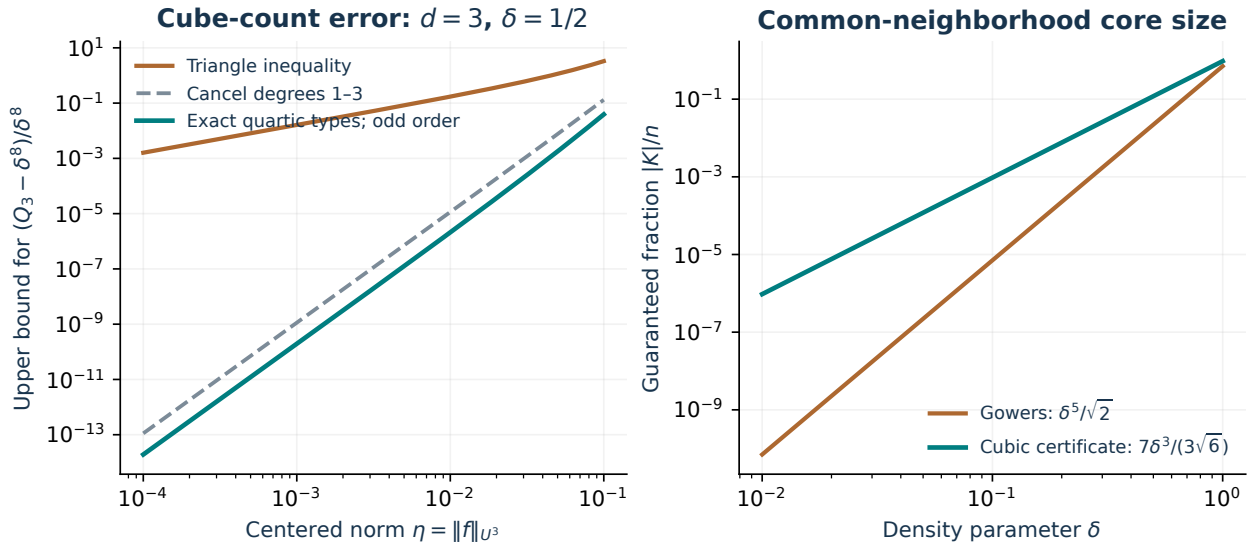


Figure 1: Evaluations of proved local bounds. Left: relative cube-count error bounds for $d = 3$, $\delta = 1/2$, comparing the triangle inequality, the cancellation of degrees one through three, and the exact quartic types on an odd-order group. Right: guaranteed core fractions at the same threshold $\delta^2 m/2$ and the same 90% good-pair requirement. These are upper or lower bounds as labelled, not measurements of extremal examples or claims that each curve is attainable.

V.12.1 An explicit abstract recurrence

The following elementary calculation explains which parameters must be tracked before any global conclusion can be stated. It is a conditional recurrence theorem, not a substitute for the analytic density increment theorem needed to supply its hypotheses.

Proposition V.12.1 (Density and length bookkeeping). *Let $c > 0$, $p > 1$, $C > 0$, $q \geq 0$, and $0 < \beta < 1$. Suppose positive numbers $\delta_i \leq 1$, $N_i > 0$ satisfy, for $0 \leq i < R$,*

$$\delta_{i+1} \geq \delta_i + c\delta_i^p, \quad N_{i+1} \geq C\delta_i^q N_i^\beta.$$

Then

$$R \leq \frac{(1+c)^p}{c(p-1)} (\delta_0^{1-p} - 1). \quad (\text{V.12.1})$$

Writing $B = \max\{0, -\log(C\delta_0^q)\}$, one also has

$$\log N_r \geq \beta^r \log N_0 - B \frac{1 - \beta^r}{1 - \beta} \quad (0 \leq r \leq R). \quad (\text{V.12.2})$$

Consequently, for an integer $T \geq R$ and $H \geq 1$, the condition

$$\log N_0 \geq \beta^{-T} \left(\log H + \frac{B}{1 - \beta} \right) \quad (\text{V.12.3})$$

ensures $N_r \geq H$ at every $r \leq R$.

Proof. The potential u^{1-p} decreases at a controlled rate. Since $\delta_i \leq 1$,

$$\begin{aligned} \delta_i^{1-p} - \delta_{i+1}^{1-p} &\geq (p-1) \int_{\delta_i}^{\delta_i + c\delta_i^p} t^{-p} dt \\ &\geq \frac{(p-1)c\delta_i^p}{(\delta_i + c\delta_i^p)^p} \geq \frac{(p-1)c}{(1+c)^p}. \end{aligned}$$

Summing and using $\delta_R^{1-p} \geq 1$ gives (V.12.1). The densities are increasing, so $\log N_{i+1} \geq \beta \log N_i - B$. Iterating this affine inequality gives (V.12.2). Under (V.12.3), $\beta^r \log N_0$ is at least $\log H + B/(1 - \beta)$ for $r \leq T$, which proves the last assertion. $\square$

For fixed constants, the familiar double-exponential length budget is visible in (V.12.3): the number of iterations occurs in the exponent of β^{-T} , which already bounds a logarithm of the starting size. Improving C or q changes the additive length cost B . Improving the density growth exponent p can change the scale of the number of iterations. These are different effects. To extract a better global Szemerédi bound from this report one would have to identify, and prove throughout the later sections, the corresponding updated values or functions of all these parameters. We have not carried out that full replacement of the higher-order inverse and localization arguments.

The same distinction applies to the exact quartic identity. It is stronger information about cubes with a prescribed balanced Gowers norm. It does not imply that every set lacking arithmetic progressions has a large ordinary Fourier coefficient. Gowers's motivating examples address precisely why higher-order structure is needed. The role of the improved count is to make the uniformity branch and its constants more precise.

V.13 The Lean development and a concrete formalization route

Source 03, delivered Section 10.

V.13.1 Pinned source and status of the inspection

All repository comparisons refer to revision

3a25020a654d196f7f1aa2e37649752c000d0e65

of ProveIt, inspected on 6 October 2026. The requested paper is at `Combinatorics/Ramsey/Papers/sz-thm-gowers-proof`; the Lean files are at `Combinatorics/Ramsey/Lean/GowersSzemerédi`. The adjacent source inventory, `gowers-proof-status.json`, reports 88 numbered statements with exact companion declarations and 32 open statements at that revision [22].

[write] **6 October 2026.** *Stale since the pin.* At HEAD (25df8755d, 6 October 2026) the inventory records 98 exact companions and 22 open statements. The ten statements closed since 3a25020a6 are `corollary_5_6`, `corollary_5_7`, `lemma_5_9`, `lemma_5_10`, `corollary_5_11`, `lemma_5_14`, `lemma_16_1`, `lemma_16_3`, `lemma_16_5` and `lemma_16_6` (compared at the write). `theorem_1_3`, `theorem_18_1` and `theorem_18_2` are still open, so the conclusion of the next paragraph stands. Crosswalk: Section V.1.

Those counts are a source inventory, not a certification of the whole development. The facade exposes numbered results as propositions, usually with companion theorems whose names end in `_holds`. Defining a proposition neither proves it nor adds it as an axiom. The inventory explicitly separates exact companions from conditional, restricted, or otherwise related results. In particular, the quantitative headline `theorem_1_3` and final statements `theorem_18_1` and `theorem_18_2` are marked open there. This inspection therefore does not support describing the repository as a completed formal proof of the full quantitative theorem.

The files directly relevant to our comparisons contain substantive companion proofs. We inspected their definitions and proof scripts; we did not compile the Lean project. The new mathematical statements in this report have written proofs and independent mathematical reviews, together with the finite checks described in the appendix. No new machine-checked Lean proof is claimed or supplied.

[write] **6 October 2026.** The “independent mathematical reviews” mentioned here and in Section V.15 are not part of the delivery and are not recorded anywhere in the repository; this report relies on the printed proofs, the intake’s checks and the write’s notes, not on them.

V.13.2 Normalization map

The following conversions prevent a common source of false improvements. Let $N = |G|$, and take $d \geq 1$.

Quantity	This report	Inspected repository convention
Fourier transform	$\mathbb{E}_x f(x) \overline{\chi(x)}$	Unnormalized sum, equal to N times this transform
Gowers norm	$\ f\ _{U^d}$	<code>gowersNorm</code> d $f = N^{(d+1)/2^d} \ f\ _{U^d}$
Uniformity parameter	$\ f_A\ _{U^d}^{2^d} \leq \alpha$	<code>UniformSetOfDegree</code> A α $(d-1)$
Cube count	$Q_d(\mathbf{1}_A)$	<code>cubeCount</code> $= N^{d+1} Q_d(\mathbf{1}_A)$
Cube arguments	$x + \sum_i v_i h_i$	A minus convention in <code>cubeArgument</code> ; negating all h_i gives the same average

Both cube conventions count all parameter tuples, including degenerate cubes. This is why the rounding lemma can bound the exceptional tuples by a collision probability instead of changing the definition of the norm. Changing to unlabelled, injective, or symmetry-quotiented cubes would require an additional counting theorem.

V.13.3 Useful existing declarations

The source provides the following starting points in namespace `LeanProofs.GowersSzemerédi`. These are concrete interfaces to reuse or generalize, rather than assertions that the proposed additions have already been formalized.

Partition discrepancy and Fourier increment.

`Proofs02DensityIncrement.lean` proves `corollary_2_5_holds`. Its private `cor25_large_scale` already obtains the stronger $\rho/4$ increment and $\sqrt{\rho N/(128\pi)}$ length before weakening its conclusion. General finite weighted sums, followed by Theorem I.14.1, would expose the asymmetric information without repeating the Fourier argument.

Multilinear norms and cube expansion.

`Proofs03GowersCS.lean` contains the public homogeneous theorem `gowers_cauchy_schwarz`, which already allows functions beyond the unit disc. `Proofs03Minkowski.lean` contains the triangle inequality and private distributive expansion helpers. `Proofs03CubeUpper.lean` proves `lemma_3_10_holds` from that triangle inequality. Exporting a general expansion indexed by vertex subsets is a natural first step toward the exact quartic theorem.

Selection from many cells.

`Proofs05_10.lean` contains `lemma_5_15_holds`. Its assumptions and conclusion match the symmetric discrepancy extraction discussed in Section I.14. The general theorem should be stated with real nonnegative cell weights and asymmetric cell-sum bounds, then specialized to cardinalities.

Dependent random choice.

`Proofs07DRC.lean` proves `lemma_7_4_holds` using ordered samples with replacement, indexed by `Fin5`. Its internal score bound already has constant $11/16$ multiplying $\delta^{10}n^2$. The main new ingredient for the cubic theorem is a pointwise polynomial minorant, followed by the exact three-sample counting identity. Merely exposing the internal five-sample constant would give a smaller change.

V.13.4 Recommended order of implementation

First formalize the finite weighted selection theorem over an arbitrary finite index type. Prove positivity of its denominators once, and keep the pointwise inequalities separate from the division steps. This produces immediately reusable specializations to Lemma 5.15 and balanced indicators.

Second formalize the cubic certificate. Its good-region proof is a polynomial identity followed by nonnegativity; its bad-region proof may use concavity or a direct monotonicity argument. Keep the sampling identity generic in the number of witnesses. Specialization to three witnesses then replaces only the numerical part of the current DRC proof.

Third export the general cube expansion and prove the single-coordinate cancellation lemma. Nonempty supports of at most three selected vertices, or four vertices with nonzero XOR, leave one variable occurring in exactly one centered factor after a coordinate reflection. The corresponding reindexing is a bijection and uses no division by two. The first quartic upper bound then follows directly from Gowers–Cauchy–Schwarz, before any Fourier classification is needed.

Fourth formalize the two nonzero four-vertex types. One may begin over $\mathbb{Z}/N\mathbb{Z}$ to match existing files, then generalize to finite abelian groups. The extra quotient energy requires explicit handling

of order-two characters; assuming an invertible doubling map is appropriate only for the odd-order specialization. The combinatorial formulas for R_d, T_d are separate finite enumeration theorems.

Finally integrate the progression-count telescope, the proper progression partition, and the four-edge compression statement. Each has a clear finite interface. The generalized von Neumann theorem should state invertibility of slope differences explicitly; the progression partition should carry its lower and upper integer lengths in its output. This order establishes useful lemmas early while avoiding dependence on the unfinished quantitative headline.

V.14 Further research questions and topics

Source 03, delivered Section 11.

The following questions are proposed continuations of the proved results. Some are deliberately scoped finite problems, while others concern the larger proof architecture. A question proposed here is not being presented as a previously published open problem unless a source is identified.

Research question V.14.1 (The odd-order leading constant). For fixed $d \geq 3$, determine

$$c_d^{\text{odd}} = \sup_{\substack{G \text{ finite abelian, } |G| \text{ odd} \\ 0 \neq f: G \rightarrow \mathbb{R}, \mathbb{E}f=0}} \frac{\|f\|_{U^2}^4}{\|f\|_{U^d}^4}.$$

The exact quartic theorem reduces the best leading coefficient on odd-order groups to $R_d c_d^{\text{odd}}$. Is the supremum attained or approached by a real part of a character, by several Fourier modes, or by a different construction?

The universal bound $c_d^{\text{odd}} \leq 1$ follows from monotonicity. Our cosine example gives a positive explicit lower bound. The order-two character that attains the unrestricted coefficient is unavailable here, so the sharp universal result does not settle this restricted problem. A practical first project is a systematic search over small odd cyclic groups, followed by an inequality explaining any apparent extremizer. Numerical optimization would be evidence, not a proof of the supremum.

[write] 6 October 2026. *Settled for $d = 3$ (cross-source finding).* Source 04's Theorem III.3.4 gives $c_3^{\text{odd}} = 2^{-1/2}$, attained exactly by the real parts $2\text{Re}(c\xi)$ of nontrivial characters (the question's first alternative); proof in the note after that theorem. The best odd-order leading coefficient for $d = 3$ is $R_3 c_3^{\text{odd}} = 6\sqrt{2}$, as source 04 proves directly. For $d \geq 4$ the question is open; it coincides with source 04's Question 3 (Section V.2, item III.1).

Research question V.14.2 (Classify the fifth and sixth centered coefficients). For each $r = 5, 6$, classify the supports $S \subseteq \{0, 1\}^d$ of size r for which the character system (III.2.27) can have solutions with no trivial character. Derive multiplicities as functions of d and identify the necessary quotient correlations and Fourier phase invariants.

The coordinate-reflection test removes every support with a coordinate occupied by exactly one selected vertex or by all but one. For surviving supports, Smith invariant factors identify possible small-torsion obstructions. A successful classification should distinguish actual vanishing of a centered correlation from mere rational independence of its defining forms. The fifth-order examples discussed at the end of Section III.2 give concrete starting cases.

Research question V.14.3 (The sharp fixed-threshold DRC profile). Let $F(\delta)$ be the largest guaranteed fraction $|K|/n$ under the exact hypotheses and success criterion of Lemma 7.4. Determine

the correct order of $F(\delta)$ as $\delta \downarrow 0$. Can the exponent 3 in Theorem II.5.2 be improved using constraints specific to intersection matrices?

The scalar optimization proves only that a particular two-witness first-moment certificate cannot give a positive uniform estimate at the original parameters. Actual overlap matrices are symmetric positive semidefinite Gram matrices of indicators and obey further incidence constraints. It is a real additional problem to determine whether those constraints rule out the scalar bad distributions. Conversely, a lower bound on the necessary exponent requires realizable families of sets, not just a distribution of formal overlap values.

Research question V.14.4 (Optimize the entire elementary extraction). Optimize the three parameters $\gamma, \varepsilon, \beta$ in Proposition II.6.2, and compare alternative walk lengths and popularity thresholds. Which size–difference-set tradeoffs can be achieved while retaining only elementary finite inequalities?

The meaningful objective is the pair of conclusions, including the normalization of the difference-set bound by $|A|$ or by $|A'|$. Optimizing one displayed exponent while allowing the selected set to shrink excessively can conceal a loss. Comparison with [12] should remain explicit; the purpose of this project would be a simple formalizable proof or a better constrained tradeoff, not to mistake an elementary refinement for an improvement of the strongest known theorem.

Research question V.14.5 (Allocate phase oscillation by local mass). Suppose a phase partition can be refined at a cost depending on the desired oscillation in each cell. Can one minimize the total refinement cost subject to

$$\sum_j \int_{P_j} |f| |\psi_j - z_j| d\mu \leq \lambda \rho$$

by using local information on $|f|$, rather than a single global oscillation tolerance?

Lemma I.14.3 already provides the exact analytic constraint. The unresolved work is geometric: prove a refinement theorem with a cost function sharp enough for this optimization to matter. One can start with linear phases and known cell masses, then study polynomial phases. The global density factor in Theorem I.15.2 is the simplest instance of retaining this information.

Research question V.14.6 (Local counting that rectifies automatically). Develop a version of Theorem III.1.2 for interval-supported functions or localized norms in which the conclusion directly counts ordinary integer progressions and includes all boundary errors.

The elementary second-difference test gives a clean sufficient condition for rectification, but the global modular norm does not enforce that condition. The target is an explicit statement whose hypotheses reflect both uniformity and support. Any proposed bound should be tested on sets concentrated near the ends of the interval and on small composite moduli before it is fed into the density increment iteration.

Research question V.14.7 (Propagate improvements through a verified dependency graph). Replace the improved local lemmas in the later sections of the proof, record every dependence of density, progression length, dimension, and correlation on earlier parameters, and obtain a complete explicit iteration theorem.

Proposition V.12.1 specifies the final bookkeeping for one common type of increment. The hard work is to prove the hypotheses with the newly propagated functions. An initial useful deliverable would be a symbolic parameter dependency graph accompanied by Lean theorems for each monotone substitution. This would reveal which improvements survive and which are absorbed by later losses, even if the final asymptotic class stays unchanged.

Research question V.14.8 (Quantitative rounding and efficient derandomization). Optimize the group-size requirement for the sharpness constructions at a given error tolerance, including the exact-density version of Corollary III.2.12. Can one derandomize the rounding with an efficient finite algorithm?

The existence problem at a compatible prescribed rational density is already solved by the slow perturbation and small alteration in Corollary III.2.12. Its size requirement is deliberately generous. Sampling directly with a fixed total, a sharper norm estimate for the alteration, or a deterministic conditional-expectation argument might improve it. Independence no longer applies unchanged after conditioning on cardinality, so that approach needs its own estimate.

Research question V.14.9 (A reusable formal library of centered configuration counts). Build a finite-abelian-group library whose basic objects are systems of integer linear forms, centered multilinear averages, character constraint sets, and explicit reindexing bijections. Recover the cube and progression results as specializations.

The key interface should retain integer torsion data. A proof over $\mathbb{Q}$ alone does not control evaluation on all finite groups. At the same time, the library should make cancellation by a single free variable easy, so simple cases do not require a full Smith normal form calculation. This is an achievable software and formalization project with reusable mathematical outputs, independent of completing the entire Gowers proof.

V.15 Reproducibility and verification

Source 03, delivered Appendix A.

The accompanying archive contains the complete LaTeX source and compiled PDF, the figure in vector PDF form, scripts for the figure and finite checks, and the recorded check output. No original third-party paper is repackaged. The repository revision and source paths are recorded so that the comparisons can be repeated against the same input.

V.15.1 What the scripts check

`verify_cubes.py` uses only the Python standard library and exact rational arithmetic. It enumerates every four-vertex support of the Boolean cube for dimensions two through five. In dimension five this means all $\binom{32}{4} = 35960$ supports. It compares a classification from minors with a separate classification from XOR and coordinate patterns. The counts are exactly those in Lemma III.2.5. It also computes the coefficients of the full cube polynomial directly, without using the proposed quartic formula, on cyclic groups of orders two through six in dimensions two and three. The low-degree cancellations, torsion correction, and parity polynomial all pass.

`verify_estimates.py` checks the sharp selection extremizers with exact rational weights, evaluates the cubic certificate including its boundary and tangent cases, constructs proper phase

partitions at several finite scales, and enumerates all subsets of $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/7\mathbb{Z}$ for the progression bounds at lengths three and four. The norm powers in the latter check are rational; taking roots and comparing the bounds uses floating-point arithmetic with an explicit tolerance. The phase oscillation checks are also numerical. These are clearly distinguished from the exact cube and scalar checks.

Independent written reviews checked the asymmetric selection extremizers, finite progression geometry, Cauchy–Schwarz induction, quartic support classification and Bernoulli rounding, cubic minorant, and four-label counting injection. These reviews and finite computations are useful error checks; the proofs in the article, rather than the tested finite range, justify the theorems for arbitrary parameters.

V.15.2 Building the document

From the extracted directory, run

```
python3 verify_cubes.py
python3 verify_estimates.py
pdflatex -interaction=nonstopmode -halt-on-error gowers_local_refinements.tex
pdflatex -interaction=nonstopmode -halt-on-error gowers_local_refinements.tex
```

The figure is already included, so compiling the document does not require Python or a plotting library. To regenerate the figure, run `make_figure.py` in an environment with NumPy and Matplotlib. The LaTeX source uses standard packages including AMS mathematics, Palatino fonts, `microtype`, `booktabs`, and `hyperref`.

V.16 Quantitative comparisons and what they enable

Source 04, delivered Section 9.

The preceding results address different sources of loss. Their benefits should therefore be read at the interface where each estimate is used. The following table records those interfaces without combining unrelated parameters into a single claimed global improvement.

The field-size requirement in the restriction theorem has a particularly simple interpretation:

$$|B| \geq \Lambda_d(\alpha\eta)^{-2d}.$$

The ambient density does not otherwise enter this sufficient condition. Two elementary decisions are responsible. A rank-two exceptional relation leaves at most $|B|^{2d-2}$ tuples, because its free coordinates can be chosen from B . These contributions are then summed with their nonnegative Fourier weights, whose total is at most one. Thus the number of possible coefficient vectors does not enter the error bound. Repeated entries are handled separately. These refinements are useful even before any attempt to optimize the kernel.

For an illustrative comparison, take $\gamma = \eta = \beta = 1/2$. The retained fraction of $|B|^{15}$ in the source display is $2^{-8388608}$, whereas the convenient bound here is 2^{-222} . The corresponding sufficient field-size condition here is $p \geq 2^{231}$. These are deliberately transparent evaluations of the proved formulas, not practical finite-size estimates. They show the change in parameter dependence and also the cost of the conservative exceptional-tuple estimate.

Interface	Coarse source estimate	Strengthened estimate here
Graph moment, input $\gamma B ^3$	$\gamma^7\beta^{21}p^{15}$	$\gamma^7\beta^{15}p^{15}$; arbitrary moment order and equality cases
Almost-homomorphism restriction, order eight	$(\gamma^7\beta^6\eta/4)^{2^{19}} B ^{15}$	$2^{-95}\gamma^{112}\eta^{15} B ^{15}$, under the explicit size hypothesis in Corollary II.9.5
Three-cube upper bound, $u = \ f\ _{U^3}$	$(\delta + u)^8$	Excess begins with $6\sqrt{2}\delta^4u^4$, with explicit higher terms
k -term progression counting	Error bounded by $2^k u$	Error bounded by $u \sum_{j=0}^{k-3} \delta^j$
Section 17 local coefficient	$2^{-2(k+1)^3}$	$2^{-k^2 - k \log_2 k - O(k)}$, with an explicit constant

Table 11: Comparisons with the specified local estimates in [1, 23]. The variables and hypotheses are those of the corresponding sections. The improved restriction theorem includes a large-field condition, and the phase theorem includes a characteristic condition.

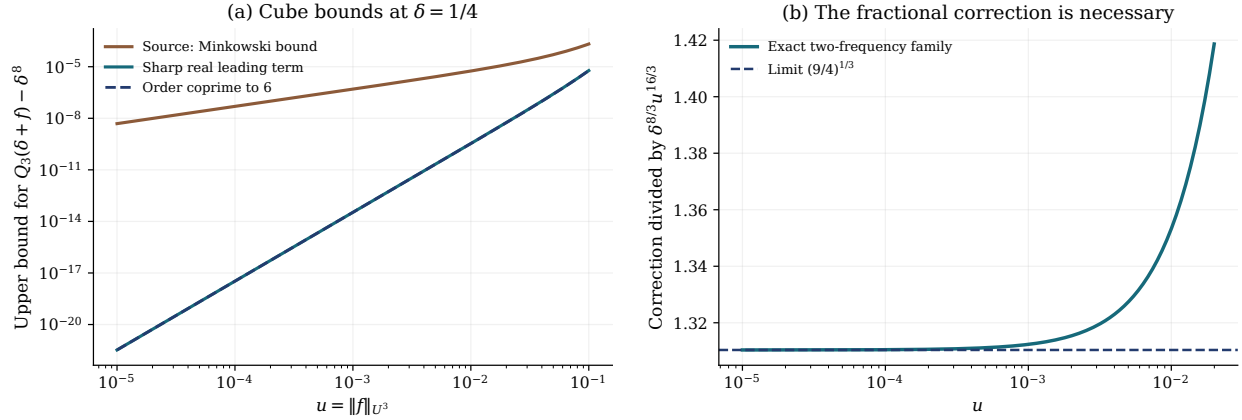


Figure 2: Proved bounds and an exact extremizing model at $\delta = 1/4$. Left: the excess above δ^8 in the source triangle-inequality bound and the two refined bounds of Theorems III.3.6 and III.3.8. The refined curves nearly coincide at small u because they have the same sharp leading term. Right: after subtracting that leading term, the explicit two-frequency family of Proposition III.3.9, divided by $\delta^{8/3}u^{16/3}$, tends to $(9/4)^{1/3}$. The figure evaluates formulas; it is not a statistical fit.

The cube results answer two precise local extremal questions. The best universal leading coefficient in the real centered, odd-order setting is $6\sqrt{2}$. In the setting without 2- or 3-torsion, the first possible correction after that leading term has optimal exponent $16/3$. What remains undetermined is the exact coefficient of this correction and the existence of a universal asymptotic expansion for the extremal envelope. These are local conclusions with explicit models, independent of any claim about the best quantitative bound for arithmetic-progression-free sets.

V.17 Further research questions and a proposed program

Source 04, delivered Section 10.

The questions below are proposed research targets arising from the calculations. The list distinguishes exact optimization problems from larger integration projects. It is not a claim that each formulation is already recognized as an open problem in the literature; a more extensive priority review should accompany work toward publication.

V.17.1 Extremal cube counts

Q1. Does the secondary extremal coefficient exist, and what is it? Fix $0 < \delta < 1$, and let $M_\delta(u)$ be the supremum of $Q_3(W)$ over prime cyclic groups of order greater than 16 and weights $0 \leq W \leq 1$ satisfying $\mathbb{E}W = \delta$ and $\|W - \delta\|_{U^3} = u$. Theorems III.3.8 and Proposition III.3.9 imply

$$(9/4)^{1/3} \leq \liminf_{u \downarrow 0} \frac{M_\delta(u) - \delta^8 - 6\sqrt{2}\delta^4 u^4}{\delta^{8/3} u^{16/3}} \leq \limsup_{u \downarrow 0} \frac{M_\delta(u) - \delta^8 - 6\sqrt{2}\delta^4 u^4}{\delta^{8/3} u^{16/3}} \leq 6. \quad (\text{Q1})$$

Determine whether the limit exists. Is the two-frequency value $(9/4)^{1/3}$ optimal? A useful first step is a sharper estimate of the quintic correlation in terms of the exact loss in the norm-gap inequality, instead of the separate bounds used above.

Q2. What is the correct stability statement for cube extremizers? Corollary III.3.5 controls the fourth-power Fourier mass outside one conjugate pair, equivalently a U^2 remainder. This does not alone control an L^2 remainder: many small Fourier coefficients can have little fourth-power mass but substantial second-power mass. Determine the strongest stability conclusion for bounded weights that nearly attain the leading cube bound, and identify any additional hypothesis needed for a stronger metric. For near optimal secondary behavior, test whether concentration on frequencies $\{\pm\xi, \pm 2\xi\}$ follows.

Q3. What replaces the sharp norm gap in higher dimensions? For $d \geq 4$, determine the best constant in the comparison of $\|f\|_{U^d}^{2^d}$ and $\|f\|_{U^2}^{2^d}$ for real centered functions, specifying the allowed group orders. Combine it with the exact coefficient P_d of Proposition III.3.2. The Boolean support matroid gives a finite organizing device for the next correlations, but their values and sharp norm bounds require additional harmonic analysis. Separate large-characteristic behavior from the extra identities produced by small torsion.

[write] **6 October 2026.** For $d = 3$ the comparison is Theorem III.3.4; for $d \geq 4$ this is source 03's Research question V.14.1 (which asks for the constant over odd-order groups). Merged in Section V.2, item III.1.

V.17.2 Restriction and moment estimates

- Q4. Can the size threshold be substantially lowered?** Theorem II.9.1 requires $|B| \geq \Lambda_d(\alpha\eta)^{-2d}$. The proof already controls exceptional relations in expectation using their Fourier weights. Can their contribution be made to reflect the additive structure of B , rather than the uniform rank-two bound? Can a kernel with a more favorable coefficient distribution reduce the required size without weakening the retained tuple count? Any proposed improvement must handle repeated entries as well as extra relations among distinct entries.
- Q5. What is the true optimal retention law?** The bound $\kappa_d \alpha^{2d} \eta^{2d-1} |B|^{2d-1}$ has the natural parameter powers for the single-kernel expectation budget in Proposition II.9.6. That proposition does not establish optimality among all selection procedures. Construct upper-bound examples, or find a different selection method with better powers. Also ask for useful simultaneous guarantees on $|B'|$, its ordinary additive energies, and the density of respected relations.
- Q6. How far does restriction extend beyond prime cyclic groups?** The interpolation theorem already applies to arbitrary finite abelian groups. The restriction theorem uses a field to turn two independent linear equations into a uniform bound on the number of solutions. Develop versions for $\mathbb{Z}/N\mathbb{Z}$, finite-field vector spaces, and maps into other groups. The appropriate hypotheses should quantify kernels of the relevant coefficient maps, rather than silently treat nonzero determinants as invertible. A weighted formulation would also allow nonuniform input measures.
- Q7. Can the cross-section profile be propagated through the proof?** For arrangements the gain is governed by $\sigma = \mathbb{E}_r \beta_r^3$, not just by the global density. Track this profile through subsequent restrictions and changes of variables. Determine which operations preserve useful control of σ , and whether a collection of higher cross-section moments gives a stronger interface than a single density parameter. Equality models in Section II.10 provide tests against overly optimistic exponents.

V.17.3 Patterns, counting, and integration

- Q8. Determine the asymptotics of the fractional pattern counts.** The proved estimates imply

$$\frac{1}{2} \leq \liminf_{k \rightarrow \infty} \frac{\log_2 F_k}{k^2} \leq \limsup_{k \rightarrow \infty} \frac{\log_2 F_k}{k^2} \leq 1, \quad (\text{Q2})$$

and the same bounds for A_k . Determine the leading constant, or improve the gap in these bounds. Relate the particular affine arrangements here to established results on threshold patterns and subset-sum arrangements before assigning novelty. Exact certified counts for $k = 4$ are a concrete computational next step; exploiting symmetries is likely to matter more than merely using a faster numerical solver.

[write] 6 October 2026. *Answered by source 01* (cross-source finding of the intake, checked at the write). The F_k here is source 01's C_k and the A_k here is source 01's A_k : the definitions are identical. Source 01's Theorem IV.2.3 gives $k^2 - k \log_2 k - O(k) \leq \log_2 C_k \leq k^2 + (\log_2 e - 1)k$. Its Theorem IV.3.1 gives $C_k \leq A_k$ and $\log_2 A_k \leq k^2 + O(k)$. Hence both limits in (Q2) exist and equal 1. The lower half uses one external input, the classical threshold-function count (Zuev; Kahn–Komlós–Szemerédi, as recorded by Baldi–Vershynin [8]), which source 01 cites and does not reprove. Source 01's Proposition IV.5.1 certifies $F_4 = C_4 = 154$, the “concrete computational next step” asked for. A_4 is

not certified (an uncertified recount gives 770), and the secondary term is source 01's first question; both remain open (Section V.2, items IV.2 and IV.1).

- Q9. Which global and local profiles are actually realizable?** The phase-extraction proof uses the uniform bound 2^k for the profiles at a fixed slope and an arrangement bound for all slopes. The slopes there come from a short discrete interval. Determine the joint distribution of profiles in that arithmetic setting, including small moduli M , and test whether an average multiplicity can replace the worst-case local multiplicity. The exact modular formula of Theorem IV.6.3 isolates one source of collisions.
- Q10. Can progression counting use more of the mixed information?** The dynamic program in Section III.4 finds the best bound within a specified telescoping family. It is not an optimality theorem for all counting arguments. Investigate whether mixed uniformity norms, additional cancellations, or information about particular pairs of factors can lower that bound. Any extension to nonadmissible slopes should display the relevant subgroup averages explicitly; the $\mathbb{Z}/2\mathbb{Z}$ example rules out simply dropping the invertibility assumptions.
- Q11. What survives complete propagation through the density increment?** Create a dependency graph for the quantitative parameters of the original proof. Substitute the improved moment, restriction, and profile estimates, and recompute every subsequent threshold with the same hypotheses. The output should be an explicit final function of k and density, with each loss traceable to a proved lemma. Compare the resulting bound with both the original argument and later work [2, 3]; an improvement in the historical proof need not improve the stronger modern theorem.
- Q12. Can the refinements be verified by a proof assistant?** Formalize the lowest-risk interfaces first: exact moment interpolation, the density-weighted telescope, and the finite support enumeration. Then formalize the selection theorem, where the main obligations are finite Fourier orthogonality, independent sampling, and the exceptional rank count. A fully checked dependency chain would clarify which quantitative gains are compatible with the later induction, and would separate a correct theorem statement from a completed formal proof. Appendix V.19 gives a proposed implementation order.

Questions 1, 4, and 8 offer relatively contained starting points: a sharp analytic constant, a finite selection threshold, and a concrete arrangement census. Questions 7 and 11 are integration projects, for which local success must be accompanied by careful propagation of hypotheses. The distinction helps keep a proposed research program testable.

V.18 Closing perspective

Source 04, delivered Section 11.

The strongest conclusions of this investigation are explicit local theorems with complete proofs. Exact moment accounting feeds a polynomial restriction theorem; centered Fourier structure yields sharp cube coefficients and a fractional correction exponent; and simultaneous floor-pattern counting replaces a cubic exponential local loss by a quadratic one. The sharpness examples identify which aspects have already reached their natural limit and which constants remain to be determined.

The next substantial step is to turn these interfaces into a fully checked quantitative dependency chain. That would establish how much of the local improvement survives later stages of Gowers's

argument, while the extremal and pattern questions above offer independent research directions with precise targets.

V.19 A map to the Lean development

Source 04, delivered Appendix A.

This appendix identifies interfaces in the inspected snapshot [23]. It records a plan for additional work; it is not a report that new Lean declarations have been proved. A declaration defining a proposition, a theorem with a proof term, and a theorem successfully checked by the Lean kernel are different pieces of evidence.

Inspected interface	Proposed strengthening and principal proof obligations
<code>Proofs09Moments.lean;</code> <code>lemma_9_2_holds</code>	Keep the exact second moment before applying the higher-moment inequality. State the arbitrary-order graph version, with $ B $ in the denominator. The proof already computes an exact cardinality before weakening it.
<code>Proofs03CubeUpper.lean;</code> <code>lemma_3_10_holds</code>	Add centered cancellation and the real odd-order norm gap as separate lemmas. Prove the twelve quartic and eight quintic support classifications with explicit finite certificates or a transparent finite case split.
<code>Proofs03ProgressionAverage.lean;</code> <code>theorem_3_2_holds</code>	Generalize the required changes of variables to admissible slopes. Make multiplication automorphisms explicit, and reuse the Gowers–Cauchy–Schwarz infrastructure.
<code>Proofs03ProgressionCount.lean;</code> <code>corollary_3_3_holds</code>	Replace the full subset expansion by a finite telescoping identity. Retain density products and cancel the exact two-factor average.
<code>Proofs09Restriction.lean;</code> <code>lemma_9_3_holds</code> and <code>corollary_9_4_holds</code>	Introduce a prime-specialized version of the strengthened restriction theorem. Formalize the rank-two completion bound, the nonnegative Fourier weights, and the random-selection expectation.
<code>Sections17_18.lean</code>	Add definitions of fractional patterns and normalized profiles, followed by the hyperplane bound and the local phase-extraction theorem with explicit characteristic assumptions.

Table 12: Suggested interfaces for formalizing the proved refinements.

One signature issue deserves explicit review. The inspected definitions of `lemma_9_3` and `corollary_9_4` in `Sections08_09.lean` quantify over nonzero moduli without an explicit primality typeclass. The selection theorem in this article uses primality essentially. It must therefore be formalized as a separate prime-specialized statement, or accompanied by a justified revision of the intended source interface. It is not automatically a proof of the broader existing proposition. The repository already has proofs of those statements through another route; the present claim is a stronger prime-field interface. By contrast, the exact moment inequality itself does hold over arbitrary finite abelian groups.

[write] **6 October 2026.** Checked at HEAD: `lemma_9_3` and `corollary_9_4` still quantify over all $N \geq N_0$ without primality and have exact companions, as stated. The table omits one interface that was already formal at source 04’s pin: Lemma III.4.5 is `hasNatAP_of_short_modular_sequence` (note there). The other proposed strengthenings are not formalized.

For a practical implementation order, first isolate a normalized and an unnormalized Fourier

convention with explicit conversion lemmas. Next prove the exact moment and telescoping statements, which need little specialized finite enumeration. Then introduce a library of finite Boolean support certificates for the cube identities. Finally add the probabilistic selection and floor-pattern arguments. Each step should compile against the pinned toolchain of the repository, and the final verification report should name the checked declarations and the commit. None of those future verification steps is represented as complete here.

V.20 Reproducibility and supplementary checks

Source 04, delivered Appendix B.

The source archive contains the complete L^AT_EX manuscript, its compiled PDF, the figure source and PDF, a provenance manifest, and three independent verification programs. The programs are deliberately supplementary to the proofs. In particular, testing finitely many functions cannot prove the norm-gap theorem, and checking small groups cannot prove the restriction theorem for all sufficiently large primes.

V.20.1 Energy, cube, and coefficient checks

`verify_refinements.py` uses the Python standard library. The recorded run checked all 8 470 nonempty graphs of partial maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ for $n = 2, 3, 4, 5$, giving 59 290 exact integer instances of the energy inequality for moment orders $2, \dots, 8$. It also enumerated the three-cube support ranks over the rationals and over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, and counted parallelograms through dimension seven. These computations exhibit the different support count in characteristic two.

The kernel checks use exact rational arithmetic for the selected coefficient ratios and the central-binomial inequality, and exact integer arithmetic for the convenient powers of two in Corollary II.9.5. The complete two-frequency polynomial in Proposition III.3.9 is independently recovered with rational coefficients.

Floating-point checks are identified separately in the results file. They test the Fourier identity and norm gap on 40 real centered functions on small odd cyclic groups, the cosine examples, and 36 density-weighted progression identities, including an admissible example modulo 25. The largest recorded Fourier-identity discrepancy is less than 1.1×10^{-16} , and the progression identity discrepancy is less than 3.4×10^{-17} . The asserted numerical tolerance is 2×10^{-11} . These are consistency checks, not rational certificates for arbitrary functions.

The supplied `verify_selection.py` checks the selection expectation bounds directly by exact rational Fourier expansion in 10 examples over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, covering 40 584 ordered additive tuples. It treats each distinct selected point only once and includes 504 distinct unrespected six-tuples. These checks use the degree-one kernel and supplement the proof of the general weighted selection lemma; the large-degree constants are checked separately.

V.20.2 Exact floor-pattern certificates

`pattern_census.py` verifies the supplied `pattern_census.json` using only rational arithmetic from the Python standard library. It obtains

$$(F_1, F_2, F_3) = (1, 2, 10), \quad (A_1, A_2, A_3) = (2, 6, 38).$$

Every branch in the hyperplane enumeration has one of two certificates: a rational point satisfying all relevant strict inequalities, or nonnegative rational weights whose combination proves that those strict inequalities are infeasible. The verifier checks both sides of every split, so it checks completeness as well as feasibility. There are 450 certified branches in the supplied record. Generating the certificates uses a numerical linear-programming solver to propose candidates, but accepting them and verifying the saved census does not use numerical tolerances.

V.20.3 Build and verification commands

From the extracted archive directory, the following commands rebuild the article and check its supplementary records:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  gowers_quantitative_refinements.tex
python3 -S verify_refinements.py --output verification_results.json
python3 -S verify_selection.py --report selection_verification.json
python3 -S pattern_census.py --verify pattern_census.json
```

The prebuilt vector figure is included, so rebuilding the article does not require Python plotting libraries. To regenerate the figure, run `python3 make_figures.py`; that optional step requires NumPy and Matplotlib. Certificate generation additionally requires SciPy. The source manifest records the inspected repository commit and the relevant Git object identifiers. The archive does not redistribute the original paper or the full repository.

V.21 Relation to the Lean development

Source 05, delivered Section 7.

V.21.1 Inspected source and a conservative integration plan

The relevant files are `Proofs02Partition.lean`, `Proofs02PhasePartition.lean`, and `Proofs02DensityIncrement.lean`; the first and third were inspected in detail in the ranges relevant to these results. The existing density-increment file imports the phase-partition file. The repository uses `NatAP`, its carrier and properness predicate, and a finite partition relation. Its partition construction already groups residue classes into blocks with lengths at least L and less than $2L$, matching the integer mechanism used here [25].

The efficient formalization route is to keep that construction and expose a parameterized theorem, rather than duplicate it for each numerical constant. The mathematical contract should assume $2L(L-1)M \leq rs$ in the modular version and conclude a partition with cell lengths in $[L, 2L-1]$ and image diameter at most s . The proof then invokes the existing small-image-step lemma with $Q = \lfloor r/L \rfloor$.

The sharp capped result becomes a separate integer calculation with $U = \lfloor \sqrt{rs/M} \rfloor$ and $K = \lceil U/2 \rceil$. The optimality example can be formalized independently using the cardinality of a disjoint union of two-element cells. It does not require complex analysis.

For the density theorem, the main additional interfaces are the exact identity $\sum |\mathbf{1}_A - \delta| = 2\delta(1-\delta)N$, a midpoint or disk-center phase estimate, and the capacity-sensitive positive-mass

lemma. The existing file already proves the zero-sum identity, density bounds, positive/negative mass equality, and nonzero Fourier-coefficient bounds by both the set and its complement [26].

V.21.2 Recommended separation of proof layers

There are four natural layers. The first is integer combinatorics: proper progressions, residue classes, lengths, and disjointness. The second is phase approximation on $\mathbb{R}/\mathbb{Z}$ or $\mathbb{Z}\text{Mod } N$. The third is finite weighted real and complex inequalities. The fourth specializes to Fourier coefficients and density increments. Keeping these layers separate permits the same weighted lemmas to be used with later polynomial-phase partitions.

For a minimum-risk first contribution, formalize the sharp $1/3$ modular theorem and the strengthened Corollary 2.4 before introducing the spectral-energy material. The simultaneous box lemma is another independent milestone. The algebraic obstruction is mathematically self-contained, but its Lean implementation would additionally need algebraic-number and resultant or norm infrastructure; it need not block the upper partition bounds.

V.21.3 Verification status

No new Lean file in this package is claimed to compile, and no new theorem has been kernel-checked here. The package supplies ordinary mathematical proofs, an exact-arithmetic implementation of the finite partition constructions, and a formalization roadmap. We did not perform a whole-repository proof audit or infer the status of the complete Gowers development from the inspected modules.

The observed `main` reference during this work was

327773149bd272239f1f6c0015f76d5097512c88.

The accompanying source-audit file records the content hashes of the compared files. The partition module was also fetched explicitly at that commit. The other inspected contents are identified by their returned blob hashes; the moving branch is not treated as a permanent identifier.

[write] **6 October 2026.** Checked at HEAD: the cited blobs are correct and unchanged (291b149 for `Proofs02Partition.lean`, 85b5c3a for `Proofs02DensityIncrement.lean`, 41085a0 for the transcription). One defect of the shipped audit file `05-phase-partitions-SOURCE_AUDIT.md`: it says the commit-pinned read of `Proofs02Partition.lean` “included lines 580–780”, but that file has 702 lines; the blob and the construction it describes (blocks of lengths at least L and less than $2L$) are correct. The audit file is shipped unchanged (README).

V.22 Reproducible computation and the limits of testing

Source 05, delivered Section 8.

The companion implementation `code/phase_partitions.py` uses Python integers and rational fractions for all partition decisions. It implements the exact scale (I.16.2), the capped $1/3$ theorem, the simultaneous ceiling condition (I.18.2), and the tolerance allocation (I.19.10). An independent checker verifies complete coverage, no repeated points, proper arithmetic progressions, one common difference, the claimed cell lengths, the number-of-cells bound, and exact circular arc widths.

The completed exact tests comprise 13,325 one-phase cases, 11,440 capped cases, 320 simultaneous cases, and 200 tolerance-allocation cases. The one-phase tests vary the interval length from 1 to

41, all slopes with denominators from 2 to 11, and five rational widths. Of the simultaneous cases, 166 have minimum length greater than one. The seeded random tests use dimensions from one to four. These checks test rounding and implementation; they are not substitutes for the proofs in Sections I.16 and I.18.

Fourier diagnostics use 80 decimal digits. There are 60 correlation-and-popularity cases and 96 spectral-energy cases. Twenty-four spectral cases have a non-singleton guaranteed minimum length, including six genuinely multifrequency cases at larger sizes. The larger examples are modular pullbacks of intervals, with their Fourier coefficients computed by a geometric-series formula independently of the partition construction. All tested inequalities passed with positive observed margins. These are numerical diagnostics, not exact certificates for transcendental inequalities.

The verification files distinguish exact structural tests from high-precision diagnostics and explicitly record that new Lean verification was not performed. To reproduce the results, run the commands in the package README. No source paper, repository file, or font file is redistributed.

V.23 Further research questions

Source 05, delivered Section 9.

The following questions arise naturally from the proved results. Unless explicitly stated otherwise, they are research directions formulated here; we do not assert that they are all previously recognized open problems.

1. Exact finite partition profiles. The universal constant $1/3$ with the fixed upper cap is settled above, but the exact best lower length for each triple (r, s, M) and each slope can be much larger. Determine an efficient characterization of the feasible pairs of minimum and maximum lengths, taking account of residue-class sizes rather than only a worst-case blocking lemma. The explicit obstruction at total size nine explains one universal constant, not the full profile.

2. Best constants in the simultaneous minimax theorem. The exponent and the symmetric dependence on ε are sharp, but the upper constant obtained from the elementary algebraic norm argument is very poor. Determine the optimal leading constants, first for one or two phases, and the extent to which requiring a common difference across the whole partition changes them. One must specify the order of the limits in N and ε .

3. Anisotropic lower bounds. The lower construction depends on $\prod_i \varepsilon_i$. Is there a matching minimax upper bound in all highly anisotropic regimes, up to dimension-dependent constants and an additive one? A coordinate with a very loose tolerance can cease to be a genuine constraint, so a naive product formula may require a regime decomposition. Weighted Diophantine approximation is the natural language, but no such theorem is assumed here.

4. Optimal density-sensitive correlation transfer. The disk radius and the capacity inequality are separately sharp, but their simultaneous extremizers need not arise from an indicator and a linear phase. Determine the exact feasible region of block positive mass, Fourier correlation, density, and phase width. Even a finite-dimensional extremal formulation could improve the trade-off (I.17.9) and identify which constant losses are genuine.

5. Adaptive selection of frequencies and tolerances. The continuous allocation problem is solved by Proposition I.19.3, but selection of the frequency subset and optimization of the exact ceiling formula are not. Develop an algorithm that balances captured energy, coefficient imbalance, dimension, and the final integer progression length. Its correctness should be stated relative to a certified Fourier input rather than floating-point estimates alone.

6. Higher-degree and nilsequence interfaces. Apply the residual-variance inequality to valid higher-degree localization theorems. The challenge is to track density-sensitive oscillation budgets together with geometric complexity and exceptional cells. A useful target is an explicit theorem where all losses are parameterized, so that an improvement of a partition result propagates mechanically through the density argument. This would still need a separate analysis before it implied any global Szemerédi improvement.

7. Formalizing sharpness as well as existence. A Lean formalization should include the parity obstruction proving optimality of $1/3$, not just the existence theorem. This makes the quantitative API resistant to later accidental strengthening. The next independent target is the exact residual inequality (I.19.4); it isolates a reusable finite-probability fact without depending on the full higher-order proof.

8. Interval-localized spectral extraction. On a proper interval, restricted characters have a nontrivial Gram matrix. Corollary I.19.4 provides an explicit scalar-norm correction, but a large value of κ can erase the gain from several correlations. Establish useful certified conditioning bounds for specific frequency configurations, and optimize the linear combination using more of the Gram matrix than its largest eigenvalue. This would turn the abstract extension into an effective local Fourier theorem.

V.24 Conclusion

Source 05, delivered Section 10.

The finite localization step admits stronger statements than the corrected repository currently exports. The lower constant $1/3$ in the capped partition theorem is exactly optimal; the new Corollary 2.4 retains the same upper cell size with a better lower size and correlation retention; and the density increment uses the true mass $2\delta(1 - \delta)$ instead of a unit bound. Simultaneous localization has a clean, sharp exponent, and a residual-variance inequality turns an appropriate Fourier-energy hypothesis into a multiplicative increment with a density-sensitive tolerance budget.

These are local quantitative results with complete proofs and reproducible computational checks. Their place in the larger program is to provide stronger interfaces and clearer bottlenecks, not to bypass the higher-order structural work needed for a new global theorem.

References

Merged from the five sources' bibliographies. A work cited by several sources appears once, with each source's annotations; the repository entries are kept per source, because each records that source's pin.

- [1] W. T. Gowers, *A new proof of Szemerédi's theorem*, Geometric and Functional Analysis **11** (2001), 465–588. doi:10.1007/s00039-001-0332-9 (02, 03, 04). Public PDF: <https://www.cs.umd.edu/~gasarch/TOPICS/vdw/sz-thm-gowers-proof.pdf> (01, 03, 04; source 01 gives the path `users/gasarch`). Source 01: relevant components are Corollaries 2.4–2.5, Lemma 17.4, Corollaries 17.5–17.6, Proposition 17.7, and the partition interface of Section 18; the public PDF was consulted. Source 05: the theorem numbering used for comparison is that of this paper; finite corrections in the repository are identified separately ([repo05*] below).
- [2] J. Leng, A. Sah and M. Sawhney, *Improved bounds for Szemerédi's theorem*, arXiv:2402.17995, version 2, 29 February 2024. <https://arxiv.org/abs/2402.17995> (all five sources). Source 05: the comparison uses Theorem 1.1 and the simultaneous-localization discussion in Sections 1–2; it does not import their quantitative theorem into any proof.
- [3] J. Leng, A. Sah and M. Sawhney, *Quasipolynomial bounds on the inverse theorem for the Gowers $U^{s+1}[N]$ -norm*, arXiv:2402.17994 (version 3, 10 August 2024, per source 02). <https://arxiv.org/abs/2402.17994> (02, 03, 04).
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- [5] B. Green and T. Tao, *Yet another proof of Szemerédi's theorem*, arXiv:1002.2254 (2010). <https://arxiv.org/abs/1002.2254> (source 03).
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- [11] J. Fox and B. Sudakov, *Dependent random choice*, arXiv:0909.3271 (2009). <https://arxiv.org/abs/0909.3271> (source 03).
- [12] C. Reiher and T. Schoen, *Note on the theorem of Balog, Szemerédi, and Gowers*, Combinatorica **44** (2024), 691–698. doi:10.1007/s00493-024-00092-5; <https://arxiv.org/abs/2308.10245> (source 03).
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- [14] F. Manners, *True complexity and iterated Cauchy–Schwarz*, 2021. <https://arxiv.org/abs/2109.05731> (source 04).

- [15] [source 01] V. Reshetnikov, *ProveIt: Combinatorics/Ramsey*, repository snapshot 3a25020a654d196f7f1aa2e37649752c000d0e65, inspected 6 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/3a25020a654d196f7f1aa2e37649752c000d0e65/Combinatorics/Ramsey>.
- [16] [source 01] *Definitions.lean*, in the snapshot [15], under *Combinatorics/Ramsey/Lean/GowersSzemerédi/*. Inspected definitions: partition, density, balanced function, unnormalized Fourier transform, natural progression, and average cell size.
- [17] [source 01] *Proofs02DensityIncrement.lean*, in the snapshot [15]. Inspected header and preparatory density and positive-mass lemmas.
- [18] [source 01] *Proofs17FinitePatterns.lean*, in the snapshot [15]. Inspected header and finite-dimensional sign-family infrastructure.
- [19] [source 01] *Sections17_18.lean*, in the snapshot [15]. Inspected floor-pattern specifications, affine patterns, partition uniformity, and the editorial dependency warning attached to Theorem 18.1.
- [20] [source 02] V. Reshetnikov, *ProveIt repository*, inspected October 6, 2026. Paper transcription at <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Combinatorics/Ramsey/Papers/sz-thm-gowers-proof>; Lean development at <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Combinatorics/Ramsey/Lean/GowersSzemerédi>. Specific inspected blobs are listed in source 02’s Appendix A (Section V.11). These links follow the moving branch *main*; the blob identifiers are the permanent record.
- [21] [source 03] V. Reshetnikov, *ProveIt: Gowers paper transcription and Lean development*, repository revision 3a25020a654d196f7f1aa2e37649752c000d0e65, inspected 6 October 2026. [Pinned paper directory](#); [pinned Lean directory](#).
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- [24] [source 05] V. Reshetnikov, *ProveIt repository*, edited transcription *sz-thm-gowers-proof.tex*, in *Combinatorics/Ramsey/Papers/sz-thm-gowers-proof*. Inspected 6 October 2026; returned blob identifier 41085a0efde8c1932e86e80c791984f52841c22e. [Repository paper directory](#).
- [25] [source 05] V. Reshetnikov, *ProveIt repository*, *Proofs02Partition.lean*, in *Combinatorics/Ramsey/Lean/GowersSzemerédi*. Returned blob identifier 291b1494ab5af839456dcac35f4bb447d4c90844. [Commit-pinned partition module](#).
- [26] [source 05] V. Reshetnikov, *ProveIt repository*, *Proofs02DensityIncrement.lean*, in the same directory. Inspected portions include the balanced-function identities and the large-scale proof of Corollary 2.5. Returned blob identifier 85b5c3ad8769b9769e1939ae5c05f0757f0678b5. [Density-increment module](#).